A NEW DEVELOPMENT FOR:

ELKS COMMUNITY ENRICHMENT CENTER

262 WILBERFORCE AVE
OAK RIDGE, TN 37830

LIST OF DRAWINGS:

PROJECT INFORMATION:

PROJECT DESCRIPTION
A NEW 6,235 S.F. COMMUNITY ENRICHMENT CENTER FOR ATOMIC ELKS LODGE.

JURISDICTION
CITY OF OAK RIDGE
PLANNING AND DEVELOPMENT
200 SOUTH TULANE AVENUE
OAK RIDGE, TN 37830
PHONE NUMBER: (865) 425-3532

DESIGN CODES
2024 INTERNATIONAL BUILDING CODE
2023 INTERNATIONAL ELECTRICAL CODE
2024 INTERNATIONAL FIRE CODE
2024 INTERNATIONAL MECHANICAL CODE
2024 INTERNATIONAL PLUMBING CODE
2018 INTERNATIONAL ENERGY CONSERVATION CODE W/ 2009 CHART

TYPE OF CONSTRUCTION	TYPE VB, SPRINKLERED
OCCUPANCY:	MULTIPLE OCCUPANCY (A2, B)
NUMBER OF STORIES:	ALLOWED: 2 ACTUAL: 1
BUILDING AREA ALLOWABLE:	24,000 S.F. ACTUAL: 6,235 S.F.
OCCUPANT LOAD / FLOOR:	170 OCC.
REQUIRED EXIT WIDTH / FLOOR:	170 OCC. *.2 = 34" MINIMUM
NUMBER OF PARKING SPACES:	REQUIRED: 36 PROPOSED: 40 ACCESSIBLE: 4

FIRE PROTECTION HOURLY RATING FOR ALL STRUCTURAL COMPONENTS AND SEPARATION OF HAZARDS COMPONENTS REQUIRED BY THE APPLICABLE BUILDING CODE

0 COLUMNS	0 BEAMS	0 WALLS
0 FLOOR/CEILING	0 ROOF/CEILING	0 ROOF COVERING
0 CORRIDORS	N/A SHAFT ENCLOSURES	N/A STAIR ENCLOSURE
N/A TENANT SEPARATIONS		
N/A OCCUPANCY SEPARATIONS		

SPRINKLER SYSTEM TYPE - NFPA 13
STANDPIPE SYSTEM - N/A
FIRE/ SMOKE ALARM SYSTEM - N/A

PROJECT DIRECTORY:

OWNER:
OHRA DEVELOPMENT CORPORATION
MARIA CATRON
10 VAN HICKS ROAD
OAK RIDGE, TN 37830
(865) 482-1006 ext. 125
MCATRON@ORHA.NET

CIVIL ENGINEER:
SPRINKLE ENGINEERING, LLC
MATT SPRINKLE
P.O. BOX 4013
MARYVILLE, TN 37802
(865) 599-9192
MATT@SPRINKLEENGINEERING.COM

ELECTRICAL ENGINEER:
FACILITY SYSTEMS CONSULTANTS, LLC
NATHAN BROWN, P.E.
713 SOUTH CENTRAL STREET, SUITE 101
KNOXVILLE, TN 37902
(865) 246-0164
NBROWN@FACILITYSYSTEMS.ORG

ARCHITECT:
THE ARCHITECTURE COLLABORATIVE
DAN BRITTAIN
6215 KINGSTON PIKE
KNOXVILLE, TN 37919
(865) 342-7505
DAN@TAC-45.COM

STRUCTURAL ENGINEER:
ELAM STRUCTURAL ENGINEERING
JOSH ELAM, PE
PO BOX 30799
KNOXVILLE, TN 37930
(865) 607-3517
ELAM@ELAMSE.COM

MECHANICAL/PLUMBING ENGINEER:
FACILITY SYSTEMS CONSULTANTS, LLC
PRESTON HALL, P.E.
713 SOUTH CENTRAL STREET, SUITE 101
KNOXVILLE, TN 37902
(865) 246-0164
PHALL@FACILITYSYSTEMS.ORG

TAC
The Architecture Collaborative

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THOMAS DANIEL BRITTAIN
REGISTERED ARCHITECT
AGRICULTURE
COMMERCE
No. 102274
STATE OF TENNESSEE
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

Drawing Title:
COVER SHEET

Date: 11/26/2025

Project No.
25026

Sheet No.
.CVR

GENERAL NOTES

1. EXISTING CONDITIONS AS DEPICTED ON THESE PLANS ARE GENERAL AND ILLUSTRATIVE IN NATURE. IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO EXAMINE THE SITE AND BE FAMILIAR WITH EXISTING CONDITIONS PRIOR TO BIDDING ON THIS PROJECT. IF CONDITIONS ENCOUNTERED DURING EXAMINATION ARE SIGNIFICANTLY DIFFERENT FROM THOSE SHOWN, THE CONTRACTOR SHALL NOTIFY THE ENGINEER IMMEDIATELY.
2. SITE BOUNDARY AND TOPOGRAPHIC INFORMATION FROM SURVEY FILES PROVIDED BY PROFESSIONAL LAND SYSTEMS DATED 08/13/25. SPRINKLE ENGINEERING IS NOT RESPONSIBLE FOR THE ACCURACY OF THE INFORMATION.
3. THE CONTRACTOR SHALL VERIFY LOCATION AND ELEVATION OF ALL EXISTING UTILITIES (INCLUDING THOSE LABELED PER RECORD DATA) PRIOR TO THE BEGINNING OF CONSTRUCTION OR EARTH MOVING OPERATIONS. INFORM ENGINEER OF ANY CONFLICTS DETRIMENTAL TO THE DESIGN INTENT.
4. 72 HOURS BEFORE DIGGING IS TO COMMENCE, THE CONTRACTOR SHALL NOTIFY THE FOLLOWING AGENCIES: THE TENNESSEE UTILITY PROTECTION SERVICES, AND ALL OTHER AGENCIES THAT MAY HAVE UNDERGROUND UTILITIES INVOLVING THIS PROJECT AND ARE NON-MEMBERS OF TENNESSEE UNDERGROUND PROTECTION, INC.
5. THE CONTRACTOR AND SUBCONTRACTORS SHALL BE RESPONSIBLE FOR COMPLYING WITH APPLICABLE FEDERAL, STATE AND LOCAL REQUIREMENTS, TOGETHER WITH EXERCISING PRECAUTIONS AT ALL TIMES FOR THE PROTECTION OF PERSONS (INCLUDING EMPLOYEES) AND PROPERTY. IT IS THE SOLE RESPONSIBILITY OF THE CONTRACTOR AND SUBCONTRACTORS TO INITIATE, MAINTAIN AND SUPERVISE ALL SAFETY REQUIREMENTS, PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK.
6. THE CONTRACTOR SHALL INDEMNIFY AND HOLD HARMLESS THE OWNER AND OWNER'S REPRESENTATIVE FOR ANY AND ALL INJURIES AND/OR DAMAGES TO PERSONNEL, EQUIPMENT AND/OR EXISTING FACILITIES OCCURRING IN THE COURSE OF THE DEMOLITION AND CONSTRUCTION DESCRIBED IN THE PLANS AND SPECIFICATIONS.
7. CONTRACTOR SHALL OBTAIN A PERMIT FOR ALL CONSTRUCTION ACTIVITIES IN ACCORDANCE WITH LOCAL, STATE, & FEDERAL REGULATIONS.
8. THE CONTRACTOR SHALL COMPLY WITH ALL LOCAL CODES, OBTAIN ALL APPLICABLE PERMITS, AND PAY ALL REQUIRED FEES PRIOR TO BEGINNING WORK.
9. ANY WORK PERFORMED IN THE STATE RIGHT OF WAYS SHALL BE IN ACCORDANCE WITH THE APPLICABLE STATE REQUIREMENTS. IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY TO OBTAIN THE NECESSARY PERMITS FOR THE WORK, SCHEDULE NECESSARY INSPECTIONS, AND PROVIDE THE NECESSARY TRAFFIC CONTROL MEASURES AND DEVICES, ETC., FOR WORK PERFORMED IN THE RIGHT OF WAYS.
10. THE CONTRACTOR IS TO PERFORM ALL INSPECTIONS AS REQUIRED BY TDEC FOR THE NATIONAL POLLUTANT DISCHARGE ELIMINATION SYSTEM (NPDES) PERMIT AND FURNISH OWNERS REPRESENTATIVE WITH WRITTEN REPORTS.
11. CONTRACTOR SHALL IMPLEMENT ALL SOIL AND EROSION CONTROL, PRACTICES REQUIRED BY OAK RIDGE AND TDEC.
12. ALL GROUND SURFACE AREAS THAT HAVE BEEN EXPOSED OR LEFT BARE AS A RESULT OF CONSTRUCTION AND ARE TO FINAL GRADE AND ARE TO REMAIN SO, SHALL BE SEEDED AND MULCHED AS SOON AS PRACTICAL IN ACCORDANCE WITH SPECIFICATIONS.
13. THE CONTRACTOR SHALL REFER TO OTHER PLANS WITHIN THIS CONSTRUCTION SET FOR OTHER PERTINENT INFORMATION. IT IS NOT THE ENGINEER'S INTENT THAT ANY SINGLE PLAN SHEET IN THIS SET OF DOCUMENTS FULLY DEPICT ALL WORK ASSOCIATED WITH THE PROJECT.
14. BEFORE INSTALLATION OF STORM OR SANITARY SEWER, OR OTHER UTILITY, THE CONTRACTOR SHALL VERIFY ALL CROSSINGS, BY EXCAVATION WHERE NECESSARY, AND INFORM THE OWNER AND THE ENGINEER OF ANY CONFLICTS. THE ENGINEER WILL BE HELD HARMLESS IN THE EVENT HE IS NOT NOTIFIED OF DESIGN CONFLICTS PRIOR TO CONSTRUCTION.
15. ADJUST/RECONSTRUCT ALL EXISTING CASTINGS, CLEANOUTS, ETC. WITHIN PROJECT AREA TO GRADE AS REQUIRED.
16. CONTRACTOR TO REMOVE & REPLACE PAVEMENT AS SPECIFIED.

DEMOLITION NOTES

1. ALL EXISTING ABOVE AND BELOW GROUND STRUCTURES WITHIN THE LIMITS OF CONSTRUCTION SHALL BE REMOVED UNLESS NOTED OTHERWISE WITHIN THIS CONSTRUCTION SET AND/OR PROJECT SPECIFICATIONS. THIS INCLUDES FOUNDATION SLABS, WALLS AND FOOTINGS. CAVITIES LEFT BY STRUCTURE REMOVAL SHALL BE BACKFILLED WITH SATISFACTORY MATERIALS AND COMPACTED TO THE GEOTECHNICAL ENGINEER'S RECOMMENDATION.
2. CLEARING LIMITS SHALL BE PHYSICALLY MARKED IN THE FIELD BY THE CONTRACTOR.
3. NO TREES SHALL BE REMOVED, NOR VEGETATION DISTURBED BEYOND THE LIMITS OF CONSTRUCTION WITHOUT THE EXPRESS WRITTEN APPROVAL OF THE OWNER'S REPRESENTATIVE.
4. TREE PROTECTION FENCING SHALL BE IN ACCORDANCE WITH OAK RIDGE STANDARDS - OR - IN ACCORDANCE WITH THE DETAILED DRAWINGS. DO NOT OPERATE OR STORE EQUIPMENT, NOR HANDLE OR STORE MATERIALS WITHIN THE DRIP LINES OF THE TREES SHOWN TO REMAIN.
5. PROTECTION OF EXISTING TREES AND VEGETATION: PROTECT EXISTING TREES AND OTHER VEGETATION INDICATED TO REMAIN IN PLACE AGAINST UNNECESSARY CUTTING, BREAKING OR SKINNING OF ROOTS, SKINNING OR BRUISING OF BARK, SMOTHERING OF TREES BY STOCKPILING CONSTRUCTION MATERIALS OR EXCAVATED MATERIALS WITHIN DRIP LINE, EXCESS FOOT OR VEHICULAR TRAFFIC, OR PARKING OF VEHICLES WITHIN DRIP LINE. PROVIDE TEMPORARY GUARDS TO PROTECT TREES AND VEGETATION TO BE LEFT STANDING.
6. ALL DEMOLITION WASTE AND CONSTRUCTION DEBRIS SHALL BECOME THE PROPERTY OF THE CONTRACTOR UNLESS OTHERWISE DESIGNATED AND SHALL BE REMOVED BY THE CONTRACTOR AND DISPOSED OF OFFSITE IN A STATE APPROVED WASTE SITE AND IN ACCORDANCE WITH ALL LOCAL AND STATE CODES AND PERMIT REQUIREMENTS. TAKE CARE TO PROTECT UTILITIES THAT ARE TO REMAIN. REPAIR DAMAGE ACCORDING TO THE APPROPRIATE UTILITY COMPANY STANDARDS AND AT THE CONTRACTOR'S EXPENSE.
7. ALL UTILITY DISCONNECTION, REMOVAL, RELOCATION, CUTTING, CAPPING AND/OR ABANDONMENT SHALL BE COORDINATED WITH THE APPROPRIATE UTILITY COMPANY / AGENCY. UTILITY CONTACTS ARE LISTED ON THE COVER SHEET.
8. THE BURNING OF CLEARED MATERIAL AND DEBRIS SHALL NOT BE ALLOWED UNLESS CONTRACTOR OBTAINS PRIOR WRITTEN AUTHORIZATION FROM THE LOCAL AUTHORITIES.
9. EROSION & SEDIMENT CONTROL MEASURES AROUND AREAS OF DEMOLITION SHALL BE PROPERLY INSTALLED AND FUNCTION PROPERLY PRIOR TO INITIALIZATION OF DEMOLITION ACTIVITIES.
10. ASBESTOS OR HAZARDOUS MATERIALS ARE NOT EXPECTED, IF FOUND ON SITE, SUCH MATERIALS SHALL BE REMOVED BY A LICENSED HAZARDOUS MATERIALS CONTRACTOR. CONTRACTOR SHALL NOTIFY OWNER IMMEDIATELY IF HAZARDOUS MATERIALS ARE ENCOUNTERED.
11. CONTRACTOR SHALL ADHERE TO ALL LOCAL, STATE, FEDERAL AND OSHA REGULATIONS DURING ALL DEMOLITION ACTIVITIES.
12. CONTRACTOR SHALL PROTECT ALL CORNER PINS, MONUMENTS, PROPERTY CORNERS AND BENCHMARKS DURING DEMOLITION ACTIVITIES. IF DISTURBED, CONTRACTOR SHALL HAVE DISTURBED ITEMS RESET BY A LICENSED SURVEYOR AT NO ADDITIONAL COST TO THE OWNER.
13. CONTRACTOR SHALL PROTECT ALL EXISTING UTILITIES, STRUCTURES, AND FEATURES TO REMAIN. ANY ITEMS TO REMAIN THAT HAVE BEEN DISTURBED OR DAMAGED AS A RESULT OF CONSTRUCTION SHALL BE REPAIRED OR REPLACED BY THE CONTRACTOR AT CONTRACTOR'S EXPENSE.
14. CONTRACTOR SHALL PROVIDE AND MAINTAIN TRAFFIC CONTROL MEASURES IN ACCORDANCE WITH STATE DEPARTMENT OF TRANSPORTATION REGULATIONS AND AS REQUIRED BY LOCAL AGENCIES WHEN WORKING IN AND/OR ALONG STREETS, ROADS, HIGHWAYS, ETC.. IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY TO OBTAIN APPROVAL AND COORDINATE WITH LOCAL AND/OR STATE AGENCIES REGARDING THE NEED, EXTENT AND LIMITATIONS ASSOCIATED WITH INSTALLING AND MAINTAINING TRAFFIC CONTROL MEASURES.
15. PROVIDE NEAT, STRAIGHT, FULL DEPTH, SAW CUTS OF EXISTING PAVEMENT WHERE INDICATED ALONG LIMITS OF PAVEMENT DEMOLITION.
16. ALL UTILITY AND STRUCTURE REMOVAL, RELOCATION, CUTTING, CAPPING AND/OR ABANDONMENT SHALL BE COORDINATED AND PROPERLY DOCUMENTED BY A CERTIFIED PROFESSIONAL, WHEN APPLICABLE, WITH THE APPROPRIATE UTILITY COMPANY, MUNICIPALITY AND/OR AGENCY. DEMOLITION OF REGULATED ITEMS MAY INCLUDE, BUT ARE NOT LIMITED TO WELLS, ASBESTOS, UNDER GROUND STORAGE TANKS, SEPTIC TANKS AND ELECTRIC TRANSFORMERS. DEMOLITION CONTRACTOR SHALL REFER TO ANY ENVIRONMENTAL STUDIES FOR DEMOLITION RECOMMENDATIONS AND GUIDANCE. AVAILABLE ENVIRONMENTAL STUDIES MAY INCLUDE, BUT ARE NOT LIMITED TO PHASE I ESA, PHASE II, WETLAND AND STREAM DELINEATION AND ASBESTOS SURVEY. ALL APPLICABLE ENVIRONMENTAL STUDIES SHALL BE MADE AVAILABLE UPON REQUEST.
17. ALL PAVEMENT, BASE COURSES, SIDEWALKS, CURBS, BUILDINGS, FOUNDATIONS, ETC., WITHIN THE AREA TO BE DEMOLISHED SHALL BE REMOVED TO FULL DEPTH. EXISTING BASE COURSE MATERIALS MAY BE WORKED INTO THE NEW PAVEMENT OR BUILDING SUBGRADE IF THE GRADATION, CONSISTENCY, COMPACTION, SUBGRADE CONDITION, ETC., ARE IN ACCORDANCE WITH THE SPECIFICATIONS AND RECOMMENDATIONS OF THE REPORT OF GEOTECHNICAL INVESTIGATION. BASE COURSE MATERIALS SHALL NOT BE WORKED INTO THE SUBGRADE AREAS TO RECEIVE LANDSCAPING.
18. THE CONTRACTOR SHALL USE SUITABLE METHODS TO CONTROL DUST AND DIRT CAUSED BY THE DEMOLITION ACTIVITIES.

GRADING NOTES

1. ALL PROPOSED GRADES SHOWN ARE FINAL GRADES, TOP OF GROUND LEVEL, OR TOP OF PAVEMENT, OR GRATE ELEVATION AT THE DRAWDOWN POINT, UNLESS INDICATED OTHERWISE.
2. REFER TO AND FOLLOW THE RECOMMENDATIONS OF THE GEOTECHNICAL REPORT.
3. ALL ELEVATIONS SHOWN ARE FINISHED GRADE ELEVATIONS.
4. CONTRACTOR SHALL STRICTLY ADHERE TO THE EROSION & SEDIMENT CONTROL PLAN PREPARED FOR THIS PROJECT.
5. EARTHWORK SHALL INCLUDE CLEARING AND GRUBBING, STRIPPING AND STOCKPILING TOPSOIL, MASS GRADING, EXCAVATION, FILLING, UNDER CUT AND REPLACEMENT, IF REQUIRED, AND COMPACTION.
6. CONTRACTOR TO REFILL UNDERCUT AREAS WITH SUITABLE MATERIAL AND COMPACT AS RECOMMENDED BY THE GEOTECHNICAL ENGINEER.
7. PLACE TOPSOIL OVER THE SUBGRADE OF UNPAVED, DISTURBED AREAS TO A DEPTH INDICATED ON THE LANDSCAPE PLANS (6” MINIMUM).
8. ALL AREAS NOT PAVED SHALL BE STABILIZED IN ACCORDANCE WITH THE EROSION & SEDIMENT CONTROL PLAN, UNLESS NOTED OTHERWISE.
9. ALL EXCESS SOIL MATERIALS SHALL BECOME THE PROPERTY OF THE CONTRACTOR UNLESS OTHERWISE DESIGNATED SHALL BE REMOVED BY THE CONTRACTOR AND DISPOSED OF OFFSITE AT NO ADDITIONAL COST TO THE OWNER IN ACCORDANCE WITH ALL LOCAL AND STATE CODES AND PERMIT REQUIREMENTS.
10. THE CONTRACTOR IS RESPONSIBLE FOR BALANCING THE SITE EARTHWORK BY IMPORTING OR EXPORTING AS NECESSARY TO ACHIEVE DESIGN GRADES AND SPECIFICATIONS.

EROSION PREVENTION AND SEDIMENT CONTROL NOTES:

1. A SPECIFIC INDIVIDUAL SHALL BE DESIGNATED TO BE RESPONSIBLE FOR EROSION AND SEDIMENT CONTROLS ON PROJECT SITE. THIS INDIVIDUAL MUST HAVE COMPLETED THE "FUNDAMENTALS OF EROSION PREVENTION AND SEDIMENT CONTROL" COURSE OR AN EQUIVALENT COURSE.
2. REFER TO THE TENNESSEE EROSION AND SEDIMENT CONTROL HANDBOOK FOR DESIGN CRITERIA AND GUIDELINES FOR EROSION CONTROL MEASURES.
3. CLEARING AND GRUBBING MUST BE HELD TO THE MINIMUM NECESSARY FOR GRADING AND EQUIPMENT OPERATION.
4. CONSTRUCTION MUST BE SEQUENCED TO MINIMIZE THE EXPOSURE TIME OF CLEARED SURFACE AREA.
5. CONSTRUCTION STAGING AND PHASING IS CRITICAL TO REDUCING SEDIMENT RUNOFF FROM SITE.
6. EROSION CONTROL MEASURES MUST BE IN PLACE AND FUNCTIONAL BEFORE EARTH MOVING OPERATIONS BEGIN, AND MUST BE PROPERLY CONSTRUCTED AND MAINTAINED THROUGHOUT THE CONSTRUCTION PERIOD.
7. ALL EROSION CONTROL MEASURES SHALL BE CHECKED TWICE WEEKLY AND AFTER EACH RAINFALL. CHECK DAILY DURING PROLONGED RAINFALL.
8. CONSTRUCTION DEBRIS MUST BE KEPT FROM ENTERING THE STORM MANAGEMENT SYSTEM.
9. STOCKPILED SOIL SHALL BE PROTECTED AND LOCATED FAR ENOUGH FROM STREAMS AND DRAINAGEWAYS SO THAT RUNOFF CANNOT CARRY SEDIMENT DOWNSTREAM.
10. VEGETATIVE GROUND COVER SHALL NOT BE DESTROYED, REMOVED OR DISTURBED MORE THAN 10 CALENDAR DAYS PRIOR TO GRADING.
11. TEMPORARY SOIL STABILIZATION WITH APPROPRIATE ANNUAL VEGETATION SHALL BE APPLIED ON AREAS THAT WILL REMAIN UNFINISHED FOR MORE THAN 14 CALENDAR DAYS.
12. PERMANENT SOIL STABILIZATION WITH PERENNIAL VEGETATION SHALL BE APPLIED AS SOON AS PRACTICAL AFTER FINAL GRADING. CONTRACTOR SHALL INSPECT THE SITE PERIODICALLY TO REPAIR AND RE-ESTABLISH VEGETATION TO DAMAGED AREAS.
13. STAKED AND ENTRENCHED SILT FENCE MUST BE INSTALLED ALONG THE BASE OF ALL FILLS AND CUTS, ON THE DOWNHILL SIDES OF STOCKPILED SOIL, AND ALONG STREAM BANKS IN CLEARED AREAS TO PREVENT EROSION INTO STREAMS. SILT FENCE MAY BE REMOVED AT THE BEGINNING OF THE WORK DAY, BUT MUST BE REPLACED AT THE END OF THE WORK DAY OR PRIOR TO FORECASTING RAIN EVENTS.
14. WHERE APPROPRIATE, SURFACE WATER FLOWING TOWARD CONSTRUCTION AREA SHALL BE DIVERTED AROUND THE CONSTRUCTION AREA USING DIKES, TO REDUCE EROSION POTENTIAL.
15. PLACEMENT AND MAINTENANCE OF CHECK DAMS SHALL BE AS SPECIFIED ON PLANS AND AS REQUIRED IN THE TENNESSEE EROSION AND SEDIMENT CONTROL HANDBOOK.
16. ALL ROCK SHALL BE CLEAN, HARD ROCK CONTAINING NO SAND, DUST, OR ORGANIC MATERIAL.
17. REFER TO THE TENNESSEE EROSION CONTROL HANDBOOK FOR MAINTENANCE REQUIREMENTS OF EROSION AND SEDIMENT CONTROL MEASURES.
18. CONTRACTOR SHALL MAINTAIN SILT FENCES AND OTHER EROSION CONTROL DEVICES FOR THE DURATION OF THE PROJECT, TO ENSURE EFFECTIVENESS, UNTIL ACCEPTED BY THE OWNER, AT NO ADDITIONAL EXPENSE TO THE OWNER. IF CONSTRUCTION ACTIVITIES CEASE DUE TO WEATHER RELATED CAUSES, THEN THE CONTRACTOR WILL ENSURE THAT THE SITE IS PROPERLY STABILIZED AND ALL EROSION CONTROL DEVICES ARE MAINTAINED AND FUNCTIONAL DURING THOSE PERIODS OF INACTIVITY.
19. CONSTRUCTION EXIT - CONTRACTOR SHALL INSTALL TEMPORARY CONSTRUCTION EXIT PRIOR TO ANY EARTHWORK OPERATIONS, CONSTRUCTION EXIT SHALL BE LOCATED AS SHOWN. CONSTRUCTION EXIT SHALL BE MAINTAINED IN A CONDITION WHICH WILL PREVENT TRACKING OR FLOW OF MUD TO PUBLIC RIGHTS-OF-WAYS. ALL MATERIAL SPILLED, DROPPED, WASHED OR TRACKED FROM VEHICLES OR SITE ONTO ADJACENT ROADWAYS SHALL BE REMOVED IMMEDIATELY FROM THE ROADWAY.
20. CONTRACTOR IS RESPONSIBLE FOR CLEANING OUT AND PROPER DISPOSAL OF ALL DEBRIS WITHIN THE STORM DRAINAGE STRUCTURES, INCLUDING SILT FROM FLUMES, PIPES, ETC., PRIOR TO COMPLETION OF THE PROJECT.
21. ADDITIONAL PROTECTION IN ADDITION TO THE ABOVE, SHALL BE PROVIDED THAT WILL PREVENT SILT FROM LEAVING THE SITE DUE TO UNFORESEEN CONDITIONS OR ACCIDENTS.
22. MEASURES SHOWN FOR SEDIMENT AND EROSION CONTROL REPRESENT THE MINIMUM ANTICIPATED. ADDITIONAL PROTECTION SHALL BE PROVIDED AS NECESSARY THAT WILL PREVENT SEDIMENT FROM LEAVING THE SITE DUE TO UNFORESEEN CONDITIONS OR ACCIDENTS.
23. THE GRADING CONTRACTOR AND BUILDING CONTRACTOR WILL REFRAIN FROM DOING ANY WORK OUTSIDE OF THE DELINEATED LIMITS OF DISTURBANCE.
24. ALL SILT FENCE IS TO BE TYPE A EXCEPT WHERE SPECIFIED DIFFERENTLY.
25. EROSION CONTROL MATTING TO BE JUTE MESH(OR APPROVED EQUAL) AND INSTALLED ACCORDING TO MANUFACTURER'S RECOMMENDATIONS.

LAYOUT NOTES

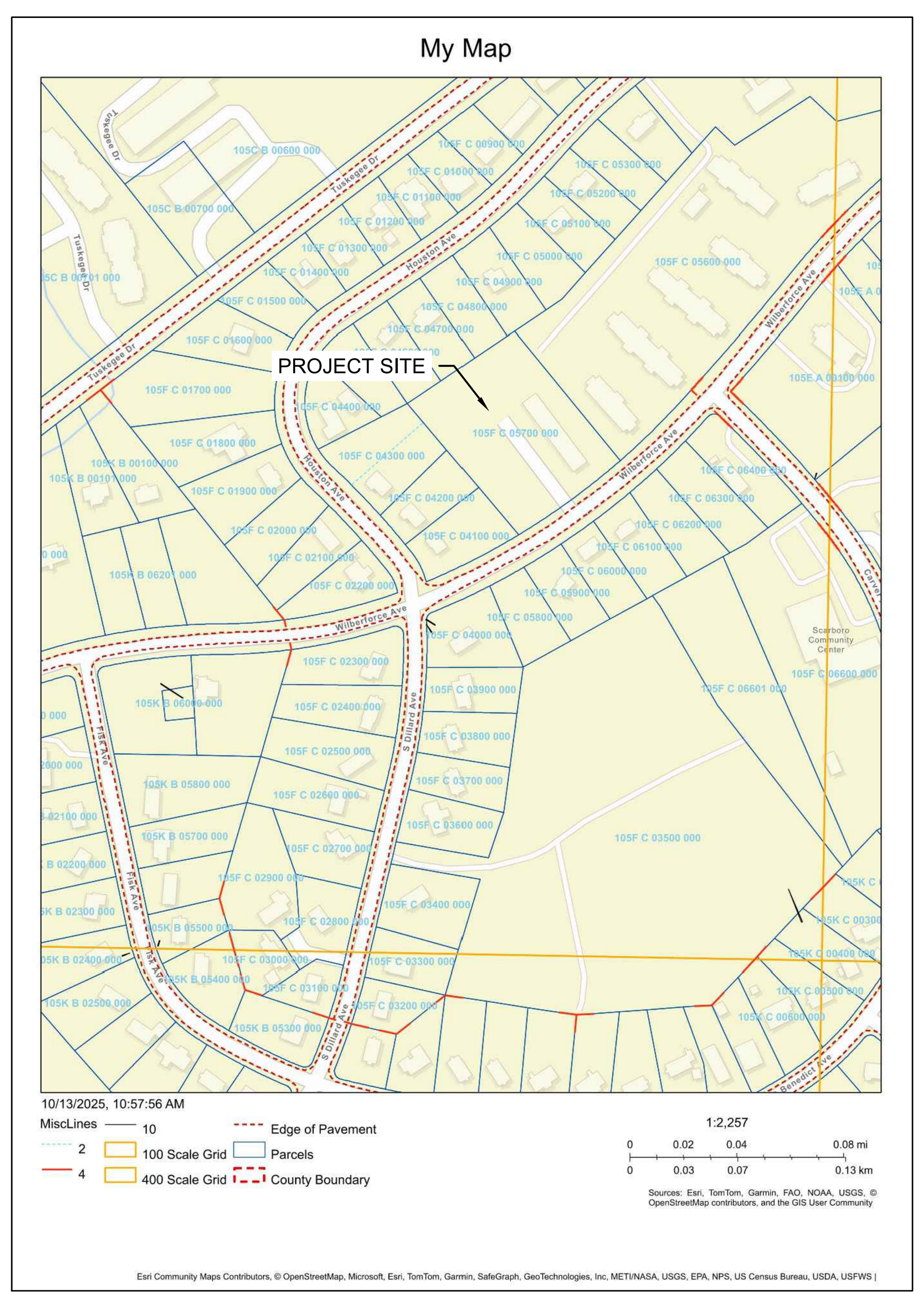
1. THE CONTRACTOR SHALL CHECK EXISTING GRADES, DIMENSIONS, AND INVERTS IN THE FIELD AND REPORT ANY DISCREPANCIES TO THE OWNER'S REPRESENTATIVE PRIOR TO BEGINNING WORK.
2. THE CONTRACTOR SHALL VERIFY THE EXACT LOCATION OF ALL EXISTING UTILITIES, INCLUDING IRRIGATION LINES. TAKE CARE TO PROTECT UTILITIES THAT ARE TO REMAIN. RELOCATE EXISTING UTILITIES AS INDICATED, OR AS NECESSARY FOR CONSTRUCTION.
3. PROVIDE A SMOOTH TRANSITION BETWEEN EXISTING PAVEMENT AND NEW PAVEMENT. FIELD ADJUSTMENT OF FINAL GRADES MAY BE NECESSARY. INSTALL ALL UTILITIES, INCLUDING IRRIGATION SLEEVING, PRIOR TO INSTALLATION OF PAVED SURFACES.
4. THE CONTRACTOR SHALL PROTECT ALL TREES TO REMAIN IN ACCORDANCE WITH THE SPECIFICATIONS.
5. ALL DAMAGE TO EXISTING PAVEMENT TO REMAIN, WHICH RESULTS FROM THE CONTRACTOR'S OPERATIONS SHALL BE REPLACED WITH LIKE MATERIALS AT THE CONTRACTOR'S EXPENSE.
6. SITE DIMENSIONS SHOWN ARE TO THE FACE OF CURB, OR EDGE OF PAVEMENT UNLESS OTHERWISE NOTED.
7. COORDINATES ARE FOR BUILDING COLUMNS, EXTERIOR BUILDING WALL, CENTER OF DRIVEWAYS, CENTER OF SANITARY SEWER MANHOLES, AND CENTER OF STRUCTURE PLACED SIX INCHES INSIDE FACE OF CURB FOR DRAIN INLETS, UNLESS OTHERWISE NOTED.
8. CONTRACTOR SHALL MAINTAIN ONE SET OF AS-BUILT / RECORD DRAWINGS ON-SITE DURING CONSTRUCTION FOR DISTRIBUTION TO THE OWNER AND/OR OWNER'S REPRESENTATIVE UPON COMPLETION.
9. REFER TO THE ARCHITECTURAL, PLUMBING & ELECTRICAL DRAWINGS FOR EXACT DIMENSIONS AND LOCATIONS OF UTILITY SERVICE ENTRY LOCATIONS AND PRECISE BUILDING DIMENSIONS.
10. THIS SITE LAYOUT IS SPECIFIC TO THE APPROVALS NECESSARY FOR THE CONSTRUCTION IN ACCORDANCE WITH THE CITY OF OAK RIDGE. NO CHANGES TO THE SITE LAYOUT ARE ALLOWED WITHOUT THE WRITTEN APPROVAL OF THE ENGINEER. CHANGES MADE TO THE SITE LAYOUT WITHOUT APPROVAL IS SOLELY THE RESPONSIBILITY OF THE CONTRACTOR. CHANGES INCLUDE BUT ARE NOT LIMITED TO, INCREASED IMPERVIOUS PAVEMENT, ADDITION / DELETION OF PARKING SPACES, MOVEMENT OF CURB LINES, CHANGES TO DRAINAGE STRUCTURES AND PATTERNS, LANDSCAPING, ETC.

STORM DRAINAGE NOTES

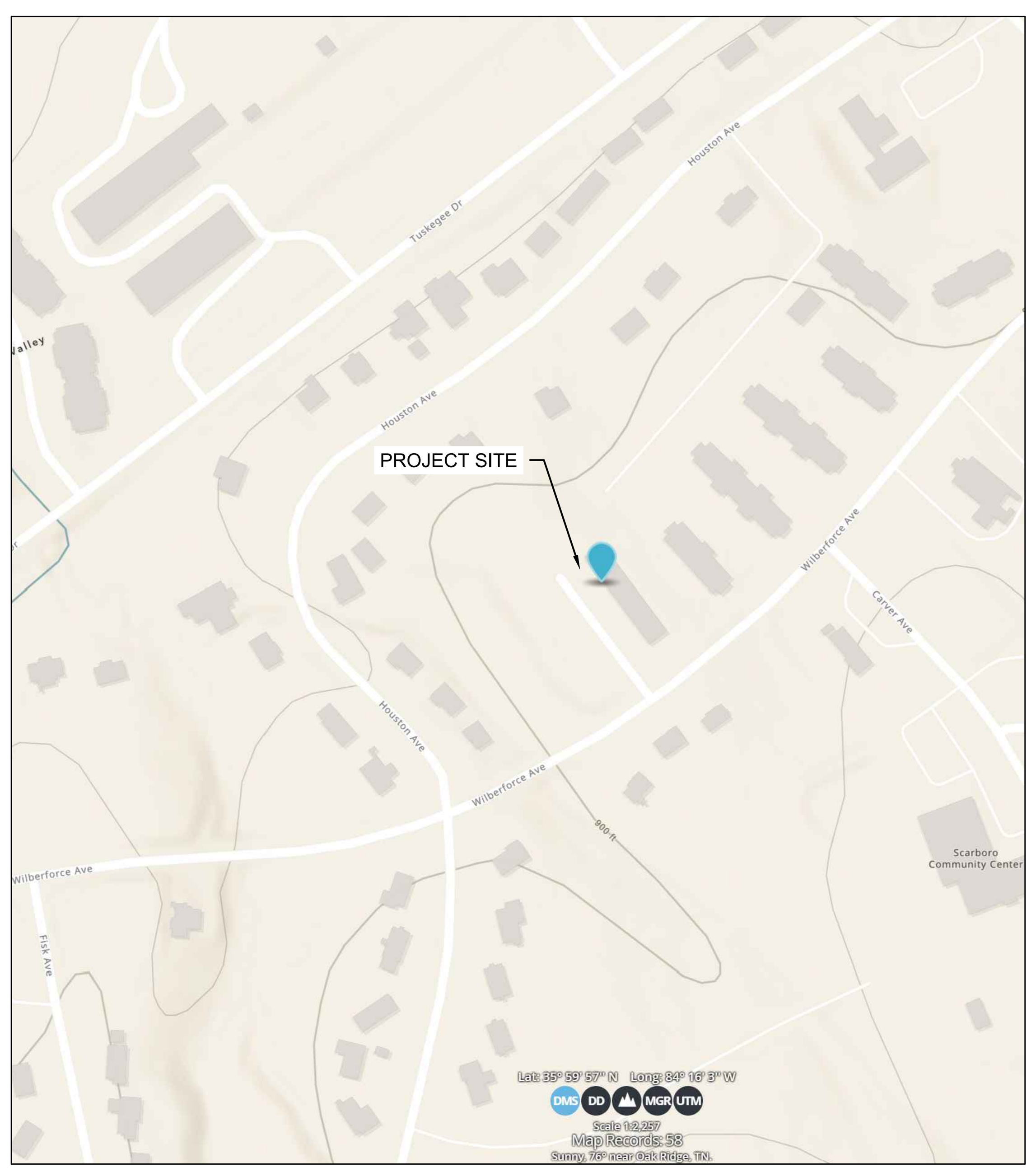
1. DISTANCES SHOWN ON PIPING ARE HORIZONTAL DISTANCES FROM CENTER OF STRUCTURE TO CENTER OF STRUCTURE, UNLESS OTHERWISE NOTED.
2. THE CONTRACTOR SHALL BE RESPONSIBLE FOR ALL COSTS ASSOCIATED WITH THE INSTALLATION, INSPECTION, TESTING AND FINAL ACCEPTANCE(INCLUDING AS-BUILT SURVEY PLANS, IF REQUIRED) OF ALL NEW STORMWATER MANAGEMENT FACILITIES CONSTRUCTION. CONTRACTOR SHALL COORDINATE WITH ALL APPLICABLE REGULATING AGENCIES CONCERNING INSTALLATION, INSPECTION AND APPROVAL OF THE STORM DRAINAGE SYSTEM CONSTRUCTION.
3. ALL STORMWATER MANAGEMENT FACILITIES, INCLUDING COLLECTION AND CONVEYANCE STRUCTURES SHALL BE INSTALLED IN ACCORDANCE WITH ALL APPLICABLE LOCAL AND STATE CODES AND REGULATIONS.
4. ANY FIELD TILE CUT IN EXCAVATION, WHICH DRAINS AN OFFSITE AREA, MUST BE TIED INTO THE STORM DRAINAGE SYSTEM.
5. ALL PROPOSED STORM SEWERS, SURFACE OR OTHER DRAINAGE FACILITIES ARE TO BE PRIVATE AND MAINTAINED BY THE OWNER.
6. ALL STORM STRUCTURES ARE TDOT TYPES UNLESS OTHERWISE INDICATED.
7. ALL CATCH BASINS AND MANHOLES WITH A DEPTH GREATER THAN 4' SHALL BE PROVIDED WITH STEPS.

UTILITY NOTES

1. ALL PROPOSED UTILITY LINES AND EXTENSIONS ARE TO BE CONSTRUCTED IN ACCORDANCE WITH THE LOCAL UTILITY SPECIFICATIONS. CONTRACTOR SHALL COORDINATE UTILITY DISCONNECTIONS WITH THE APPROPRIATE AGENCY.
2. THE CONTRACTOR IS PARTICULARLY CAUTIONED THAT THE LOCATION AND/OR ELEVATION OF THE EXISTING UTILITIES SHOWN HEREON IS BASED ON TOPOGRAPHIC SURVEYS AND RECORD DRAWINGS. THE CONTRACTOR SHALL NOT RELY UPON THIS INFORMATION AS BEING EXACT OR COMPLETE. SHOULD UNCHARTED UTILITIES BE ENCOUNTERED DURING EXCAVATION OPERATIONS, THE CONTRACTOR SHALL NOTIFY THE ENGINEER AS SOON AS POSSIBLE FOR INSTRUCTIONS. THE CONTRACTOR SHALL CALL THE APPROPRIATE UTILITY COMPANY AT LEAST 48 HOURS PRIOR TO ANY EXCAVATION AND REQUEST FIELD VERIFICATION OF UTILITY LOCATIONS. IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY TO RELOCATE EXISTING UTILITIES CONFLICTING WITH IMPROVEMENTS SHOWN HEREON IN ACCORDANCE WITH ALL LOCAL, STATE, AND FEDERAL REGULATIONS GOVERNING SUCH OPERATIONS.
3. THE CONTRACTOR SHALL OBTAIN ALL REQUIRED PERMITS PRIOR TO COMMENCEMENT OF CONSTRUCTION.
4. THE CONTRACTOR SHALL BE RESPONSIBLE FOR COORDINATING THE SEQUENCING OF CONSTRUCTION FOR ALL UTILITY LINES SO THAT WATER LINES, GAS LINES, AND UNDERGROUND ELECTRIC DO NOT CONFLICT WITH SANITARY SEWERS OR STORM SEWERS. INSTALL UTILITIES PRIOR TO PAVEMENT CONSTRUCTION.
5. ALL TRENCH SPOILS SHALL BECOME THE PROPERTY OF THE CONTRACTOR UNLESS OTHERWISE DESIGNATED SHALL BE REMOVED BY THE CONTRACTOR AND DISPOSED OF OFFSITE AT NO ADDITIONAL COST TO THE OWNER IN ACCORDANCE WITH ALL LOCAL AND STATE CODES AND PERMIT REQUIREMENTS.
6. ROOF DRAINS, FOUNDATION DRAINS AND ALL OTHER CLEAR WATER CONNECTIONS TO THE SANITARY SEWER SYSTEMS ARE PROHIBITED.
7. ADJUST ALL EXISTING UTILITY SURFACE FEATURES INCLUDING BUT NOT LIMITED TO CASTINGS, VALVE BOXES, PEDESTALS, CLEANOUTS, ETC. TO MATCH PROPOSED FINISHED GRADES, UNLESS OTHERWISE INDICATED.



VICINITY MAP



USGS TOPO QUAD

Know what's below.
Call before you dig

LEGEND

PROPERTY BOUNDARY	
EXISTING PAVEMENT	
EXISTING SANITARY MANHOLE	
EXISTING UTILITY POLE	
EXISTING HYDRANT	
WV	EXISTING WATER VALVE
EXISTING CATCH BASIN	
EXISTING ELECTRICAL BOX	
890	EXISTING CONTOUR INDEX
887	EXISTING CONTOUR INTERMEDIATE
	EXISTING WATER LINE
OHW	EXISTING OVERHEAD LINE
UG-E	EXISTING UNDERGROUND LINE
890	PROPOSED CONTOUR INDEX
887	PROPOSED CONTOUR INTERMEDIATE
	PROPOSED STORM LINE
	PROPOSED STORM STRUCTURE
	PROPOSED LIMITS OF DISTURBANCE
SF	PROPOSED SILT FENCE

OWNER: ELKS ATOMIC LODGE
PO BOX 4445
OAK RIDGE, TN 37831

DESIGN ENGINEER: SPRINKLE ENGINEERING, LLC - MATT SPRINKLE, PE
PO BOX 4013
MARYVILLE, TN 37802
PHONE NUMBER: 865-599-9192
MATT@SPRINKLEENGINEERING.COM

SITE DATA

PARCEL ID:	105 F C 057.00
ADDRESS:	262 WILBERFORCE AVE OAK RIDGE, TENNESSEE 37830
EXISTING USE:	COMMUNITY USE
EXISTING ZONING:	R-3
PROPOSED USE:	COMMUNITY USE
ACREAGE OF EX SITE:	1.94 AC TOTAL
FAR:	0.14

SPRINKLE ENGINEERING
SPRINKLE ENGINEERING, LLC
PO BOX 4013
MARYVILLE, TN 37802

TAC
The Architecture Collaborative
6215 Kingston Pike | Knoxville, TN 37919
www.tac-45.com | 865.342.7505

A NEW DEVELOPMENT FOR:
OAK RIDGE ATOMIC ELKS LODGE
262 WILBERFORCE AVE
OAK RIDGE, TN

MATTHEW S. SPRINKLE
REGISTERED ENGINEER
AGRICULTURE
COMMERCE
No. 111153
STATE OF TENNESSEE
Matthew S. Sprinkle
10/31/25

THIS DRAWING IS ISSUED FOR:
PERMITTING

Revisions:

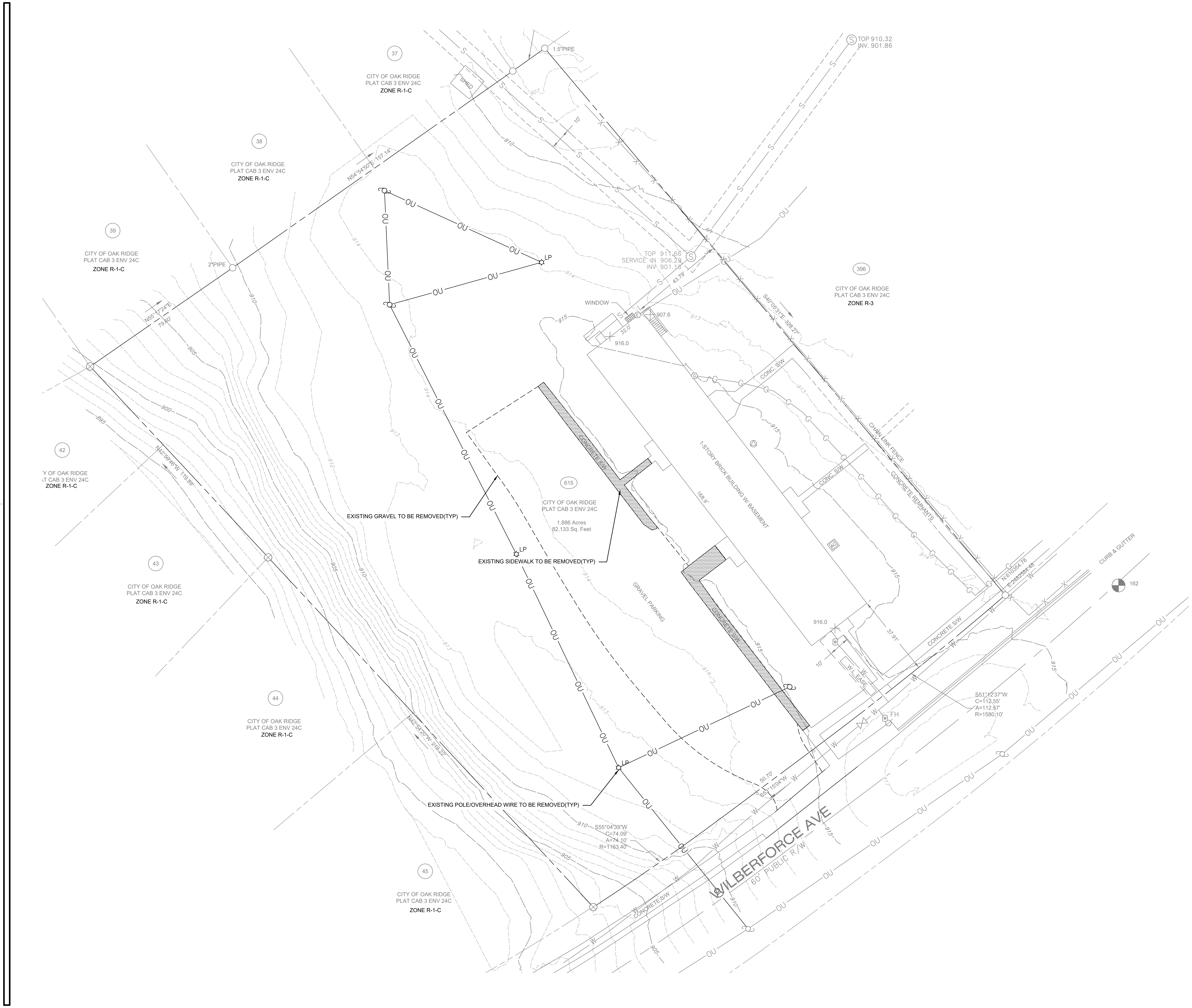
No.	Date

Drawing Title: GENERAL NOTES AND LEGEND

Date: 10/31/2025

Project No.

Sheet No. C001



BASIS OR BEARING

EXISTING BOUNDARY AND TOPOGRAPHIC INFORMATION FROM SURVEY FILES PROVIDED BY PROFESSIONAL LAND SYSTEMS DATED 08/13/25.

FLOOD NOTE

THE PROJECT SITE IS NOT LOCATED WITHIN AN AREA OF FLOOD HAZARD PER THE FEMA FLOOD INSURANCE RATE MAP NO.47001C0282F EFFECTIVE JANUARY 17, 2007.

UTILITY NOTE

THE EXACT LOCATION OF UNDERGROUND FEATURES CANNOT BE ACCURATELY, COMPLETELY AND RELIABLY DEPICTED. UTILITIES MAY EXIST THAT ARE NOT SHOWN. FIELD VERIFY SIZE, LOCATION AND TYPE PRIOR TO CONSTRUCTION.

0 20 40 Feet

SPRINKLE
ENGINEERING

SPRINKLE ENGINEERING, LLC
PO BOX 4013
MARYVILLE, TN 37802

TAC
The Architecture Collaborative

6215 Kingston Pike | Knoxville, TN 37919
www.tac-45.com | 865.342.7505

A NEW DEVELOPMENT FOR:

OAK RIDGE ATOMIC ELKS LODGE

262 WILBERFORCE AVE
OAK RIDGE, TN

MATTHEW S. SPRINKLE
REGISTERED ENGINEER
AGRICULTURE
COMMERCE
No. 111153
STATE OF TENNESSEE

Matthew S. Sprinkle
10/31/25

THIS DRAWING IS ISSUED FOR:

PERMITTING

Revisions:

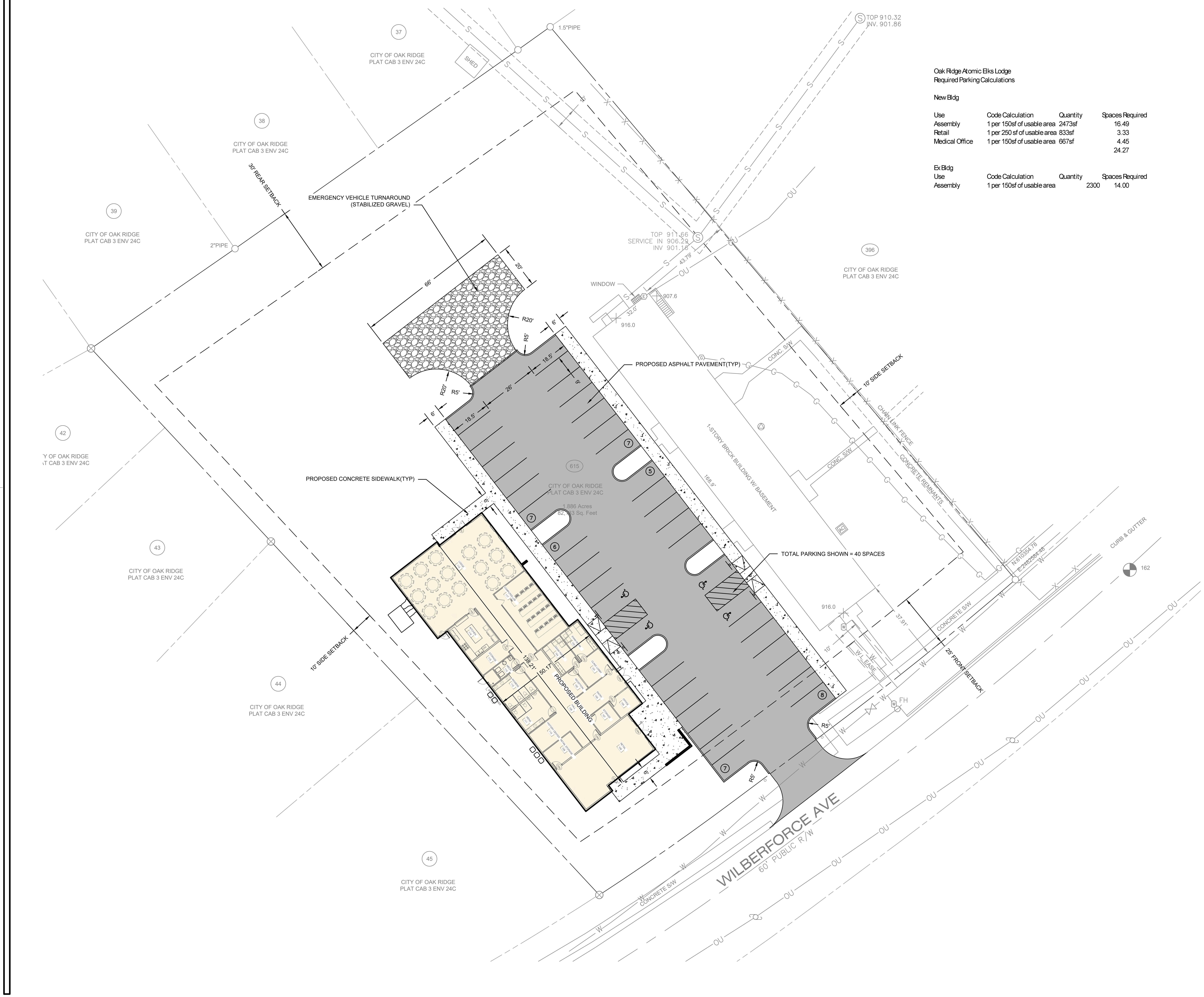
No.	Date

Drawing Title:
EXISTING CONDITIONS PLAN

Date: 10/31/2025

Project No.

Sheet No.
C100



Oak Ridge Atomic Elks Lodge
Required Parking Calculations

New Bldg

Use	Code Calculation	Quantity	Spaces Required
Assembly	1 per 150sf of usable area	2473sf	16.49
Retail	1 per 250 sf of usable area	833sf	3.33
Medical Office	1 per 150sf of usable area	667sf	4.45
			24.27

Ex Bldg

Use	Code Calculation	Quantity	Spaces Required
Assembly	1 per 150sf of usable area	2300	14.00

BASIS OR BEARING

EXISTING BOUNDARY AND TOPOGRAPHIC INFORMATION FROM SURVEY FILES PROVIDED BY PROFESSIONAL LAND SYSTEMS DATED 08/13/25.

FLOOD NOTE

THE PROJECT SITE IS NOT LOCATED WITHIN AN AREA OF FLOOD HAZARD PER THE FEMA FLOOD INSURANCE RATE MAP NO.47001C0282F EFFECTIVE JANUARY 17, 2007.

UTILITY NOTE

THE EXACT LOCATION OF UNDERGROUND FEATURES CANNOT BE ACCURATELY, COMPLETELY AND RELIABLY DEPICTED. UTILITIES MAY EXIST THAT ARE NOT SHOWN. FIELD VERIFY SIZE, LOCATION AND TYPE PRIOR TO CONSTRUCTION.

0 20 40 Feet

SPRINKLE ENGINEERING, LLC
PO BOX 4013
MARYVILLE, TN 37802

TAC
The Architecture Collaborative
6215 Kingston Pike | Knoxville, TN 37919
www.tac-45.com | 865.342.7505

A NEW DEVELOPMENT FOR:

OAK RIDGE ATOMIC ELKS LODGE

262 WILBERFORCE AVE
OAK RIDGE, TN

MATTHEW S. SPRINKLE
REGISTERED ENGINEER
AGRICULTURE
COMMERCE
No. 111153
STATE OF TENNESSEE

Matthew S. Sprinkle
10/31/25

THIS DRAWING IS ISSUED FOR:

PERMITTING

Revisions:

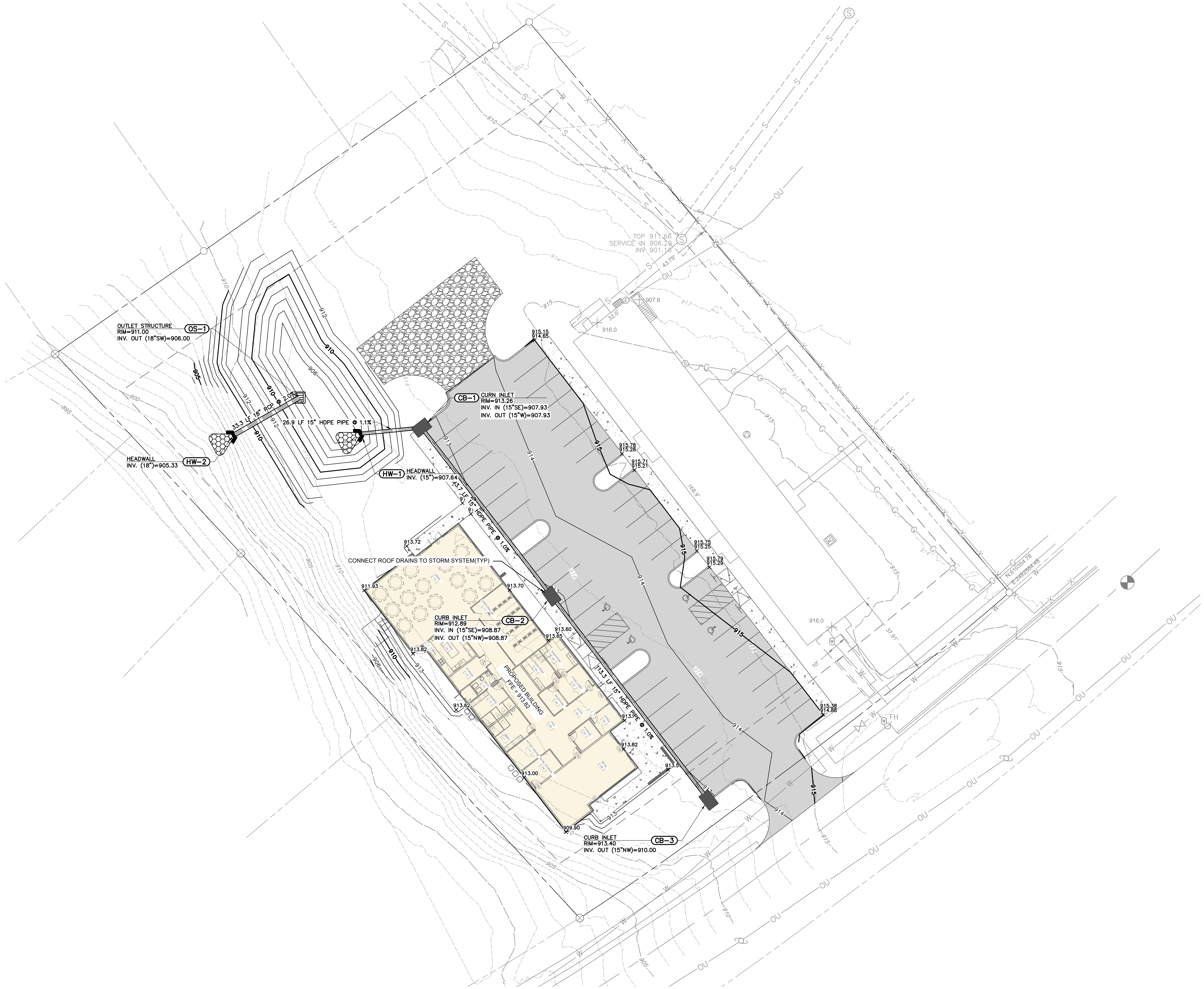
No.	Date

Drawing Title: SITE LAYOUT PLAN

Date: 10/31/2025

Project No.

Sheet No. C200



BASIS OR BEARING

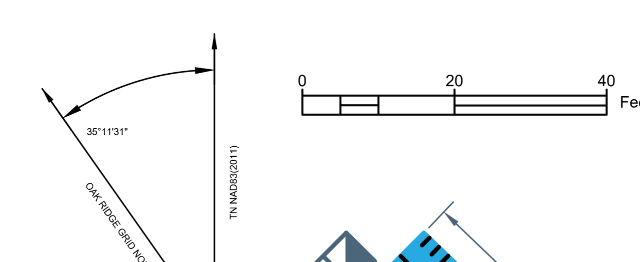
EXISTING BOUNDARY AND TOPOGRAPHIC INFORMATION FROM SURVEY FILES PROVIDED BY PROFESSIONAL LAND SYSTEMS DATED 08/13/25.

FLOOD NOTE

THE PROJECT SITE IS NOT LOCATED WITHIN AN AREA OF FLOOD HAZARD PER THE FEMA FLOOD INSURANCE RATE MAP NO.47001C0282F EFFECTIVE JANUARY 17, 2007.

UTILITY NOTE

THE EXACT LOCATION OF UNDERGROUND FEATURES CANNOT BE ACCURATELY, COMPLETELY AND RELIABLY DEPICTED. UTILITIES MAY EXIST THAT ARE NOT SHOWN. FIELD VERIFY SIZE, LOCATION AND TYPE PRIOR TO CONSTRUCTION.



SPRINKLE ENGINEERING

SPRINKLE ENGINEERING, LLC
PO BOX 4013
MARYVILLE, TN 37802

TAC
The Architecture Collaborative

6215 Kingston Pike | Knoxville, TN 37919
www.tac-45.com | 865.342.7505

A NEW DEVELOPMENT FOR:

OAK RIDGE ATOMIC ELKS LODGE

262 WILBERFORCE AVE
OAK RIDGE, TN

MATTHEW S. SPRINKLE
REGISTERED ENGINEER
AGRICULTURE
COMMERCE
No. 111153
STATE OF TENNESSEE

Matthew S. Sprinkle
10/31/25

THIS DRAWING IS ISSUED FOR:

PERMITTING

Revisions:

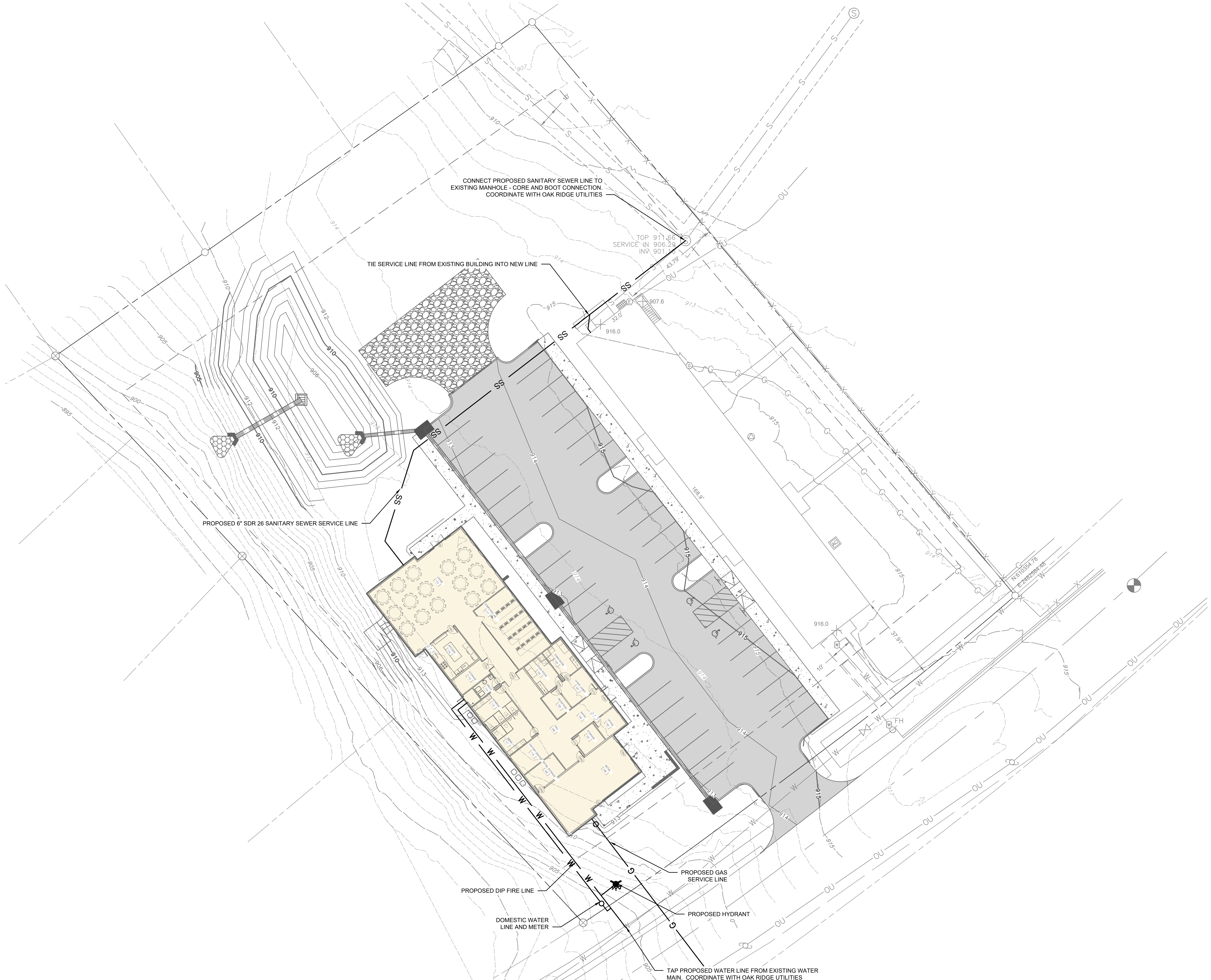
No.	Date

Drawing Title:
SITE GRADING AND DRAINAGE PLAN

Date: 10/31/2025

Project No.

Sheet No.
C300



BASIS OR BEARING

EXISTING BOUNDARY AND TOPOGRAPHIC INFORMATION FROM SURVEY FILES PROVIDED BY PROFESSIONAL LAND SYSTEMS DATED 08/13/25.

FLOOD NOTE

THE PROJECT SITE IS NOT LOCATED WITHIN AN AREA OF FLOOD HAZARD PER THE FEMA FLOOD INSURANCE RATE MAP NO.47001C0282F EFFECTIVE JANUARY 17, 2007.

UTILITY NOTE

THE EXACT LOCATION OF UNDERGROUND FEATURES CANNOT BE ACCURATELY, COMPLETELY AND RELIABLY DEPICTED. UTILITIES MAY EXIST THAT ARE NOT SHOWN. FIELD VERIFY SIZE, LOCATION AND TYPE PRIOR TO CONSTRUCTION.

0 20 40 Feet

SPRINKLE
ENGINEERING

SPRINKLE ENGINEERING, LLC
PO BOX 4013
MARYVILLE, TN 37802

TAC
The Architecture Collaborative
6215 Kingston Pike | Knoxville, TN 37919
www.tac-45.com | 865.342.7505

A NEW DEVELOPMENT FOR:

OAK RIDGE ATOMIC ELKS LODGE

262 WILBERFORCE AVE
OAK RIDGE, TN

MATTHEW S. SPRINKLE
REGISTERED ENGINEER
AGRICULTURE
COMMERCE
No. 111153
STATE OF TENNESSEE

Matthew S. Sprinkle
10/31/25

THIS DRAWING IS ISSUED FOR:

PERMITTING

Revisions:

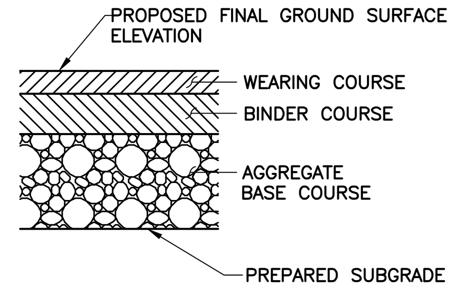
No.	Date

Drawing Title:
SITE UTILITY PLAN

Date: 10/31/2025

Project No.

Sheet No.
C500



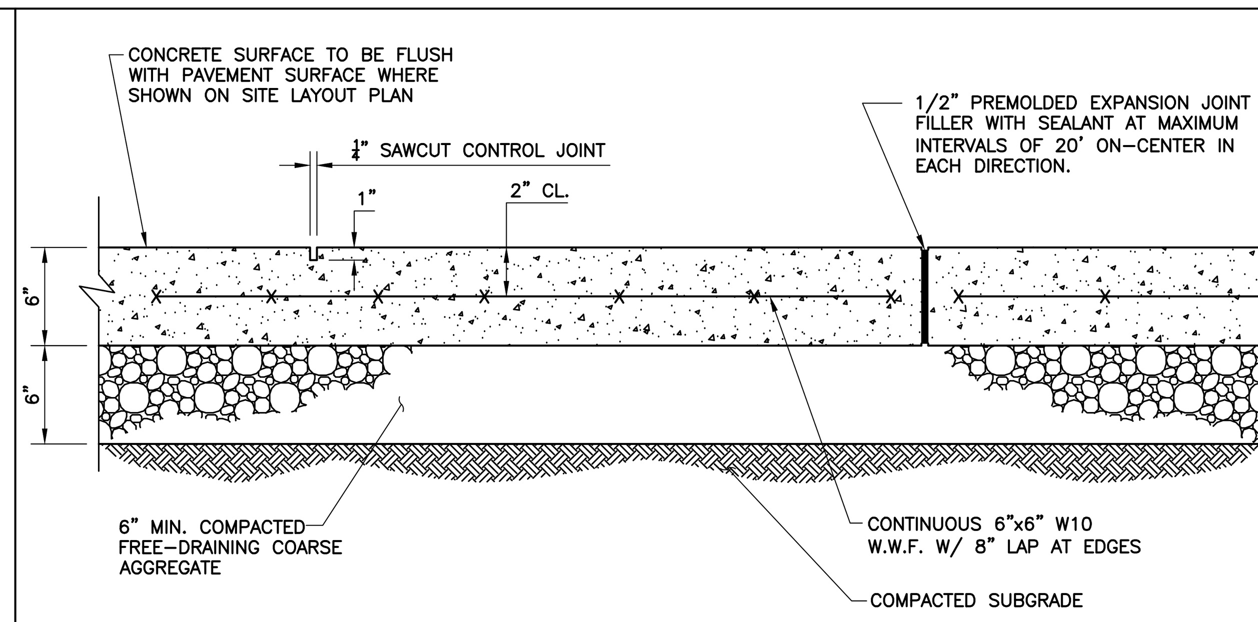
TDOT SPECIFICATION	ITEM	LIGHT DUTY PAVEMENT SECTION
SECTION 411, GRADING "E"	BITUMINOUS CONCRETE WEARING COURSE	1.5"
SECTION 307, GRADING "BM"	BITUMINOUS CONCRETE BINDER COURSE	2"
SECTION 903.05, GRADING A & D	AGGREGATE BASE COURSE	6"

NOTES:

1. THE PAVEMENT SUBGRADE SHALL BEPREPARED AS FOLLOWS:
   - A. PAVEMENT SUBGRADE SHALL BE PROOFROLLED AND COMPACTED TO AT LEAST 100% OF ITS MAXIMUM DRY DENSITY AND WITHIN ± 3% OF ITS OPTIMUM MOISTURE CONTENT, AS DETERMINED BY ASTM D1557 (MODIFIED PROCTOR). ANY SOFT AREAS SHALL BE OVEREXCAVATED TO A FIRM AND COMPETENT MATERIAL AND BACKFILLED AS DESCRIBED IN 1B.
   - B. ALL APPROVED FILL SHALL BE PLACED IN MAXIMUM 8" THICK, LOOSE LIFTS AND COMPACTED TO 100% OF ITS MAXIMUM DRY DENSITY AND WITHIN ±3% OF ITS OPTIMUM MOISTURE CONTENT. THE FILL MATERIAL'S MAXIMUM DRY DENSITY AND OPTIMUM MOISTURE CONTENT SHALL BE DETERMINED BY ASTM D1557 (MODIFIED PROCTOR).
   - C. AFTER COMPACTION, THE PAVEMENT SUBGRADE SHALL PROVIDE A FIRM UNYIELDING FOUNDATION WITH NO SUDDEN, SHARP OR ABRUPT CHANGES OR BREAKS IN GRADES. NO STANDING WATER OR EXCESS MOISTURE SHALL BE PRESENT. ALL SOFT AND YIELDING AREAS SHALL BE REWORKED BY OVEREXCAVATING TO A FIRM AND COMPETENT MATERIAL, AND BACKFILLED AS DESCRIBED IN 1B.
   - D. THE SUBGRADE SHALL BE GRADED AND SHAPED AS REQUIRED TO CONSTRUCT THE AGGREGATE BASE COURSE IN CONFORMANCE WITH THE GRADES, LINES AND THICKNESSES SHOWN ON THE DRAWINGS.
2. THE BITUMINOUS COURSES SHALL BE TAPERED INTO THE STORM SEWER INLETS WHILE MAINTAINING FULL PAVEMENT SECTION DEPTH.
3. REFER TO GEOTECHNICAL REPORT FOR ADDITIONAL PAVEMENT RECOMMENDATIONS.

ASPHALT PAVEMENT SECTION

NOT TO SCALE

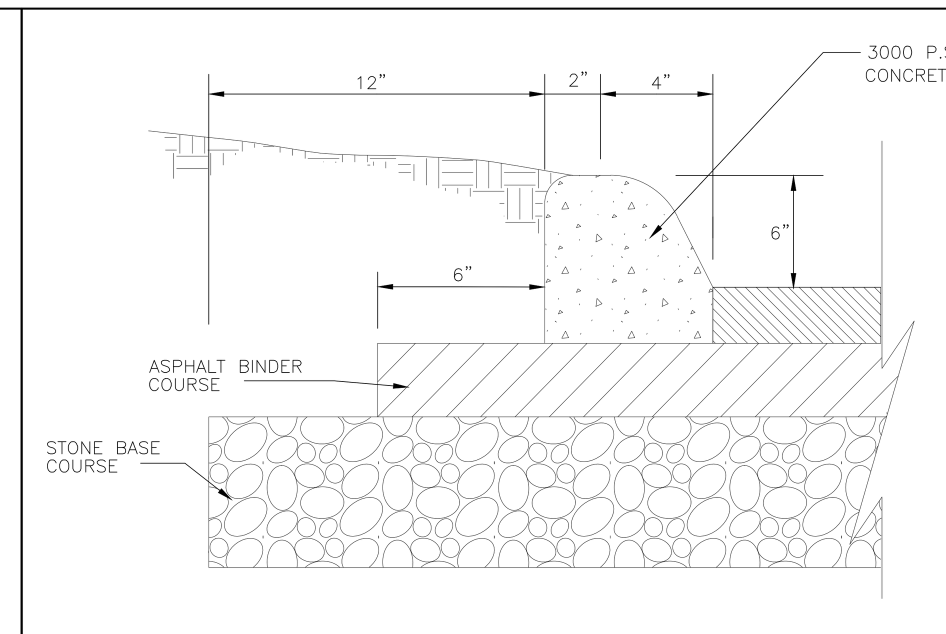


NOTES:

1. PROVIDE 1/2" EXPANSION JOINT FILLER WITH SEALANT WHERE THE CONCRETE SIDEWALK ABUTS THE BUILDING.
2. REFER TO THE PROJECT SPECIFICATIONS FOR SPECIAL FINISHES, AGGREGATETREATMENT, COLORS, PATTERN FINISHES, ETC.
3. SEE C200 FOR LOCATIONS AND WIDTHS.

CONCRETE PAVEMENT

NOT TO SCALE

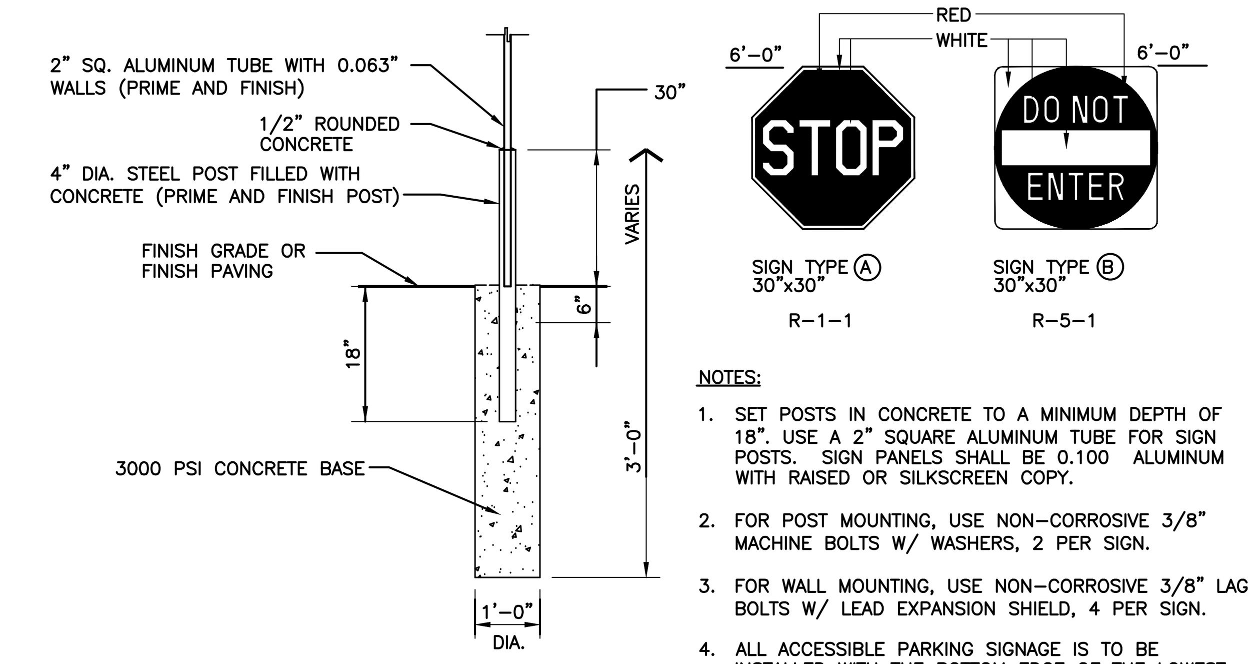


NOTES:

1. PREFORMED 1/2" EXPANSION JOINTS SHALL BE EQUALLY SPACED AT 20' MAX. CENTERS, WITH 1/4" CONTRACTION JOINTS EQUALLY SPACED AT 5' MAX. CENTERS BETWEEN EXPANSION JOINTS.
2. EXPANSION JOINT MATERIAL REQUIRED FOR GRASS MEDIANS.
3. EXACT CURB DIMENSIONS MAY BE ALTERED SLIGHTLY TO FIT STANDARD EXTRUDED CURB MACHINES, BUT SUCH VARIANCES MUST BE APPROVED BY THE ENGINEER.

TYPICAL EXTRUDED CURB DETAIL

NOT TO SCALE

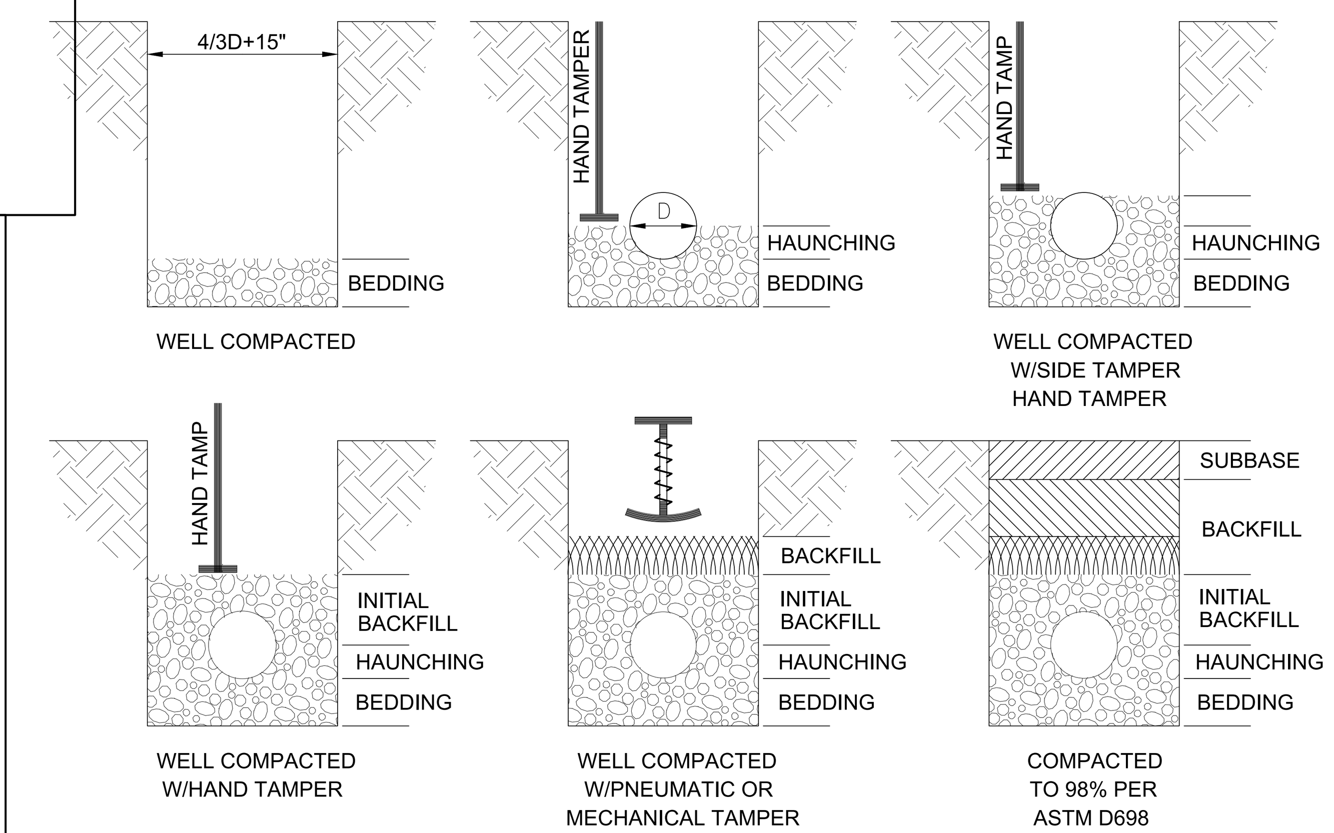


NOTES:

1. SET POSTS IN CONCRETE TO A MINIMUM DEPTH OF 18". USE A 2" SQUARE ALUMINUM TUBE FOR SIGN POSTS. SIGN PANELS SHALL BE 0.100 ALUMINUM WITH RAISED OR SILKSCREEN COPY.
2. FOR POST MOUNTING, USE NON-CORROSIVE 3/8" MACHINE BOLTS W/ WASHERS, 2 PER SIGN.
3. FOR WALL MOUNTING, USE NON-CORROSIVE 3/8" LAG BOLTS W/ LEAD EXPANSION SHIELD, 4 PER SIGN.
4. ALL ACCESSIBLE PARKING SIGNAGE IS TO BE INSTALLED WITH THE BOTTOM EDGE OF THE LOWEST SIGN AT LEAST 60" ABOVE FINISHED GRADE.

SIGNAGE

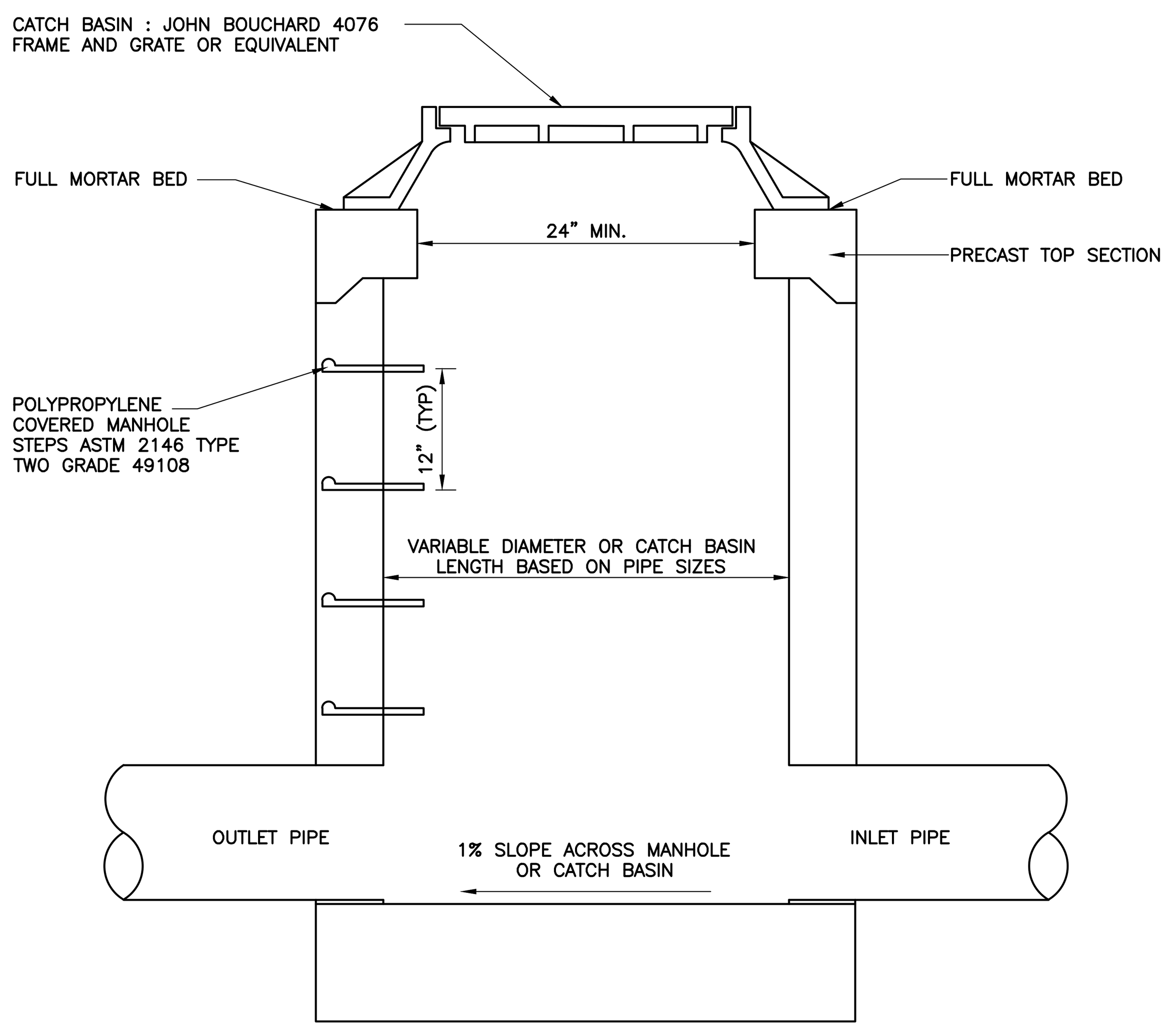
NOT TO SCALE



LAYER	DEPTH	MATERIAL
BEDDING	6" MIN.	CLASS I FOR FLEXIBLE PIPE, CLASS I-IV FOR RIGID PIPE
HAUNCHING	VARIES	CLASS I FOR FLEXIBLE PIPE, CLASS I-IV FOR RIGID PIPE, TAMPED IN 6" MAXIMUM LAYERS.
INITIAL BACKFILL	VARIES	CLASS I FOR FLEXIBLE PIPE, CLASS I-IV FOR RIGID PIPE
BACKFILL	VARIES	CLASS I-IV FOR ALL PIPE, 75% OF BACKFILL MATERIAL MAY CONTAIN BROKEN STONES NOT EXCEEDING 6" IN DIAMETER-TAMPED IN 6" MAXIMUM LAYERS
SUBBASE	12"	CLASS I-IV FOR ALL PIPE - EXCEPT USE T.D.O.T. MINERAL AGGREGATE BASE TYPE A, ACROSS OR ALONG EXISTING PAVEMENT

TRENCH BACKFILLING & COMPACTION IN UNIMPROVED AND IMPROVED AREAS

N.T.S.

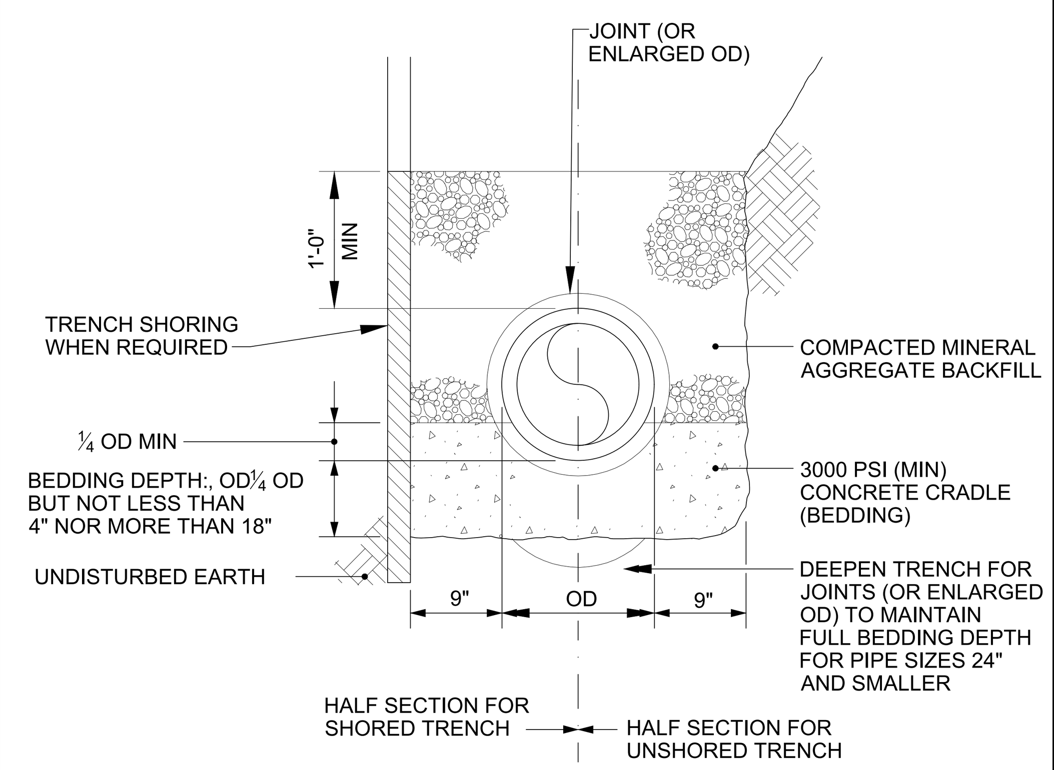


CATCH BASIN NOTE:

PRECAST CONCRETE CATCH BASIN WITH MASTIC JOINTS AND WEEP HOLES. OPENINGS FOR PIPES SHOULD BE 1" LARGER THAN THE OUTSIDE PIPE DIAMETER. CLEARANCE BETWEEN PIPE AND MANHOLE TO BE SEALED WITH SIKAFLEX 405, OR EQUIVALENT SEALANT.

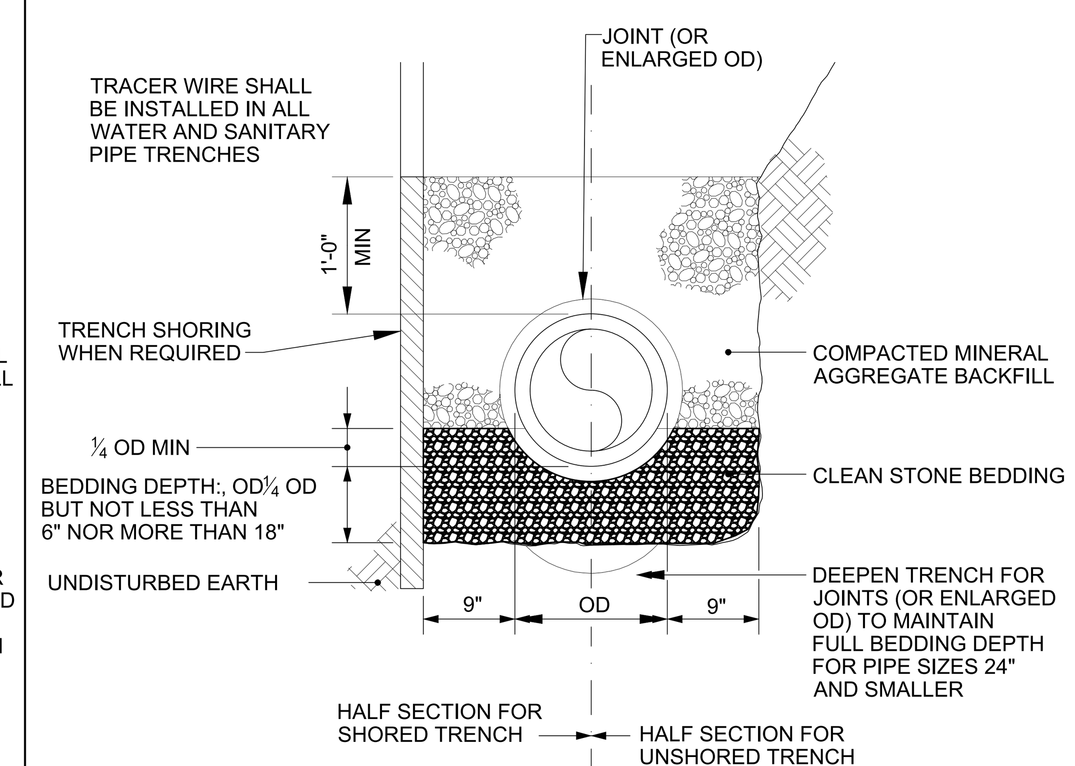
CATCH BASIN

NOT TO SCALE



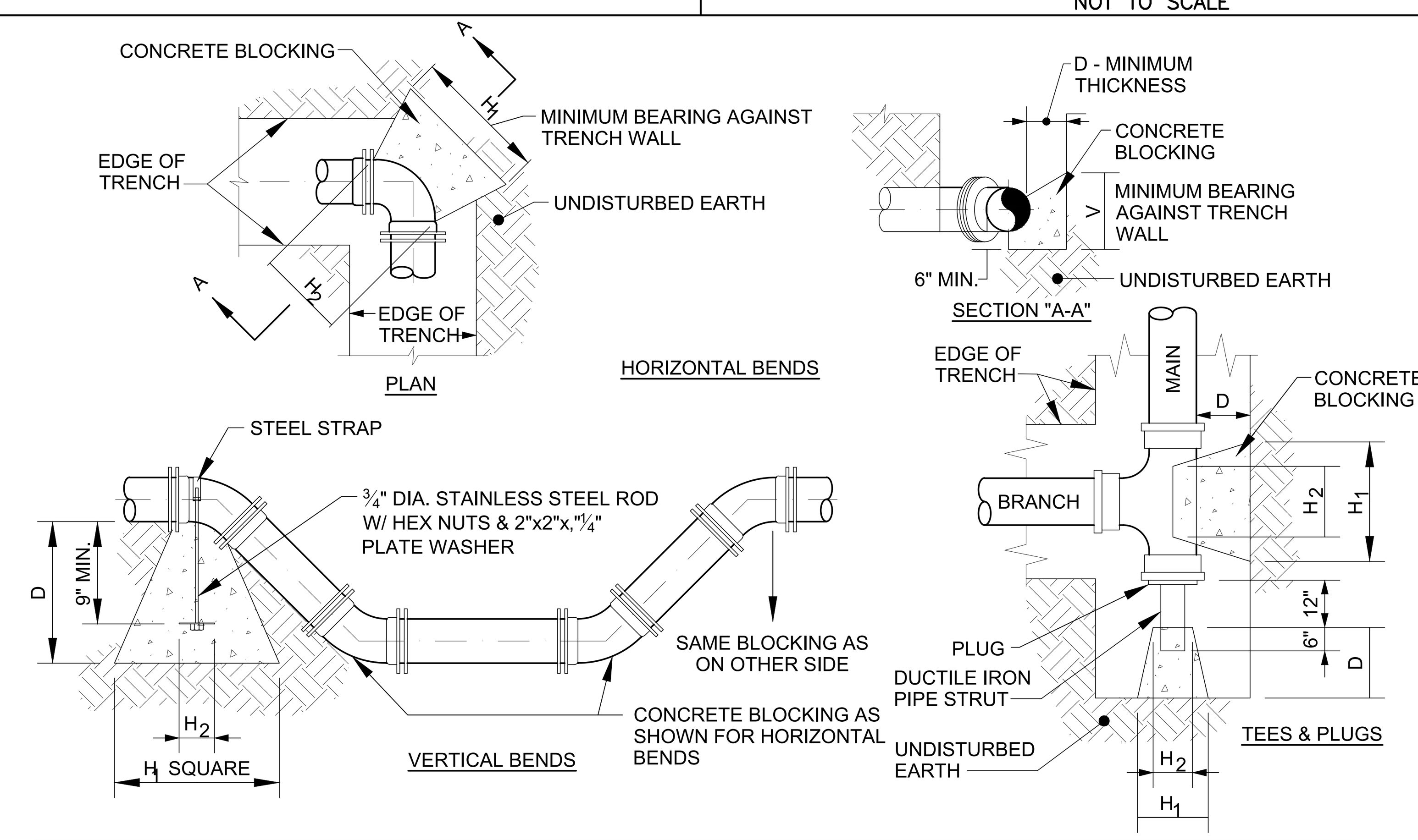
CLASS "A" PIPE BEDDING DETAIL

N.T.S.



WATER PIPE BEDDING DETAIL

NOT TO SCALE



PIPE SIZE	90° BENDS				45° BENDS				22.5° BENDS				11.25° BENDS				TEES				PLUGS				PIPE SIZE
	H_1	H_2	V	D	H_1	H_2	V	D	H_1	H_2	V	D	H_1	H_2	V	D	H_1	H_2	V	D	H_1	H_2	V	D	
4"	21"	11"	14"	18"	15"	7"	10"	18"	11"	5"	8"	18"	8"	4"	6"	18"	25"	8"	12"	18"	18"	10"	12"	18"	4"
6"	30"	15"	20"	18"	22"	10"	15"	18"	16"	7"	11"	18"	11"	5"	8"	18"	35"	10"	17"	18"	25"	14"	17"	18"	6"
8"	39"	17"	26"	18"	29"	11"	19"	18"	21"	8"	14"	18"	15"	5"	10"	18"	46"	13"	22"	18"	33"	19"	22"	18"	8"
12"	56"	25"	38"	24"	42"	17"	28"	24"	30"	11"	20"	24"	21"	7"	14"	24"	67"	18"	32"	24"	48"	27"	32"	24"	12"

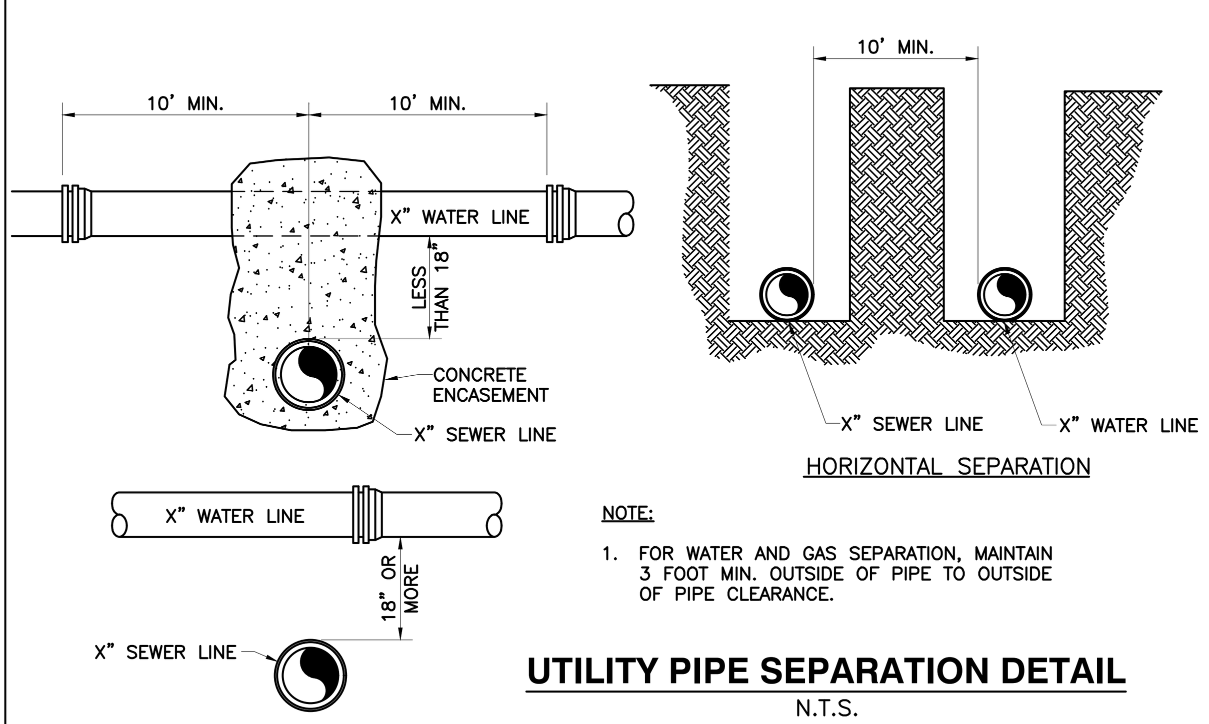
MINIMUM DIMENSIONS TABLE

NOTES:

1. THIS TABLE IS BASED ON AN INTERNAL HYDROSTATIC PRESSURE OF 200 PSI AND AN ALLOWABLE SOIL BEARING CAPACITY OF 4000 LB/SQ FT.
2. SHOULD THE INTERNAL HYDROSTATIC PRESSURE EXCEED 200 PSI AND/OR THE ALLOWABLE SOIL BEARING CAPACITY BE LESS THAN 4000 LB/SQ FT, THE THRUST BLOCK SIZE MUST BE ENLARGED. IF EITHER OR BOTH OF THESE CONDITIONS OCCUR, PROVIDE CALCULATIONS TO SUPPORT THE ENLARGED SIZE THRUST BLOCKS.
3. TEE DIMENSIONS ARE CONTROLLED BY BRANCH DIAMETER.

CONCRETE THRUST BLOCKING

N.T.S.

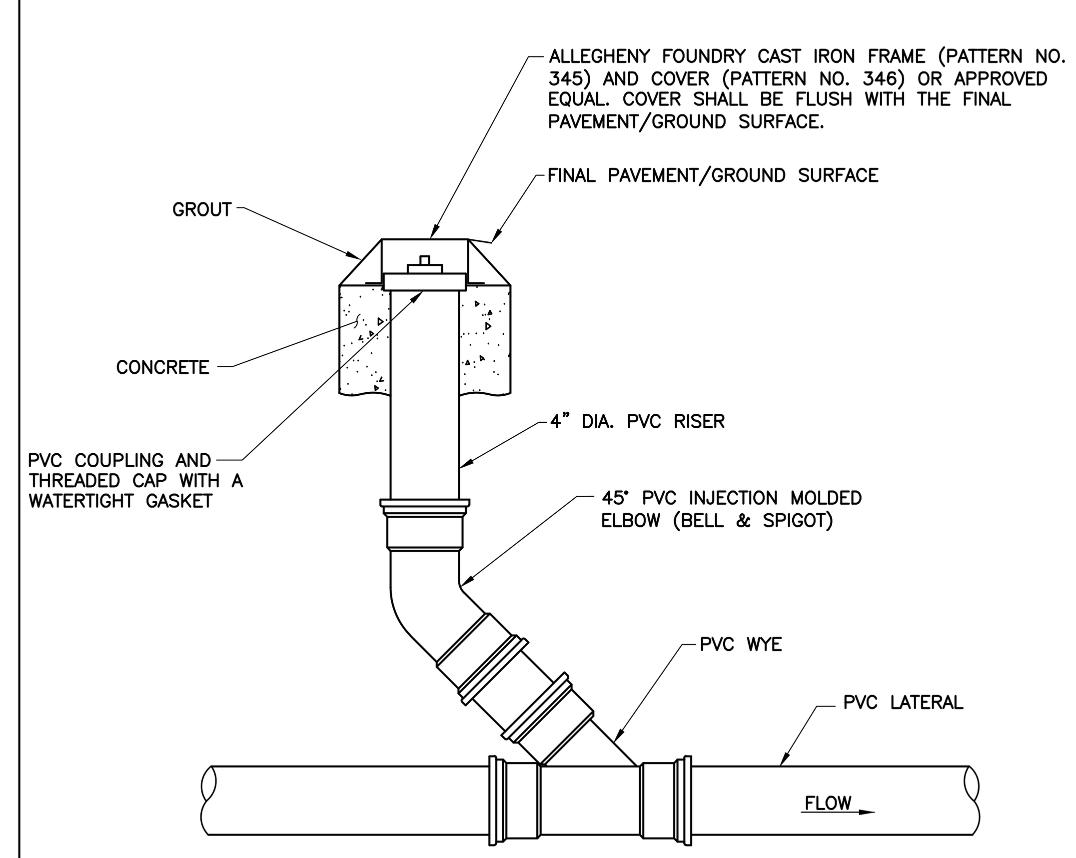


NOTE:

1. FOR WATER AND GAS SEPARATION, MAINTAIN 3 FOOT MIN. OUTSIDE OF PIPE TO OUTSIDE OF PIPE CLEARANCE.

UTILITY PIPE SEPARATION DETAIL

N.T.S.



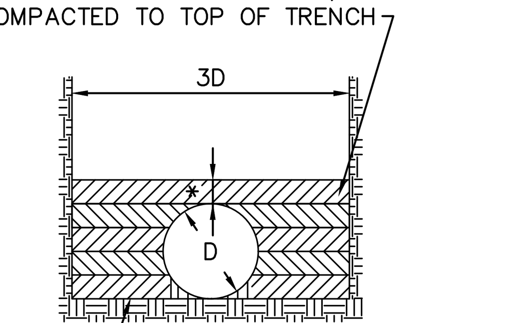
NOTES:

1. THIS DETAIL APPLIES TO STORM AND SANITARY SEWER CLEANOUTS WITH APPROPRIATE CHANGES TO WYE AND PIPE SIZES.
2. ALL PVC PIPE AND FITTINGS SHALL BE SDR 35 WITH RUBBER GASKETED JOINTS.

SANITARY SEWER CLEANOUT

NOT TO SCALE

LEVEL LIFTS OF STRUCTURE BACKFILL 6 TO 12 IN. THICK, COMPACTED TO TOP OF TRENCH



* MINIMUM COVER OF STRUCTURE BACKFILL IS D/8 OR 12 IN. WHICHEVER IS GREATER.

COMPACTION KEY DETAIL

NOT TO SCALE

SPRINKLE ENGINEERING
SPRINKLE ENGINEERING, LLC
PO BOX 4013
MARYVILLE, TN 37802

TAC
The Architecture Collaborative
6215 Kingston Pike | Knoxville, TN 37919
www.tac-45.com | 865.342.7505

A NEW DEVELOPMENT FOR:

OAK RIDGE ATOMIC ELKS LODGE

262 WILBERFORCE AVE
OAK RIDGE, TN

MATTHEW S. SPRINKLE
REGISTERED ENGINEER
AGRICULTURE COMMERCE
No. 111155
STATE OF TENNESSEE

Matthew S. Sprinkle
10/31/25

THIS DRAWING IS ISSUED FOR:
PERMITTING

Revisions:

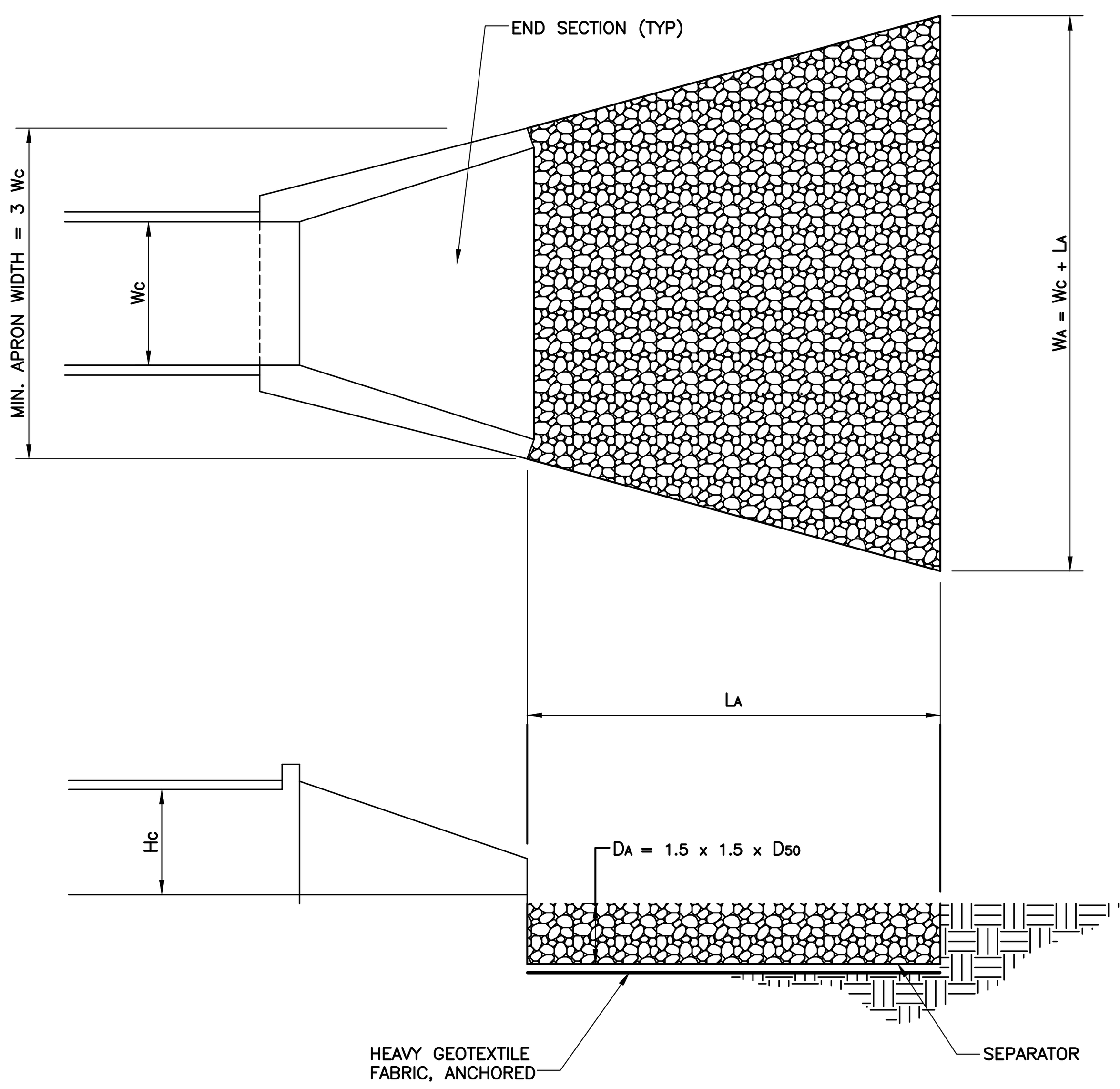
No.	Date

Drawing Title:
SITE DETAILS

Date: 10/31/2025

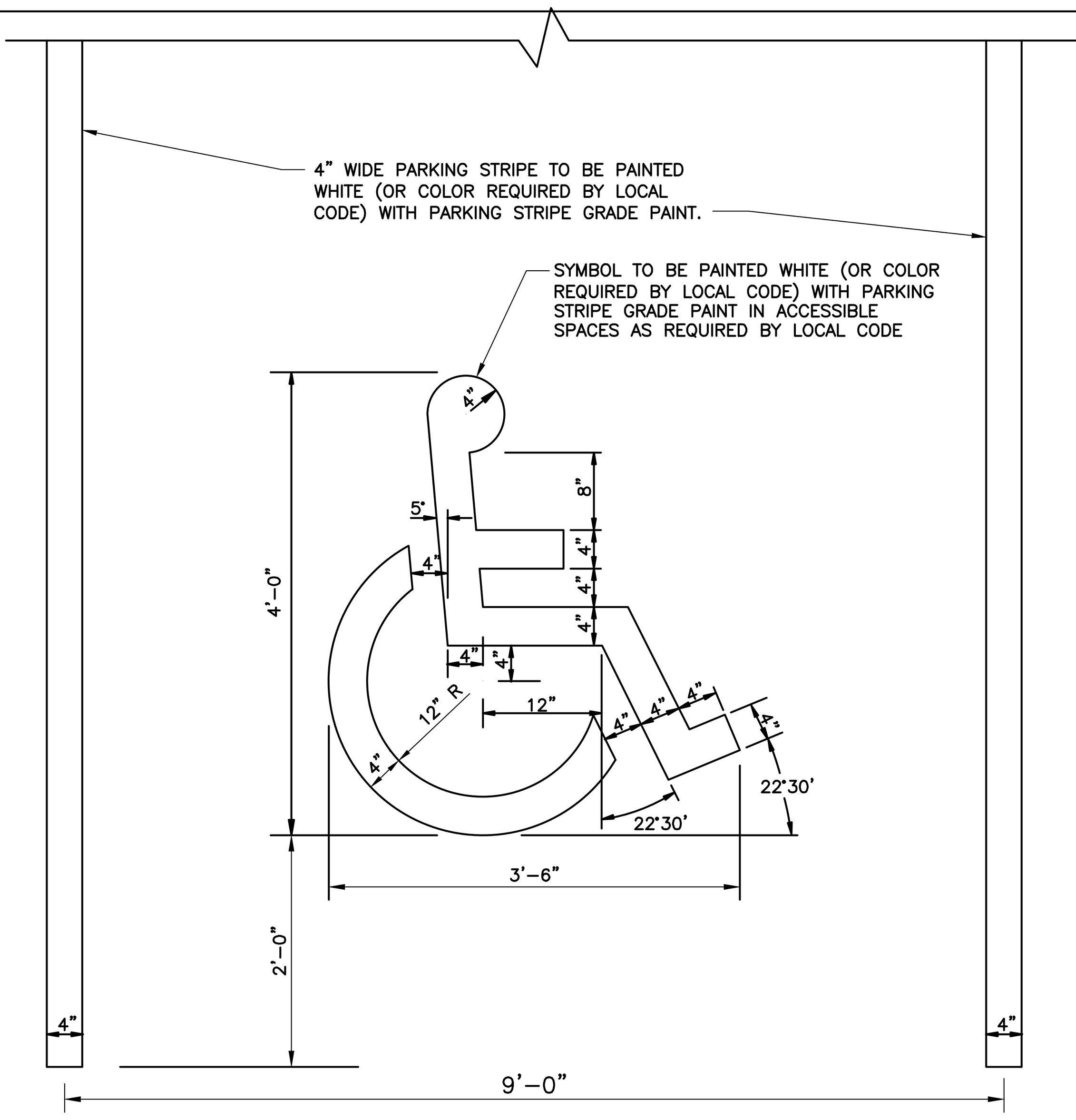
Project No.

Sheet No.
C800

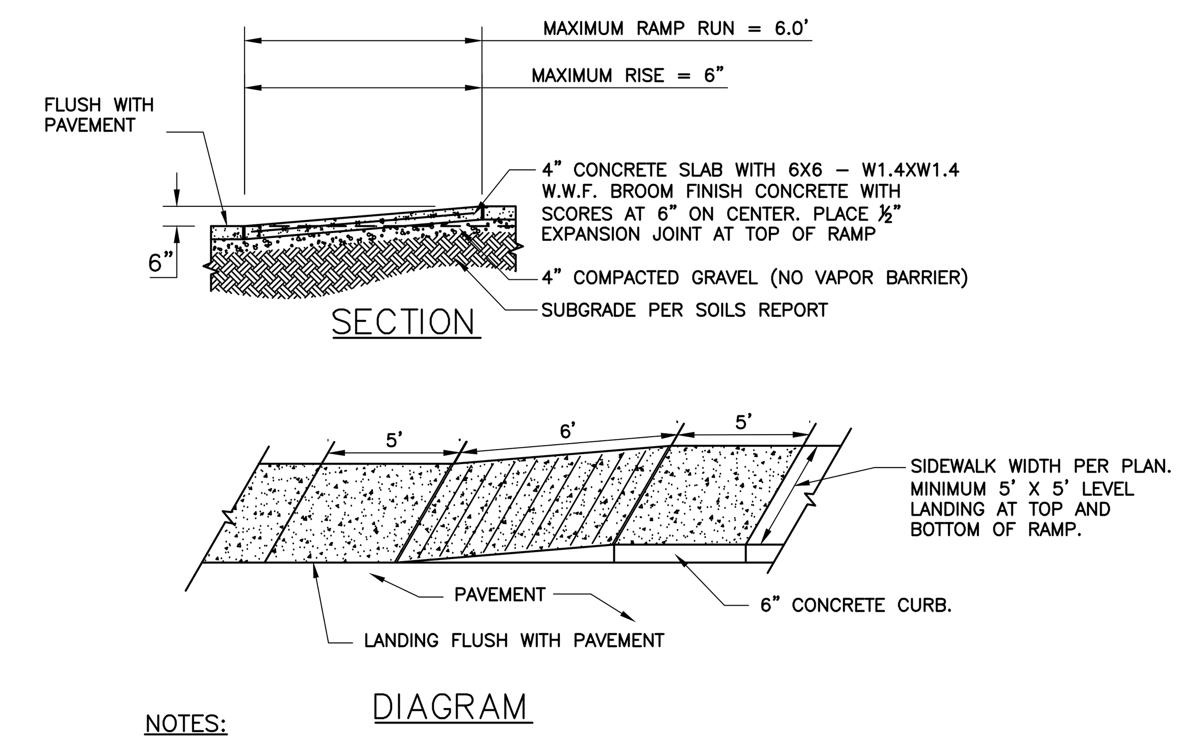


TAILWATER < 0.5 Hc AND ASSUMING FULL CULVERT FLOW (LOW TAILWATER CONDITIONS)

EROSION CONTROL PLAN LEGEND: OP

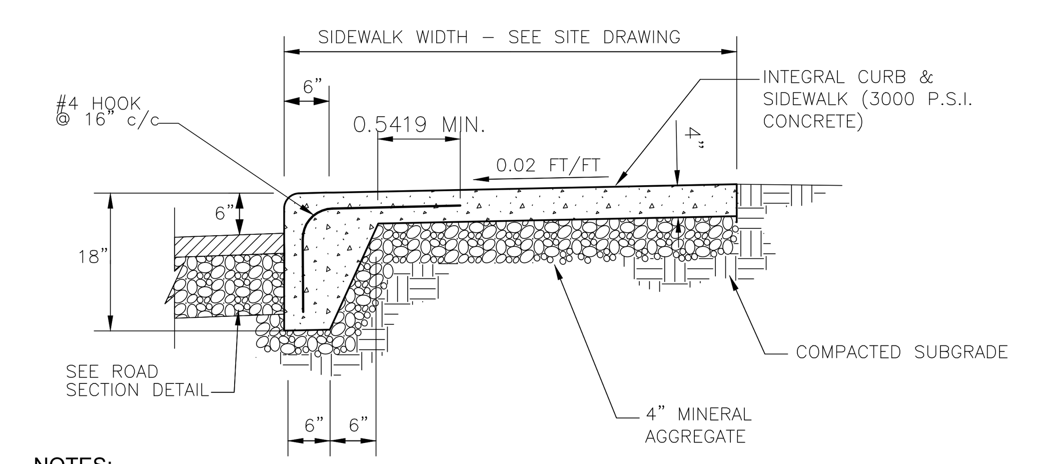


ACCESSIBLE PARKING SYMBOL
NOT TO SCALE

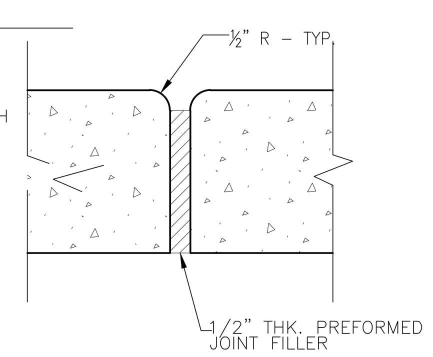


NOTES:
A. IN-LINE RAMP SHOWN WITH CURB AT SIDES SHALL HAVE A MAXIMUM ELEVATION CHANGE OF 6".
B. PROVIDE MINIMUM 5' X 5' LEVEL (<2% SLOPE IN ALL DIRECTIONS) LANDING AT TOP AND BOTTOM OF RAMP.
C. VERIFY SIZE AND STYLE OF RAMP WITH SITE IMPROVEMENT PLAN.

ACCESSIBLE PARALLEL RAMP
NOT TO SCALE

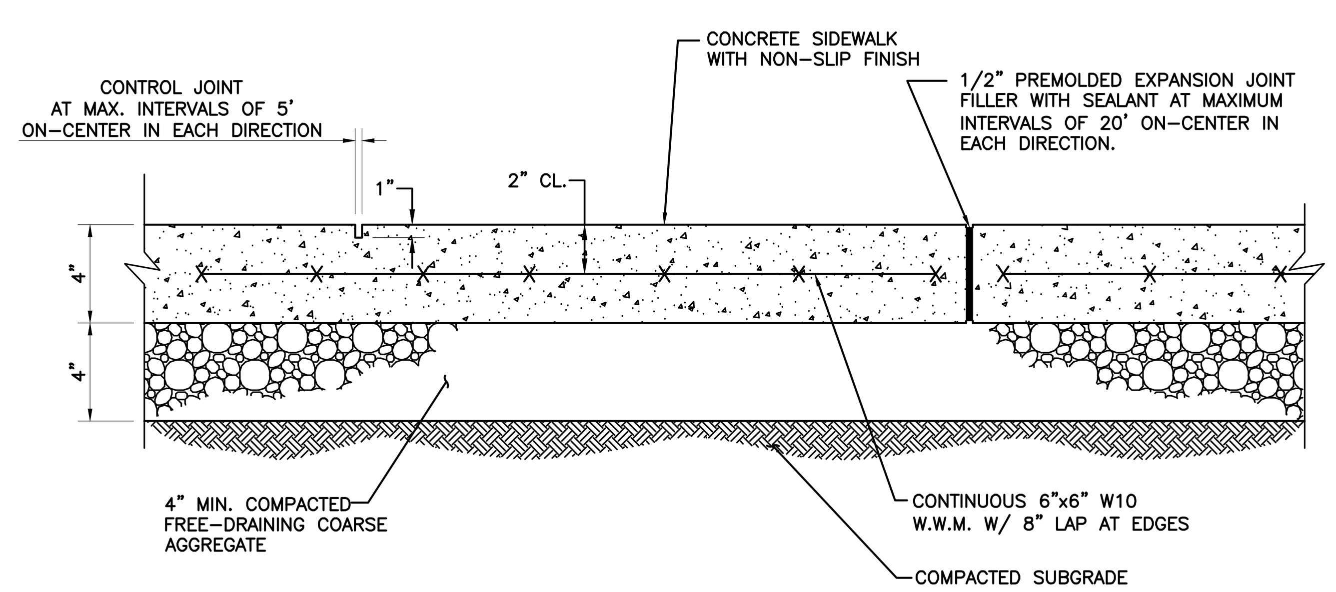


NOTES:
1. SPACE CONTRACTION JOINTS EQUAL TO WALK WIDTH BUT DO NOT EXCEED 5'. USE 1/4" RADIUS TOOLED JOINTS.
2. CONCRETE TO HAVE MINIMUM 3000 PSI COMPRESSIVE STRENGTH AT 28 DAYS.
3. SPACE 1/2" EXPANSION JOINTS NO MORE THAN 20' APART.
4. PLACE PREFORMED 1/2" EXPANSION JOINTS BETWEEN ALL FIXED OBJECTS AND SIDEWALKS, INCLUDING INTERSECTIONS WITH OTHER STRUCTURES, SIDEWALKS, FOUNDATIONS, ETC.



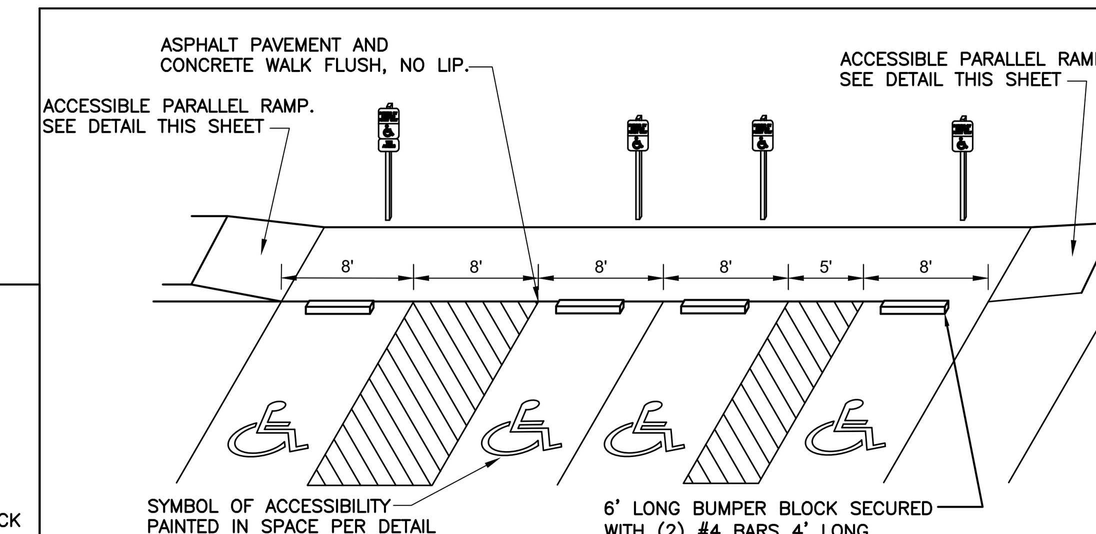
EXPANSION JOINT EA. 20 L.F.

INTEGRAL CURB & SIDEWALK
NOT TO SCALE



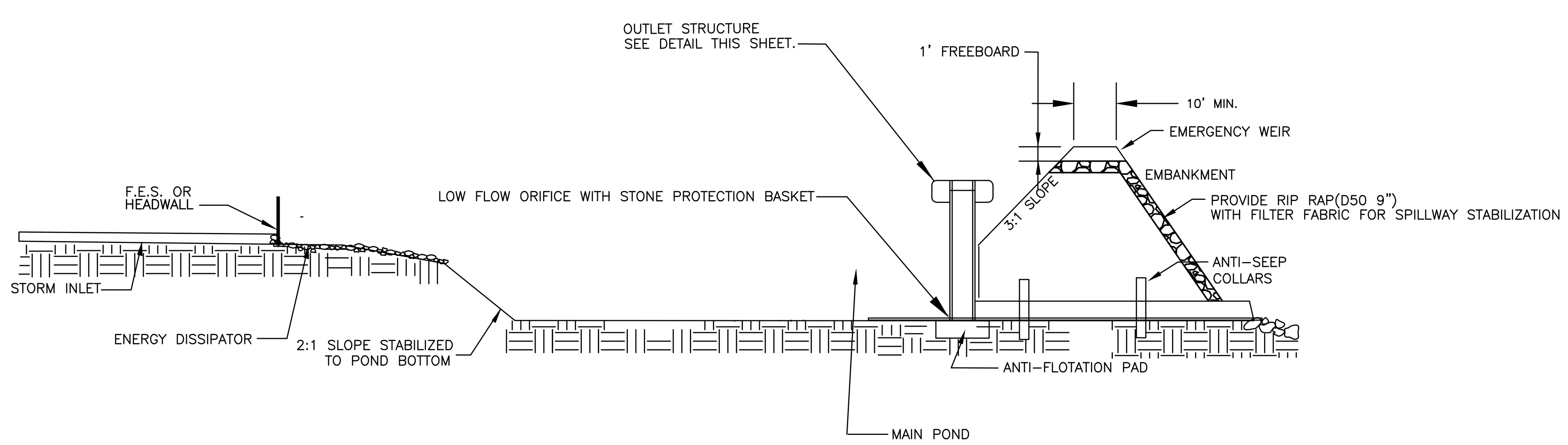
NOTES:
1. PROVIDE 1/2" EXPANSION JOINT FILLER WITH SEALANT WHERE THE CONCRETE SIDEWALK ABUTS THE BUILDING.
2. PROVIDE MEDIUM BROOM FINISH.
3. CONCRETE SHALL BE CLASS "A" 4,000 PSI UNLESS OTHERWISE NOTED.

CONCRETE SIDEWALK DETAIL
NOT TO SCALE



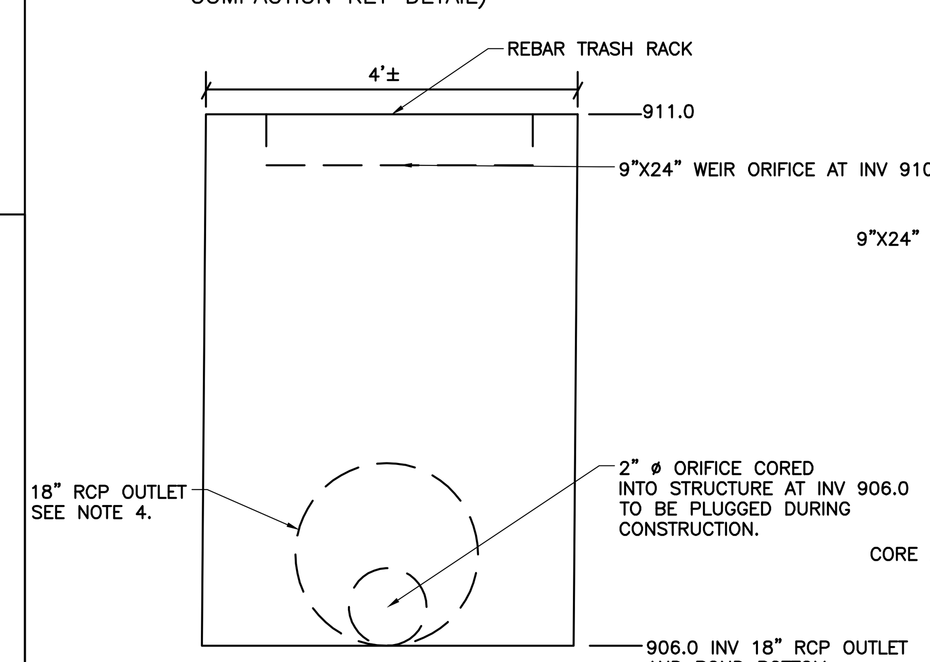
NOTES:
1. ACCESSIBLE SPACE / LOADING AREA: MAXIMUM 2.0% (1:50) SLOPE IN ANY DIRECTION.
2. ASPHALT TO CONCRETE BLENDED TRANSITION TO BE COMPLETELY FLUSH, NO LIP.
3. PLACEMENT AND STRIPING SHOWN IN DETAIL IS SCHEMATIC, SEE SITE PLAN FOR CONFIGURATION.
4. PASSENGER LOADING AREAS SHALL BE PAINT STRIPED AT 45' ANGLE , EACH 6" WIDE STRIPE AT 24" ON CENTER.

ACCESSIBLE PARKING
NOT TO SCALE

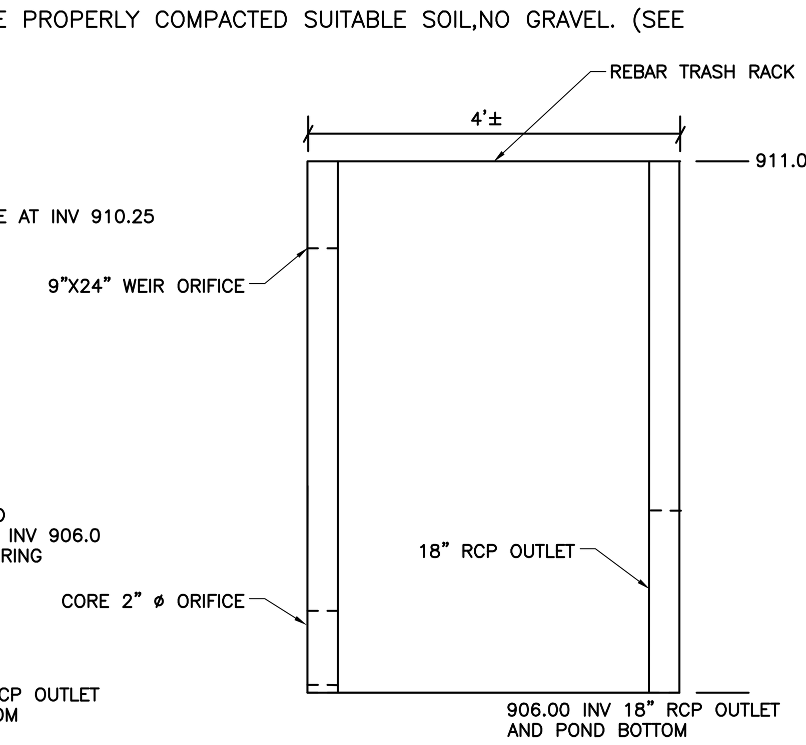


POND PROFILE

PROPOSED OUTLET STRUCTURE NOTES:
1. ALL OUTLET STRUCTURE METAL GRATES SHALL BE PAINTED WITH RUST RESISTANT ZINC PAINT.
2. INLET JOINTS TO BE SEALED WITH 1" DIAMETER FLEXIBLE BUTYL RUBBER JOINT SEALANT, USE 1/2" DIAMETER FOR FRAME AND GRATE.
3. CONTRACTOR SHALL REFER TO COMPACTION KEY DETAIL FOR OUTLET PIPE AND ANTI-SEEP COLLAR DETAIL FOR DETAILS OF THESE ITEMS.
4. BACKFILL AND BEDDING FOR OUTLET PIPE SHALL BE PROPERLY COMPACTED SUITABLE SOIL,NO GRAVEL. (SEE COMPACTION KEY DETAIL)



PROPOSED OUTLET STRUCTURE
NOT TO SCALE
FILTER CAGE NOT SHOWN FOR CLARITY



PROPOSED OUTLET STRUCTURE PROFILE
NOT TO SCALE

POND 1

SPRINKLE ENGINEERING
SPRINKLE ENGINEERING, LLC
PO BOX 4013
MARYVILLE, TN 37802

TAC
The Architecture Collaborative
6215 Kingston Pike | Knoxville, TN 37919
www.tac-45.com | 865.342.7505

A NEW DEVELOPMENT FOR:
OAK RIDGE ATOMIC ELKS LODGE
262 WILBERFORCE AVE
OAK RIDGE, TN

MATTHEW S. SPRINKLE
REGISTERED ENGINEER
STATE OF TENNESSEE
COMMERCE
No. 111153

THIS DRAWING IS ISSUED FOR:
PERMITTING

Revisions:

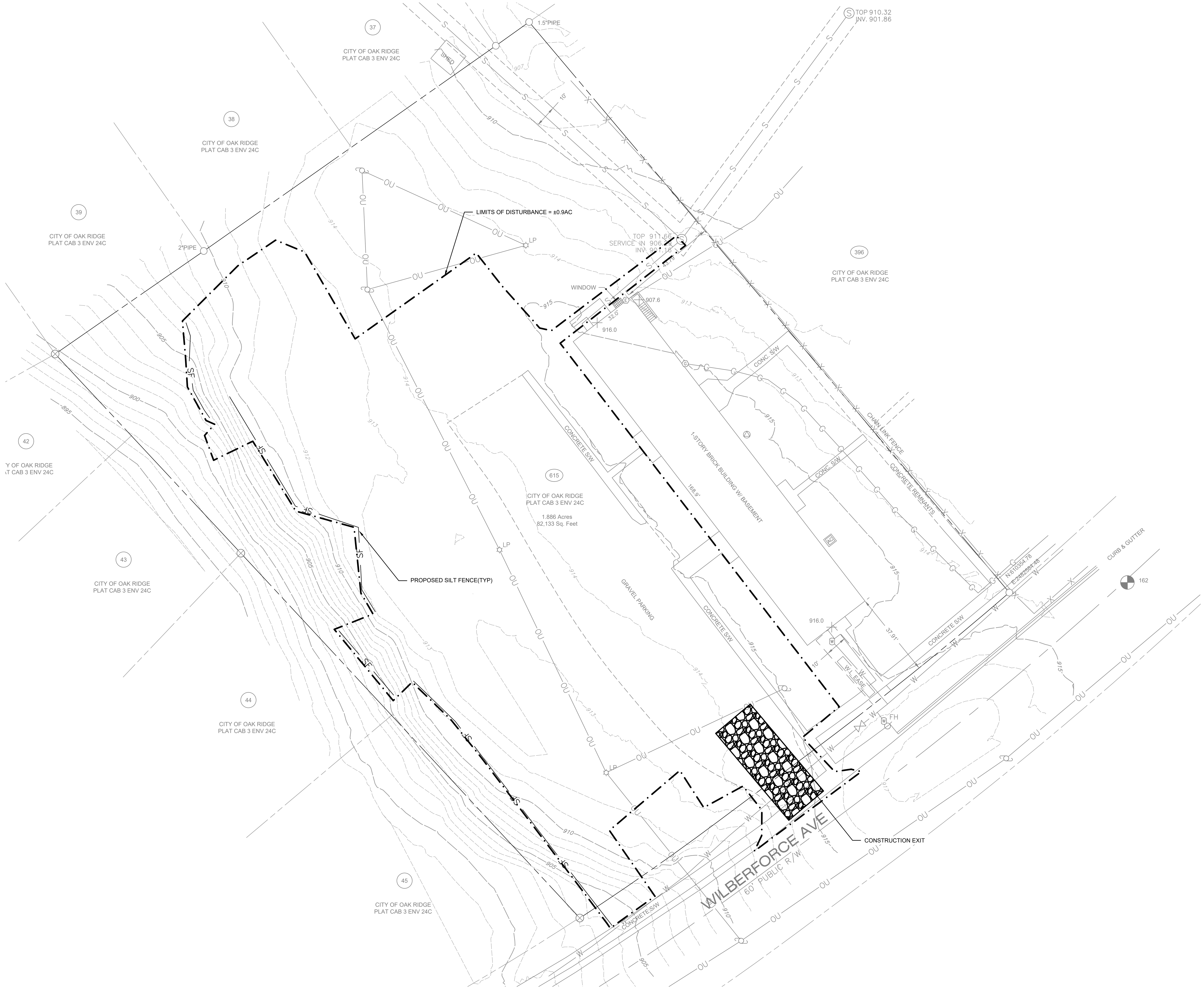
No.	Date

Drawing Title:
SITE DETAILS

Date: 10/31/2025

Project No.

Sheet No.
C801



BASIS OR BEARING

EXISTING BOUNDARY AND TOPOGRAPHIC INFORMATION FROM SURVEY FILES PROVIDED BY PROFESSIONAL LAND SYSTEMS DATED 08/13/25.

FLOOD NOTE

THE PROJECT SITE IS NOT LOCATED WITHIN AN AREA OF FLOOD HAZARD PER THE FEMA FLOOD INSURANCE RATE MAP NO.47001C0282F EFFECTIVE JANUARY 17, 2007.

UTILITY NOTE

THE EXACT LOCATION OF UNDERGROUND FEATURES CANNOT BE ACCURATELY, COMPLETELY AND RELIABLY DEPICTED. UTILITIES MAY EXIST THAT ARE NOT SHOWN. FIELD VERIFY SIZE, LOCATION AND TYPE PRIOR TO CONSTRUCTION.

35°11'31"

0 20 40 Feet

SPRINKLE ENGINEERING

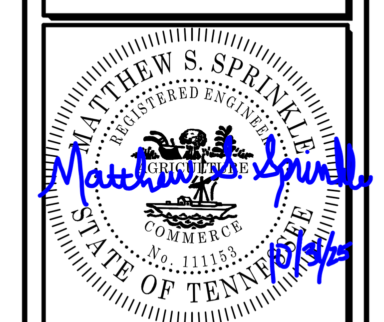
SPRINKLE ENGINEERING, LLC
PO BOX 4013
MARYVILLE, TN 37802

TAC
The Architecture Collaborative
6215 Kingston Pike | Knoxville, TN 37919
www.tac-45.com | 865.342.7505

A NEW DEVELOPMENT FOR:

OAK RIDGE ATOMIC ELKS LODGE

262 WILBERFORCE AVE
OAK RIDGE, TN



THIS DRAWING IS ISSUED FOR:

PERMITTING

Revisions:

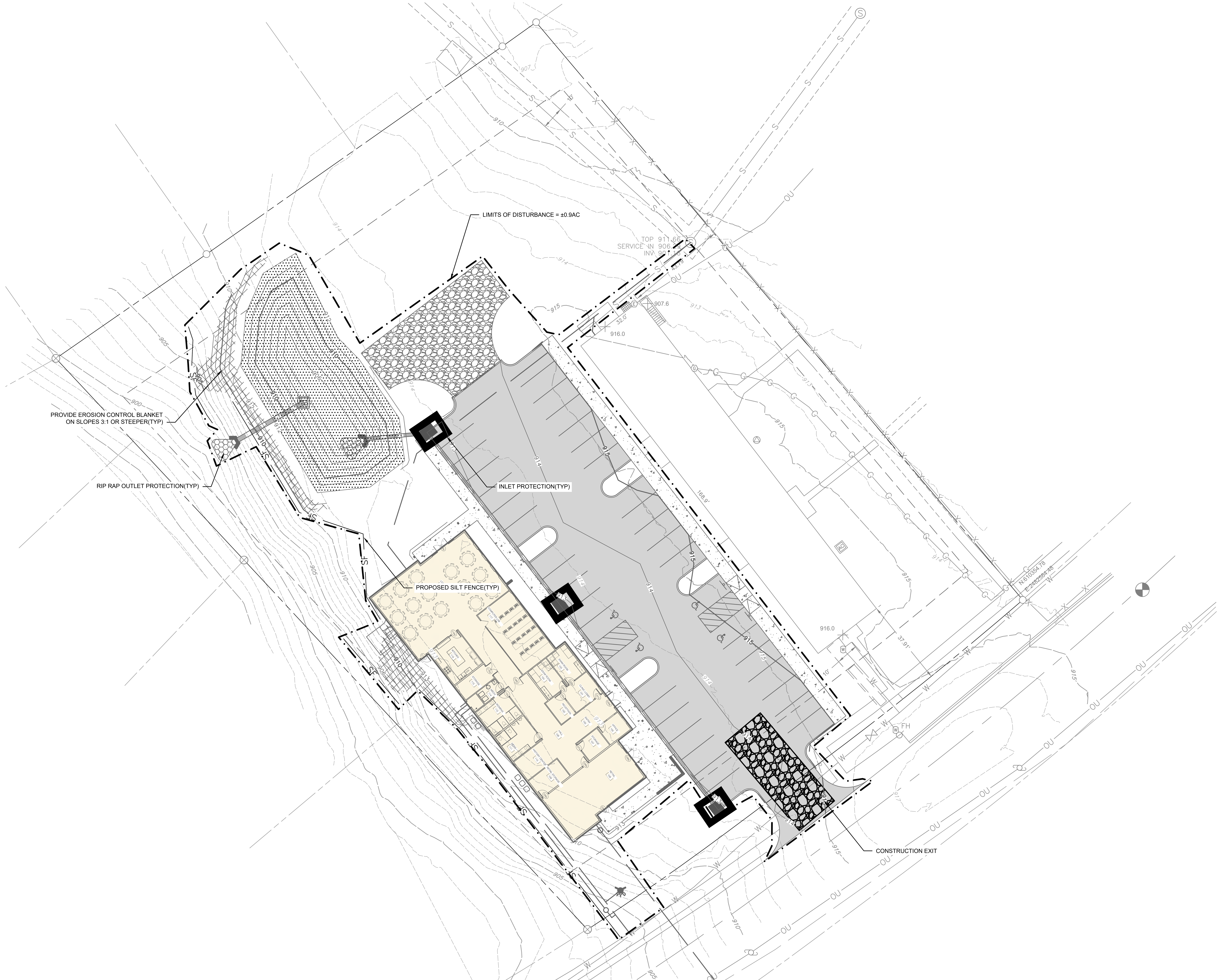
No.	Date

Drawing Title:
INITIAL EPSC PLAN

Date: 10/31/2025

Project No.

Sheet No.
C900



BASIS OR BEARING

EXISTING BOUNDARY AND TOPOGRAPHIC INFORMATION FROM SURVEY FILES PROVIDED BY PROFESSIONAL LAND SYSTEMS DATED 08/13/25.

FLOOD NOTE

THE PROJECT SITE IS NOT LOCATED WITHIN AN AREA OF FLOOD HAZARD PER THE FEMA FLOOD INSURANCE RATE MAP NO.47001C0282F EFFECTIVE JANUARY 17, 2007.

UTILITY NOTE

THE EXACT LOCATION OF UNDERGROUND FEATURES CANNOT BE ACCURATELY, COMPLETELY AND RELIABLY DEPICTED. UTILITIES MAY EXIST THAT ARE NOT SHOWN. FIELD VERIFY SIZE, LOCATION AND TYPE PRIOR TO CONSTRUCTION.

35°11'31"

0 20 40 Feet

SPRINKLE ENGINEERING

SPRINKLE ENGINEERING, LLC
PO BOX 4013
MARYVILLE, TN 37802

TAC
The Architecture Collaborative
6215 Kingston Pike | Knoxville, TN 37919
www.tac-45.com | 865.342.7505

A NEW DEVELOPMENT FOR:

OAK RIDGE ATOMIC ELKS LODGE

262 WILBERFORCE AVE
OAK RIDGE, TN

MATTHEW S. SPRINKLE
REGISTERED ENGINEER
AGRICULTURE
COMMERCE
No. 111153
STATE OF TENNESSEE

Matthew S. Sprinkle
10/31/25

THIS DRAWING IS ISSUED FOR:

PERMITTING

Revisions:

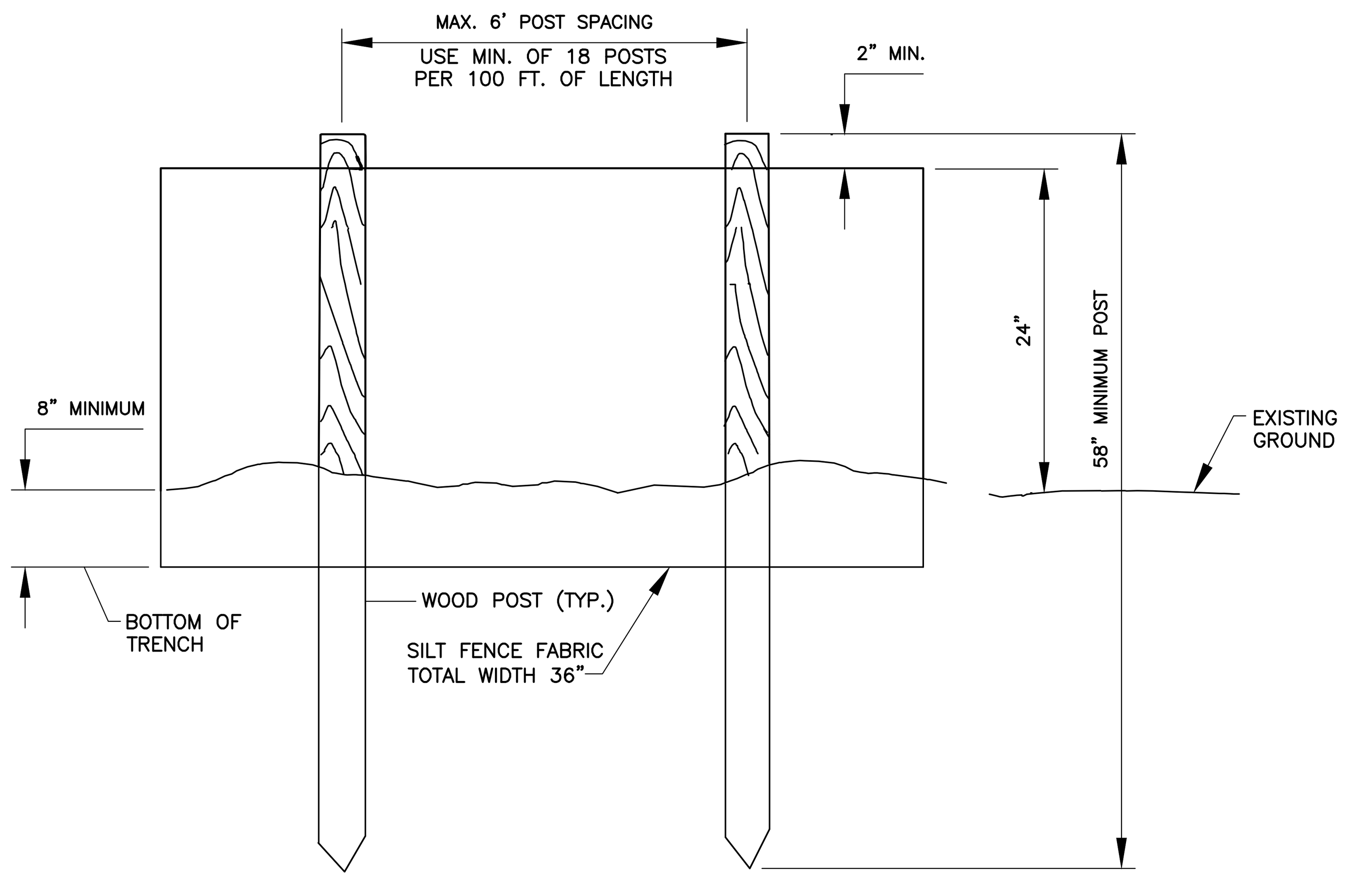
No.	Date

Drawing Title:
FINAL EPSC PLAN

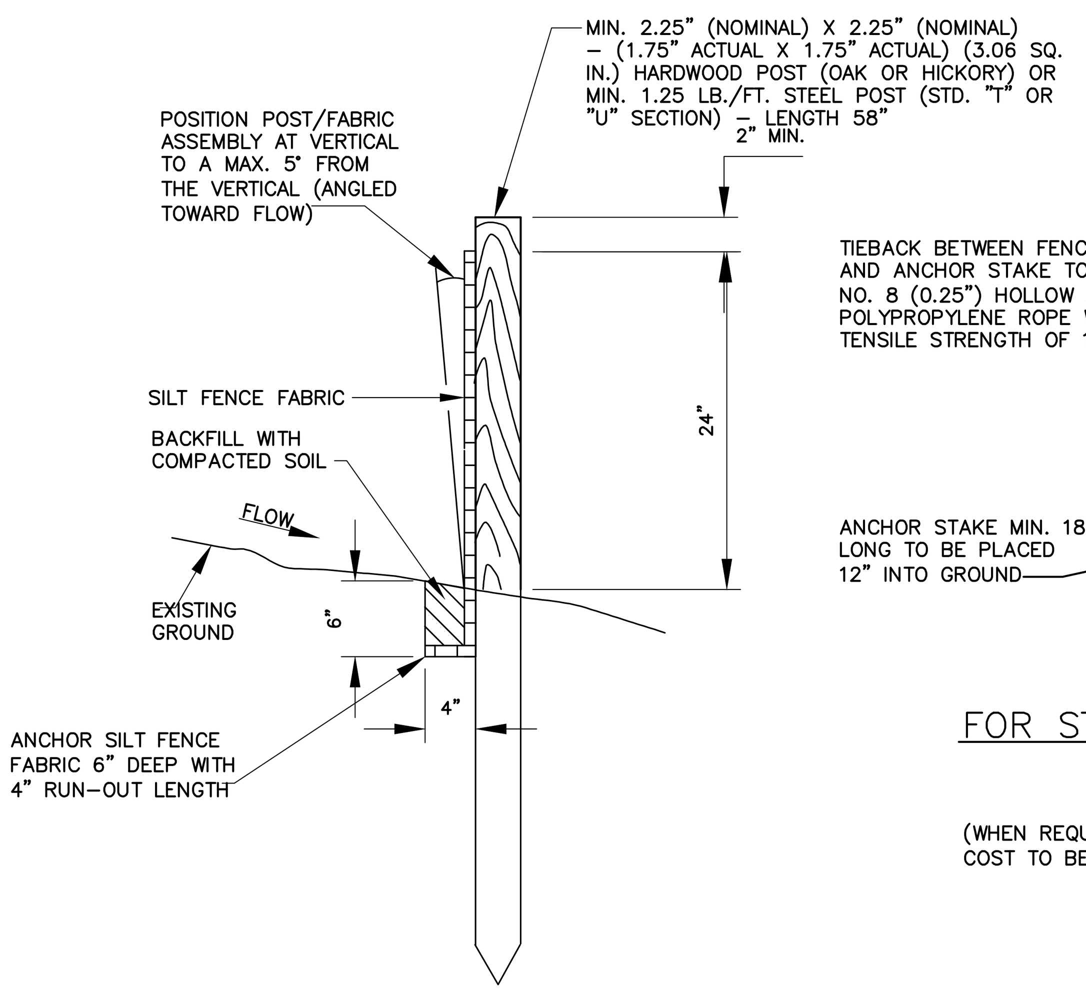
Date: 10/31/2025

Project No.

Sheet No.
C901

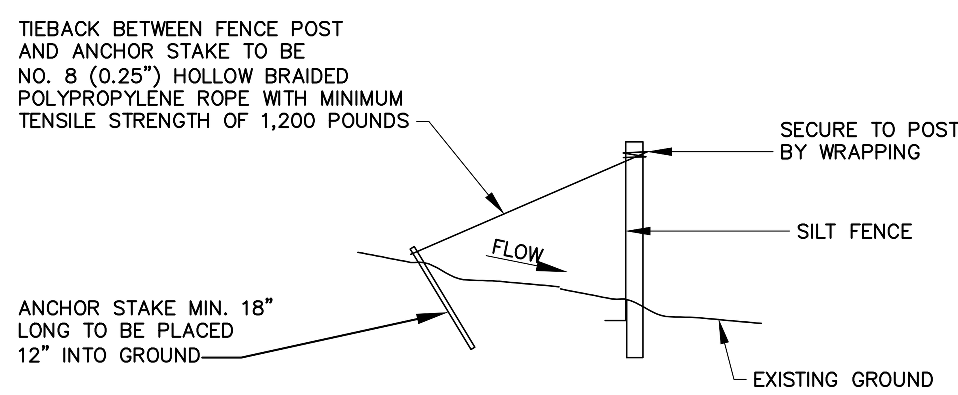


ELEVATION VIEW



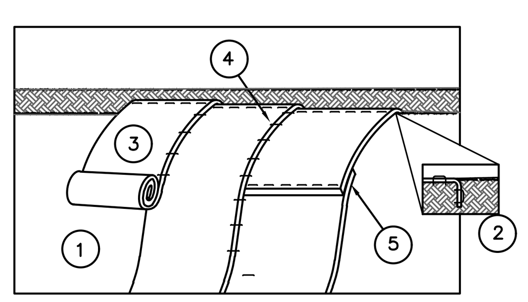
SECTIONAL VIEW

EROSION CONTROL PLAN LEGEND: SILT FENCE



SILT FENCE TIEBACK
FOR STEEL POSTS OR WOOD POSTS

(WHEN REQUIRED BY THE ENGINEER OR NOTED IN THE PLANS. COST TO BE INCLUDED IN THE ITEMS FOR SILT FENCE)

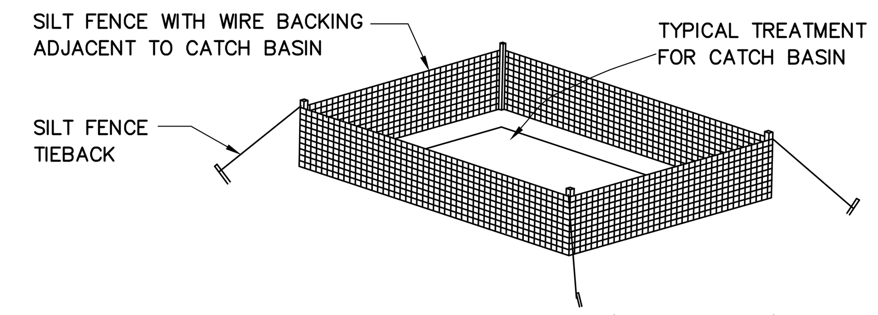


EROSION CONTROL BLANKET

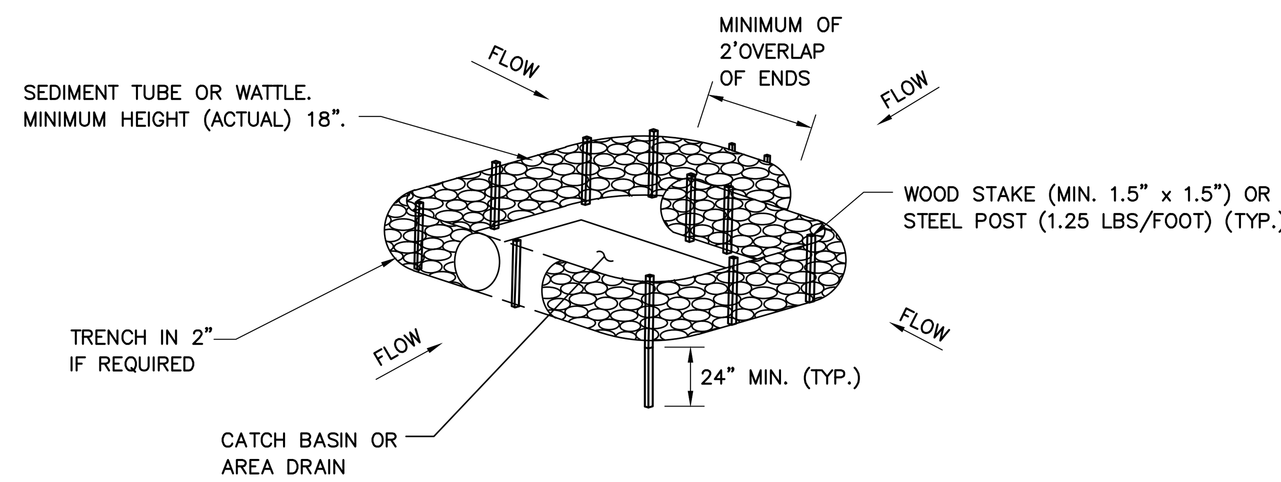
NOT TO SCALE

1. PREPARE SOIL BEFORE INSTALLING BLANKETS, INCLUDING APPLICATION OF LIME, FERTILIZER AND SEED. SEE LANDSCAPING NOTES, M16190.
2. BEGIN AT THE TOP OF THE SLOPE BY ANCHORING THE BLANKET IN A 6" DEEP X 6" WIDE TRENCH. BACKFILL AND COMPACT THE TRENCH AFTER STAPLING.
3. ROLL THE BLANKETS DOWN THE SLOPE IN THE DIRECTION OF THE WATER FLOW.
4. THE EDGES OF PARALLEL BLANKETS MUST BE STAPLED WITH APPROXIMATELY 2" OVERLAP.
5. WHEN BLANKETS MUST BE SPLICED DOWN THE SLOPE, PLACE BLANKETS END OVER END (SHINGLE STYLE) WITH APPROXIMATELY 6" OVERLAP. STAPLE THROUGH OVERLAPPED AREA APPROXIMATELY 12" APART. REFER TO GENERAL STAPLE PATTERN GUIDE FOR CORRECT STAPLE PATTERN RECOMMENDATIONS FOR SLOPE INSTALLATIONS.

CATCH BASIN PROTECTION (TYPE E)



CATCH BASIN PROTECTION (TYPE D)



PERMANENT SEEDING MIXTURES

SEEDING DATES	GRASS SEED	PERCENTAGES
FEBRUARY 1 TO JULY 1	KENTUCKY 31 FESCUE	88%
	ENGLISH RYE	12%
JUNE 1 TO AUGUST 15	KENTUCKY 31 FESCUE	60%
	ENGLISH RYE	25%
	GERMAN MILLET	15%
AUGUST 1 TO DECEMBER 1	KENTUCKY 31 FESCUE	70%
	ENGLISH RYE	20%
	WHITE CLOVER	10%
DECEMBER 1 TO FEBRUARY 1	KENTUCKY 31 FESCUE	83%
	ENGLISH RYE	17%

SOURCE: TDOT STANDARDS SPECIFICATIONS

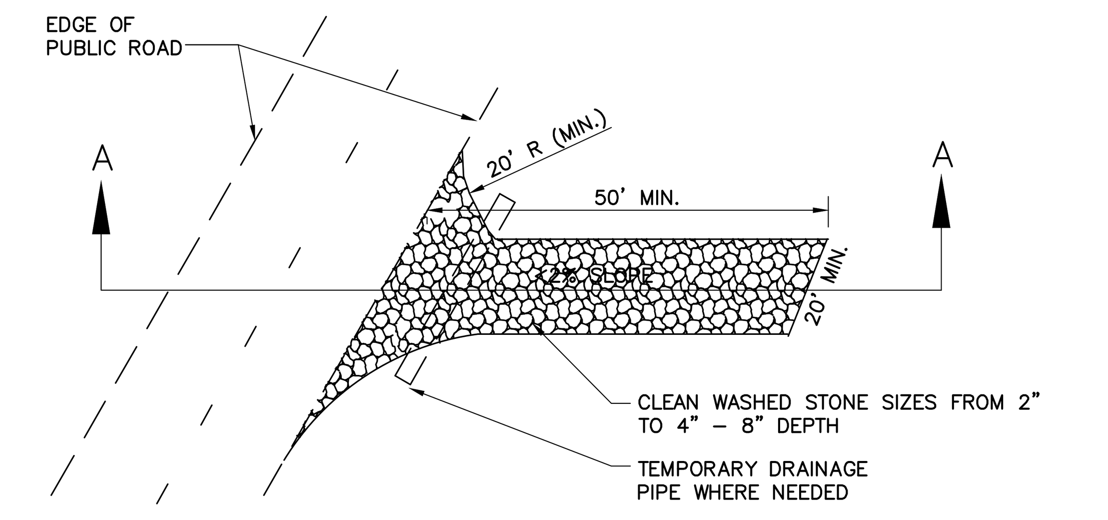
TEMPORARY SEEDING MIXTURES

SEEDING DATES	GRASS SEED	PERCENTAGES
JANUARY 1 TO MAY 1	ITALIAN RYE	50%
	SUMMER OATS	50%
MAY 1 TO JULY 15	SUDAN-SORGHURM	100%
MAY 1 TO JULY 15	STARR MILLET	100%
JULY 15 TO JANUARY 1	BALBOA RYE	67%
	ITALIAN RYE	33%

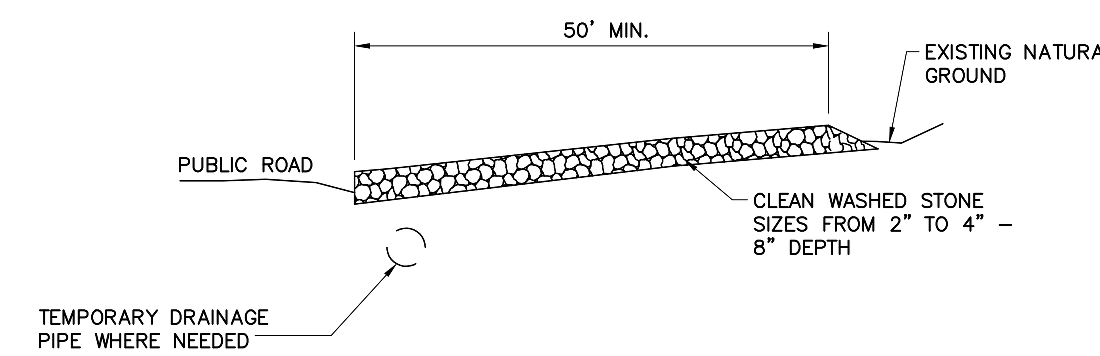
SOURCE: TDOT STANDARDS SPECIFICATIONS

SEEDING SCHEDULE

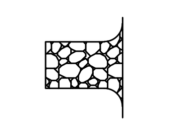
NOT TO SCALE



PLAN VIEW OF TEMPORARY CONSTRUCTION ROAD



SECTION A–A



CONSTRUCTION EXIT

SPRINKLE ENGINEERING

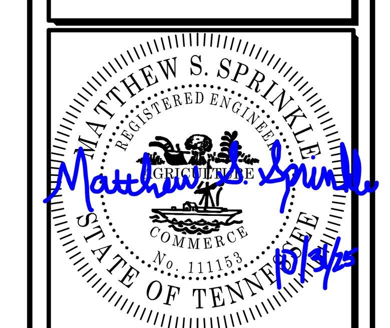
SPRINKLE ENGINEERING, LLC
PO BOX 4013
MARYVILLE, TN 37802

TAC The Architecture Collaborative
6215 Kingston Pike | Knoxville, TN 37919
www.tac-45.com | 865.342.7505

A NEW DEVELOPMENT FOR:

OAK RIDGE ATOMIC ELKS LODGE

262 WILBERFORCE AVE
OAK RIDGE, TN



THIS DRAWING IS ISSUED FOR:
PERMITTING

Revisions:

No.	Date

Drawing Title:
EPSC DETAILS

Date: 10/31/2025

Project No.

Sheet No.
C902

PLANT SCHEDULE

FLOWERING TREES	BOTANICAL NAME	COMMON NAME	CALIPER
4	CERCIS CANADENSIS	EASTERN REDBUD	2-2.5" CAL
3	ILEX VOMITORIA	YAUPON HOLLY	2-2.5" CAL
DECIDUOUS SHRUBS	BOTANICAL NAME	COMMON NAME	CONTAINER
7	DIRCA PALUSTRIS	LEATHERWOOD	3 GAL
3	AMORPHA FRUTICOSA	INDIGO BUSH	3 GAL
1	HYPERICUM PROLFICUM	SHURBBY ST. JOHN'S WORT	2 GAL

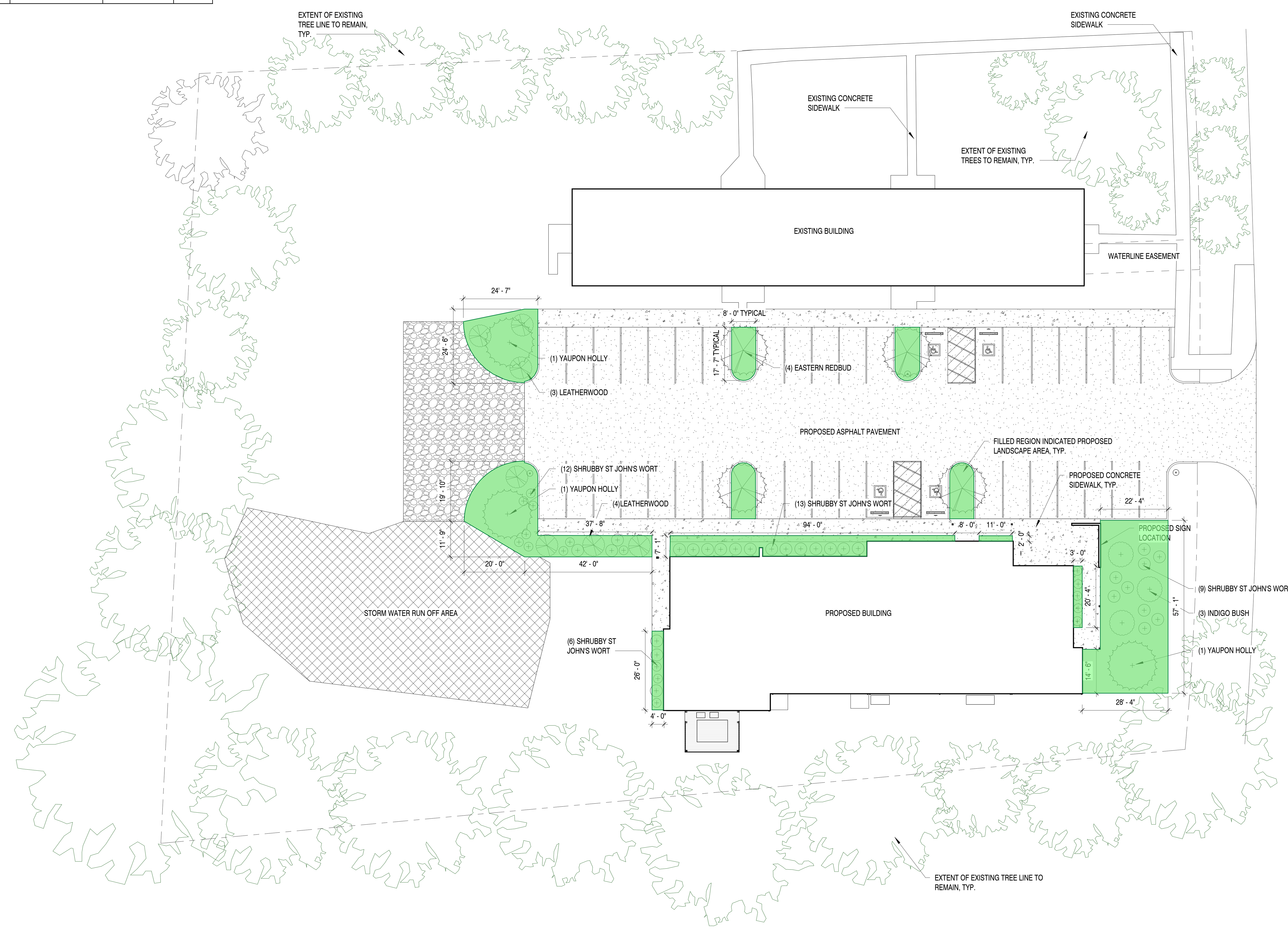
AREA CALCULATIONS

TOTAL ACREAGE:	1.886 ACRES
OPEN SPACE:	0.943 ACRES (50% OF TOTAL AREA)
PAVED AREA:	13,594 SF (17% OF TOTAL AREA)
ACTUAL LANDSCAPE AREA:	2406.4 SF (18% OF PAVED AREA)
REQUIRED LANDSCAPE AREA:	1359.4 SF MIN. (13,594*0.1)
PLANTED TREES: (1359.4/200 = 6.795 MINIMUM)	7 TREES PROPOSED

LANDSCAPING GENERAL NOTES

1. ALL TREES AT THE TIME OF PLANTING SHALL BE A MINIMUM CALIPER OF 1 1/2" OR A MINIMUM HEIGHT OF FIVE (5) FEET.
2. ALL LANDSCAPE PLANTING AREAS SHALL BE STABILIZED AND MAINTAINED WITH SEED, SOD, GROUND COVERS, MULCHES, OR OTHER APPROVED MATERIALS TO PREVENT SOIL EROSION AND ALLOW RAINWATER INFILTRATION.
3. NOTHING EXCEPT SMALL TREES, SHRUBS OR GROUNDCOVER SHALL BE PLANTED OR INSTALLED WITHIN ANY UNDERGROUND OR OVERHEAD UTILITY, DRAINAGE, OR GAS EASMEMENT WITHOUT THE WRITTEN CONSENT OF THE UTILITY PROVIDER, EASEMENT HOLDER OR CITY.
4. CONTRACTOR SHALL LOCATE AND PROTECT ALL UTILITIES PRIOR TO BEGINNING WORK. ANY DAMAGES TO PROPERTY OR UTILITIES ARE AT THE CONTRACTOR'S EXPENSE.
5. ALL PLANT MATERIAL TO BE UNDER 1 YEAR WARRANTY FROM DATE OF SUBSTANTIAL COMPLETION.
6. PROVIDE AND INSTALL ONLY PLANTS THAT ARE FREE FROM DAMAGE, DISEASE AND PESTS, AND THAT COMPLY WITH THE LATEST EDITION OF PUBLICATION ANSI Z60.1, "AMERICAN STANDARDS FOR NURSERY STOCK," BY THE AMERICAN ASSOCIATION OF NURSERYMEN.
7. TREES AND SHRUBS SHALL BE PLANTED AS SHOWN IN THE LANDSCAPE PLAN AND DETAILS.
8. ADD FERTILIZER AS REQUIRED FOR OPTIMUM PLANT PERFORMANCE AND ADD PRE-EMERGENT HERBICIDE UNDERNEATH AND ON TOP OF MULCH IN LANDSCAPE BEDS.
9. TOPSOIL TO HAVE 4% ORGHANIC MATERIAL AND BE FREE OF 1" DIAMETER OR LARGER, ROOTS, PLANTS, CLAY LUMPS, AND OTHER MATERIALS HARMFUL TO PLANT GROWTH.
10. ALL PLANTS SHALL BE WATERED THOROUGHLY TWICE DURING THE FIRST 24 HOUR PERIOD. ALL PLANTS SHALL THEN BE WATERED TWICE WEEKLY BY THE CONTRACTOR UNTIL PLANTING AND CONSTRUCTION ARE COMPLETE, AND CONTRACTORS ARE GONE FROM SITE.
11. THE LANDSCAPING PLAN SUBMITTED IS THE APPROVED LANSCAPE PLAN. ANY CHANGES HAVE TO BE SUBMITTED IN WRITING TO THE CITY OF OAK RIDGE.
12. ALL EXTERIOR LIGHTING IS REQUIRED TO BE DARK SKY CERTIFIED AND SHALL COMPLY WITH THE INTERNATIONAL DARK SKY ASSOCIATION.

SITE				
	Description	Label	Quantity	Unit
	SITE	CONCRETE	3,877.84	sf



1 LANDSCAPING PLAN
1" = 20'-0"

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THOMAS DANIEL BRITT
REGISTERED ARCHITECT
COMMERCE
No. 102274
STATE OF TENNESSEE
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

Drawing Title:
LANDSCAPING PLAN AND DETAILS

Date: 11/26/2025

Project No. 25026

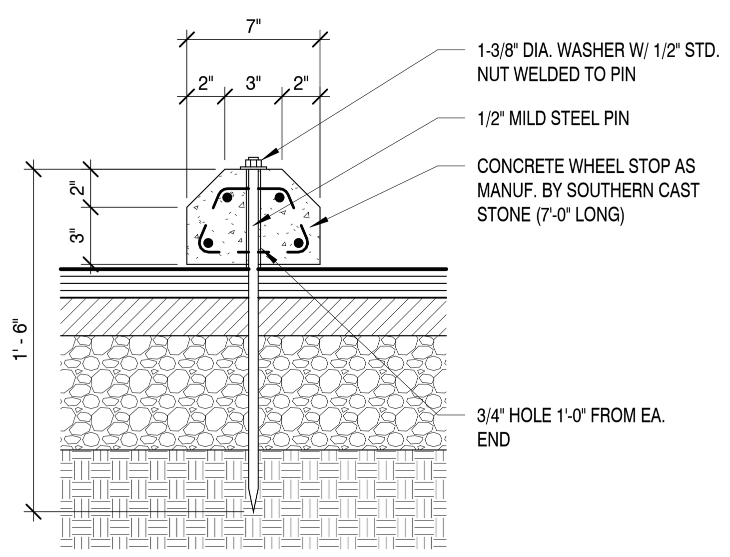
Sheet No. AS1.1

SITE KEY NOTES

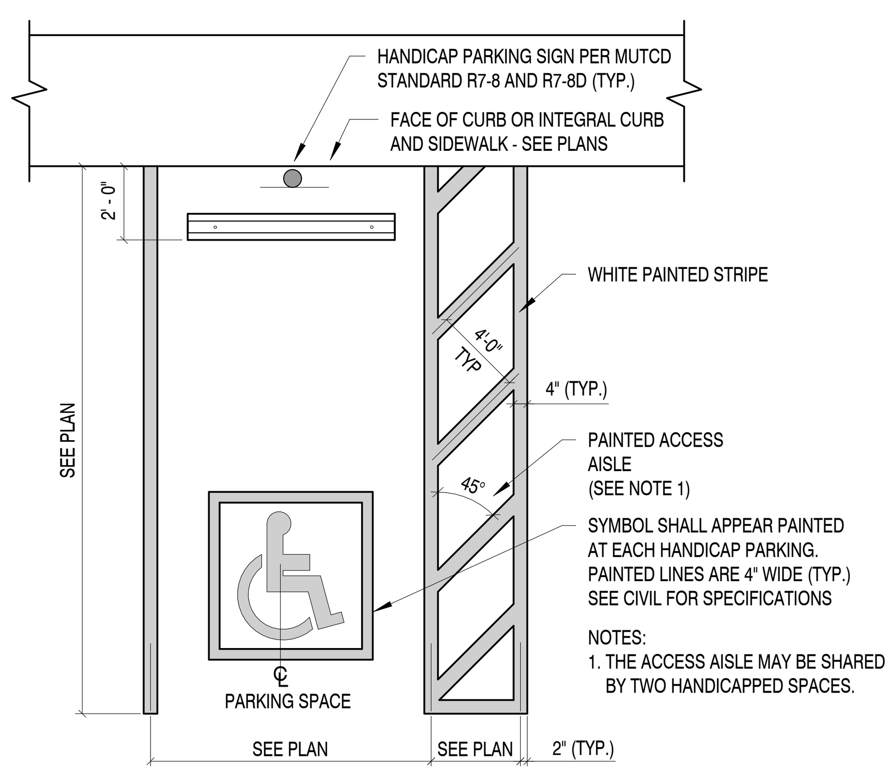
1. NEW CONCRETE WALKWAY, SEE CIVIL.
2. NEW CONCRETE ADA COMPLIANT RAMP, SEE CIVIL.
3. NEW VAN ACCESSIBLE PARKING SPACE WITH ASSOCIATED ACCESS AISLE.
4. NEW POLE MOUNTED HANDICAP SIGN.
5. NEW PAINTED HANDICAP SYMBOL, SEE CIVIL.
6. NEW HVAC UNITS PER MECHANICAL DRAWINGS. PROVIDE CONCRETE PAD FOR UNIT PLACEMENT.
7. NEW PACKAGED HVAC UNIT PER MECHANICAL DRAWINGS. PROVIDE DECK FOR UNIT PLACEMENT.
8. NEW LANDSCAPING IN THIS AREA. SEE LANDSCAPING PLAN.
9. NEW 6" MACHINE FORMED CONCRETE CURB.
10. NEW GRASS ISLAND WITH 6" CONCRETE CURB.
11. NEW BICYCLE PARKING AREA WITH STEEL BIKE RACK.
12. EXISTING SIDEWALK TO REMAIN.
13. NEW COMPACTED GRAVEL.
14. NEW 6' CONCRETE BUMPER BLOCK SECURED WITH (2) #4 BARS 4' LONG, SEE CIVIL.
15. NEW SITE AREA LIGHT POLE, SEE ELECTRICAL.

SITE				
	Description	Label	Quantity	Unit
	SITE	GRAVEL	2,071.04	sf
	SITE	LD ASPHALT	13,427.72	sf

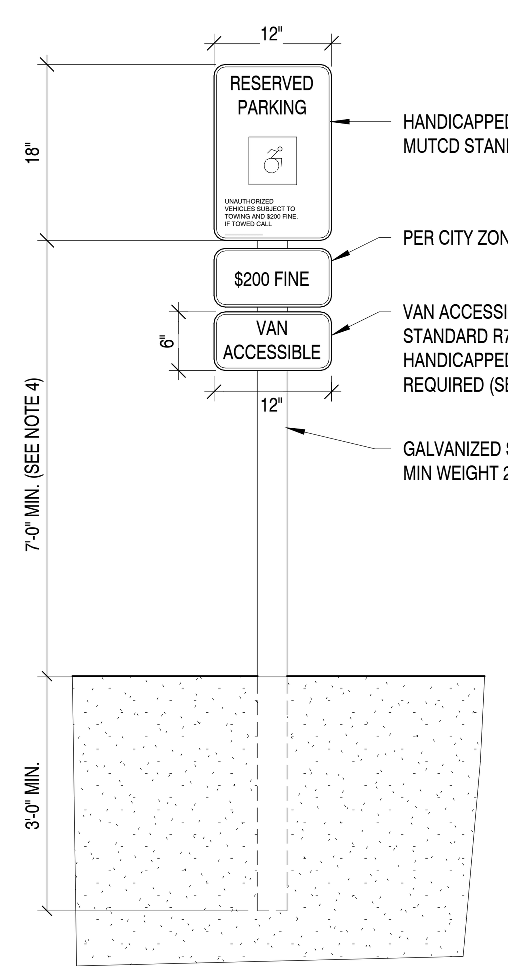
CONCRETE				
	Description	Label	Quantity	Unit
	CONCRETE	SIDEWALK	40.74	cu yd



2 WHEEL STOP DETAIL
1 1/2" = 1'-0"

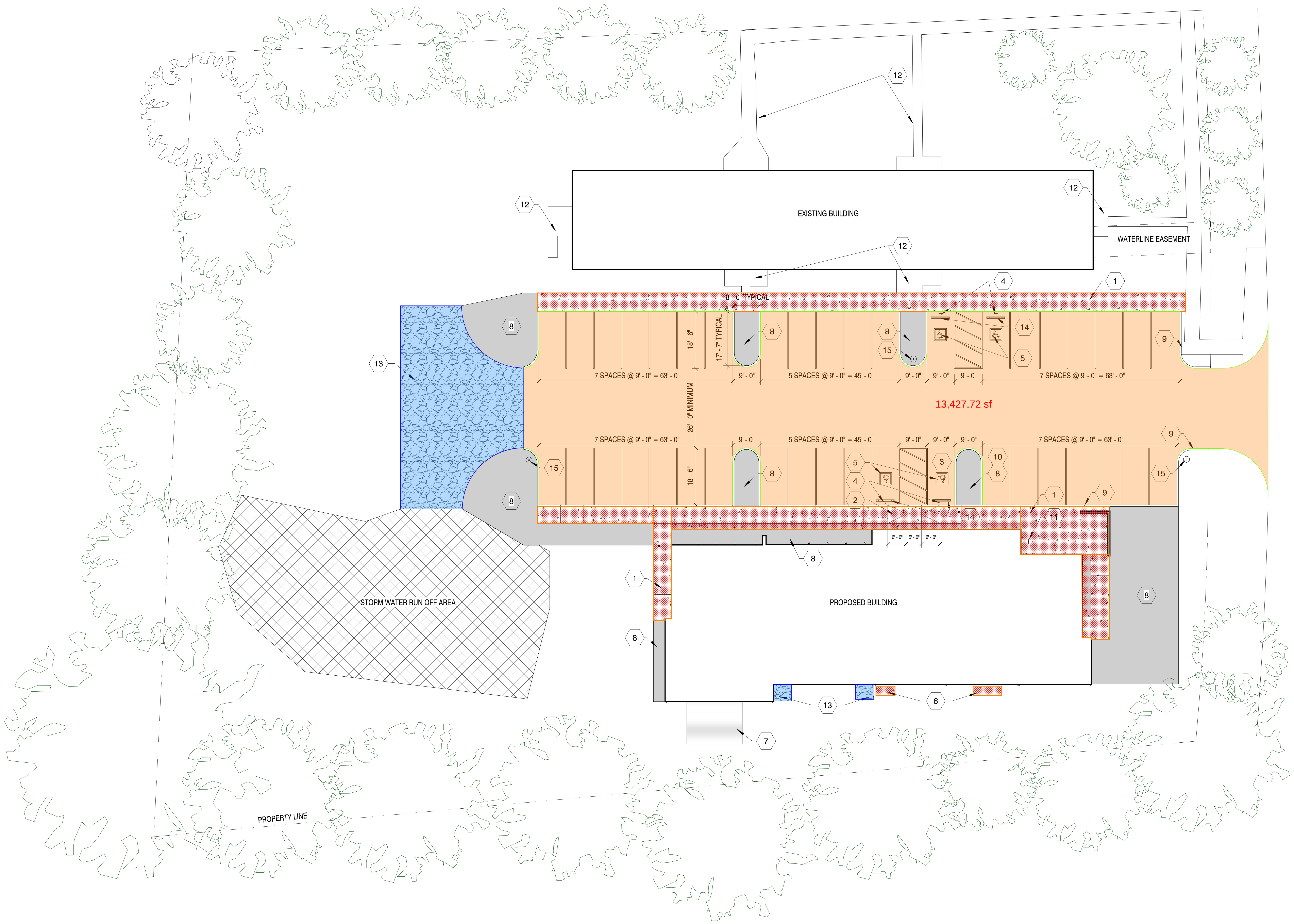


3 ADA PARKING SPACE
3/4" = 1'-0"



NOTES:
1. CONTRACTOR TO PROVIDE NEW SIGN AT ALL ACCESSIBLE PARKING STALLS.
2. CONTRACTOR TO VERIFY NAME AND NUMBER OF CONTACT PERSON OR COMPANY PRIOR TO FABRICATION OF SIGNAGE.
3. AT LEAST (1) PARKING SPACE SHALL BE VAN ACCESSIBLE PER SECTION 11.03(f)
4. MOUNTING HEIGHT FOR SIGN MAY BE REDUCED TO 5'-0" ABOVE GROUND LEVEL IF IT IS LOCATED IN A NON-PEDESTRIAN AREA, SUCH AS ON A BUILDING OR IN A LANDSCAPED AREA.

4 STANDARD HANDICAP SIGN
1" = 1'-0"



1 LANDSCAPING PLAN
1" = 20'-0"

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THOMAS DANIEL BRITT
REGISTERED ARCHITECT
STATE OF TENNESSEE
No. 102274
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

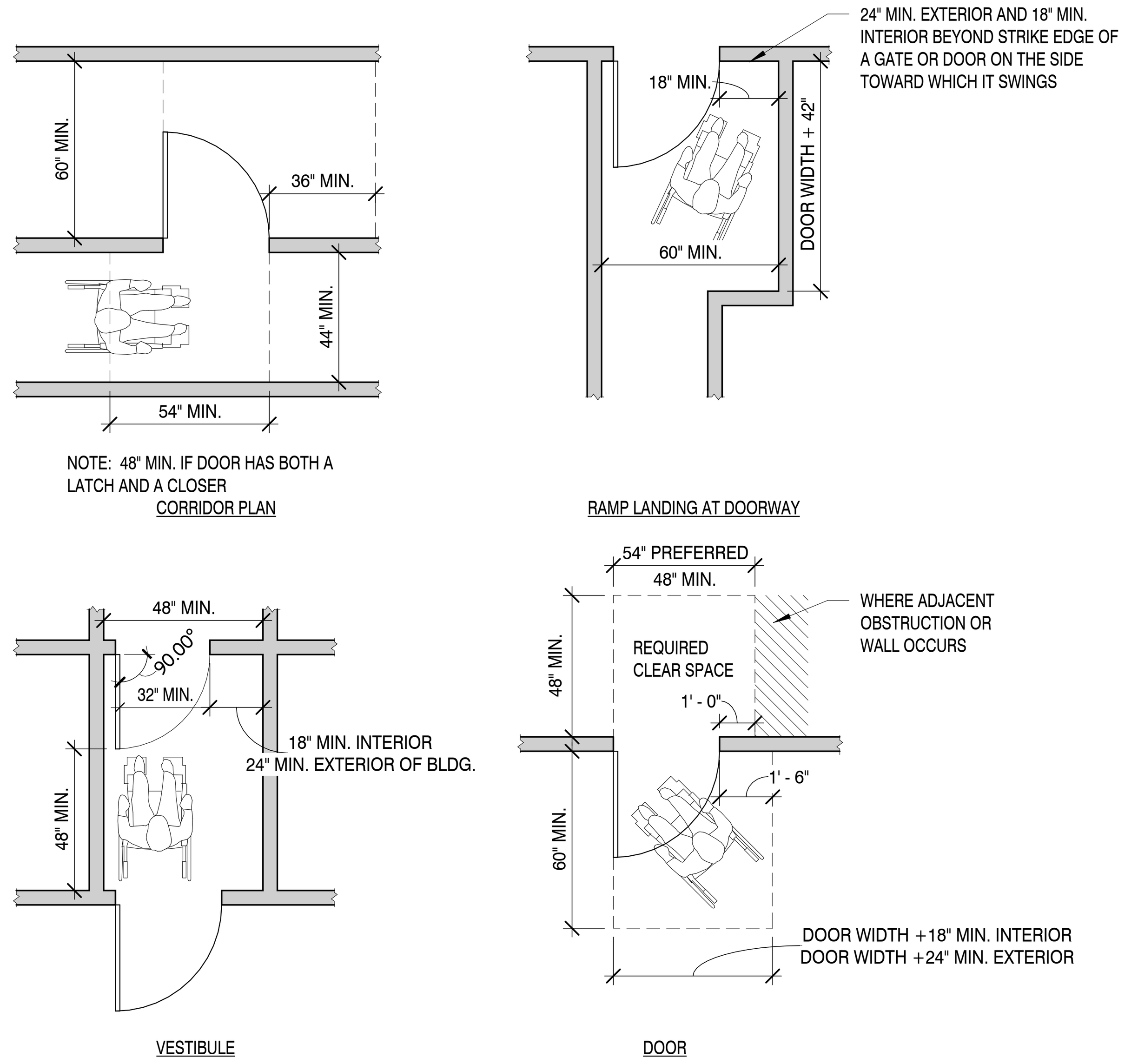
Drawing Title:
SITE PLAN AND DETAILS

Date: 11/26/2025

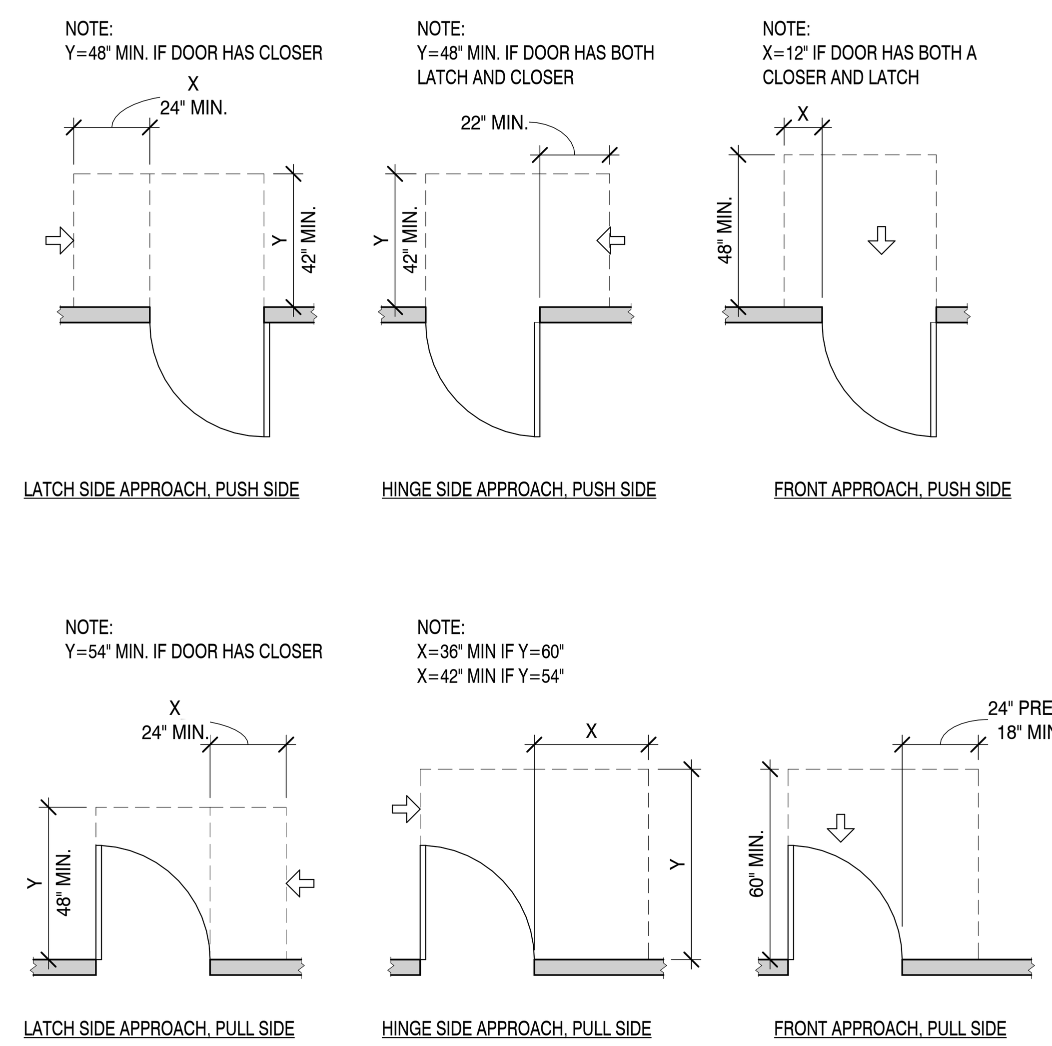
Project No.
25026

Sheet No.
AS2.1

NOTES:
1. CLEAR SPACES MUST BE LEVEL TO PREVENT WHEELCHAIRS FROM ROLLING WHEN THE OCCUPANT RELEASES THE WHEEL GRIPS TO REACH FOR THE DOOR. 1/4" SLOPE PER FOOT IS ALLOWED FOR DRAINAGE.
2. WHERE DOORS OPEN ONTO BUT NOT INTO A CORRIDOR. THE REQUIRED LEVEL AREA BEYOND THE DOORS MAY BE A MINIMUM OF 48" A.F.F.

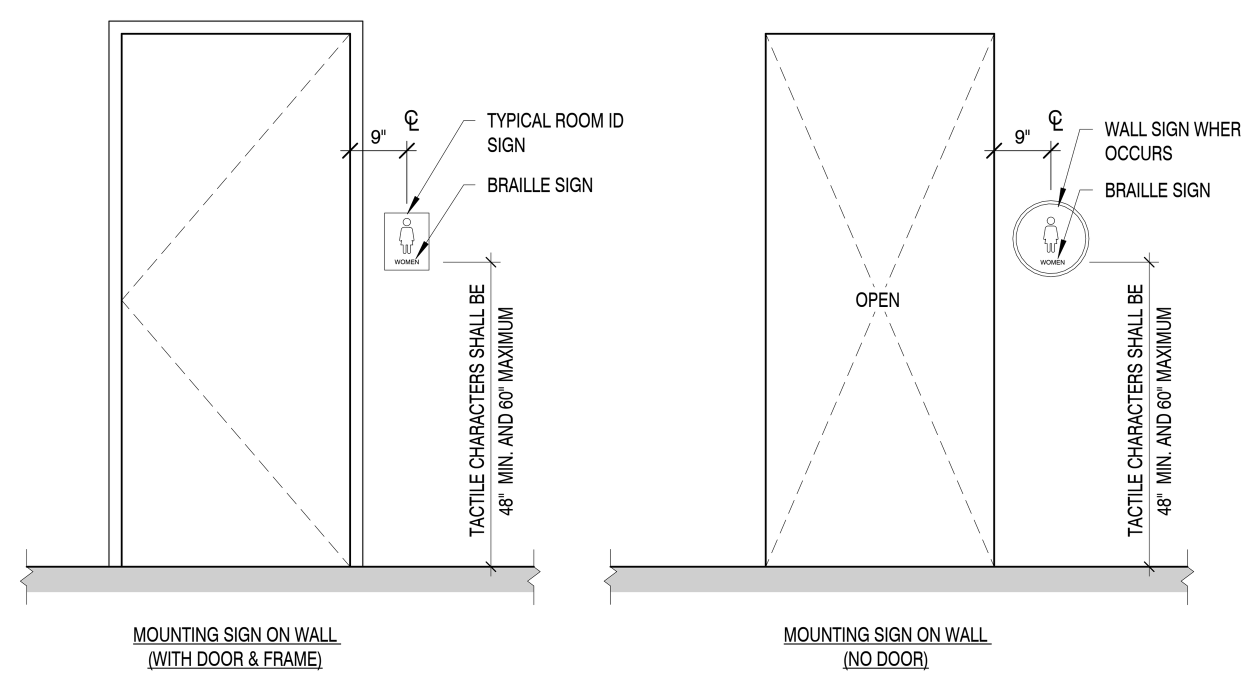


1 CORRIDOR CLEARANCES
1/4" = 1'-0"

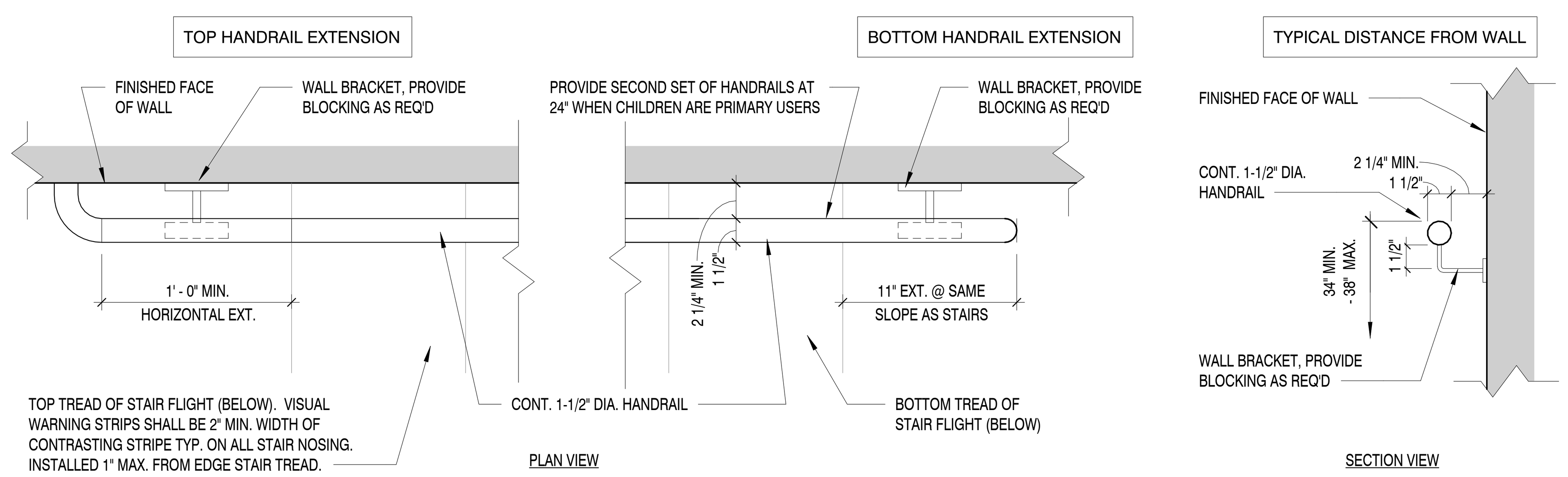


2 DOOR CLEARANCES
1/4" = 1'-0"

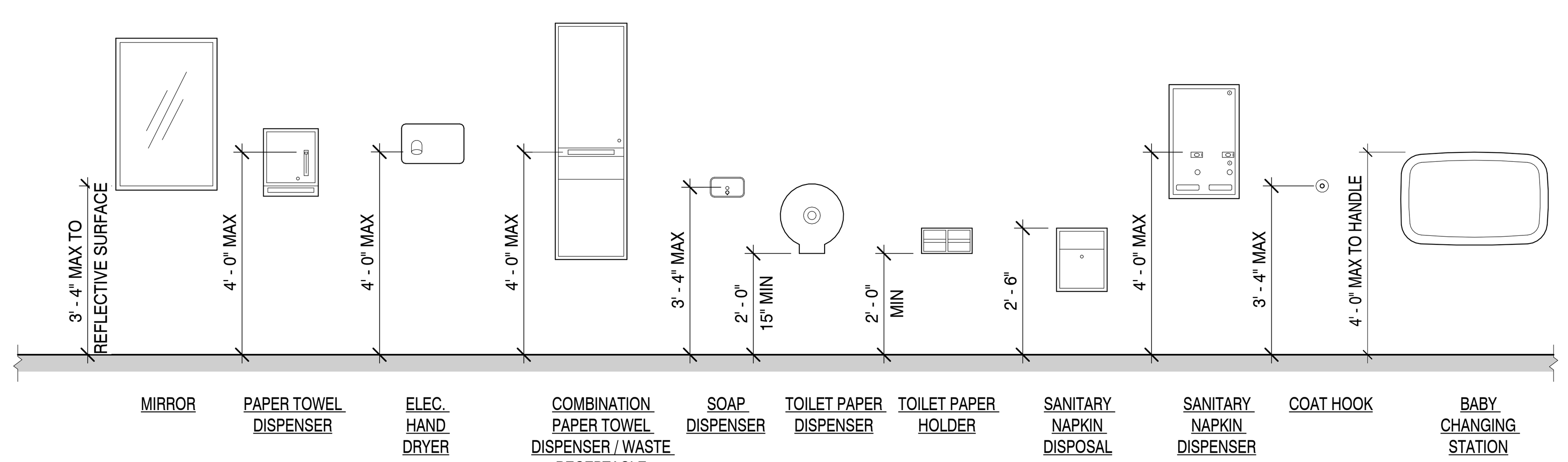
NOTES:
1. WHERE A SIGN CONTAINING RAISED CHARACTERS AND BRAILLE IS PROVIDED AT A DOOR, THE SIGN SHALL BE ALONGSIDE THE DOOR AT THE LATCH SIDE. WHERE A SIGN CONTAINING RAISED CHARACTERS AND BRAILLE IS PROVIDED AT DOUBLE DOORS WITH ONE ACTIVE LEAF, THE SIGN SHALL BE LOCATED ON THE INACTIVE LEAF. WHERE A SIGN CONTAINING RAISED CHARACTERS AND BRAILLE IS PROVIDED AT DOUBLE DOORS WITH TWO ACTIVE LEAVES, THE SIGN SHALL BE TO THE RIGHT OF THE RIGHT-HAND DOOR. WHERE THERE IS NO WALL SPACE ON THE LATCH SIDE OF A SINGLE DOOR, OR TO THE RIGHT SIDE OF DOUBLE DOORS, SIGNS SHALL BE ON THE NEAREST ADJACENT WALL. SIGNS CONTAINING RAISED CHARACTERS AND BRAILLE SHALL BE LOCATED SO THAT A CLEAR FLOOR AREA 18 INCHES (455 MM) MINIMUM BY 18 INCHES (455 MM) MINIMUM, CENTERED ON THE RAISED CHARACTERS IS PROVIDED BEYOND THE ARC OF ANY DOOR SWING BETWEEN THE CLOSED POSITION AND 45 DEGREE OPEN POSITION.
2. RAISED CHARACTERS SHALL BE 48 INCHES (1220 MM) MINIMUM ABOVE THE FLOOR, MEASURED TO THE BASELINE OF THE LOWEST RAISED CHARACTER AND 60 INCHES (1525 MM) MAXIMUM ABOVE THE FLOOR, MEASURED TO THE BASELINE OF THE HIGHEST RAISED CHARACTER.



3 SIGNAGE NOTES AND DETAILS
1/2" = 1'-0"

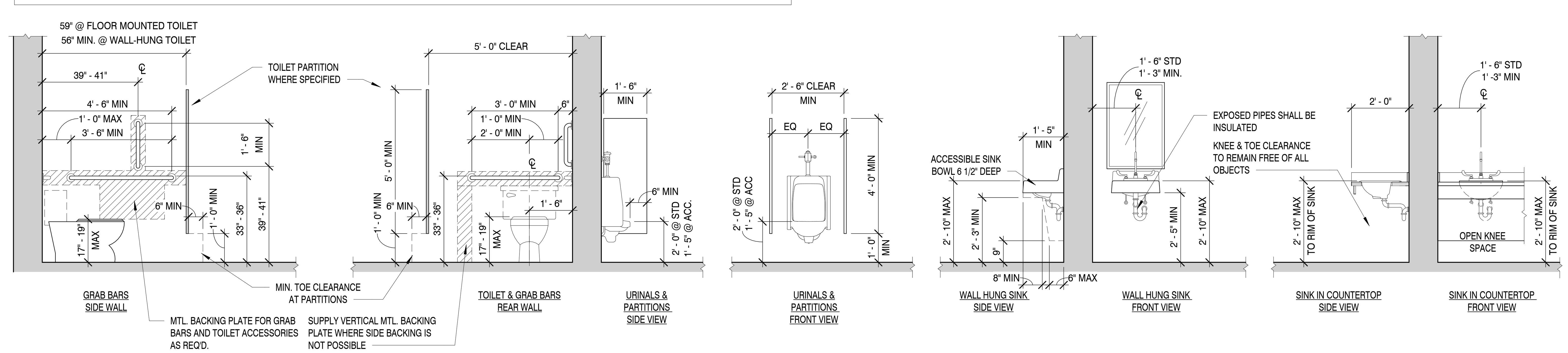


4 STAIR HANDRAIL DETAILS AND NOTES
1 1/2" = 1'-0"



5 RESTROOM ACCESSORY MOUNTING HEIGHTS
3/8" = 1'-0"

NOTES:
1. ALL GRAB BARS SHALL BE 1 1/4" - 1 1/2"Ø W/ 1 1/2" CLEAR BETWEEN GRAB BAR & FINISH SURFACE OF WALL.
2. FIXED SIDE WALL GRAB BARS SHALL BE 42" MIN. IN LENGTH, LOCATED 12" MAX. FROM REAR WALL AND EXTENDING 54" MIN. FROM REAR WALL. IN ADDITION, A VERT. GRAB BAR 18" MIN. IN LENGTH SHALL BE MOUNTED WITH THE BOTTOM OF THE BAR BETWEEN 39" - 41" A.F.F. AND THE CENTER LINE OF THE BAR LOCATED BETWEEN 39" - 41" FROM REAR WALL. [ANSI 604.5.1]
3. REAR WALL GRAB BAR SHALL BE 36" MIN. IN LENGTH, AND EXTEND FROM THE CENTERLINE OF THE WATER CLOSET 12" MIN. ON SIDE CLOSEST TO WALL, AND 24" MIN. ON THE TRANSFER SIDE. [ANSI 604.5.2]
4. ALLOWABLE STRESS SHALL NOT BE EXCEED FOR MATERIALS USED WHERE A VERT. OR HORZ. FORCE OF 250 POUNDS IS APPLIED AT ANY POINT ON THE GRAB BAR, FASTENER MOUNTING DEVICE, OR SUPPORTING STRUCTURE. [ANSI 609.8]
5. GRAB BARS SHALL BE INSTALLED IN A HORIZ. POSITION, 33" MIN - 36" MAX. A.F.F. MEASURED TO THE TOP OF THE GRIPPING SURFACE. [ANSI 609.4]



6 PLUMBING FIXTURES DETAILS
3/8" = 1'-0"

GENERAL NOTES

1. THE ARCHITECT HAS MADE EVERY EFFORT TO SET FORTH IN THE CONTRACT DOCUMENTS THE COMPLETE SCOPE OF WORK. THE CONTRACTOR IS NEVERTHELESS CAUTIONED THAT MINOR OMISSIONS AND DISCREPANCIES IN THE DRAWINGS AND SPECIFICATIONS SHALL NOT EXCUSE HIM FROM PROVIDING A COMPLETED FACILITY AND IMPROVEMENTS IN ACCORDANCE WITH THE INTENT OF THESE DOCUMENTS. IN THE EVENT OF DISCREPANCIES, CONTRACTOR SHALL PRICE THE MORE EXPENSIVE AND EXTENSIVE WORK, UNLESS DIRECTED OTHERWISE.

2. DETAILS ARE INTENDED TO SHOW END RESULT OF DESIGN. MINOR MODIFICATIONS MAY BE REQUIRED TO SUIT JOB DIMENSIONS OR CONDITIONS, AND SUCH MODIFICATIONS SHALL BE INCLUDED AS PART OF THE WORK. THE DRAWINGS, GENERAL NOTES AND SPECIFICATIONS ARE COMPLIMENTARY, AND WHAT IS CALLED FOR BY ANY WILL BE BINDING AS IF CALLED FOR BY ALL. WORK SHOWN OR REFERRED TO ON ANY DRAWING SHALL BE PROVIDED AS THOUGH SHOWN ON ALL RELATED DRAWINGS. IF THERE IS ANY CONFLICT OR DISCREPANCY WITHIN OR BETWEEN ANY OF THE CONTRACT DOCUMENTS INVOLVING THE QUALITY OR QUANTITY OF WORK REQUIRED, THE WORK OF HIGHEST QUALITY AND/OR GREATEST QUANTITY SHOWN OR SPECIFIED SHALL BE FURNISHED.

3. THE CONTRACTOR SHALL BE RESPONSIBLE FOR BECOMING FAMILIAR WITH ALL CONTRACT DOCUMENTS AND FIELD CONDITIONS, AND CONFIRM THAT THE WORK MAY BE ACCOMPLISHED AS SHOWN PRIOR TO PROCEEDING WITH CONSTRUCTION.

4. THE CONTRACTOR SHALL SUPERVISE AND DIRECT THE WORK, USING HIS BEST SKILL AND ATTENTION. HE SHALL BE SOLELY RESPONSIBLE FOR ALL CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES AND PROCEDURES AND FOR COORDINATING ALL PORTIONS OF THE WORK UNDER THE CONTRACT. THE CONTRACTOR SHALL PROVIDE AT THE PROJECT SITE A FULL SET OF CONSTRUCTION DOCUMENTS ANNOTATED WITH THE LATEST REVISIONS AND CLARIFICATIONS FOR THE USE BY ALL.

5. CONDUCT OPERATIONS IN SUCH A MANNER AS TO MINIMIZE INTERFERENCE WITH USE OF PUBLIC WAYS AND ADJACENT USED FACILITIES. DO NOT CLOSE, BLOCK OR OTHERWISE OBSTRUCT USE OF PUBLIC WAYS OR FACILITIES WITHOUT WRITTEN CONSENT OF AUTHORITIES HAVING JURISDICTION. PROVIDE ALTERNATE ROUTES TO CLOSED OR OBSTRUCTED FACILITIES AS REQUIRED BY LOCAL REGULATIONS.

6. EXCEPT WHERE OTHERWISE SPECIFIED, THE CONTRACTOR SHALL AT ALL TIMES PROVIDE PROTECTION AGAINST WEATHER TO MAINTAIN ALL WORK, MATERIALS, APPARATUS, AND FIXTURES FROM INJURY OR DAMAGES. AT THE END OF THE DAY'S WORK, ALL NEW WORK LIKELY TO BE DAMAGED SHALL BE COVERED OR OTHERWISE PROTECTED AS REQUIRED.

7. THE CONTRACTOR SHALL BE RESPONSIBLE FOR PROVIDING HIS OWN TELEPHONE AND TOILET FOR ALL SCOPE OF WORK. THE CONTRACTOR SHALL BE RESPONSIBLE FOR PROVIDING ALL TAPS, EXTENSIONS, VALVES, OR OTHER DEVICES NECESSARY TO RUN POWER TOOLS AND EQUIPMENT. SUCH MODIFICATIONS TO EXISTING UTILITIES MUST BE REMOVED AT COMPLETION OF THE PROJECT.

8. THE CONTRACTOR SHALL LIMIT THE INGRESS AND EGRESS OF WORKERS AND EQUIPMENT TO THE CONSTRUCTION SITE TO AUTHORIZED PERSONS ONLY. DAMAGE TO ANY EXISTING INTERIOR OR EXTERIOR CONSTRUCTION SHALL BE REPAIRED TO "LIKE NEW" CONDITION UNDER THIS CONTRACT.

9. THE CONTRACTOR SHALL MAINTAIN AT ALL TIMES ADEQUATE SAFETY BARRICADES FOR PROTECTION OF JOB PERSONNEL AND THE PUBLIC AND CLEAR ACCESS IN AND OUT OF THE WORK SITE SO AS TO FACILITATE DAILY TRAFFIC MOVEMENT, DELIVERIES, AND SAFETY. REMOVE BARRICADES WHEN NO LONGER REQUIRED.

10. REMOVE DEBRIS, RUBBISH, AND OTHER SUBSTANCES FROM SITE. LEGALLY TRANSPORT AND DISPOSE OF SUCH MATERIALS OFF-SITE. BURYING OR BURNING OF "TO BE REMOVED" MATERIALS ON THE PROJECT SITE IS FORBIDDEN.

11. COOPERATE WITH THE APPLICABLE CITY OR OTHER GOVERNMENT OFFICIALS AND INSPECTORS AT ALL TIMES. IF SUCH OFFICIAL OR INSPECTOR DEEMS SPECIAL INSPECTION NECESSARY, PROVIDE ALL ASSISTANCE AND FACILITIES THAT WILL EXPEDITE HIS INSPECTION.

12. ALL DETAILS OF CONSTRUCTION SHALL CONFORM WITH THE APPLICABLE CODES (SEE PROJECT INFORMATION ON COVER SHEET)

13. PROVIDE HIGH SECURITY SURFACE MOUNTED BOX W/ TAMPER SWITCH (FIRE DEPARTMENT KEY BOX) AT THE ENTRANCE. THREE COMPLETE SETS OF KEYS MUST BE PROVIDED. KEYS MUST BE PROVIDED FOR ALL ROOMS CONTAINING FIRE AND LIFE SAFETY SYSTEM CONTROLS. PRIOR TO INSTALLATION VERIFY EXACT LOCATION AND EXACT TYPE OF BOX REQUIRED WITH LOCAL AUTHORITY HAVING JURISDICTION.

14. MOUNT FIRE EXTINGUISHERS LISTED IN SPECIFICATIONS AT LOCATIONS SHOWN AND/OR DIRECTED BY FIRE DEPARTMENT CODE OFFICIAL HAVING JURISDICTION.

15. INSTALL ALL MANUFACTURED ITEMS, MATERIALS, AND EQUIPMENT IN STRICT ACCORDANCE WITH MANUFACTURER'S RECOMMENDED SPECIFICATIONS, UNLESS OTHERWISE INDICATED OR WHERE LOCAL CODES OR REGULATIONS TAKE PRECEDENCE. ALL WORK PERFORMED AND MATERIALS INSTALLED SHALL BE IN STRICT ACCORDANCE, AS A MINIMUM STANDARD, WITH ALL APPLICABLE CODES, REGULATIONS AND ORDINANCES HAVING JURISDICTION. CONTRACTOR SHALL GIVE ALL NOTICES AND COMPLY WITH ALL LAWS, ORDINANCES, RULES, REGULATIONS AND LAWFUL ORDERS OF ANY PUBLIC AUTHORITY BEARING ON THE PERFORMANCE OF THE WORK.

16. PLANS ARE NOT TO BE SCALED. THESE PLANS ARE INTENDED TO BE A DIAGRAMMATIC OUTLINE ONLY, UNLESS NOTED OTHERWISE. THE WORK SHALL INCLUDE FURNISHING MATERIALS, EQUIPMENT AND APPURTENANCES, AND LABOR NECESSARY TO AFFECT ALL INSTALLATIONS INDICATED ON THE DRAWINGS. THE WORK SHALL ALSO INCLUDE ALL MATERIALS, DETAIL AND LABOR NECESSARY FOR THE SUCCESSFUL INSTALLATION OF THE WORK DESCRIBED HEREIN.

17. ALL DIMENSIONS ARE TO FACE OF CONC. BLOCK, CONC. PANEL, FACE OF EXISTING FINISH, OR FACE OF NEW STUD, UNLESS OTHERWISE NOTED. "CLEAR" DENOTES FINISH TO FINISH DIMENSIONS.

18. CONTRACTOR IS TO COORDINATE THE BUILDING PLANS WITH THE CIVIL AND SURVEY DRAWINGS FOR EXACT ELEVATIONS AND SLOPES OF EXTERIOR GRADES FOR INSTALLATION OF NEW EXTERIOR STAIRS, RAMPS AND SIDEWALKS. CONTRACTOR TO FIELD VERIFY EXTERIOR GRADES AT BUILDING ENTRANCES TO ALIGN WITH FINISHED FLOOR ELEVATIONS AND/OR NEW STAIR/RAMP ELEVATIONS. GRADING AT BUILDING PERIMETER TO SLOPE AWAY FROM BUILDING MIN. 1/4" PER FOOT.

19. ALL GRADES, LINES, LEVELS, AND DIMENSIONS SHOWN ON THE DRAWINGS SHALL BE FIELD VERIFIED BY THE CONTRACTOR PRIOR TO START OF CONSTRUCTION. ANY ERROR OR INCONSISTENCY SHALL BE REPORTED TO THE ARCHITECT FOR INSTRUCTIONS PRIOR TO START OF CONSTRUCTION.

20. CONTRACTOR IS TO FIELD VERIFY LOCATIONS OF ALL EXISTING UTILITIES PRIOR TO START OF CONSTRUCTION AND NOTIFY THE ARCHITECT OF ANY DISCREPANCIES OR POSSIBLE CONFLICTS.

21. CONTRACTOR IS TO FIELD VERIFY LOCATIONS AND RUNS OF ALL NEW AND EXISTING STORM SEWER PIPING AND ROOF TIE-INS. REPORT ANY DISCREPANCY TO THE ARCHITECT PRIOR TO START OF CONSTRUCTION.

22. DO NOT INTERRUPT EXISTING UTILITIES IN OCCUPIED FACILITIES UNLESS AUTHORIZED IN WRITING BY AUTHORITIES HAVING JURISDICTION. IF INTERRUPTION IS ALLOWED, PROVIDE ALTERNATE TEMPORARY SERVICES ACCEPTABLE TO GOVERNING AUTHORITIES. CONTRACTOR SHALL COORDINATE WITH ALL UTILITY COMPANIES 48 HOURS PRIOR TO ANY DEMOLITION WORK.

23. CONTRACTOR SHALL PERFORM HIGH QUALITY PROFESSIONAL WORK. JOIN MATERIALS TO UNIFORM ACCURATE FITS SO THEY MEET WITH NEAT, STRAIGHT LINES, FREE OF SMEARS OR OVERLAPS. INSTALL EXPOSED MATERIALS APPROPRIATELY LEVEL, PLUMB AND AT THE ACCURATE RIGHT ANGLES, OR FLUSH WITH ADJOINING MATERIALS. WORK OF EACH TRADE SHALL MEET ALL NATIONAL STANDARDS PUBLISHED BY THAT TRADE.

24. BEFORE ORDERING ANY MATERIALS OR DOING ANY WORK THE CONTRACTOR SHALL VERIFY ALL MEASUREMENTS AND SHALL BE RESPONSIBLE FOR THEIR CORRECTNESS. ANY DIFFERENCES BETWEEN DIMENSIONS INDICATED ON THE DRAWINGS SHALL BE SUBMITTED TO THE ARCHITECT FOR INSTRUCTIONS AND CONSIDERATIONS BEFORE PROCEEDING WITH THE WORK.

25. FURNISH AND INSTALL ALL REQUIRED BACKING FOR ALL SHELVES, CABINETS, FIXTURES, HANDRAILS AND EQUIPMENT. COORDINATING WITH OWNER AND CONTRACTOR FOR EXACT SIZE, NUMBER, AND LOCATION PRIOR TO START OF CONSTRUCTION. METAL BACKING PLATES TO BE FLAT STOCK (20 GAUGE MIN.) WHEN APPLIED TO METAL FRAMING. ALL WOOD BLOCKING, NAILERS, ETC. MUST BE FIRE RETARDANT TREATED.

26. GLAZING IN DOORS AND ADJACENT PANELS MUST BE TEMPERED. RESPONSIBILITY OF GLAZING SUBCONTRACTOR TO VERIFY & PLACE TEMPERED GLASS AS REQUIRED BY THE LOCAL BUILDING CODE & INSPECTOR.

27. SPOUT OUTLETS FOR WHEELCHAIR ACCESSIBLE DRINKING FOUNTAINS SHALL BE 36" MAX A.F.F. AND FOR STANDING PERSONS SHALL BE 38" MIN A.F.F. AND 43" MAX. A.F.F.

28. PROVIDE WATER RESISTANT & MOLD RESISTANT GYP. BD. AT ALL WET LOCATIONS.

29. FILL ALL C.M.U. CELLS BELOW FINISH FLOOR OR FINISHED GRADE, WHICHEVER IS HIGHER SHALL BE SOLID GROUTED.

30. PROVIDE ADA COMPLIANT SIGNAGE AT ALL TOILET AND BATHROOMS. APPROPRIATELY IDENTIFIED AS "MEN" AND "WOMEN".

31. ALL TOILET ROOMS AND BATHROOMS WALLS SHALL EXTEND FROM FINISH FLOOR TO FLOOR/ ROOF DECK ABOVE, PROVIDE SOUND BATT INSULATION IN ENTIRE STUD CAVITY.

32. ALL WALLS WITHIN 24" OF SERVICE SINK, URINAL AND/ OR WATER CLOSET SHALL HAVE A SMOOTH, HARD, NONABSORBENT SURFACE, TO A HEIGHT OF NOT LESS THAN 48 " A.F.F. IF TILE OR FRP IS NOT SPECIFIED PROVIDE EPOXY PAINT, COLOR TO BE SELECTED BY ARCHITECT.

33. ALL WALL BASE IN TOILET ROOMS, BATHROOMS AND KITCHENS SHALL BE COVED AND EXTEND UPWARD ONTO THE WALL A MIN. OF 4" A.F.F.

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THOMAS DANIEL BRITT
REGISTERED ARCHITECT
STATE OF TENNESSEE
No. 102274
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

Drawing Title:
GENERAL NOTES AND ACCESSIBILITY DETAILS

Date: 11/26/2025

Project No.
25026

Sheet No.
A0.0

BUILDING OCCUPANCY		
GROUP	CLASSIFICATION	CODE SECTION
PER IBC 2024 CHAPTER 3		
GROUP A-3	ASSEMBLY	SECTION 303, 303.4
GROUP B	BUSINESS	SECTION 304

MIXED USE & OCCUPANCY/INCIDENTAL USES: *IBC SECTIONS 508 / 509*

- ● ACCESSORY OCCUPANCIES *IBC SECTION 508.2*
- ● NONSEPARATED OCCUPANCIES *IBC SECTION 508.3*
- ○ SEPARATED OCCUPANCIES *IBC SECTION 508.4*
- ○ INCIDENTAL USES *IBC SECTION 509*

SPECIAL REQUIREMENTS: *IBC CHAPTERS 4, 5*

- ○ HIGH-RISE BUILDING *IBC SECTION 403*
- ○ ATRIUM *IBC SECTION 404*
- ○ HAZARDOUS MATERIALS *IBC SECTION 414*
- ○ MEZZANINE *IBC SECTION 505.2*
- ○ EQUIPMENT PLATFORM *IBC SECTION 505.3*

MEANS OF EGRESS
IBC 2024 SECTIONS 1005, 1010, 1011

DOORS: *PER IBC 2024 SECTION 1010.1.1*

- THE REQUIRED CAPACITY OF EACH DOOR OPENING SHALL (...) PROVIDE A MINIMUM CLEAR OPENING WIDTH OF 32 INCHES (813mm). THE CLEAR OPENING WIDTH OF DOORWAYS WITH SWINGING DOORS SHALL BE MEASURED BETWEEN THE FACE OF THE DOOR AND THE FRAME STOP, WITH THE DOOR OPEN 90 DEGREES (1.57 RAD). WHERE (...) A DOOR OPENING INCLUDES TWO DOOR LEAVES WITHOUT A MULLION, ONE LEAF SHALL PROVIDE A MINIMUM CLEAR OPENING WIDTH OF 32 INCHES (813mm). THE MINIMUM CLEAR OPENING HEIGHT OF DOORS SHALL NOT BE LESS THAN 80 INCHES (2030mm).

PER IBC 2024 SECTION 1010.1.2.1

- SIDE-HINGED SWINGING DOORS, PIVOTED DOORS AND BALANCED DOORS SHALL SWING IN THE DIRECTION OF EGRESS TRAVEL WHERE SERVING A ROOM OR AREA CONTAINING AN OCCUPANT LOAD OF 50 OR MORE PERSONS OR A GROUP H OCCUPANCY.

STAIRWAYS: *PER IBC 2024 SECTION 1011.2*

- THE MINIMUM WIDTH SHALL BE NOT LESS THAN 44 INCHES (1118mm).

PER IBC 2024 SECTION 1011.3

- STAIRWAYS SHALL HAVE A HEADROOM CLEARANCE OF NOT LESS THAN 80 INCHES (2032mm) MEASURED VERTICALLY FROM A LINE CONNECTING THE NOSINGS.

LIFE SAFETY SYSTEMS:
LIFE SAFETY SYSTEMS ARE PER IBC 2024 CHAPTER 9

- ● AUTOMATC SPRINKLER SYSTEM PER NFPA 13
- ○ FIRE ALARM SYSTEM PER NFPA 72
- ○ PORTABLE FIRE EXTINGUISHERS PER NFPA 10
- ○ STANDPIPE SYSTEM PER NFPA 14

EGRESS CAPACITY FACTORS
EGRESS CAPACITY FACTORS ARE PER: IBC 2012 SECTION 1005

- ● AUTOMATC SPRINKLER SYSTEM
- ○ EMERGENCY VOICE/ALARM COMMUNICATION SYSTEM

MINIMUM REQUIRED EGRESS WIDTH:
STAIRWAYS: 0.3
OTHER EGRESS COMPONENTS: 0.2

CONSTRUCTION TYPE AND ALLOWABLE BUILDING AREA
BUILDING AREAS AND CONSTRUCTION TYPE PER IBC CHAPTERS 5 & 6

CONSTRUCTION TYPE:
IBC CHAPTER 6

CLASS \ TYPE	I	II	III	IV	V
A					
B					●
C					
HT					

BUILDING AREA AND HEIGHT: *IBC TABLES 506.2, 504.4, 504.3*
MAXIMUM AREA ALLOWED:
ACTUAL BUILDING AREA:
MAXIMUM HEIGHT ALLOWED:
ACTUAL BUILDING HEIGHT:
MAXIMUM STORIES ALLOWED:
ACTUAL BUILDING STORIES:

BUILDING AREA MODIFICATIONS: *IBC CHAPTER 5*

- ○ FRONTAGE INCREASE *IBC SECTION 506.3*
- ○ UNLIMITED AREA *IBC SECTION 507*
- ● AUTOMATIC SPRINKLER SYSTEM

ADDITIONAL FIRE RESISTANCE RATING INFORMATION

DESCRIPTION	FIRE RATING	CODE
SHAFT / HOISTWAY ENCLOSURES		*PER IBC 713.4 / 3002.1*
• 4 STORIES OR GREATER	2 HR	
• LESS THAN 4 STORES	1 HR	
EXIT ENCLOSURES		*PER IBC 1023.2*
• 4 STORIES OR GREATER	2 HR	
• LESS THAN 4 STORES	1 HR	
EXIT PASSAGEWAYS	1 HR	*PER IBC 1024.3*

FIRE RESISTANCE RATING REQUIREMENTS
FIRE RESISTANCE RATING REQUIREMENTS PER IBC TABLE 601

TYPE OF CONSTRUCTION:	VB
PRIMARY STRUCT FRAME:	0 HOURS
BEARING WALLS (EXT):	0 HOURS
BEARING WALLS (INT):	0 HOURS
NONBEARING WALLS AND PARTITIONS (EXT):	PER IBC TABLE 705.5
NONBEARING WALLS AND PARTITIONS (INT):	0 HOURS
FLOOR CONSTRUCTION:	0 HOURS
ROOF CONSTRUCTION:	0 HOURS

IBC TABLE 705.5: EXTERIOR WALL FIRE RATING

SEPERATION DISTANCE	CONS TYPE	OCCUPANCY GROUP H	F-1, M, S-1	A, B, E, F-2, I, R, S-2, U
X < 5	ALL	3	2	1
5 ≤ X ≤ 10	IA, IVA	3	2	1
	OTHERS	2	1	1
10 ≤ X ≤ 30	IA, IB, IVA, IVB	2	1	1
	IIB, VB	1	0	0
	OTHERS	1	1	1
X ≥ 30	ALL	0	0	0

MIN # OF EXITS WHEN VALUES EXCEED TABLE 1006.2.1 PER IBC 2024 SECTIONS 1006.2.1 & 1006.2.1.1

OCCUPANT LOAD	MIN NUMBER OF EXITS PER STORY
1-500	2
501-1000	3
> 1000	4

SPACES WITH ONE EXIT OR EXIT ACCESS DOORWAY PER IBC 2024 TABLE 1006.2.1

OCCUPANCY	MAXIMUM OCCUPANT LOAD	MAX. COMMON PATH OF EGRESS TRAVEL DIST. W/O AUTO. SPRINKLER SYSTEM (FEET) OL ≤ 30	W/O AUTO. SPRINKLER SYSTEM (FEET) OL > 30	W/ AUTO. SPRINKLER SYSTEM (FEET)
A, E, M	49	75	75	75
B	49	100	75	100
F	49	75	75	100
H-1, H-2, H-3	3	NP	NP	25
H-4, H-5	10	NP	NP	75
I-1, I-2, I-4	10	NP	NP	75
I-3	10	NP	NP	100
R-1	10	NP	NP	75
R-2	10	NP	NP	125
R-3	10	NP	NP	125
R-4	10	75	75	125
S	29	100	75	100
U	49	100	75	75

EGRESS DOOR SCHEDULE

DOOR MARK	DOOR WIDTH REQUIRED	DOOR WIDTH PROVIDED	EGRESS OCCUPANT LOAD	EGRESS MAXIMUM LOAD
FIRST FLOOR				
100a	6"	67.5"	30	337
108a	0.6"	67.5"	3	337
113a	14.9"	67.5"	74.5	337
113c	11.1"	33"	55.5	165
108b	0.6"	33"	3	165
113b	3"	67.5"	15	337

EXIT ACCESS TRAVEL DISTANCE

OCCUPANCY	W/ SPRINKLER
PER IBC 2024 TABLE 1017.2	
A, E, F-1, M, R, S-1	250 FT
B	300 FT

CORRIDOR FIRE RESISTANCE RATING

OCCUPANCY	OCCUPANT LOAD	REQ'D RATING (HRS) W/ SPRINKLER
PER IBC 2024 TABLE 1020.2		
A, B, E, F, M, S, U	>30	0

DEAD END CORRIDOR

OCCUPANCY	MAX DISTANCE W/ SPRINKLER
PER IBC 2024 SECTION 1020.5	
GROUP A-3	20 FT
GROUP B	50 FT

LIFE SAFETY LOAD CALCULATIONS

AREA NO.	AREA NAME	AREA SQ.FT.	OCCUPANT LOAD FACTOR	OCCUPANT LOAD CALCULATED
2024				
FIRST FLOOR				
GSF				
111	MENS BATH	202 SF	0	
112	WOMENS BATH	204 SF	0	
106	HALL	712 SF	0	
100	ENTRY	227 SF	0	
118	SPRINKLER RISER	21 SF	0	
109	BATHROOM	64 SF	0	
120	MECHANICAL	91 SF	300	1
115	STORAGE	133 SF	300	1
102	RECEPTION	88 SF	150	1
107	STORAGE	102 SF	300	1
105	OFFICE	102 SF	150	1
114	MECH	81 SF	300	1
110	HISTORY CENTER	177 SF	150	2
101	CLINIC LOBBY	172 SF	150	2
104	EXAM ROOM	155 SF	150	2
103	EXAM ROOM	153 SF	150	2
116	KITCHEN	308 SF	200	2
108	UNFINISHED	835 SF	150	6
NSF				
117	COMPUTER LAB	575 SF	15	39
113	BANQUET HALL	1622 SF	15	109
GRAND TOTALS		6022 SF		170

LIFE SAFETY CODE SPACE FUNCTIONS

ABBREV	SPACE FUNCTION	SF PER PERSON	GROSS / NET SF
PER IBC 2024 TABLE 1004.5			
AUT	IBC 2024 - ASSEMBLY: WITHOUT FIXED SEATS: 3) UNCONCENTRATED (TABLES AND CHAIRS)	15	NSF
B	IBC 2024 - BUSINESS AREAS	150	GSF
K	IBC 2024 - KITCHENS, COMMERCIAL	200	GSF
N/A	IBC 2024 - CORE	0	GSF
SME	IBC 2024 - ACCESSORY STORAGE AREAS, MECHANICAL EQUIPMENT ROOM	300	GSF

LIFE SAFETY PLAN LEGEND

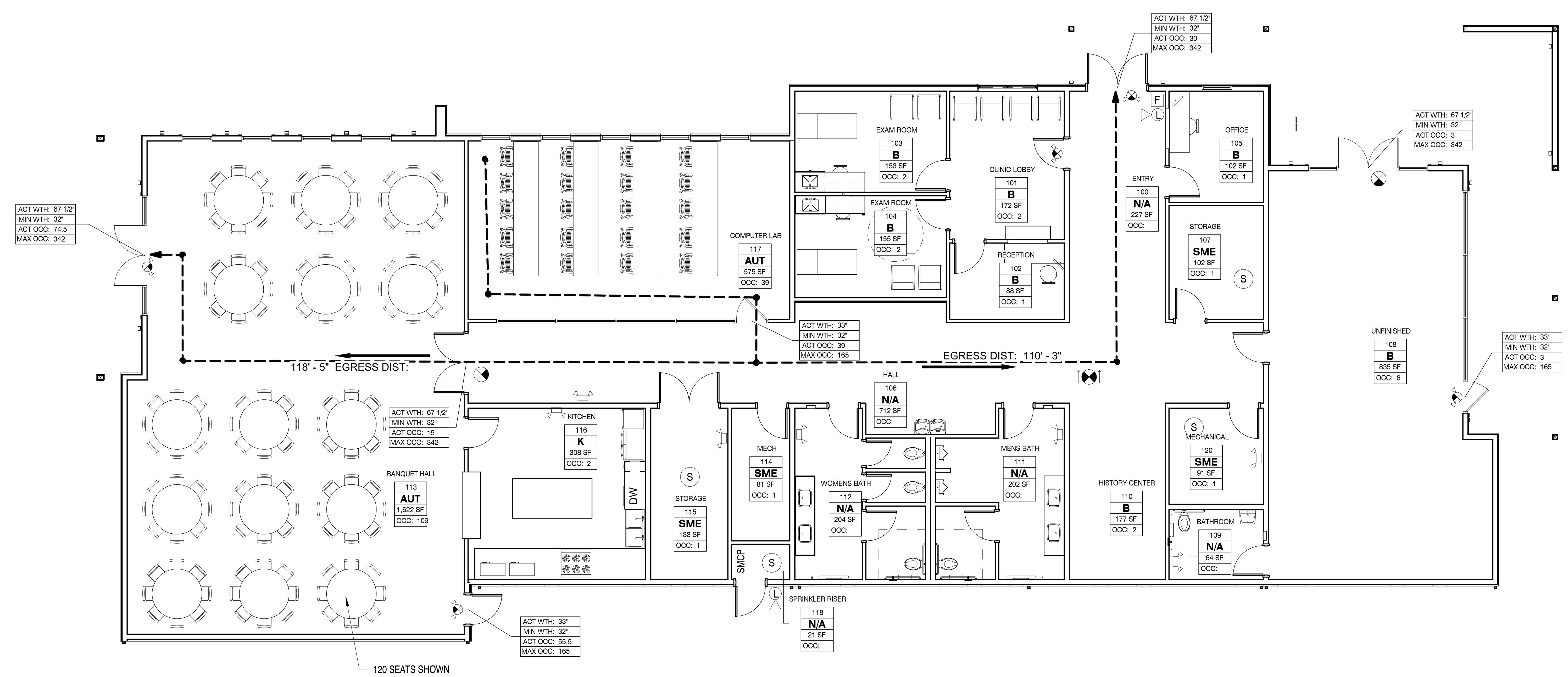
AREA OCCUPANCY TAG
AREA NAME — 101 — AST — 1,000 SF — OCC: 1
AREA NAME
AREA NUMBER
AREA OCCUPANT TYPE
AREA SQUARE FOOTAGE
AREA OCCUPANT LOAD

DOOR / STAIR EGRESS TAG
ACT WTH: XX" — ACTUAL WIDTH
MIN WTH: XX" — MIN WIDTH
ACT OCC: XX" — ACTUAL LOAD
MAX OCC: XX" — MAX LOAD

WALL MOUNTED FIRE ALARM COMBINATION AUDIO/VISUAL SPEAKER STROBE DEVICE
EXIT SIGN
EXIT SIGN AND EMERGENCY LIGHT COMBO
CEILING MOUNTED SMOKE DETECTOR DEVICE
SMCP — SPRINKLER MONITOR CONTROL PANEL

FEC — FIRE EXTINGUISHER CABINET MTD W/ HIGHEST OPERABLE PART @ 48" AFF MAX
INDICATES DISTANCE TO EXIT
EGRESS DIST: #'-#"
PATH OF TRAVEL
0 HOUR RATED WALL

NOTE: CONTRACTOR TO COORDINATE WITH LOCAL FIRE MARSHALL FOR FIRE EXTINGUISHER PLACEMENT



1 LIFE SAFETY PLAN
1/8" = 1'-0"

TAC The Architecture Collaborative
6215 Kingston Pike Knoxville, TN 37919 | www.tac-45.com 865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THOMAS DANIEL BRITT REGISTERED ARCHITECT STATE OF TENNESSEE COMMERCE No. 102274
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

Drawing Title: LIFE SAFETY PLAN

Date: 11/26/2025

Project No. 25026

Sheet No. A0.1

FINISHES					
	Description	Label	Quantity	Unit	Wall Area
	Area Measurement		20.04	sf	
	DRYWALL	A1	131.5543	ft	2,159.68
	DRYWALL	A2	124.1642	ft	2,038.36
	DRYWALL	A5	92.6593	ft	1,521.16
	DRYWALL	A5*	322.3491	ft	3,868.19
	DRYWALL	A5G*	101.1361	ft	1,213.63
	DRYWALL	A6	171.0593	ft	2,808.22
	DRYWALL	A6*	155.1070	ft	2,546.34
	DRYWALL	A6G*	82.9554	ft	1,361.85
	DRYWALL	A7G*	28.3196	ft	339.84

OPAQUE THERMAL ENVELOPE REQUIREMENTS

ROOFS
INSULATION ENTIRELY ABOVE DECK R-30CI

WALLS, ABOVE GRADE
WOOD FRAMED AND OTHER R-13 + R-3.8CI OR R-20

FLOORS
MASS R-10CI

SLAB-ON-GRADE FLOORS
UNHEATED SLAB R-10 FOR 24 BELOW

OPAQUE DOORS
SWINGING U-0.61

FOR SI: 1 INCH = 25.4 MM. CI = CONTINUOUS INSULATION. NR = NO REQUIREMENT.

FLOOR PLAN KEYNOTES

1. CONCRETE PADS FOR HEAT PUMP CONDENSOR UNITS. PADS TO BE 6 INCHES LARGER THAN THE UNITS ON ALL SIDES, TYP.
2. WOOD DECK BENEATH PACKAGED HEAT PUMP UNIT. DECK TO BE 3 FEET LARGER THAN UNIT ON ALL SIDES.
3. PER-ENGINEERED ALUMINUM COLUMN, BY OTHERS.
4. RAILINGS TO BE 42" AFF W/ PICKETS @ 4" O.C. MAX. WHERE GRADE IS GREATER THAN 30" LESS THAN 36" HORIZONTALLY FROM WALKING SURFACE (PER IBC 1015).
5. 6' PRIVACY FENCE AROUND LENGTH OF WOOD DECK.

WALL LEGEND

A1 EJ — EXTERIOR WALL - STACKED STONE STONE OVER WEATHER BARRIER OVER 5/8" WOOD SHEATHING OVER 2x6 WOOD STUDS AT 16" O.C. WITH MINIMUM R-19 BATT INSULATION AND 5/8" GYP. BOARD IN INTERIOR SIDE.

A2 EJ — EXTERIOR WALL - FIBER CEMENT VERTICAL SIDING OVER WEATHER BARRIER OVER 5/8" WOOD SHEATHING OVER 2x6 WOOD STUDS AT 16" O.C. WITH MINIMUM R-19 BATT INSULATION AND 5/8" GYP. BOARD IN INTERIOR SIDE.

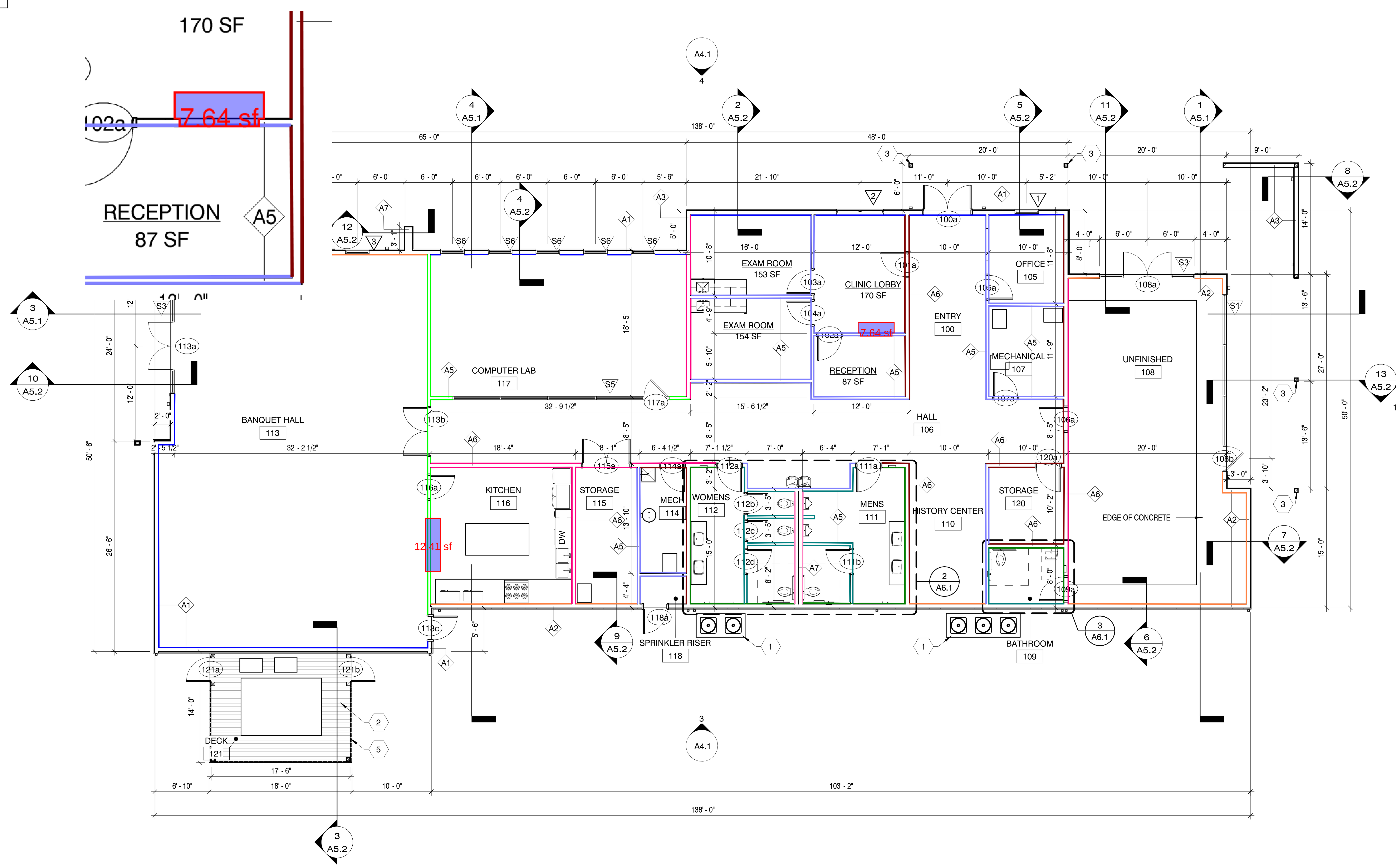
A3 EJ — EXTERIOR WALL - FIBER CEMENT HORIZONTAL SIDING OVER WEATHER BARRIER OVER 5/8" WOOD SHEATHING OVER 2x6 WOOD STUDS AT 16" O.C. WITH MINIMUM R-19 BATT INSULATION AND 5/8" GYP. BOARD IN INTERIOR SIDE.

A4 — NOT USED

A5 A6 — WALL PARTITION - 2x4(A5) OR 2x6(A6) WOOD STUDS AT 16" O.C. W/ 5/8" GYP. BD. ON EACH SIDE PLACE CONTROL JOINTS A MAXIMUM OF 30'-0" O.C.

A7 — WALL PARTITION - (2) 2x6 WOOD STUDS AT 16" O.C. SEPARATED BY 1/2" AIR GAP WITH 5/8" GYP. BD. ON EACH SIDE PLACE CONTROL JOINTS A MAXIMUM OF 30'-0" O.C.

- EJ = EXPANSION JOINT



1 FIRST FLOOR PLAN
1/8" = 1'-0"

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THOMAS DANIEL BRITT
REGISTERED ARCHITECT
COMMERCE
No. 102274
STATE OF TENNESSEE
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

Drawing Title:
FLOOR PLANS

Date: 11/26/2025

Project No. 25026

Sheet No. A1.1

WINDOW SCHEDULE					
TYPE	COUNT	WIDTH	HEIGHT	FRAME	COMMENTS
1	5	3' - 0"	4' - 0"	ALUMINUM CLAD	
2	1	6' - 0"	4' - 0"	ALUMINUM CLAD	
3	4	3' - 0"	8' - 0"	ALUMINUM CLAD	

TOTAL WINDOWS: 10

DOOR SCHEDULE

	DOORS							FRAMES			
NUMBER	FULL WIDTH	HEIGHT	THICKNESS	PANEL	D-TYPE	D-MATL	D-FINISH	F-TYPE	F-MATL	F-FINISH	REMARKS
100a	6' - 0"	8' - 0"	1 3/4"	PAIR	1	ALUM/GLASS	PRE-FINISH	1	ALUM CLAD	PRE-FINISH	INSULATED
101a	3' - 0"	8' - 0"	1 3/4"	SINGLE	4	SCW	PAINT	1	H.M.	PRE-FINISH	
102a	3' - 0"	8' - 0"	1 3/4"	SINGLE	4	SCW	PAINT	1	H.M.	PRE-FINISH	
103a	3' - 0"	8' - 0"	1 3/4"	SINGLE	4	SCW	PAINT	1	H.M.	PRE-FINISH	
104a	3' - 0"	8' - 0"	1 3/4"	SINGLE	4	SCW	PAINT	1	H.M.	PRE-FINISH	
105a	3' - 0"	8' - 0"	1 3/4"	SINGLE	4	SCW	PAINT	1	H.M.	PRE-FINISH	
106a	3' - 0"	8' - 0"	1 3/4"	SINGLE	4	SCW	PAINT	1	H.M.	PRE-FINISH	
107a	3' - 0"	8' - 0"	1 3/4"	SINGLE	4	SCW	PAINT	1	H.M.	PRE-FINISH	
108a	6' - 0"	8' - 0"	1 3/4"	PAIR	1	ALUM/GLASS	PRE-FINISH	1	ALUM CLAD	PRE-FINISH	INSULATED
108b	3' - 0"	8' - 0"	1 3/4"	SINGLE	2	ALUM/GLASS	PRE-FINISH	1	ALUM CLAD	PRE-FINISH	INSULATED
109a	3' - 0"	8' - 0"	1 3/4"	SINGLE	4	SCW	PAINT	1	H.M.	PRE-FINISH	
111a	3' - 0"	8' - 0"	1 3/4"	SINGLE	4	SCW	PAINT	1	H.M.	PRE-FINISH	
111b	3' - 0"	8' - 0"	1 3/4"	SINGLE	4	SCW	PAINT	1	H.M.	PRE-FINISH	
112a	3' - 0"	8' - 0"	1 3/4"	SINGLE	4	SCW	PAINT	1	H.M.	PRE-FINISH	
112b	2' - 6"	8' - 0"	1 3/4"	SINGLE	5	SCW	PAINT	1	H.M.	PRE-FINISH	
112c	2' - 6"	8' - 0"	1 3/4"	SINGLE	5	SCW	PAINT	1	H.M.	PRE-FINISH	
112d	3' - 0"	8' - 0"	1 3/4"	SINGLE	4	SCW	PAINT	1	H.M.	PRE-FINISH	
113a	6' - 0"	8' - 0"	1 3/4"	PAIR	1	ALUM/GLASS	PRE-FINISH	1	ALUM CLAD	PRE-FINISH	INSULATED
113b	6' - 0"	8' - 0"	1 3/4"	PAIR	1	SCW	PAINT	1	H.M.	PRE-FINISH	
113c	3' - 0"	8' - 0"	1 3/4"	SINGLE	4	H.M.	PAINT	1	H.M.	PRE-FINISH	
114a	3' - 0"	8' - 0"	1 3/4"	SINGLE	4	SCW	PAINT	1	H.M.	PRE-FINISH	
115a	6' - 0"	8' - 0"	1 3/4"	PAIR	3	SCW	PAINT	1	H.M.	PRE-FINISH	
116a	3' - 0"	8' - 0"	1 3/4"	SINGLE	4	SCW	PAINT	1	H.M.	PRE-FINISH	
117a	3' - 0"	8' - 0"	1 3/4"	SINGLE	2	ALUM/GLASS	PRE-FINISH	1	ALUM CLAD	PRE-FINISH	INSULATED
118a	3' - 0"	8' - 0"	1 3/4"	SINGLE	4	H.M.	PAINT	1	H.M.	PRE-FINISH	
120a	3' - 0"	8' - 0"	1 3/4"	SINGLE	4	SCW	PAINT	1	H.M.	PRE-FINISH	
121a	3' - 0"	6' - 0"	3/4"	SLAT	6	WOOD	STAIN		NONE	NONE	
121b	3' - 0"	6' - 0"	3/4"	SLAT	6	WOOD	STAIN		NONE	NONE	

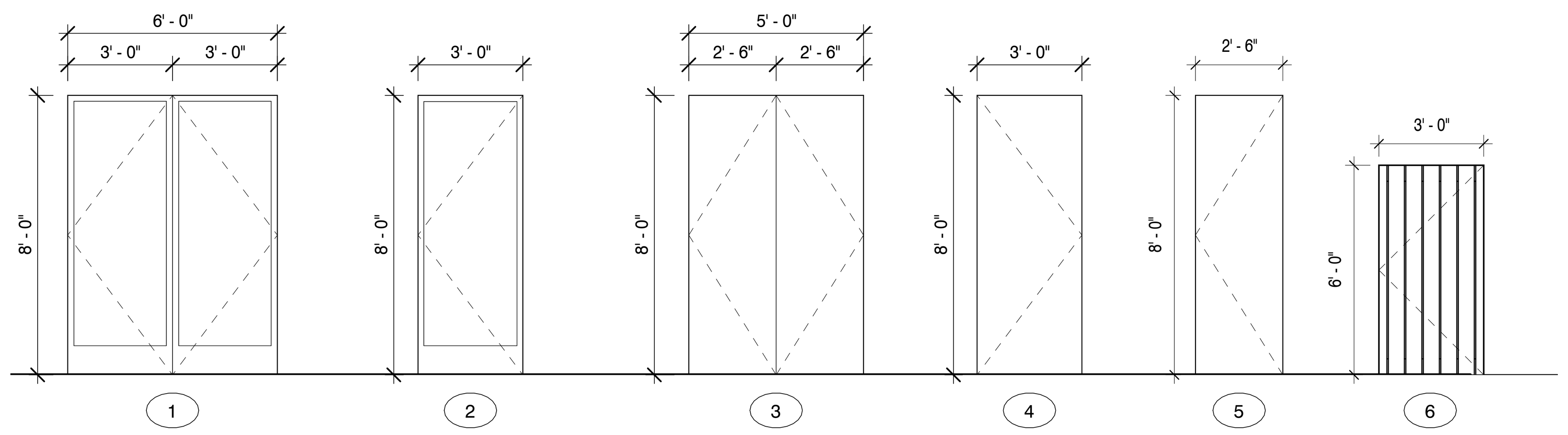
GENERAL DOOR NOTES

1. RATED DOORS ASSEMBLIES MUST HAVE RATED FRAMES, HARDWARE, CLOSERS AND OTHER RATED ACCESSORIES. ANY GLAZING SHALL BE CLEAR AND WIRELESS FIRE-RATED GLASS CERAMIC. PERMANENTLY LABEL EACH LITE W/ FIRE RATING LISTED AND LABELED BY UL.
2. CLOSERS AND POSITIVE LATCHING ARE REQUIRED ON FIRE RATED DOORS AND DOORS IN SMOKE TIGHT PARTITIONS.
3. INTERIOR WOOD DOORS TO BE FACTORY FINISH. WOOD SPECIES TO BE ROTARY CUT BIRCH.
4. EXTERIOR HOLLOW METAL DOORS ARE TO BE INSULATED.
5. EXTERIOR HOLLOW METAL DOORS AND FRAMES ARE TO BE FACTORY PRIMED AND FIELD PAINTED.

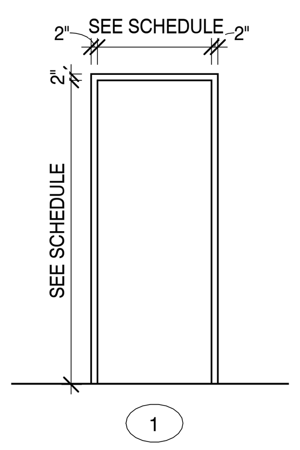
GLAZING SCHEDULE

IG — 1" THICK INSULATED GLASS WITH 1/2" AIR SPACE AND TWO 1/4" LITES

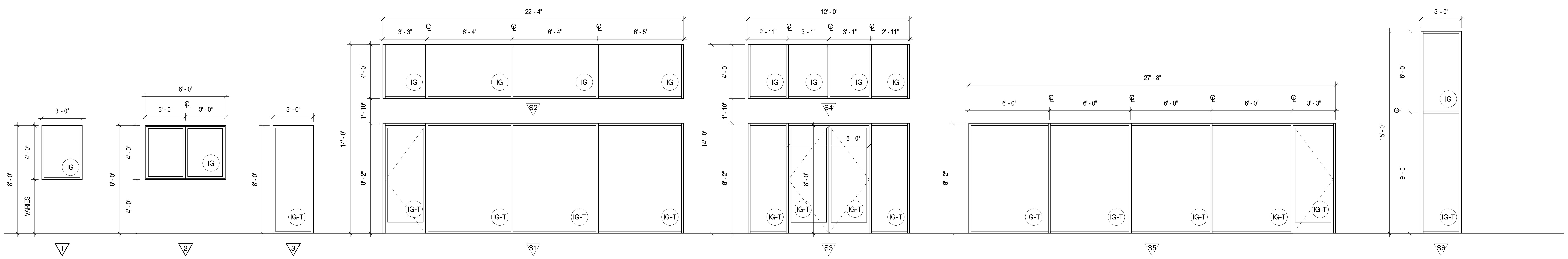
IG-T — 1" THICK INSULATED GLASS WITH 1/2" AIR SPACE AND TWO 1/4" LITES, FULLY TEMPERED



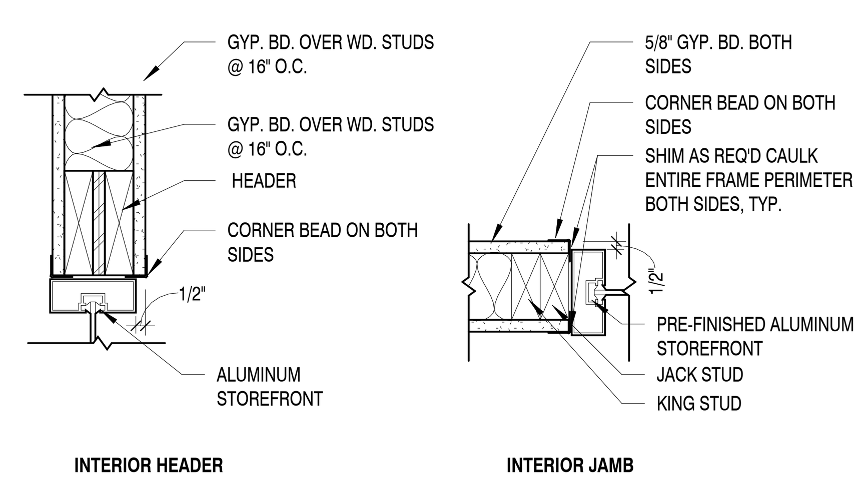
2 DOOR TYPE
1/4" = 1'-0"



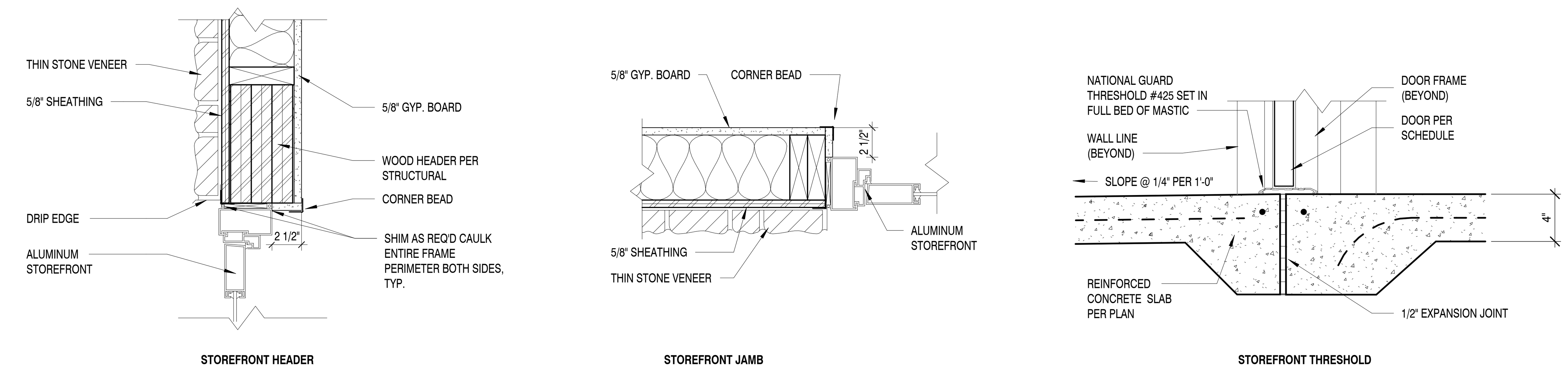
1 FRAME TYPE
1/4" = 1'-0"



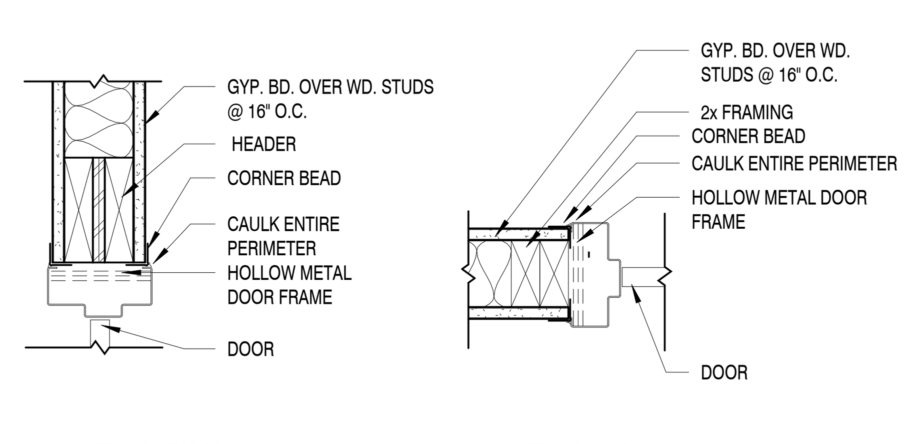
3 WINDOW/STOREFRONT ELEVATION
1/4" = 1'-0"



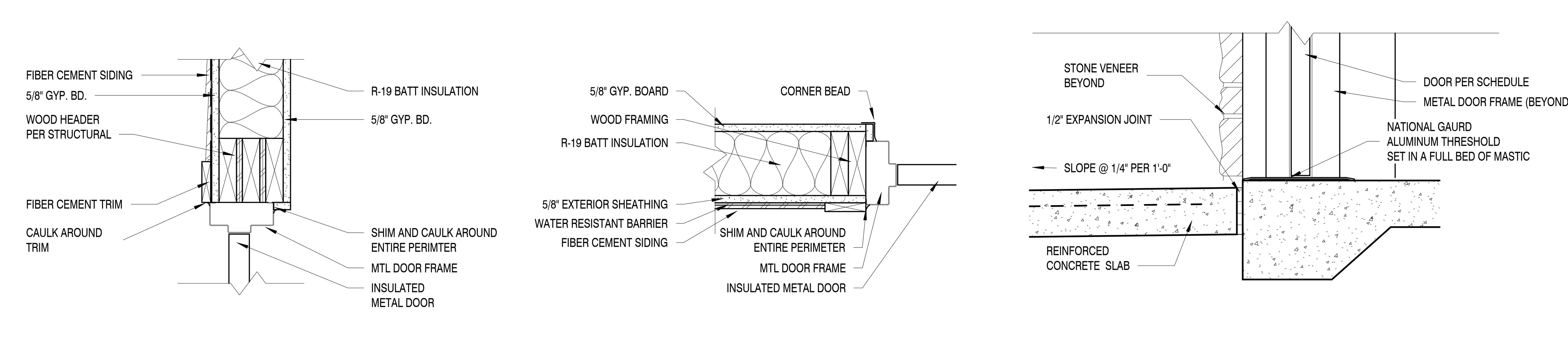
5 INTERIOR STOREFRONT DETAILS
1 1/2" = 1'-0"



4 EXTERIOR STOREFRONT DETAILS
1 1/2" = 1'-0"



7 INTERIOR DOOR DETAILS
1 1/2" = 1'-0"



6 EXTERIOR DOOR DETAILS
1 1/2" = 1'-0"

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THOMAS DANIEL BRITT
REGISTERED ARCHITECT
STATE OF TENNESSEE
No. 102274
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

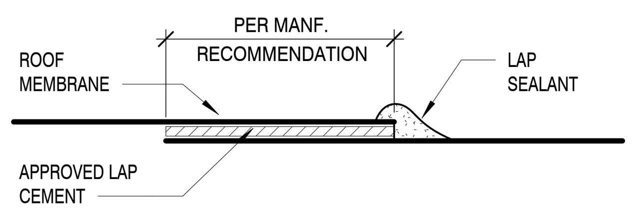
No.	Date

Drawing Title:
DOOR SCHEDULE, DOOR/FRAME ELEVATIONS & DETAILS

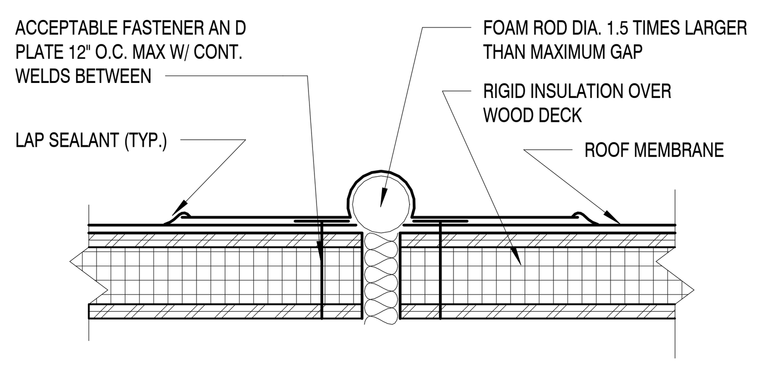
Date: 11/26/2025

Project No. 25026

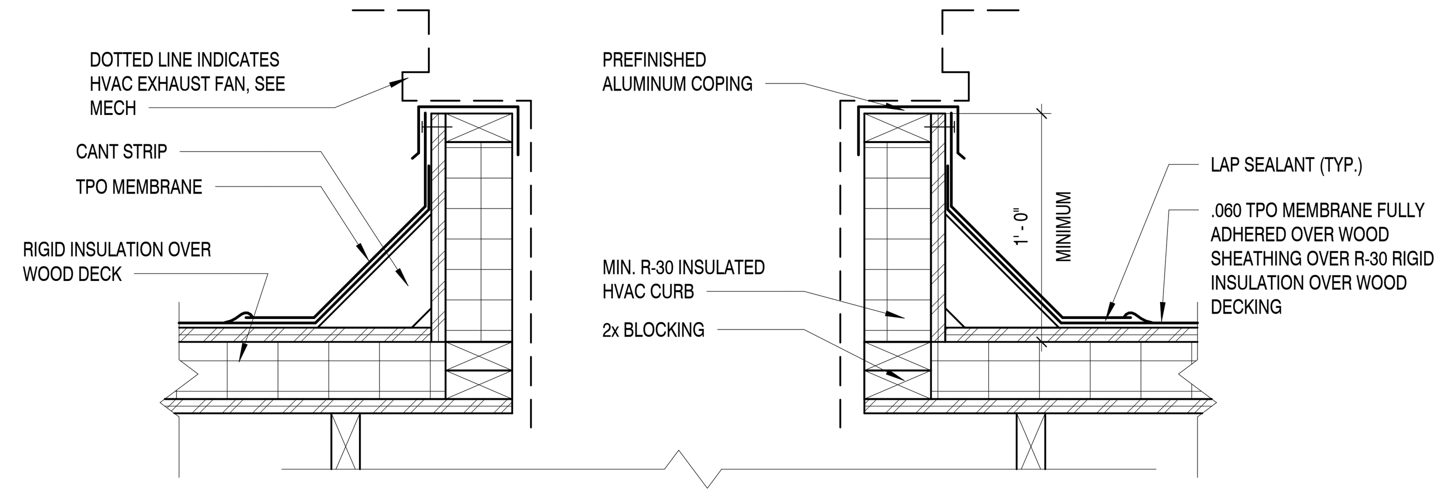
Sheet No. A2.1



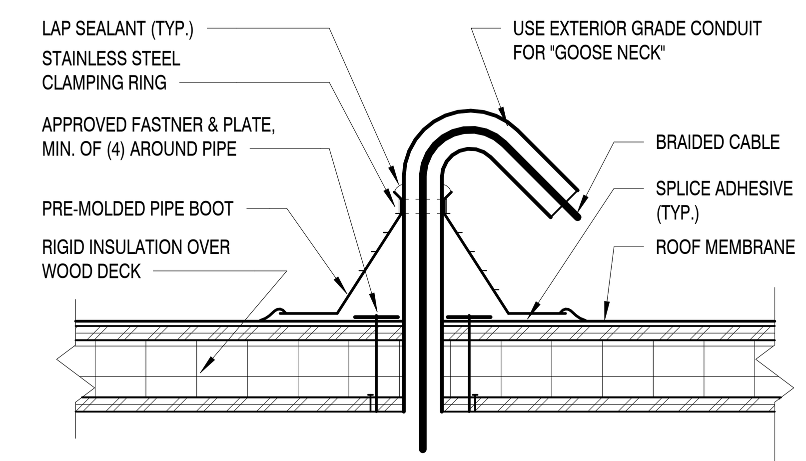
6 LAP SPLICE DETAIL
6" = 1'-0"



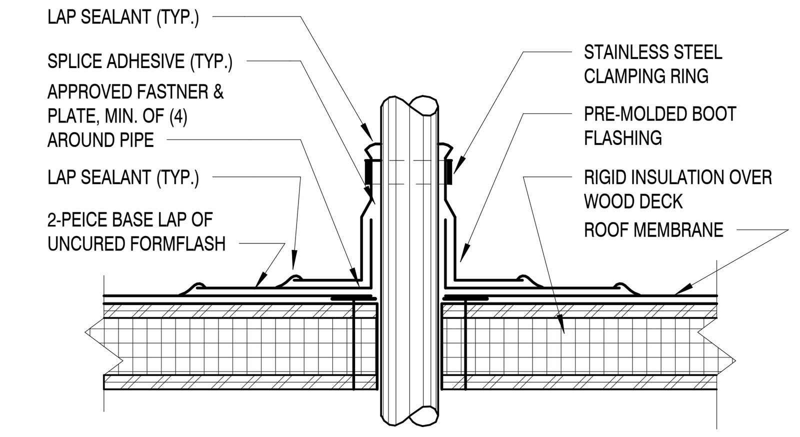
5 EXPANSION JOINT DETAIL
1 1/2" = 1'-0"



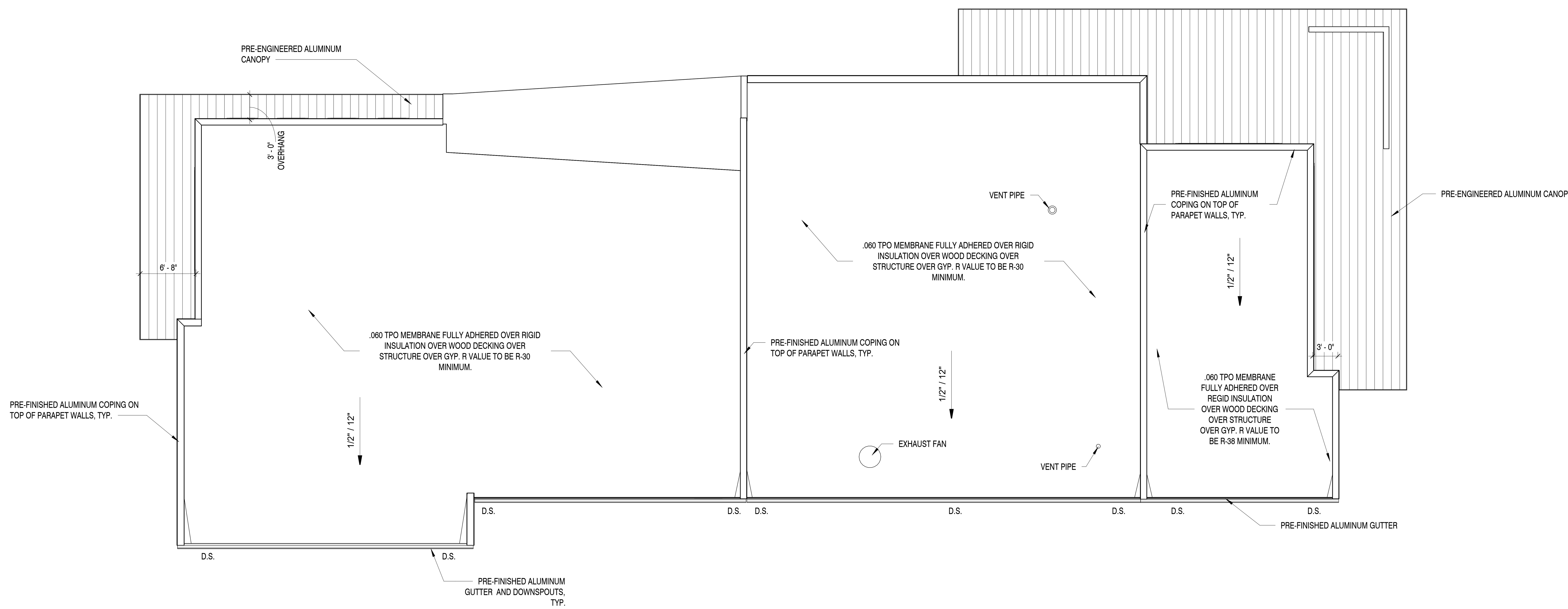
4 EXHAUST FAN PENETRATION DETAIL
1 1/2" = 1'-0"



3 ROOF CABLE PENETRATION DETAIL
1 1/2" = 1'-0"



2 VENT PIPE PENETRATION DETAIL
1 1/2" = 1'-0"



1 ROOF PLAN
1/8" = 1'-0"

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:

ELKS COMMUNITY ENRICHMENT CENTER

262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THOMAS DANIEL BRITT
REGISTERED ARCHITECT
STATE OF TENNESSEE
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

Drawing Title:
ROOF PLAN AND DETAILS

Date: 11/26/2025

Project No.
25026

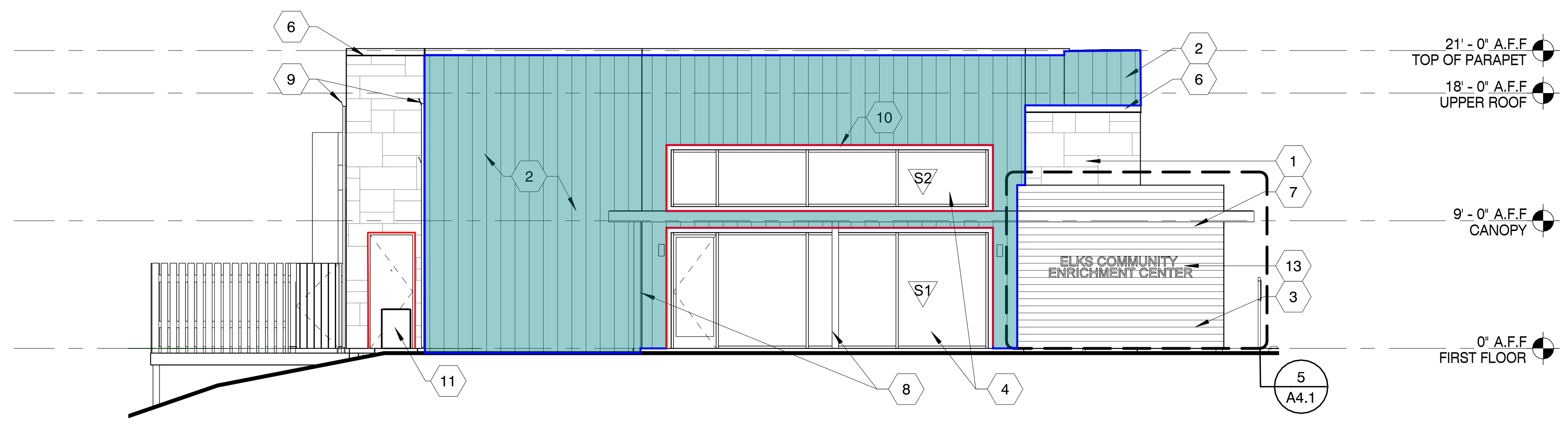
Sheet No.
A3.1

FINISHES

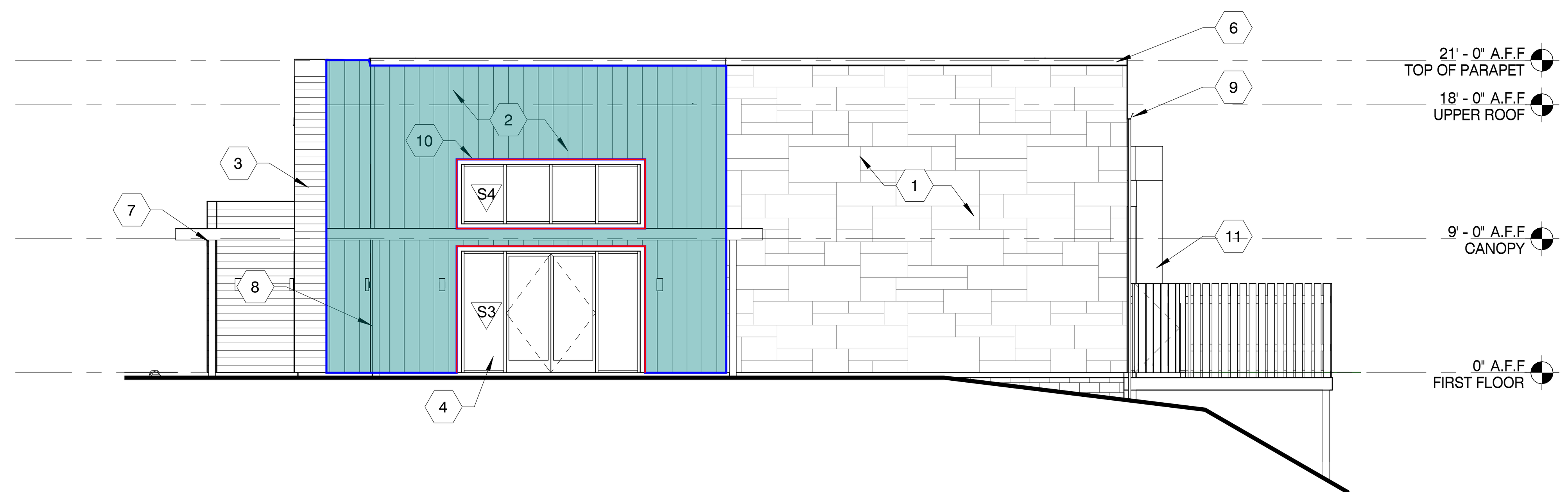
	Description	Label	Quantity	Unit
	ACT	TRIM	217.7974	ft
	SIDING	BOARD & BATEN	3,257.86	sf
	SIDING	TRIM	447.7549	ft

ELEVATION KEYNOTES

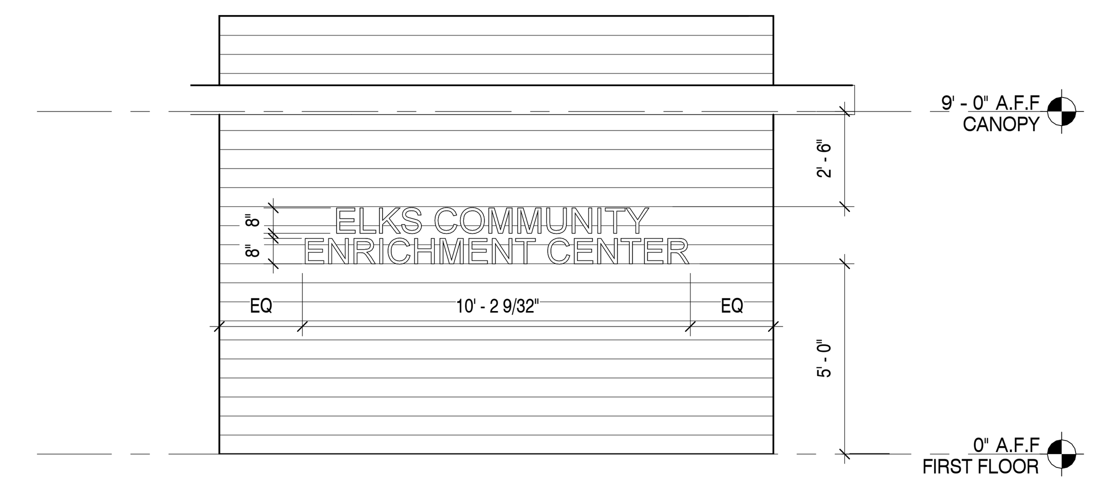
1. SYNTHETIC STONE VENEER SIDING - MANUF: ELDORADO STONE // SERIES: CUT COARSE STONE // DRYSTACKED // COLOR: CANNODADE
2. BOARD AND BATTEN FIBER CEMENT SIDING - MANUF: JAMES HARDIE // SERIES: HARDIE PANEL // TEXTURE: SMOOTH // FINISH: PRIMED FOR PAINT // COLOR: TBD
3. SHIP LAP FIBER CEMENT SIDING - MANUF: JAMES HARDIE // SERIES: HARDIE ARTISAN SIDING // TEXTURE: SHIPLAP // F INISH: PRIMED FOR PAINT // COLOR: TBD
4. PRE-FINISHED ANODIZED ALUMINUM STOREFRONT - COLOR: BLACK.
5. PRE-FINISHED ANODIZED ALUMINUM WINDOW - COLOR: BLACK
6. PRE-FINISHED ANODIZED ALUMINUM COPING - PAINTED TO MATCH SIDING
7. PRE-ENGINEERED ALUMINUM CANOPY - COLOR: BLACK
8. PRE-ENGINEERED ALUMINUM COLUMN - COLOR: BLACK
9. PRE-FINISHED ALUMINUM GUTTER AND DOWNSPOUT
10. 1x4 FIBER CEMENT TRIM AROUND WINDOWS AND STOREFRONTS - PAINT TO MATCH SIDING
11. HVAC HEAT PUMP AND PAD, SEE MECHANICAL
12. .060 TPO MEMBRANE FULLY ADHERED OVER RIGID INSULATION OVER WOOD DECKING OVER STRUCTURE OVER GYP. R VALUE TO BE R-30 MINIMUM.
13. METAL 3D LED BACKLIT SIGNAGE TO READ "ELKS COMMUNITY ENRICHMENT CENTER", SEE 5/A4.1 FOR SIZING.



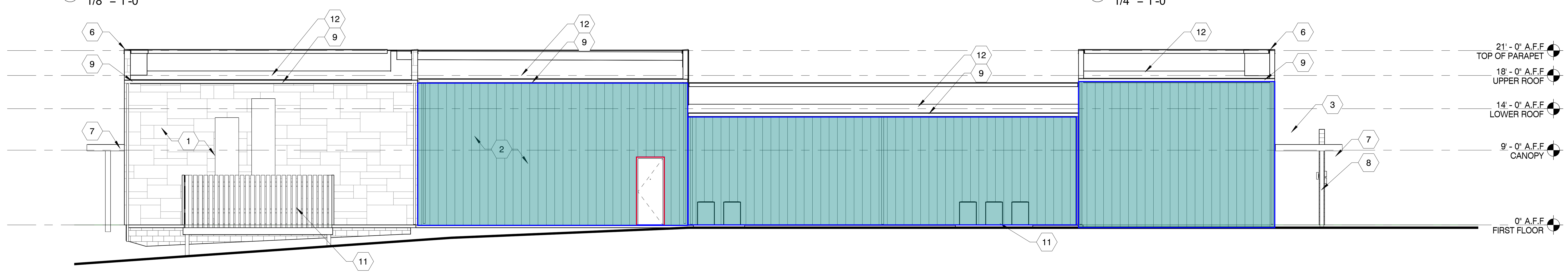
1 FRONT ELEVATION
1/8" = 1'-0"



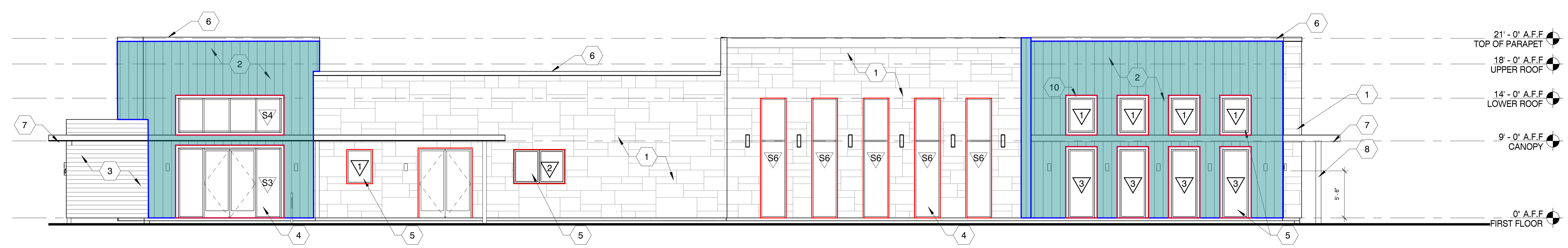
2 BACK ELEVATION
1/8" = 1'-0"



5 SIGN ELEVATION
1/4" = 1'-0"



3 LEFT ELEVATION
1/8" = 1'-0"



4 RIGHT ELEVATION
1/8" = 1'-0"

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THOMAS DANIEL BRITTAIN
REGISTERED ARCHITECT
COMMERCE
No. 102274
STATE OF TENNESSEE
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

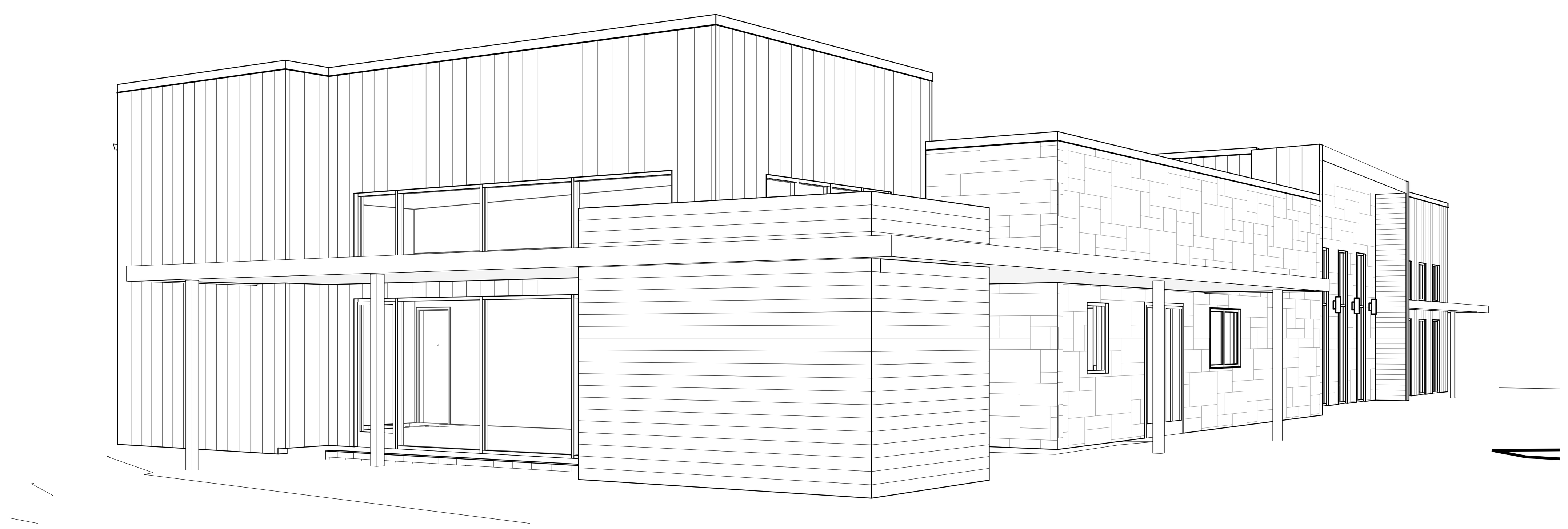
No.	Date

Drawing Title:
EXTERIOR ELEVATIONS

Date: 11/26/2025

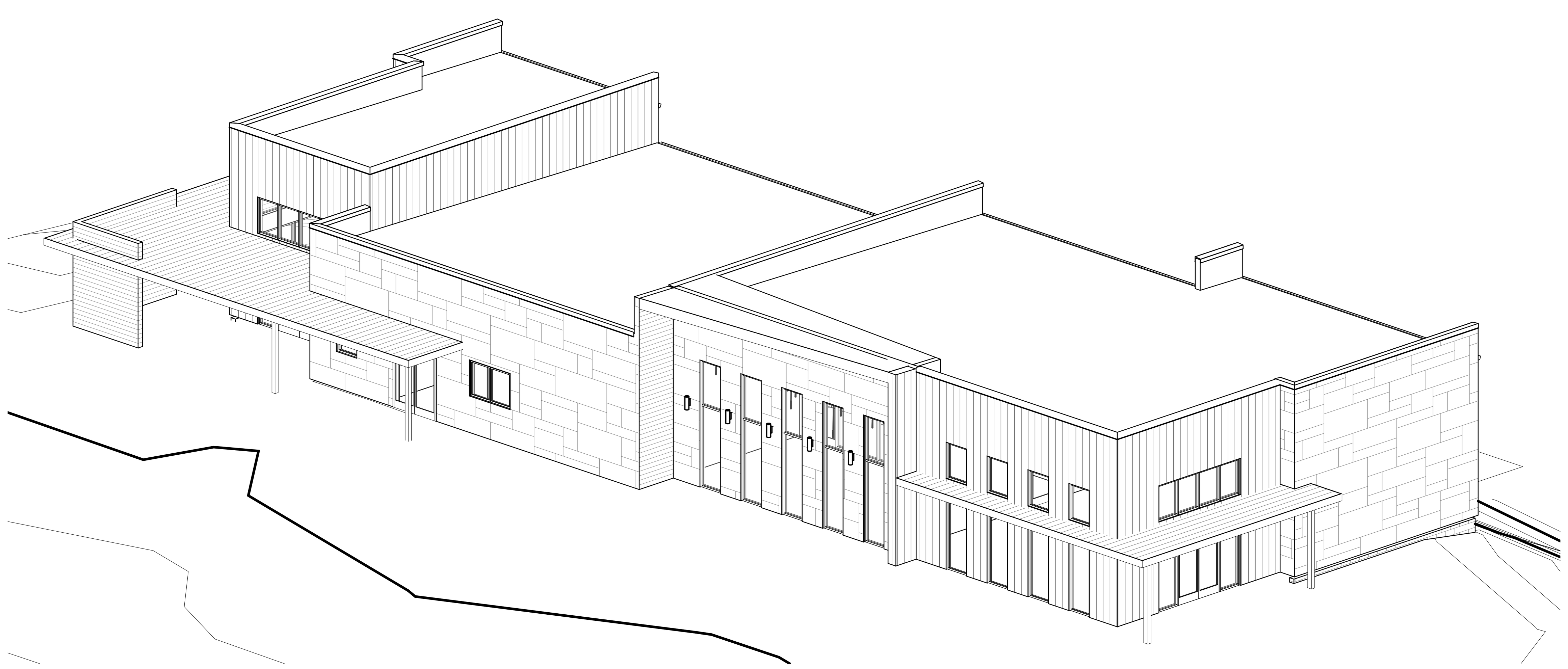
Project No.
25026

Sheet No.
A4.1

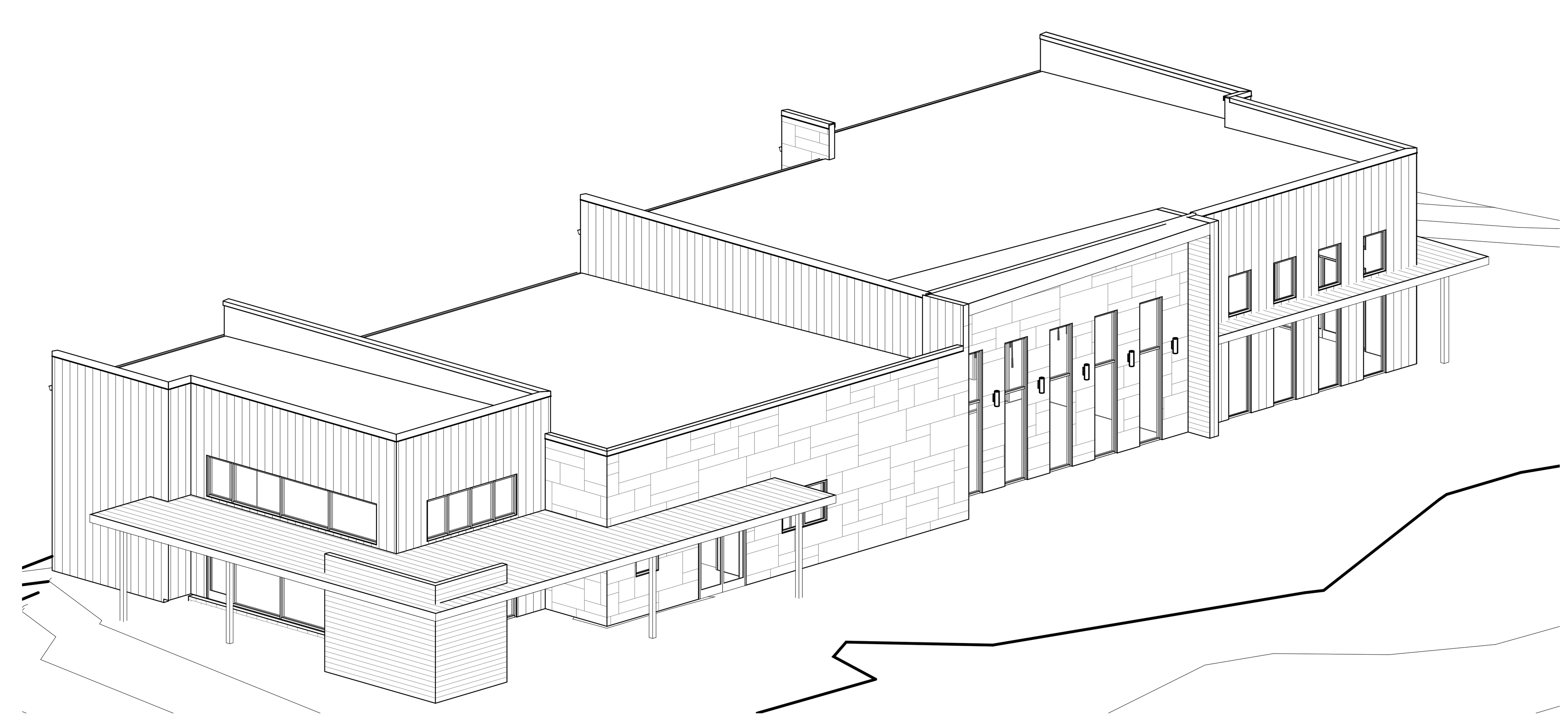


3 3D PERSPECTIVE VIEW

4 3D PERSPECTIVE VIEW



1 3D ORTHO VIEW



2 3D ORTHO VIEW

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:

ELKS COMMUNITY ENRICHMENT CENTER

262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THOMAS DANIEL BRITT
REGISTERED ARCHITECT
COMMERCE
No. 102274
STATE OF TENNESSEE
11/26/25

THIS DRAWING IS ISSUED FOR:

CONSTRUCTION DOCUMENTS

Revisions:

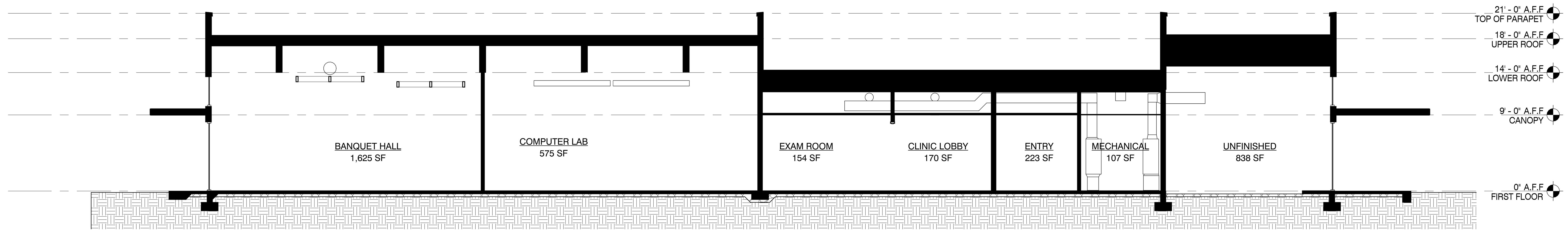
No.	Date

Drawing Title:
3D VIEWS

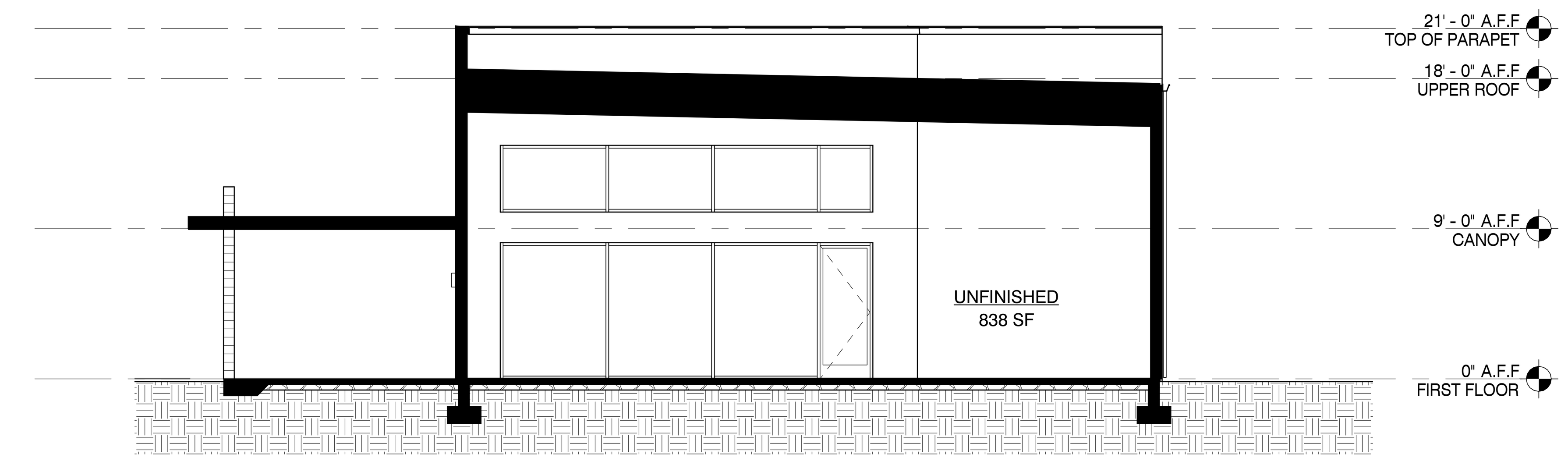
Date: 11/26/2025

Project No. 25026

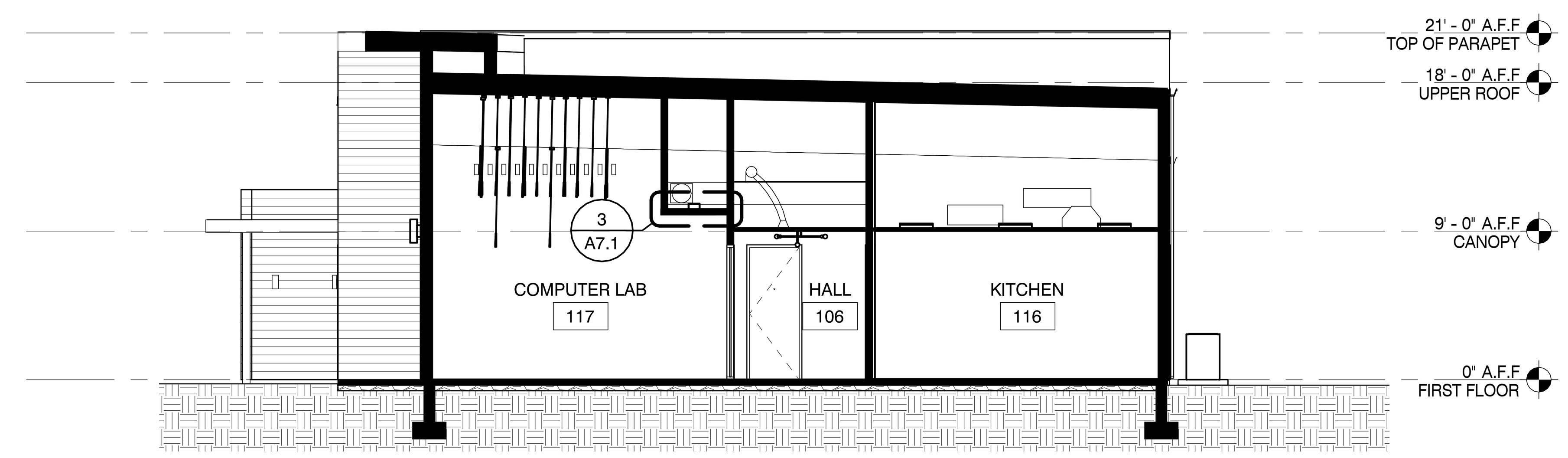
Sheet No. A4.2



3 BUILDING SECTION
1/8" = 1'-0"



1 BUILDING SECTION
1/8" = 1'-0"



4 BUILDING SECTION
1/8" = 1'-0"

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:

ELKS COMMUNITY ENRICHMENT CENTER

262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THOMAS DANIEL BRITT
REGISTERED ARCHITECT
AGRICULTURE COMMERCE
No. 102274
STATE OF TENNESSEE
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

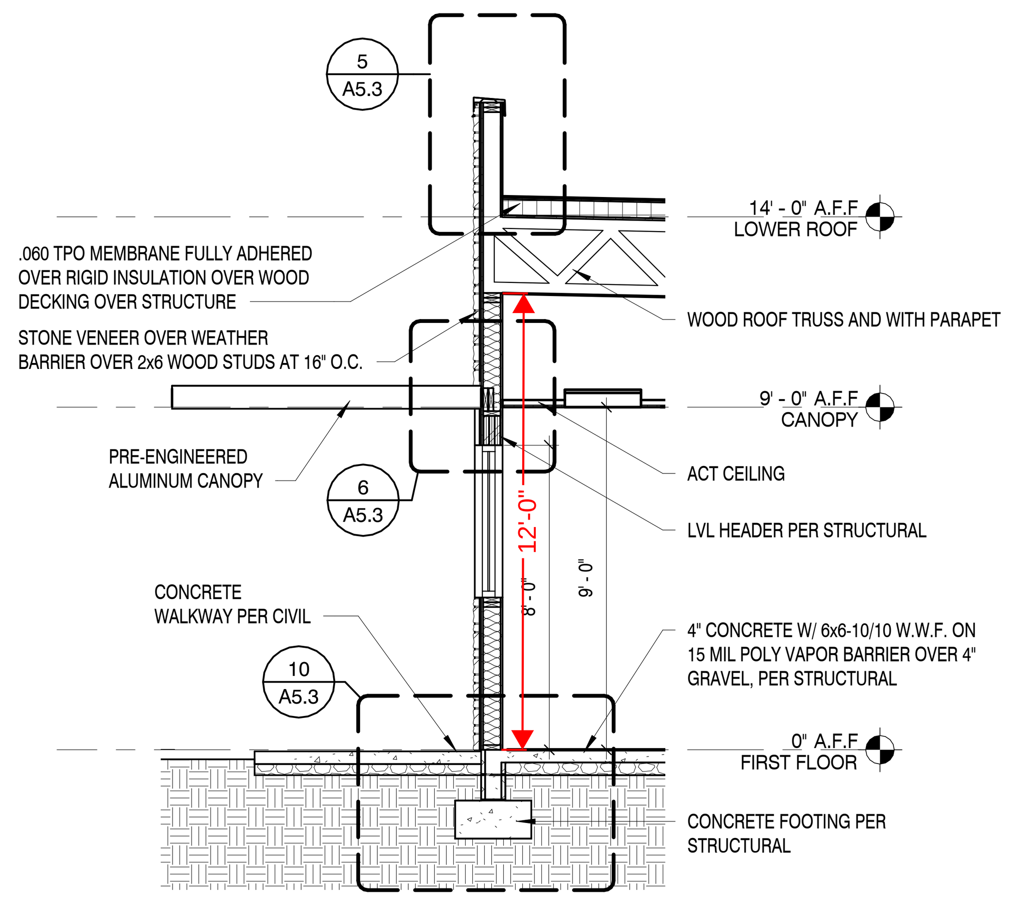
No.	Date

Drawing Title:
BUILDING SECTIONS

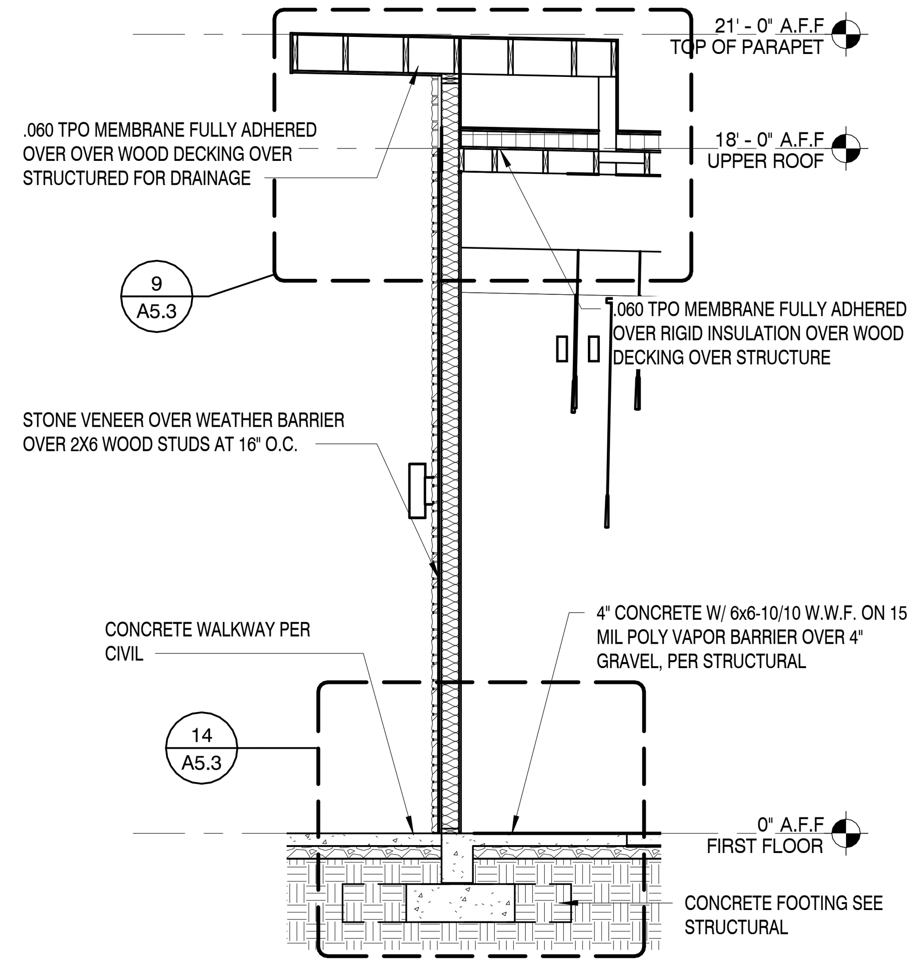
Date: 11/26/2025

Project No. 25026

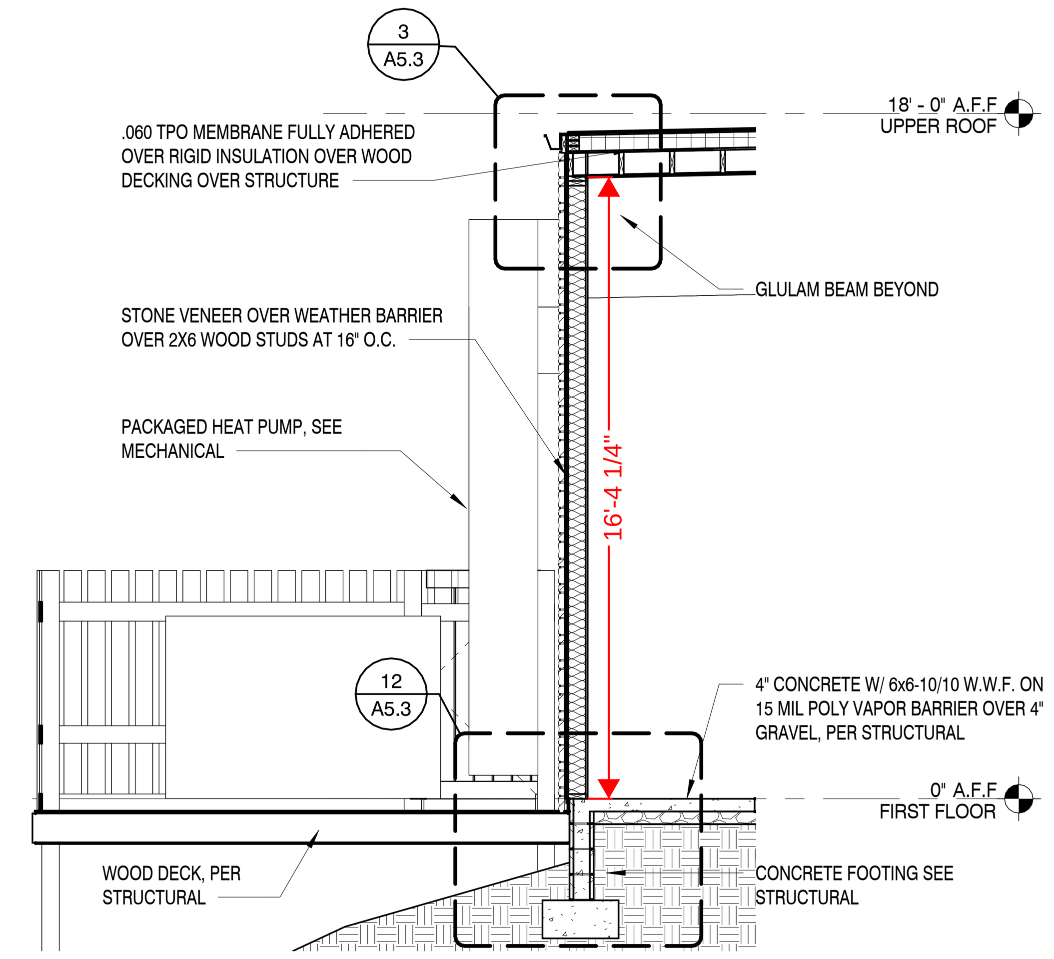
Sheet No. A5.1



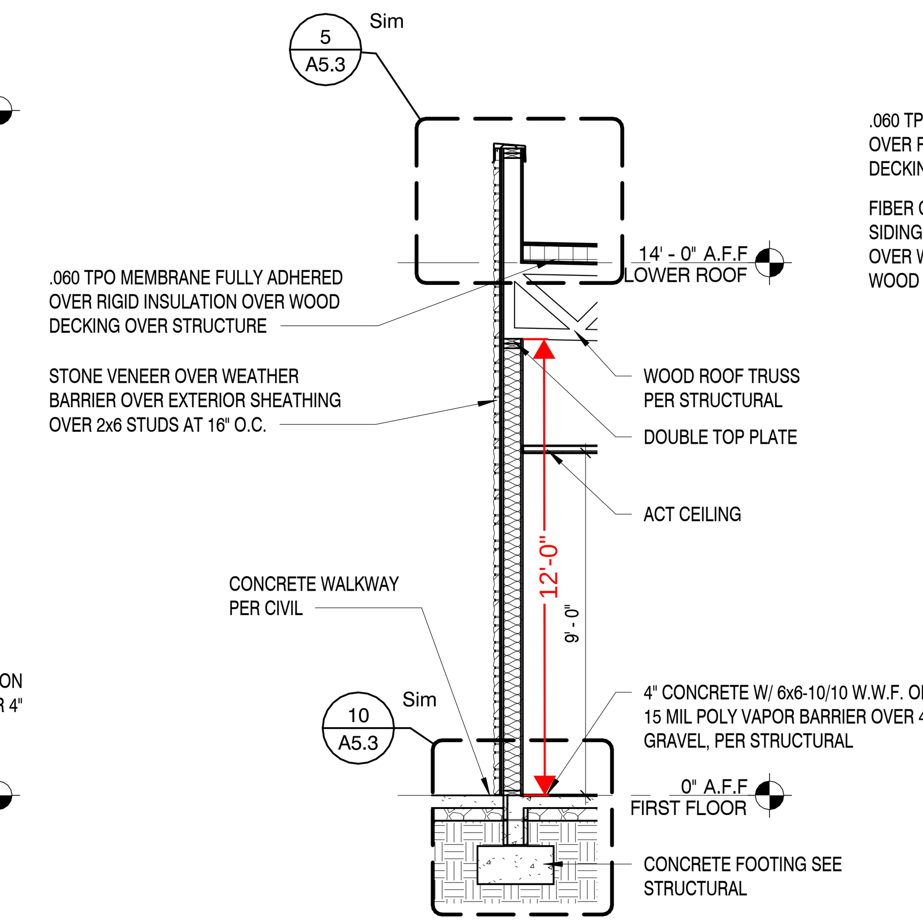
5 WALL SECTION
1/4" = 1'-0"



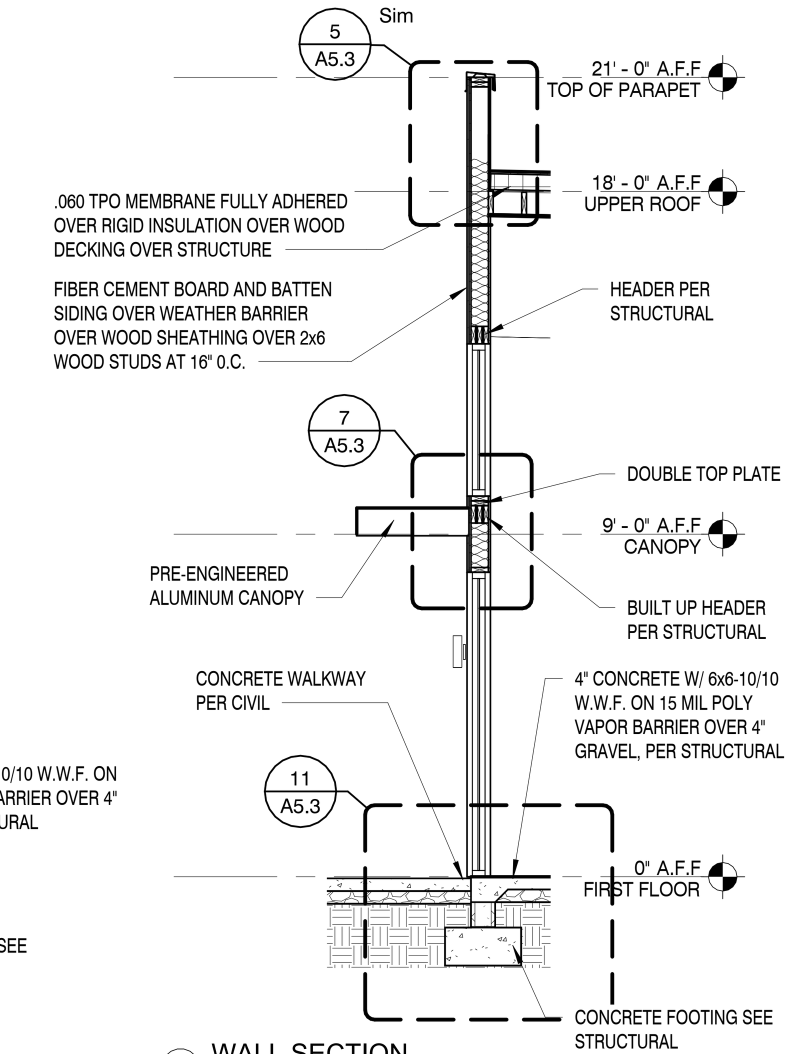
4 WALL SECTION
1/4" = 1'-0"



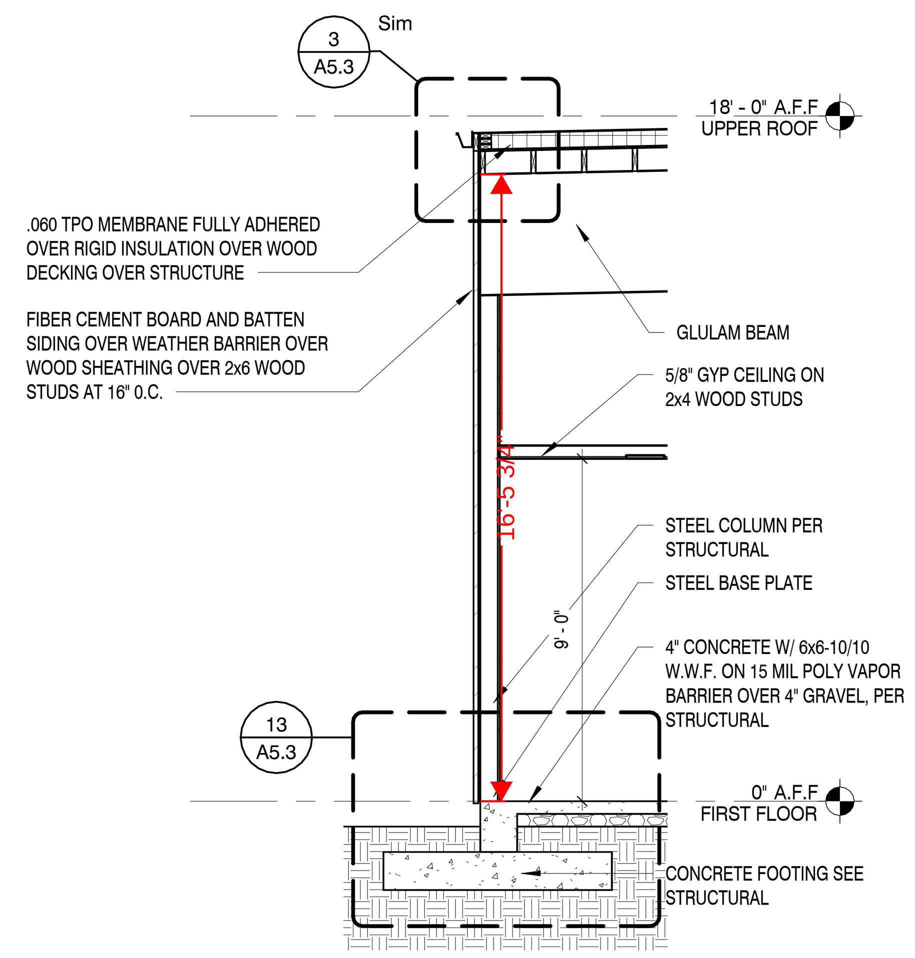
3 WALL SECTION
1/4" = 1'-0"



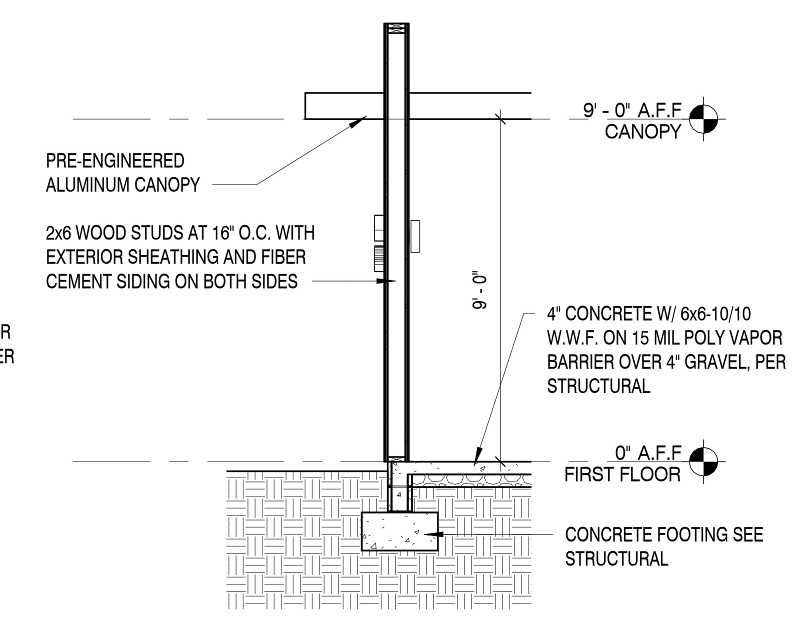
2 WALL SECTION
1/4" = 1'-0"



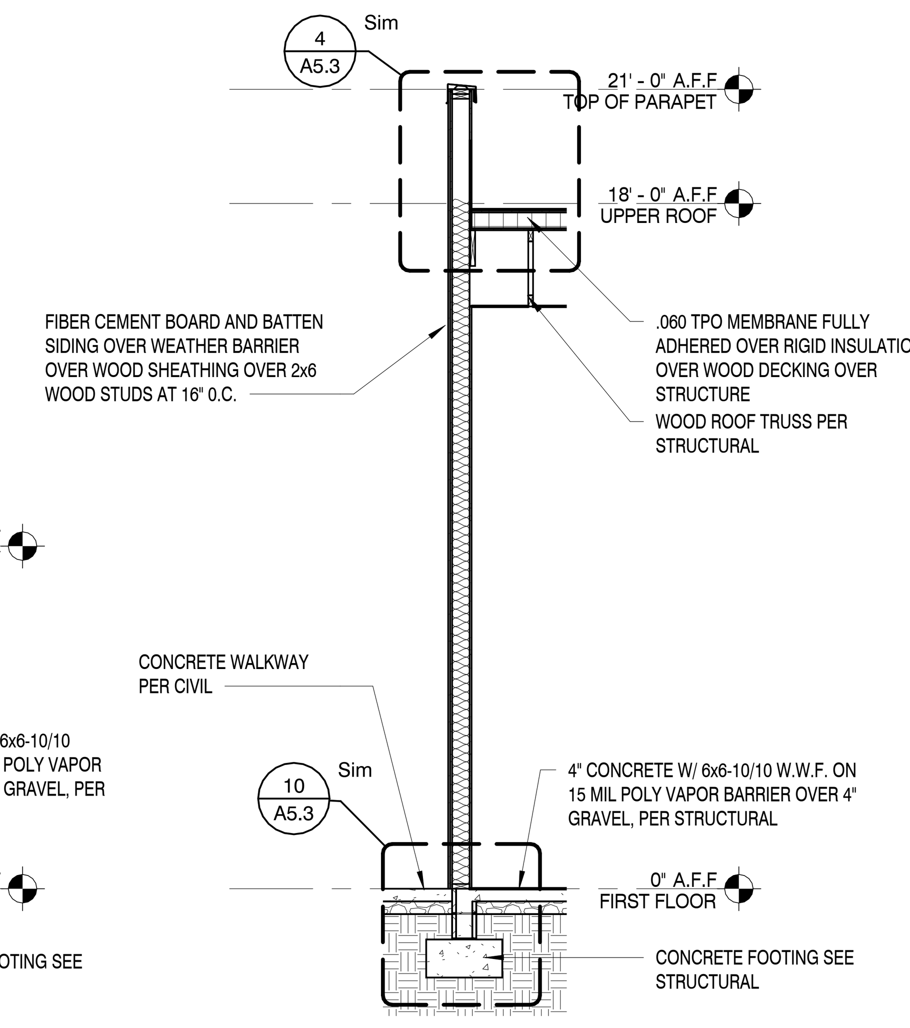
1 WALL SECTION
1/4" = 1'-0"



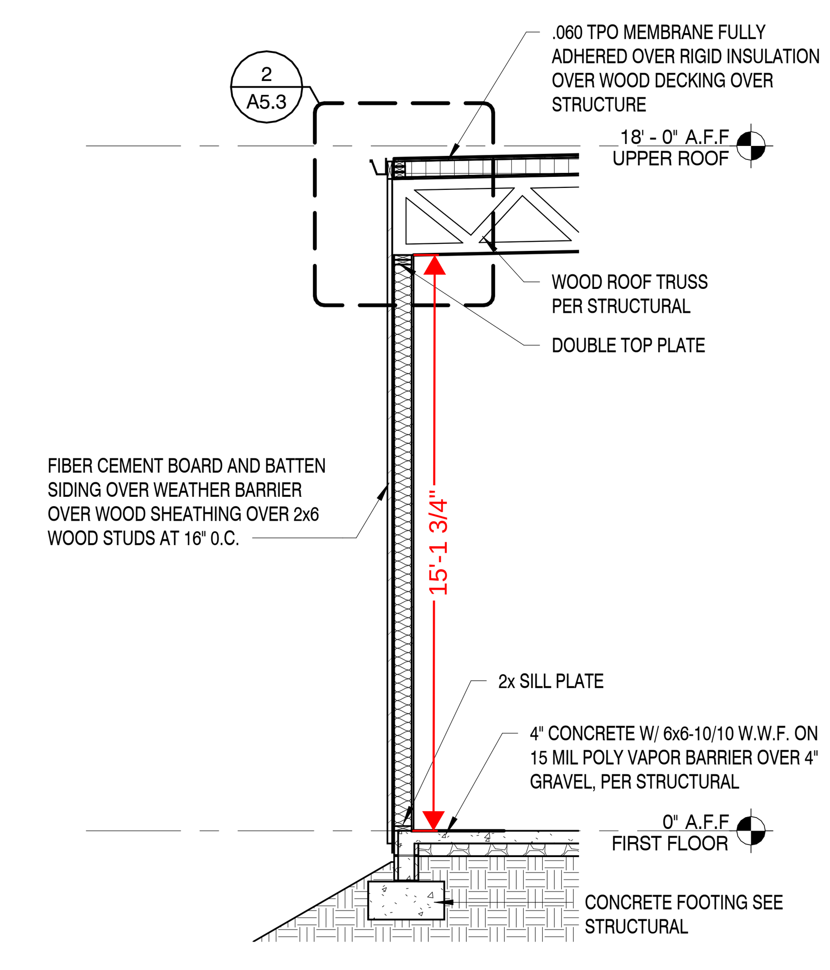
9 WALL SECTION
1/4" = 1'-0"



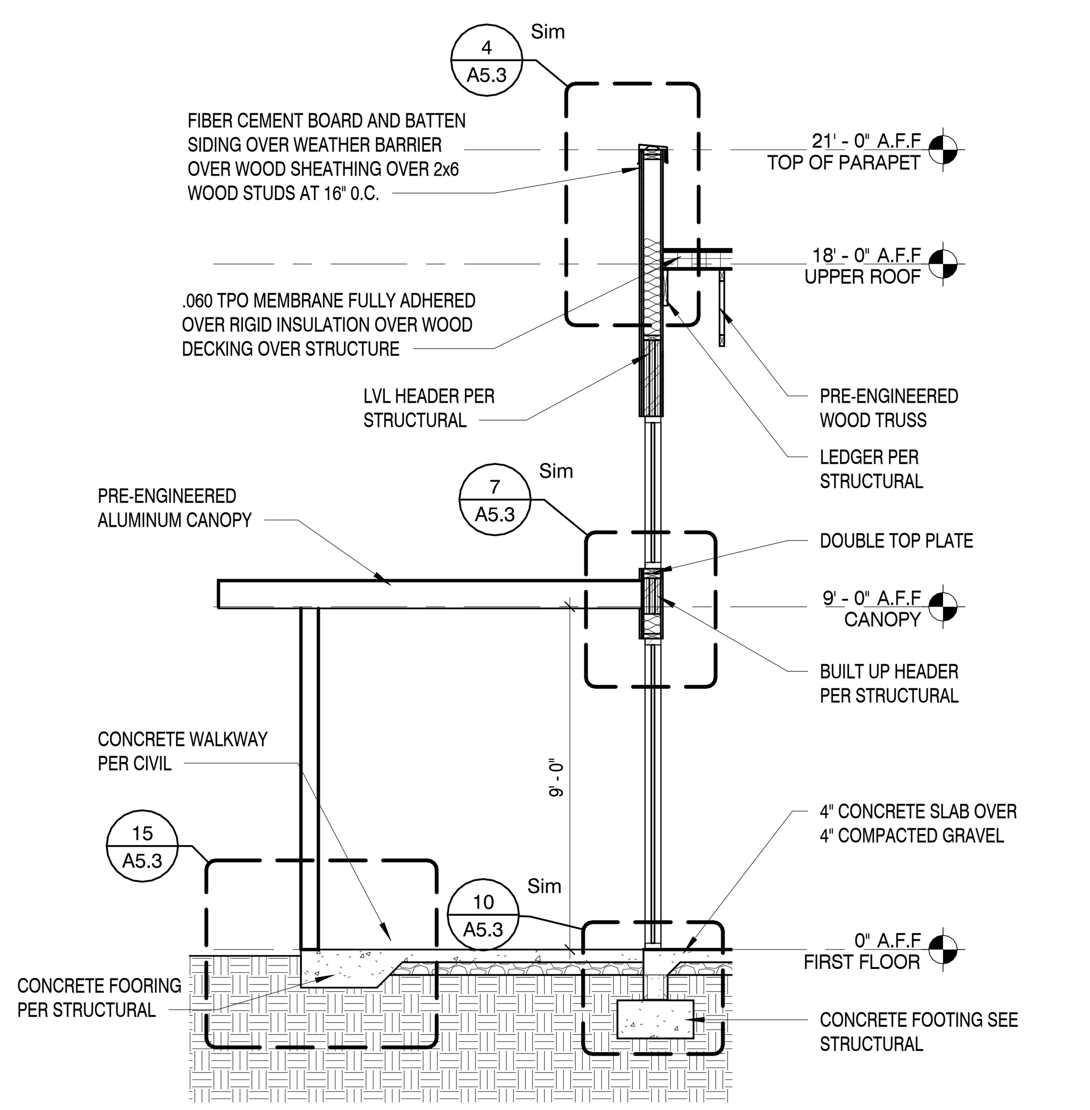
8 WALL SECTION
1/4" = 1'-0"



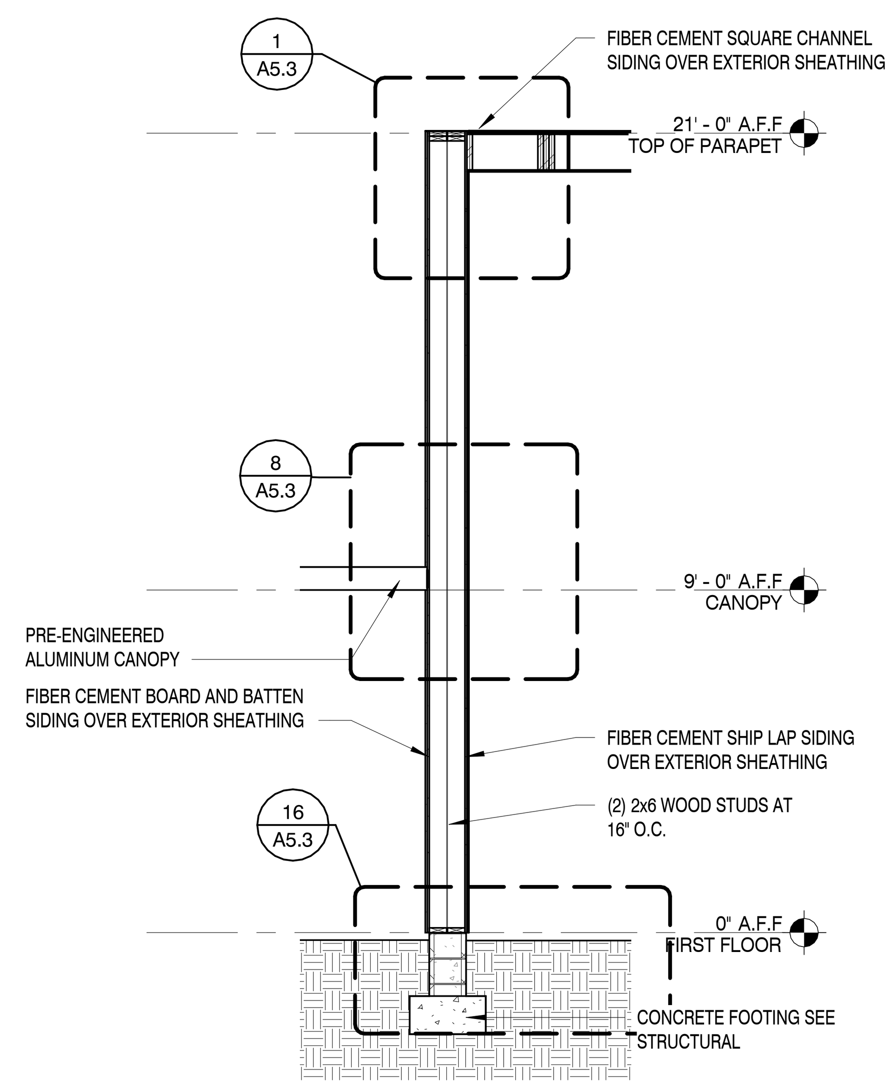
7 WALL SECTION
1/4" = 1'-0"



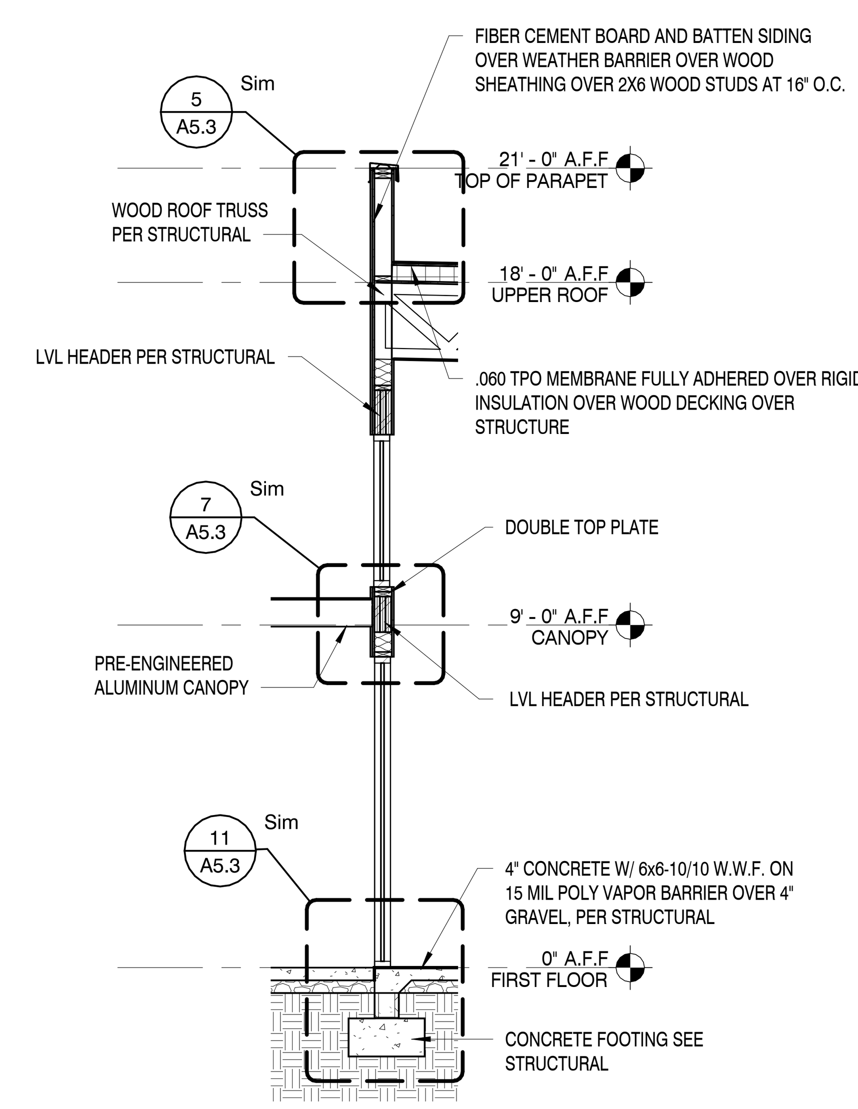
6 WALL SECTION
1/4" = 1'-0"



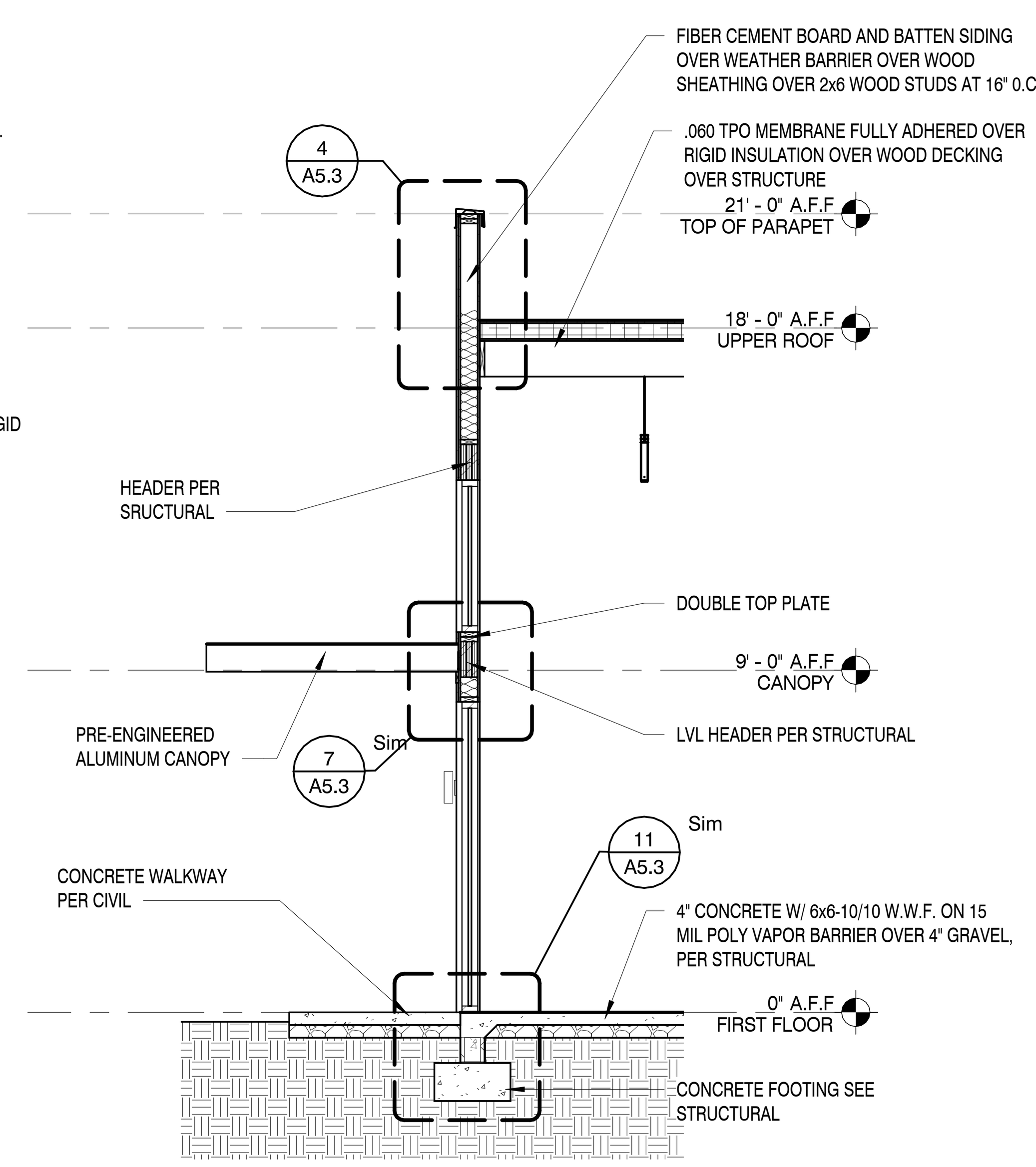
13 WALL SECTION
1/4" = 1'-0"



12 WALL SECTION
1/4" = 1'-0"



11 WALL SECTION
1/4" = 1'-0"



10 WALL SECTION
1/4" = 1'-0"

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THOMAS DANIEL BRITT
REGISTERED ARCHITECT
STATE OF TENNESSEE
No. 102271
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

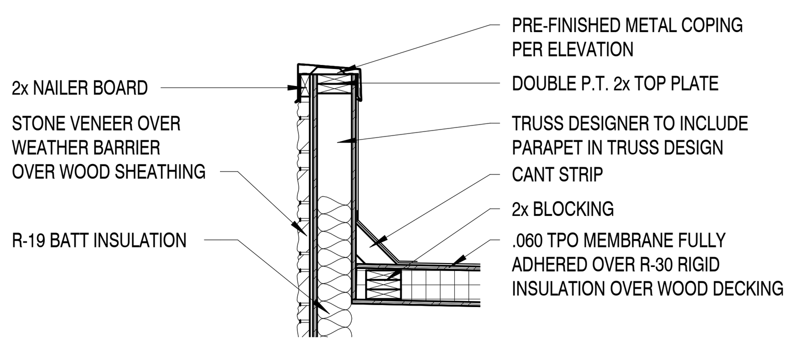
No.	Date

Drawing Title:
WALL SECTIONS

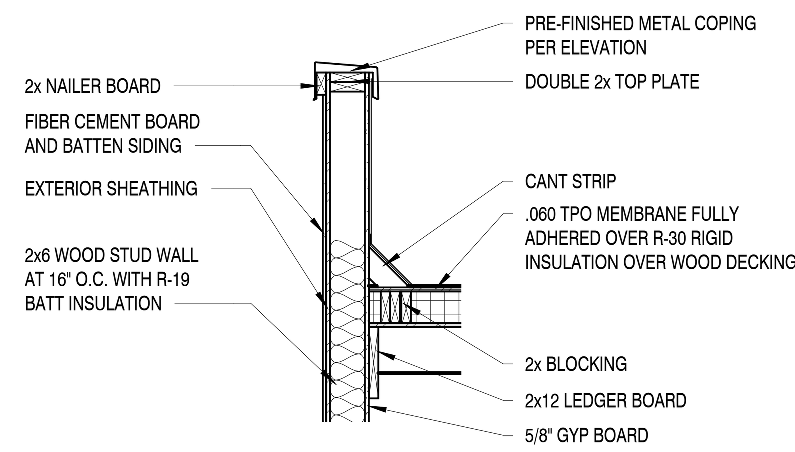
Date: 11/26/2025

Project No.
25026

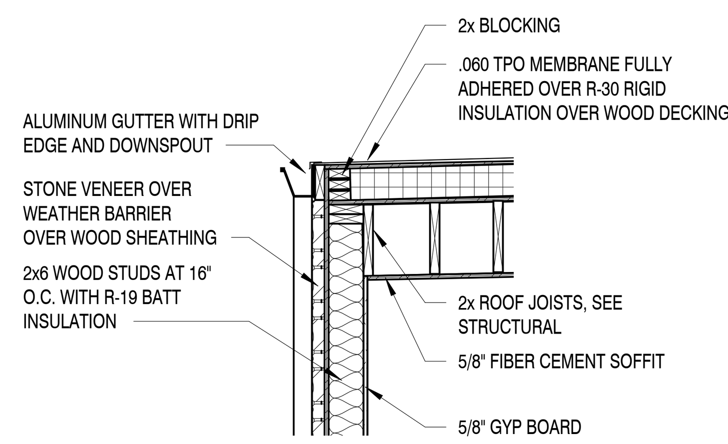
Sheet No.
A5.2



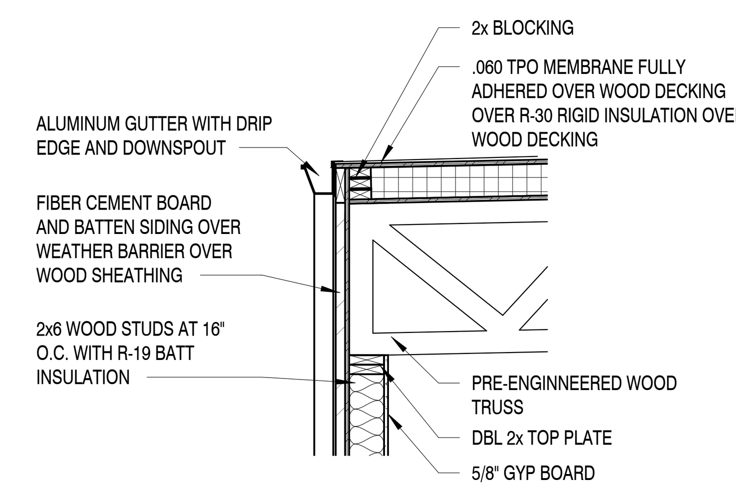
5 TRUSS PARAPET DETAIL
1/2" = 1'-0"



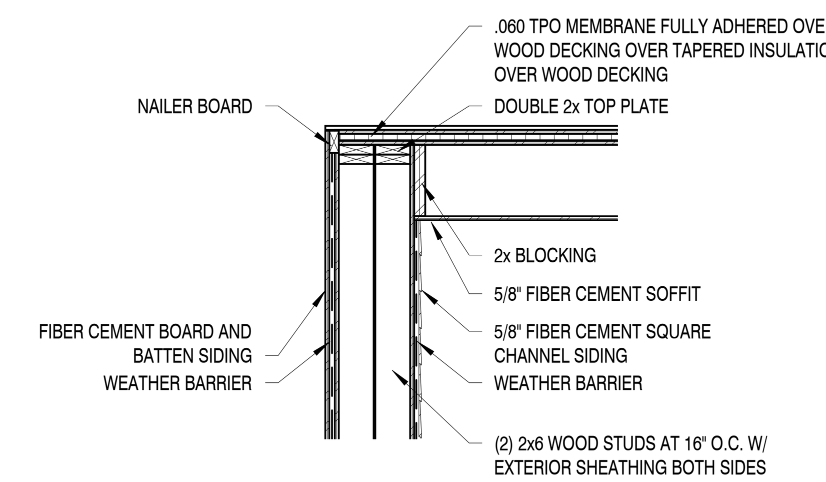
4 WALL PARAPET DETAIL
1/2" = 1'-0"



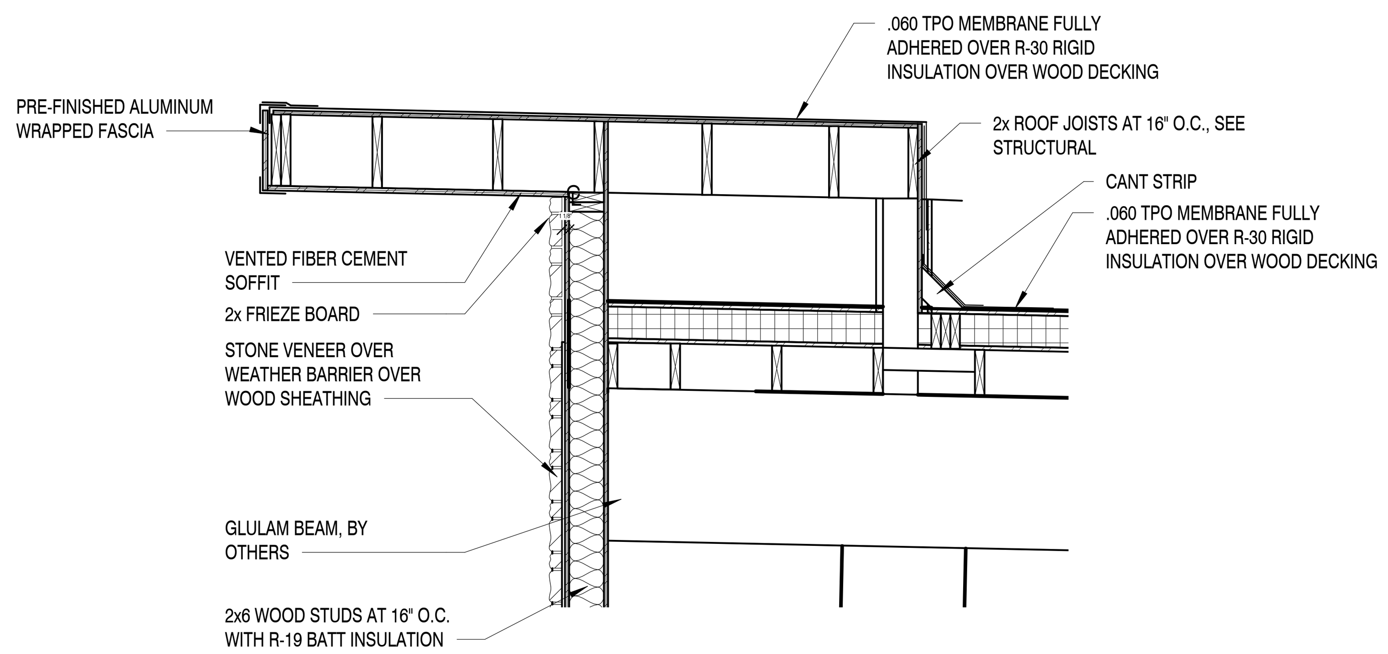
3 ROOF GUTTER DETAIL
1/2" = 1'-0"



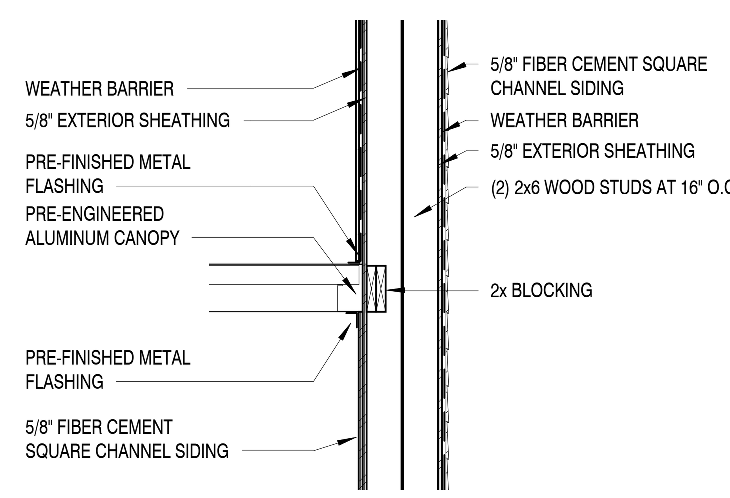
2 ROOF GUTTER DETAIL
1/2" = 1'-0"



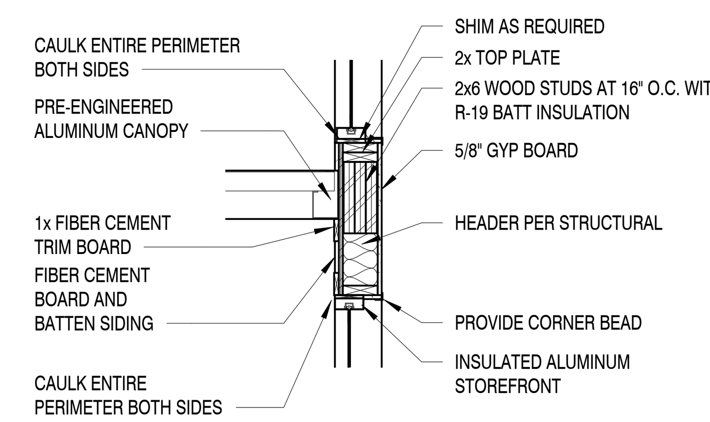
1 WING WALL CANOPY DETAIL
1/2" = 1'-0"



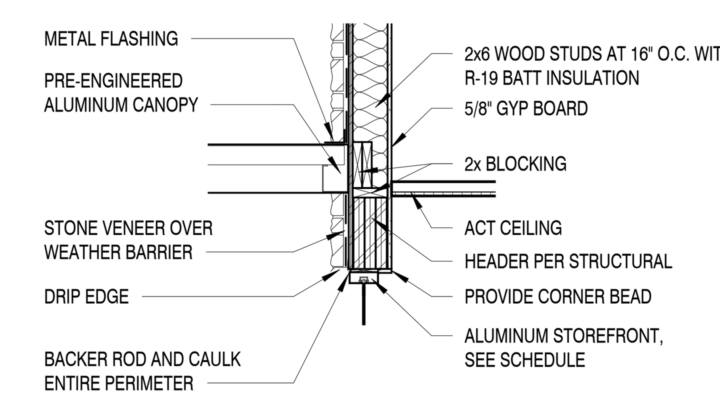
9 CANTELEVER CANOPY DETAIL
1/2" = 1'-0"



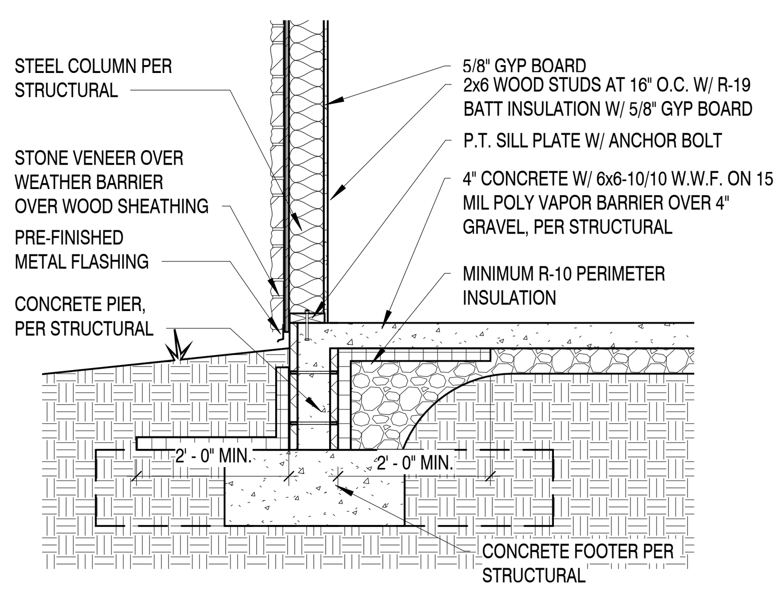
8 WING WALL AT CANOPY
1/2" = 1'-0"



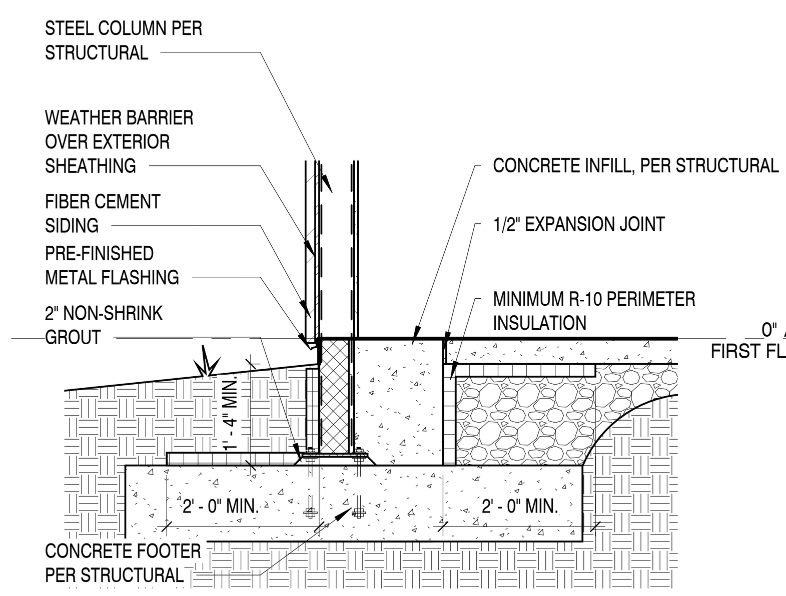
7 HEADER DETAIL AT CANOPY - FIBER CEMENT
1/2" = 1'-0"



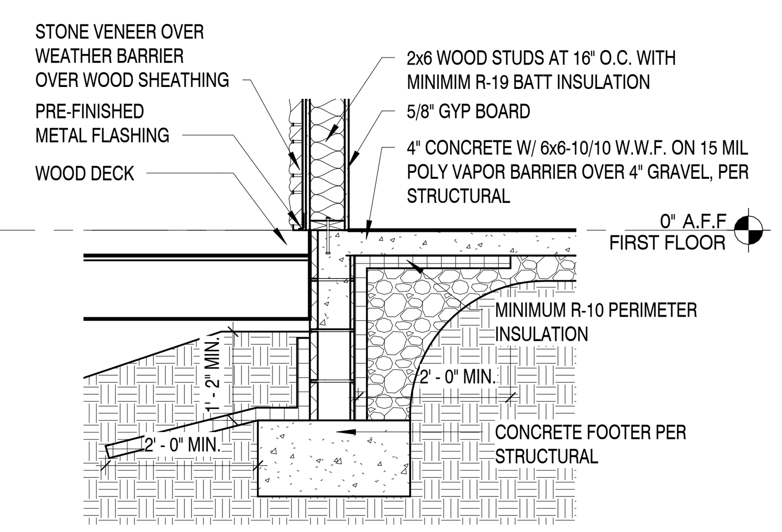
6 HEADER DETAIL AT CANOPY - STONE
1/2" = 1'-0"



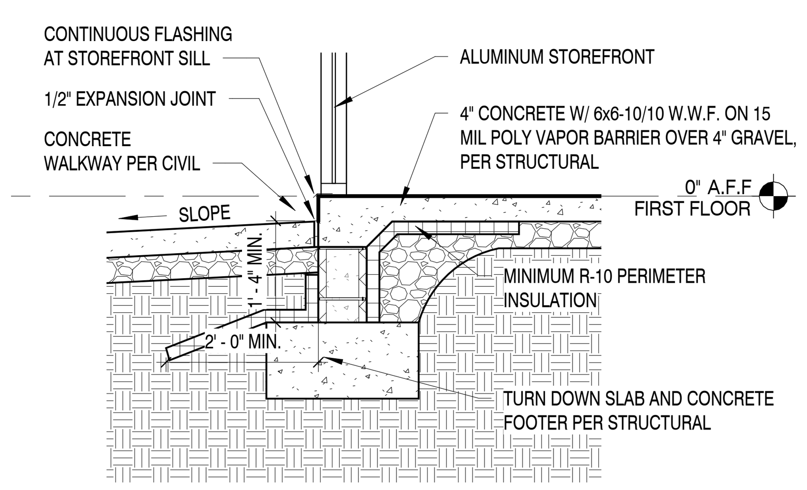
14 CONCRETE PIER DETAIL AT COLUMN - STACKED STONE
1/2" = 1'-0"



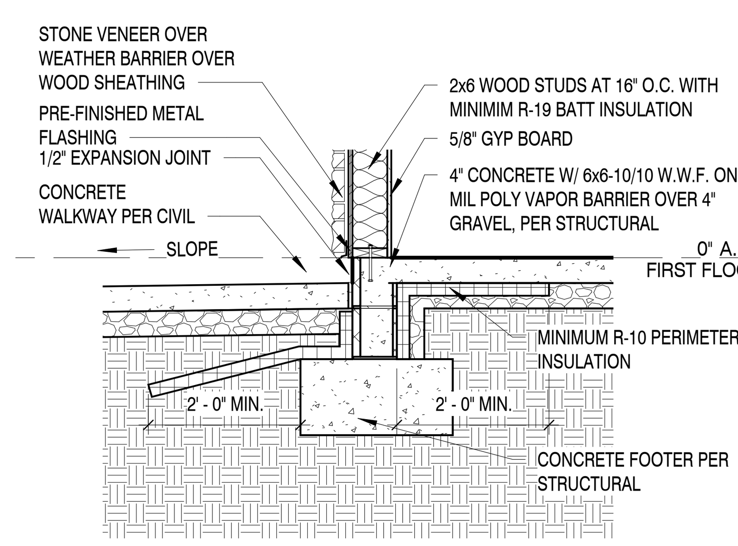
13 CONCRETE PIER DETAIL AT COLUMN - FIBER CEMENT
1/2" = 1'-0"



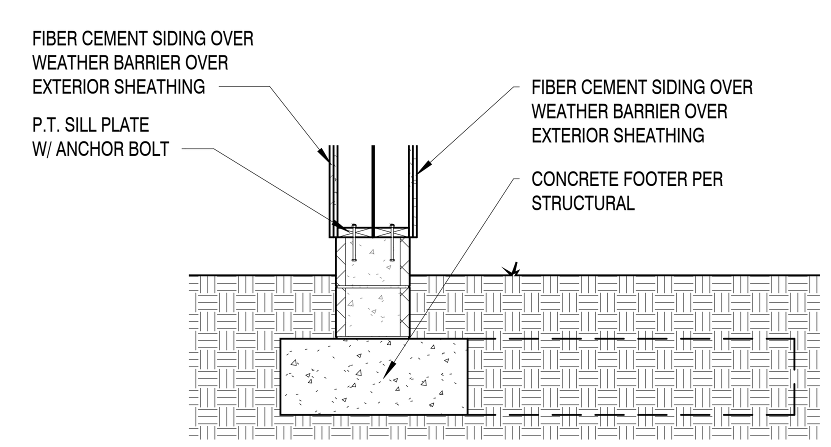
12 FOUNDATION DETAIL AT HVAC DECK
1/2" = 1'-0"



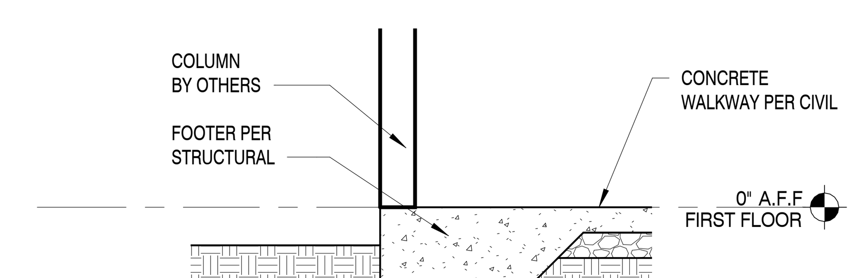
11 FOUNDATION DETAIL AT STORE FRONT
1/2" = 1'-0"



10 FOUNDATION DETAIL
1/2" = 1'-0"



16 WING WALL FOUNDATION DETAIL
1/2" = 1'-0"



15 FOUNDATION DETAIL AT COLUMN
1/2" = 1'-0"

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:

ELKS COMMUNITY ENRICHMENT CENTER

262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THOMAS DANIEL BRITT
STATE OF TENNESSEE
COMMERCE
NO. 102271
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

Drawing Title:
WALL DETAILS

Date: 11/26/2025

Project No.
25026

Sheet No.
A5.3

RESTROOM ACCESSORIES

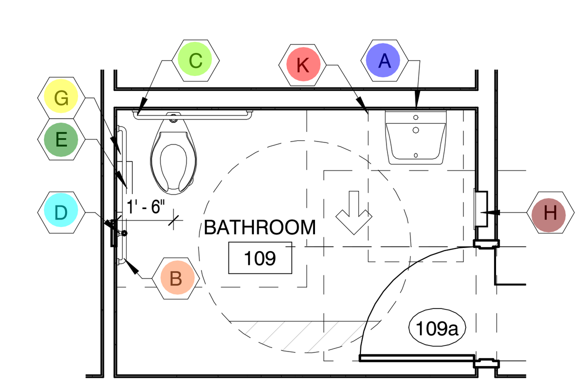
TAG	PRODUCT TYPE	MANUFACTURER	PRODUCT NAME	PRODUCT NUMBER	DESCRIPTION	LINK
A	MIRROR	BOBRICK	BOBRICK CHANNEL FRAME MIRROR	C3070155	CHANNEL FRAME MIRROR 18" W X 36" H	HTTPS://WWW.TOTALRESTROOM.COM/COLLECTIONS/BOBRICK-MIRRORS/PRODUCTS/BOBRICK-B-165-1836-CHANNEL-FRAME-MIRROR-18X36
B	42" GRAB BAR	BOBRICK	42" STRAIGHT GRAB BAR	B5806.99 X 42	FINISH/ SATIN STAINLESS STEEL	HTTPS://WWW.BUILD.COM/BOBRICK-B-5806-99X42/S1498889?UID=3511447&SEARCHID=3XRKL73X6O
C	36" GRAB BAR	BOBRICK	36" GRAB BAR	B5806 X 36	FINISH/ SATIN STAINLESS STEEL	HTTPS://WWW.BUILD.COM/PRODUCT/S1240743?UID=3162252
D	18" GRAB BAR	BOBRICK	18" GRAB BAR	B5806 X 18	FINISH/ SATIN STAINLESS STEEL	HTTPS://WWW.BUILD.COM/PRODUCT/S1247745?UID=3177618
E	TOILET PAPER HOLDER	ASI	SHELF WITH TOILET TISSUE HOLDER SURFACE MOUNTED	0697-GAL	SHELF WITH TOILET TISSUE HOLDER SURFACE MOUNTED	HTTPS://AMERICANSPECIALTIES.COM/PRODUCT/SHELF-TOILET-TISSUE-HOLDER-DOUBLE-SURFACE-MOUNTED-0697-GAL/
F	SOAP DISPENSER	ASI	LIQUID SOAP DISPENSER	0347	LIQUID SOAP DISPENSER / VERTICAL / SURFACE MOUNTED	HTTPS://AMERICANSPECIALTIES.COM/PRODUCT/LIQUID-SOAP-DISPENSERVERTICAL-SURFACE-MOUNTED/
G	SANITARY NAPKIN DISPOSAL	BOBRICK	CONTURA SERIES SANITARY NAPKIN DISPOSAL	B270	CONTURA SERIES SANITARY NAPKIN DISPOSAL WALL MOUNTED / STAINLESS STEEL	HTTPS://WWW.BUILD.COM/BOBRICK-B-270/S1217511?UID=2911155&SEARCHID=H4EH8PCCJQ
H	PAPER TOWEL DISPENSER / WASTE	BOBRICK	COMBINATION PAPER TOWEL DISPENSER/ WASTE	B-3944	RECESSED MOUNTED / STAINLESS STEEL	COMMERCIAL-PAPER-TOWEL-DISPENSER-WASTE-RECEPTACLE-RECESSEDMOUNTED-STAINLESS-STEEL?GAD_SOURCE=1 &GCLID=CJ0KCQJWJNS3BHCHARISAOXBM6QL8FQ4AUBC_VGECDYRPWCCD6BY3RJM_HFZXXWB35AAOIZYM5UO1GIAAHNWEALW_WCB
I	HOOK	AMERICAN STANDARD	STUDIO S DOUBLE ROBE HOOK	7105210.002	POLISHED CHROME FINISH	HTTPS://BUILD.COM/PRODUCT/SUMMARY/1447161?UID=3405100&JMTEST=GGGBAV2_3405100&INV2=1&&SOURCE=GG-GBA-PLA_3405100!C1674295608!A64838899357!DC!NG&GAD_SOURCE=1 &GCLID=CJ0KCQIAGJA6BHCOARISAMIL7V8R2SG4558RESKYDTI5F3UOPUEJB4S5TGQRNENVE5JWGNZEGP2MIAANL1EALW_WCB&GCLSRC=AW.DS
J	BABY CHANGING STATION	KOALA BEAR	SURFACE MOUNT BABY CHANGING STATION	KB310-SSWM	------------------------------------	HTTPS://WWW.KOALABEAR.COM/PRODUCT-CATALOG/KB310-SSWM/
K	SOAP DISPENSER	BOBRICK	WALL MOUNTED SOAP DISPENSER	B-2111	COMMERCIAL LIQUID SOAP DISPENSER, SURFACE-MOUNTED, MANUAL-PUSH, STAINLESS STEEL	HTTPS://WWW.TOTALRESTROOM.COM/PRODUCTS/BOBRICK-B-2111-COMMERCIAL-SOAP-DISPENSER-SURFACE-MOUNTED-MANUAL-PUSH-STAINLESS-STEEL?VARIANT=30926425522221 &COUNTRY=US&CURRENCY=USD&UTM_MEDIUM=PRODUCT_SYNC&UTM_SOURCE=GOOGLE&UTM_CONTENT=SAG_ORGANIC&UTM_CAMPAIGN=SAG_ORGANIC

GENERAL PLUMBING NOTES

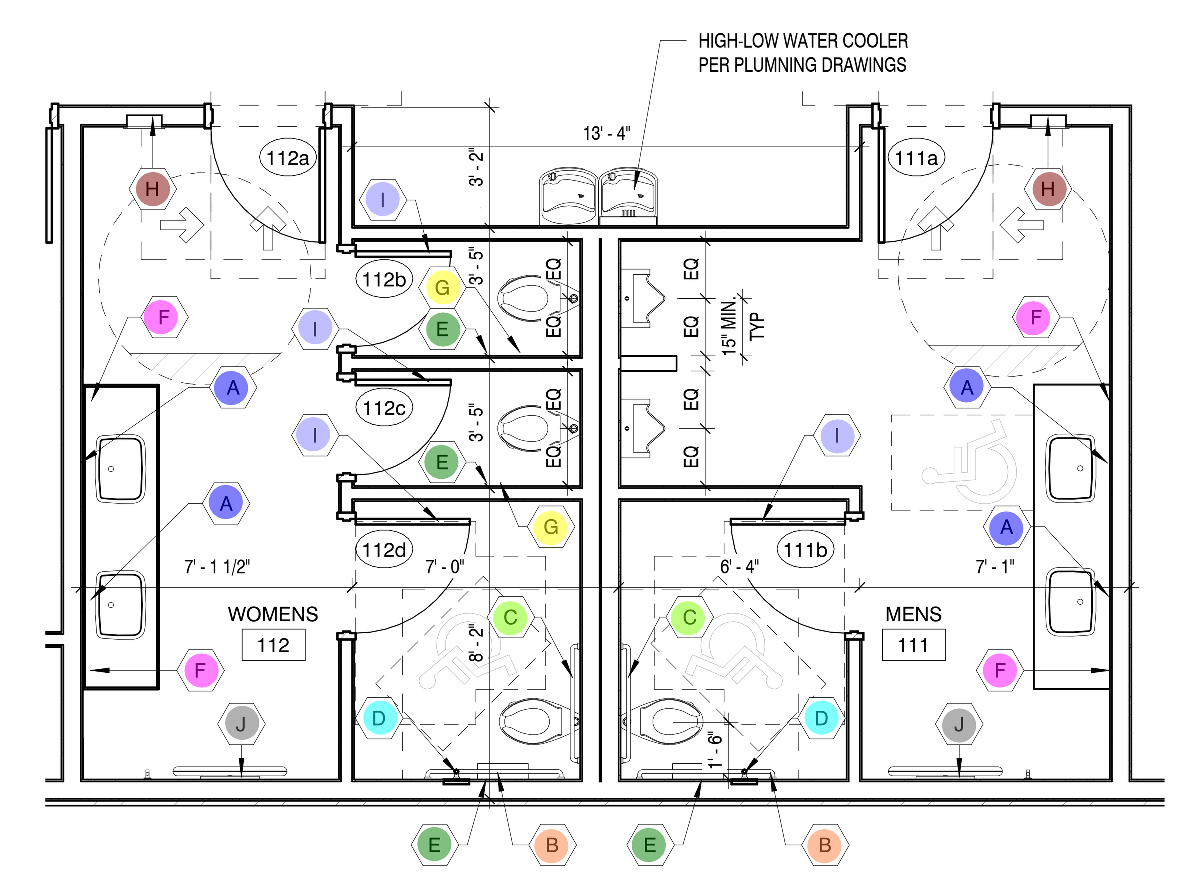
1. ALL PLUMBING MATERIAL AND INSTALLATIONS SHALL BE IN ACCORDANCE WITH ALL APPLICABLE CODES AND ORDINANCES.
2. SEE PLUMBING DRAWINGS FOR LOCATIONS AND SIZES OF ACCESS PANELS.
3. ALL FIXTURES AND ACCESSORIES SHALL COMPLY WITH THE CURRENT A.D.A., STATE OR LOCAL REGULATIONS FOR MOUNTING HEIGHTS AND CLEARANCES.
4. ALL HOT WATER AND DRAIN PIPES SHALL BE INSULATED PER A.D.A. REQUIREMENTS. MINIMUM HOT WATER SUPPLY INSULATION SHALL BE PRE-MOLDED FIBERGLASS PIPE INSULATION WITH WHITE ALL SERVICE JACKET. INSULATION THICKNESS SHALL BE MIN. 1". SEE PLUMBING DRAWINGS.
5. ALL GRAB BARS IN NEW CONSTRUCTION SHALL BE INSTALLED WITH CONCEALED ANCHOR PLATES.
6. THE FLUSH ACTIVATOR SHALL BE LOCATED ON THE WIDE CLEARANCE SIDE OF HANDICAPPED UNITS AND SHALL BE LEVER TYPE. THE FORCE TO ACTIVATE SHALL NOT EXCEED 5 POUNDS. ACTIVATION SHALL BE WITHIN 40" OF FIN. FLOOR.
7. LAVATORY FAUCET CONTROLS SHALL BE LEVER TYPE AND THE FORCE TO ACTIVATE SHALL NOT EXCEED 5 POUNDS.
8. PROVIDE BLOCKING IN WALLS AS REQ'D FOR ALL FIXTURES AND EQUIPMENT.
9. ALL DIMENSIONS ARE TO FACE OF STUD OR FACE OF FURRING UNLESS OTHERWISE NOTED. "CLEAR" DENOTES FINISH TO FINISH.
10. TOILET ROOM WALLS TO HAVE SOUND BATT INSULATION FROM FLOOR TO DECK ABOVE.
11. GYP. BD. IN ALL WET AREAS TO BE WATER RESISTANT GYP. BD.
12. CONCRETE BACKER BOARD SHALL BE PROVIDED BEHIND TILE AT WALLS.
13. ADJUST SUPPLY LINE WALL PENETRATION HEIGHTS AS NEEDED TO AVOID CONFLICTS BETWEEN FLUSH VALVES AND GRAB BAR MOUNTING HEIGHTS. GRAB BAR MOUNTING HEIGHTS ARE TO TAKE PRIORITY.
14. CONTRACTOR TO CORRIDATE WITH SPECIFIED FIXTURES AND FINISHES TO ENSURE RIM OF LAVATORIES TO BE 34" A.F.F. MAX.

BATH ACCESORIES

	Description	Label	Quantity	Unit
	ACCESORIES	A	5	Count
	ACCESORIES	B	3	Count
	ACCESORIES	C	3	Count
	ACCESORIES	D	3	Count
	ACCESORIES	E	5	Count
	ACCESORIES	F	4	Count
	ACCESORIES	G	3	Count
	ACCESORIES	H	3	Count
	ACCESORIES	I	4	Count
	ACCESORIES	J	2	Count
	ACCESORIES	K	1	Count



3 ENLARGED FLOOR PLAN - BATHROOM
1/4" = 1'-0"



2 ENLARGED FLOOR PLAN - BATHROOMS
1/4" = 1'-0"

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THOMAS DANIEL BRITT
REGISTERED ARCHITECT
COMMERCE
No. 102274
STATE OF TENNESSEE
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

Drawing Title:
ENLARGED PLANS AND DETAILS

Date: 11/26/2025

Project No. 25026

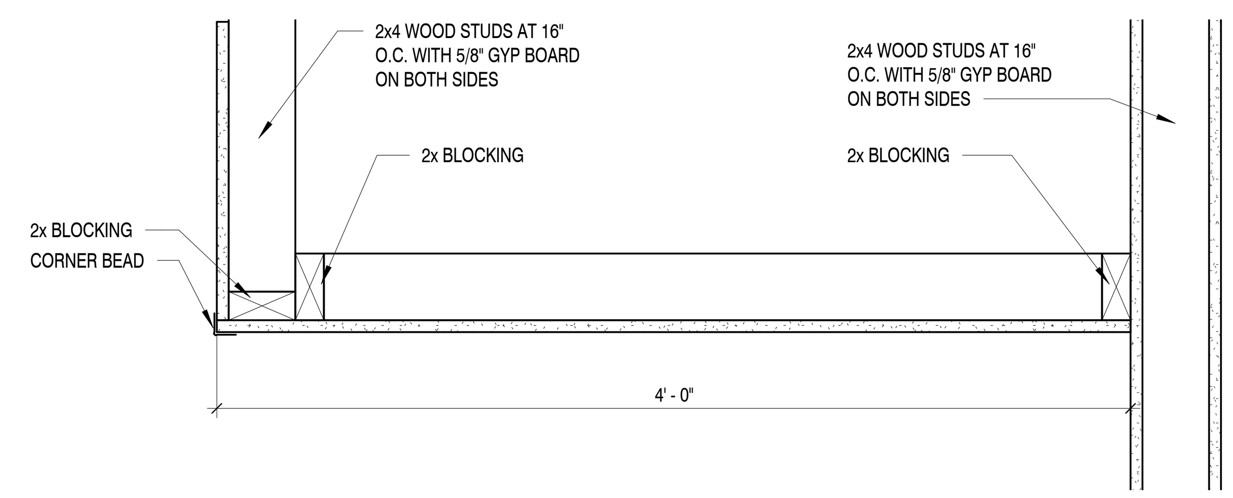
Sheet No. A6.1

RCP KEYNOTES

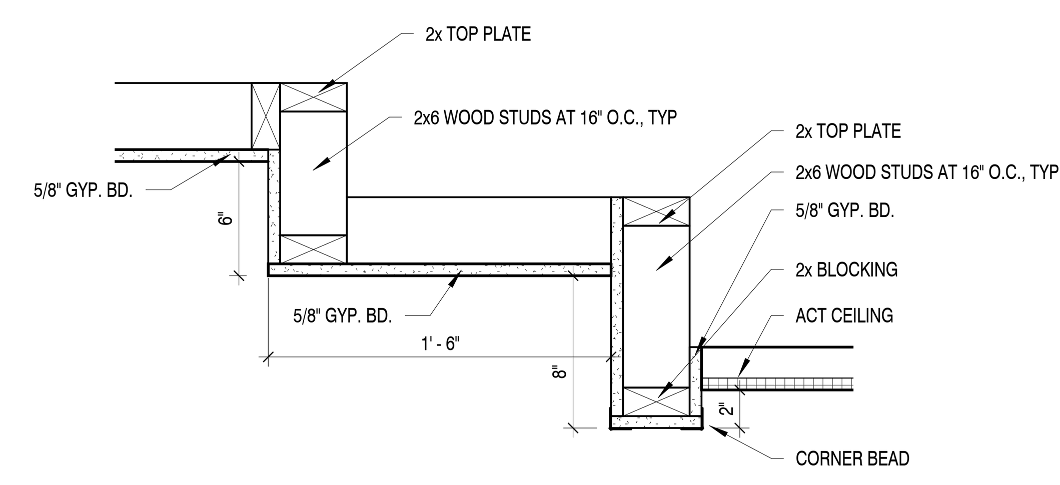
1. SEE F1.1 FOR DETAILS ON ACOUSTIC GRID.
2. SEE F1.1 FOR DETAILS ON ACOUSTIC SLATS.
3. 4' ILLUMINATED T BAR W/ DIFFUSED LENSE

REFLECTED CEILING LEGEND

- 2X2 LED FIXTURE
- 8" RECESSED CAN LIGHT
- 8" PENDANT
- 10" PENDANT
- 15" PENDANT
- DECORATIVE CHANDELIER
- HVAC SUPPLY DIFFUSER
- HVAC R/A GRILL
- EXHAUST FAN
- WALL MOUNTED TUBE SCONCE
- WALL MOUNTED DECORATIVE SCONCE
- 24"X24" ACOUSTICAL CEILING TILE (ACT2 UNLESS OTHERWISE NOTED)
- TOGUE AND GROOVE CEILING
- WOOD SLAT CEILING (TBD)
- 5/8" GYP BOARD CEILING
- 9'-0" A.F.F. — CEILING HEIGHT UNLESS OTHERWISE NOTED

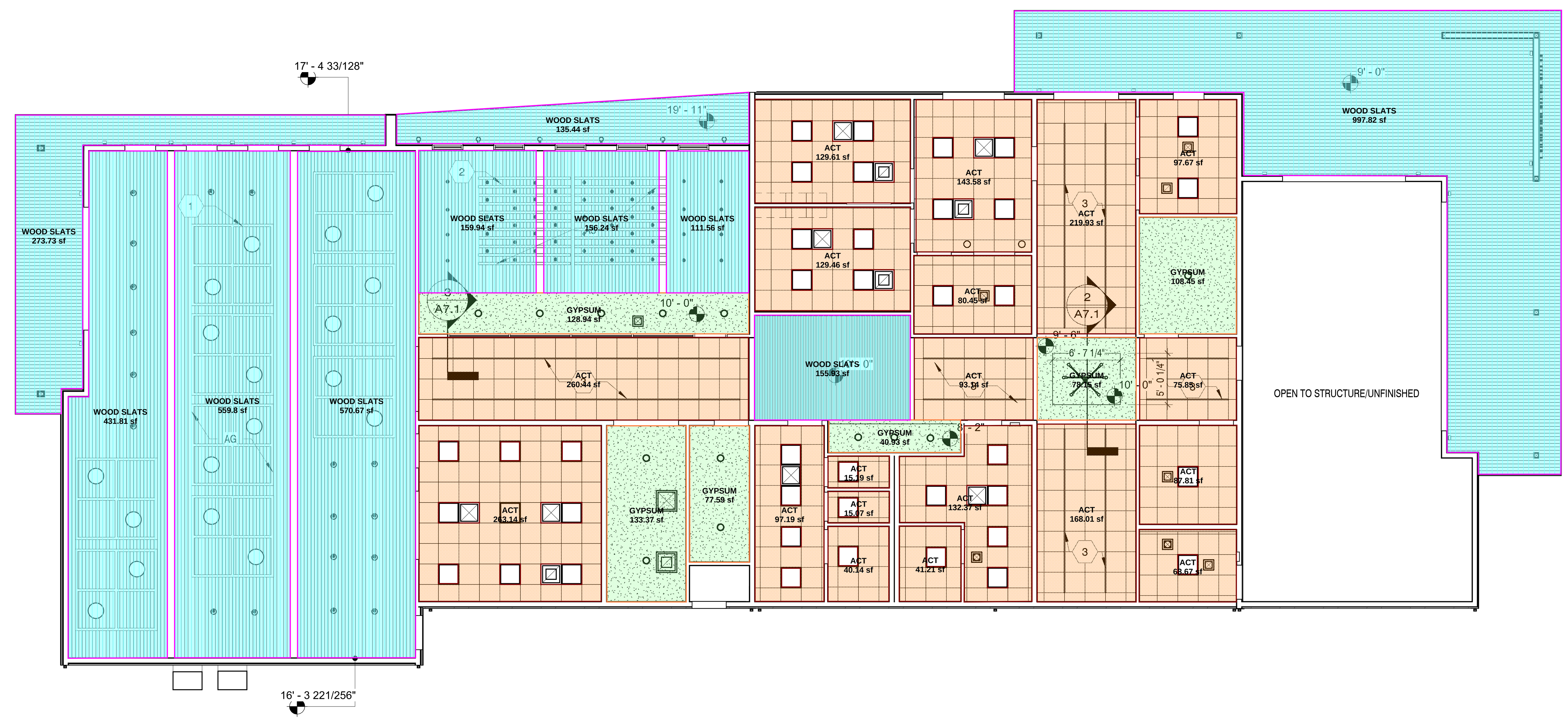


3 BULK HEAD SECTION DETAIL
1 1/2" = 1'-0"



2 SOFFIT DETAIL
1 1/2" = 1'-0"

FINISHES				
	Description	Label	Quantity	Unit
	CEILING	ACT	2,153.92	sf
	CEILING	GYPSUM	567.44	sf
	CEILING	WOOD SLATS	3,552.94	sf



1 FIRST FLOOR RCP
1/8" = 1'-0"

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

STATE OF TENNESSEE
No. 102274
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

Drawing Title:
REFLECTED CEILING PLAN AND DETAILS

Date: 11/26/2025

Project No.
25026

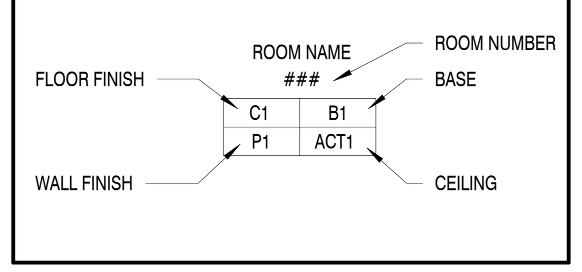
Sheet No.
A7.1

FINISH SCHEDULE

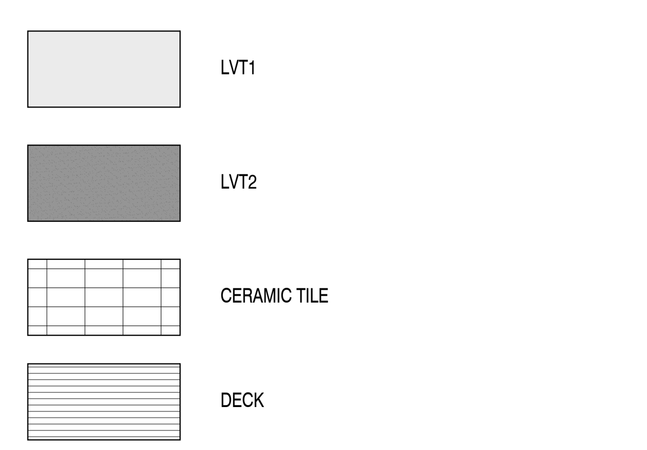
CODE	ITEM	MANUFACTURER	DESCRIPTION
	FLOORING		NOTE: LOCATE THE FLOOR FINISH CHANGE AT DOOR OPENINGS AT THE CENTERLINE OF THE DOOR LEAF
LVT1	LUXURY VINYL TILE	COBALT SURFACES	KATANGA COLLECTION/ COLOR: MUSHROOM/ SIZE: 3MM 20 MIL / SKU: K320-831
LVT2	LUXURY VINYL TILE	COBALT SURFACES	KATANGA COLLECTION/ COLOR: MAGNOLIA/ SIZE: 3MM 20 MIL / SKU: K320-833
RF	RESILIENT FLOORING	NORA FLOORING	COLLECTION: NORAMENT SATURA / COLOR: VULPECULA 5106
RF2	VINYL COMP. TILE	FORBO FLOORING	COLLECTION: SPHERA ELITE/ COLOR: 50480 SELENTINE
RF3	RESILIENT FLOORING	NORA FLOORING	COLLECTION: NORAPLAN ULTRA GRIP / COLOR: 6016
FT1	FLOOR TILE	CROSSVILLE TILE	TRAJECTORY COLLECTION/ COLOR: DIRECTION / SIZE: 12" X 24" / SKU: TRY02/ GROUT: TEC 908 DOVE GRAY
SC	SEALED CONCRETE	———	———
TR1	TRANSITION	SCHLUTER	STYLE: RENO-U EDGE-PROTECTION PROFILE FOR ABUTTING TILE TO LOWER TRANSITIONS COLOR: SATIN ANODIZED, HEIGHT: 10 MM, TILE TO LVT
TR2	TRANSITION	SCHLUTER	STYLE: VINPRO-S RESILIENT SURFACE EDGE PROTECTION MINIMAL REVEAL COLOR: BRUSHED ANTIQUE BRONZE ANODIZED. HEIGHT: 4MM
TR3	TRANSITION	SCHLUTER	STYLE: VINPRO-S RESILIENT SURFACE EDGE PROTECTION MINIMAL REVEAL COLOR: BRUSHED ANTIQUE BRONZE ANODIZED. HEIGHT: 3MM
TR4	TRANSITION	SCHLUTER	STYLE: RONDEC-DB DECORATIVE PROFILE PRONOUNCED VISIBLE SURFACE. COLOR: SATIN ANODIZED. HEIGHT 14MM
B1	BASE	SHAW CONTRACT	RUBBER COVE BASE / COLOR: EBONY
B2	BASE	CROSSVILLE TILE	COLLECTION: TRAJECTORY / COLOR: DIRECTION / SIZE: 6" X 12"
DECK	DECK	TREX	COLLECTION: ENHANCE / 1" SQUARE EDGE PROFILE / 1" x 12" x 12' FASCIA / COLOR: TBD
	WALLS		
P1	PAINT	SHERWIN WILLIAMS	COLOR: PURE WHITE / SKU: SW 7005
P2	PAINT	SHERWIN WILLIAMS	COLOR: CLOVE (DOOR PAINT) / SKU: SW 9605
P3	PAINT	SHERWIN WILLIAMS	COLOR: EVERGREEN FOG (ACCENT) / SKU: SW 9130
P4	PAINT	SHERWIN WILLIAMS	COLOR: BALANCED BEIGE (DOOR PAINT) / SKU: SW 7037
WT1	WALL TILE	PORCELANOSA	COLLECTION: NEBRASKA / COLOR: ICE NEBRASKA TEA / SIZE: 18" X 47" / INSTALLATION: VERTICAL / SKU: 100300114 / GROUT: TEC 961 SANDSTONE BEIGE
WT2	WALL TILE	PORTOBELLO	COLLECTION: ARTFACT / COLOR: STRAW / SIZE: 2.5" X 0.9" GLOSSY PRESSED/ INSTALLATION: HORIZONTAL / SKU: 900728E / GROUT: TEC 988 PEARL
WT3	WALL TILE	PORTOBELLO	COLLECTION: LIGHTHOUSE / COLOR: SEA / SIZE: 2" X 10" GLOSSY PRESSED/ INSTALLATION: VERTICAL / SKU: 204628E / GROUT: TEC 953 STARRY NIGHT
WT4	WALL TILE	CROSSVILLE TILE	COLLECTION: ARGENT 2.0 / COLOR: WHITE TIGER UPS / SIZE: 12" 24" / INSTALLATION: HORIZONTAL / SKU: ARG31.11224UPS / GROUT: TBA
WT5	WALL TILE	MSI	COLLECTION: STELLA / COLOR: OCEANO / SIZE: 2 X 10 / INSTALLATION: VERTICAL / SKU: NSTEOCE2X10G / GROUT: TEC 909 STERLING
WT6	WALL TILE	PORCELANOSA	COLLECTION: MOSAICO / COLOR: NANTES CALIZA / SIZE: 18X47 / SKU: 100239861/ INSTALLATION: HORIZONTAL / GROUT: TEC 909 STERLING
WT7	WALL TILE	FLOOR & DECOR	COLLECTION: CANVAS / COLOR: MERINGUE II / SIZE: 3X12 / SKU:100655299/ GROUT: TEC 949 SILVERADO
	PLASTIC LAMINATE	———	
PL1	P. LAM	WILSON ART	COLOR: KING LAKE / SKU: 8263
PL2	P. LAM	WILSON ART	COLOR: QUARTZ FROST / SKU: 5031
	CEILING		
ACT1	ACOUSTICAL CEILING TILE	ARMSTRONG	STYLE: FINE FISSURED SECOND LOOK II-1761, SIZE: 24" x 48" x 3/4", COLOR: WHITE, GRID: 15/16"
ACT2	ACOUSTICAL CEILING TILE	ARMSTRONG	STYLE: ULTIMA-1912 BEVELED TEGULAR, SIZE: 24" x 24" x 3/4", COLOR: WHITE FINE TEXTURE, GRID: 9/16"
GB1	GYP. BD. BULKHEAD	SHERWIN WILLIAMS	PAINT (P2) UNLESS NOTED OTHERWISE
	MISC.		
SS1	SOLID SURFACE	WILSON ART	COLOR: FROSTY WHITE / SKU: 1573SL
CB1	CABINETS	WILSON ART	SPECTRUM COLOR ALTERATION / COLOR: SW9130 EVERGREEN FOG
AP	ACOUSTIC PANELING	FELT RIGHT	STYLE: SUBWAY TILE /COLOR: INDIGO / SIZE : 2' X 2'
WS	WOOD SLATS	TERRAMAI	WHITE OAK SQUARE SLAT MODULAR PANEL/ SKU: 2864-S
FB1	CUSHION FABRIC	MOMENTUM	PATTERN: MIXER EPU / COLOR: ICE BLUE / SKU: 09554906
AS	ACOUSTIC SLATS	SOELBERG	COLLECTION: MUTO BOX / COLOR: NATURAL WALNUT
AG	ACOUSTIC GRID	SOELBERG	COLLECTION: MUTO GRID / COLOR: NATURAL ASH

NOTES:
1. ALL DOOR TRIM COLOR TO BE THE SAME AS DOOR PAINT
2. PAINT FINISHES:
   - WALLS: SATIN
   - CEILINGS: EGGSHELL
   - TRIM: SEMI-GLOSS
   - RESTROOMS: SEMI GLOSS

INTERIOR FINISH TAG



FINISH LEGEND

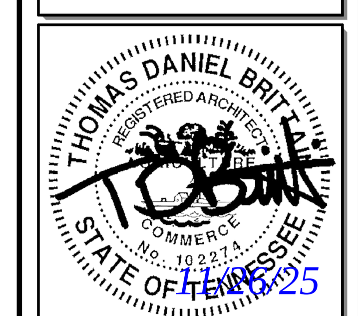


FINISHES				
	Description	Label	Quantity	Unit
	FLOORING	DECK	246.49	sf
	FLOORING	LVT1	2,785.82	sf
	FLOORING	LVT2	557.13	sf
	FLOORING	RF	308.28	sf
	FLOORING	RF2	580.11	sf
	FLOORING	RF3	307.12	sf
	FLOORING	SC	209.32	sf
	FLOORING	TILE	475.06	sf

1 FINISH FLOOR PLAN
1/8" = 1'-0"

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830



THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

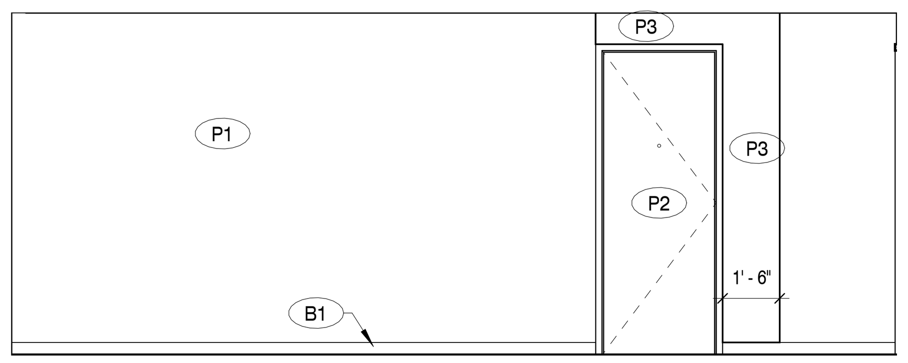
No.	Date

Drawing Title:
FINISH FLOOR PLAN

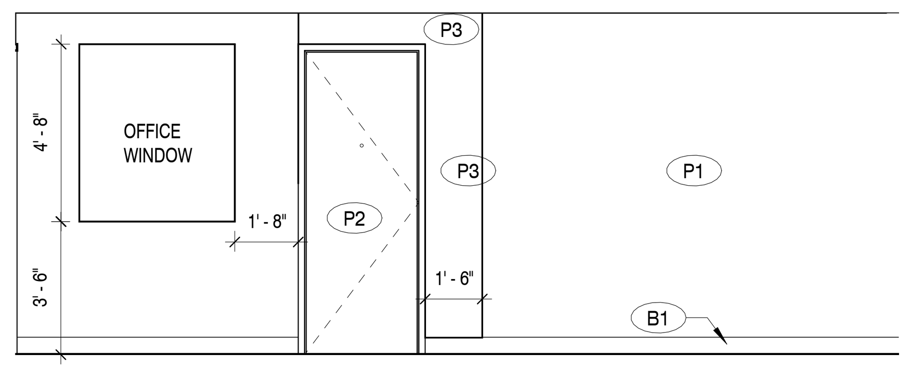
Date: 11/26/2025

Project No. 25026

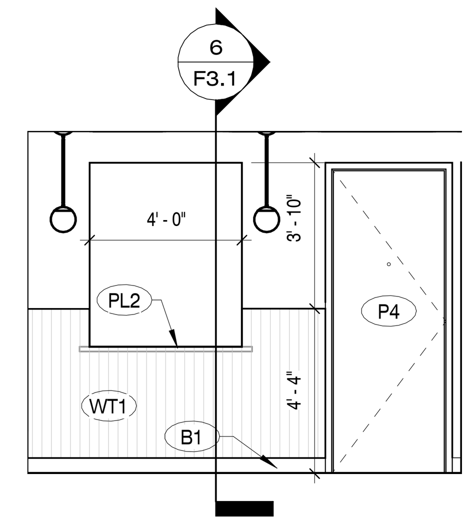
Sheet No. F1.1



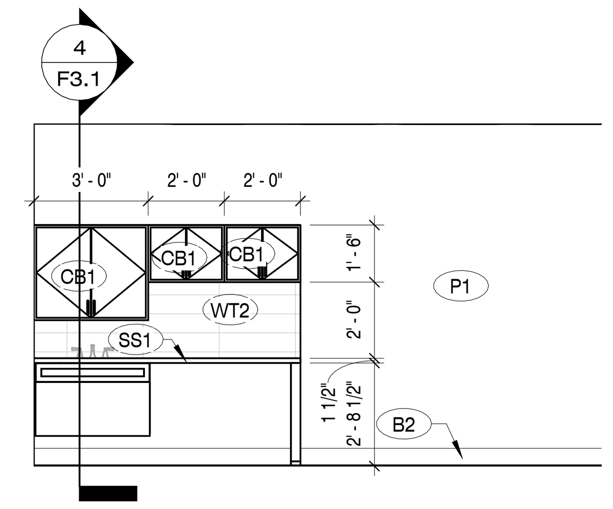
4 ENTRY WAY LEFT ELEVATION
1/4" = 1'-0"



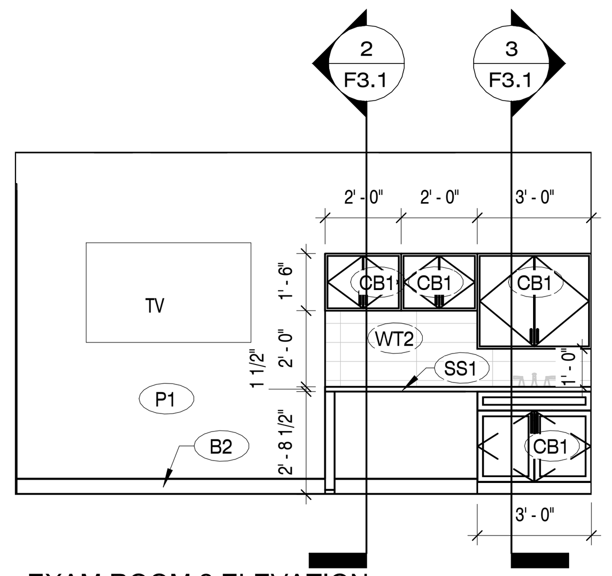
3 ENTRY WAY RIGHT ELEVATION
1/4" = 1'-0"



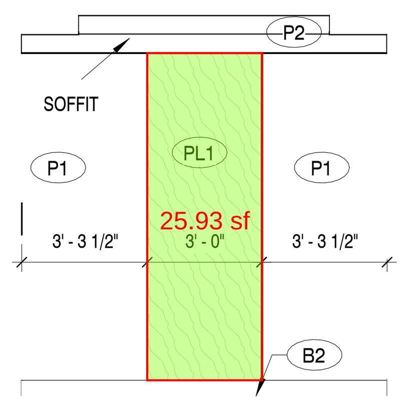
2 CLINIC RECEPTION ELEVATION
1/4" = 1'-0"



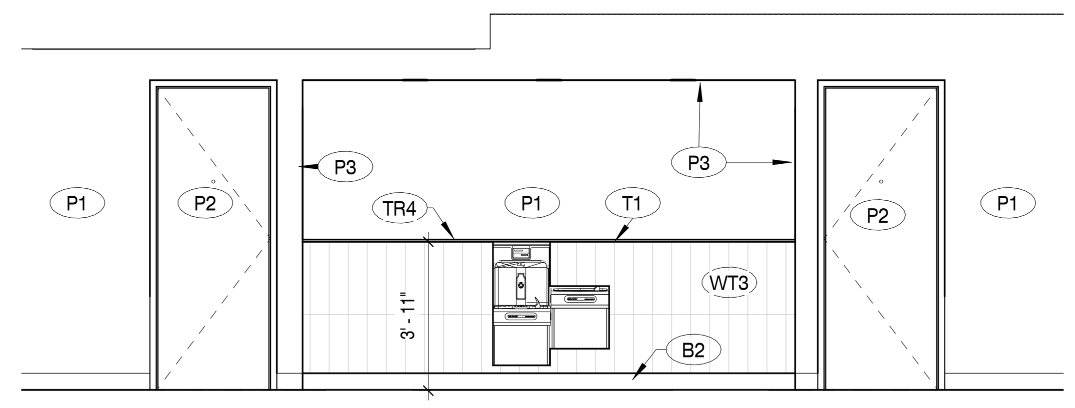
1 EXAM ROOM 1 FRONT ELEVATION
1/4" = 1'-0"



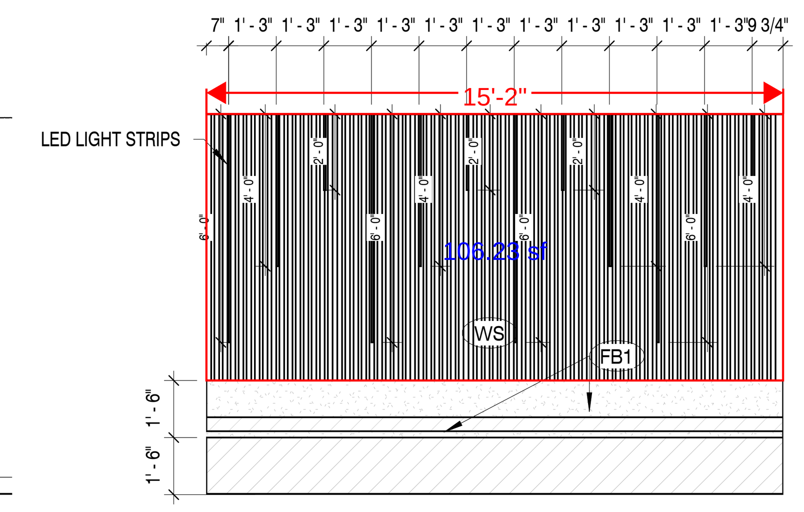
8 EXAM ROOM 2 ELEVATION
1/4" = 1'-0"



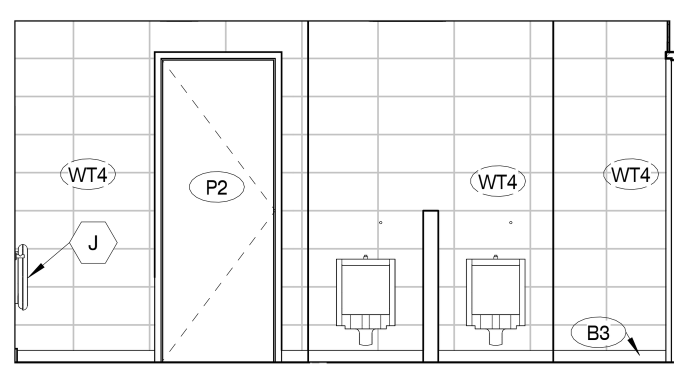
7 HISTORY CENTER ELEVATION
1/4" = 1'-0"



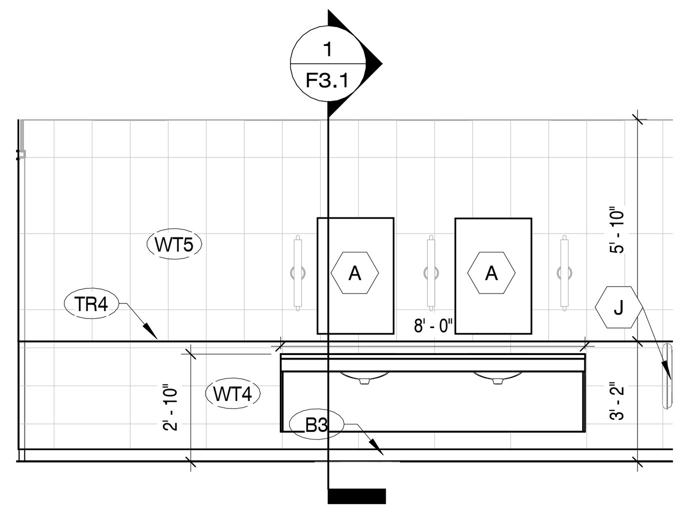
6 WATER FOUNTAIN ELEVATION
1/4" = 1'-0"



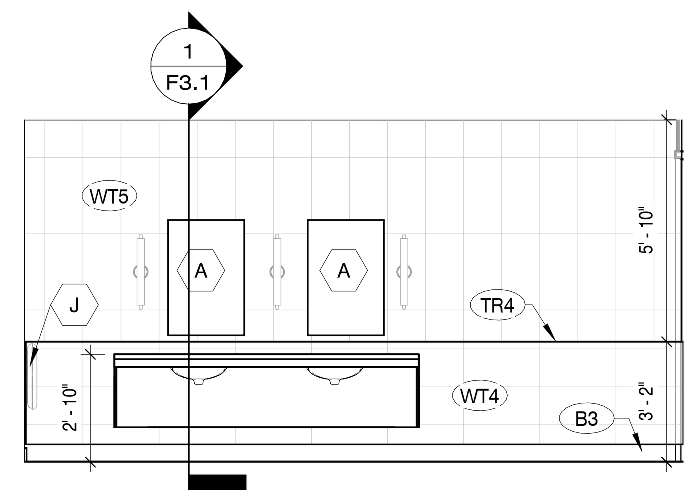
5 SEATING AREA ELEVATION
1/4" = 1'-0"



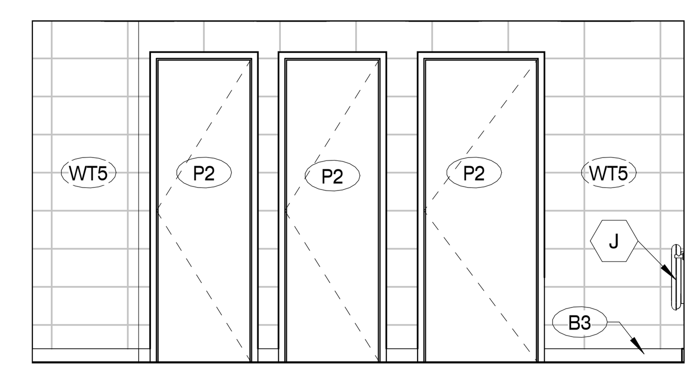
12 MENS RESTROOM LEFT ELEVATION
1/4" = 1'-0"



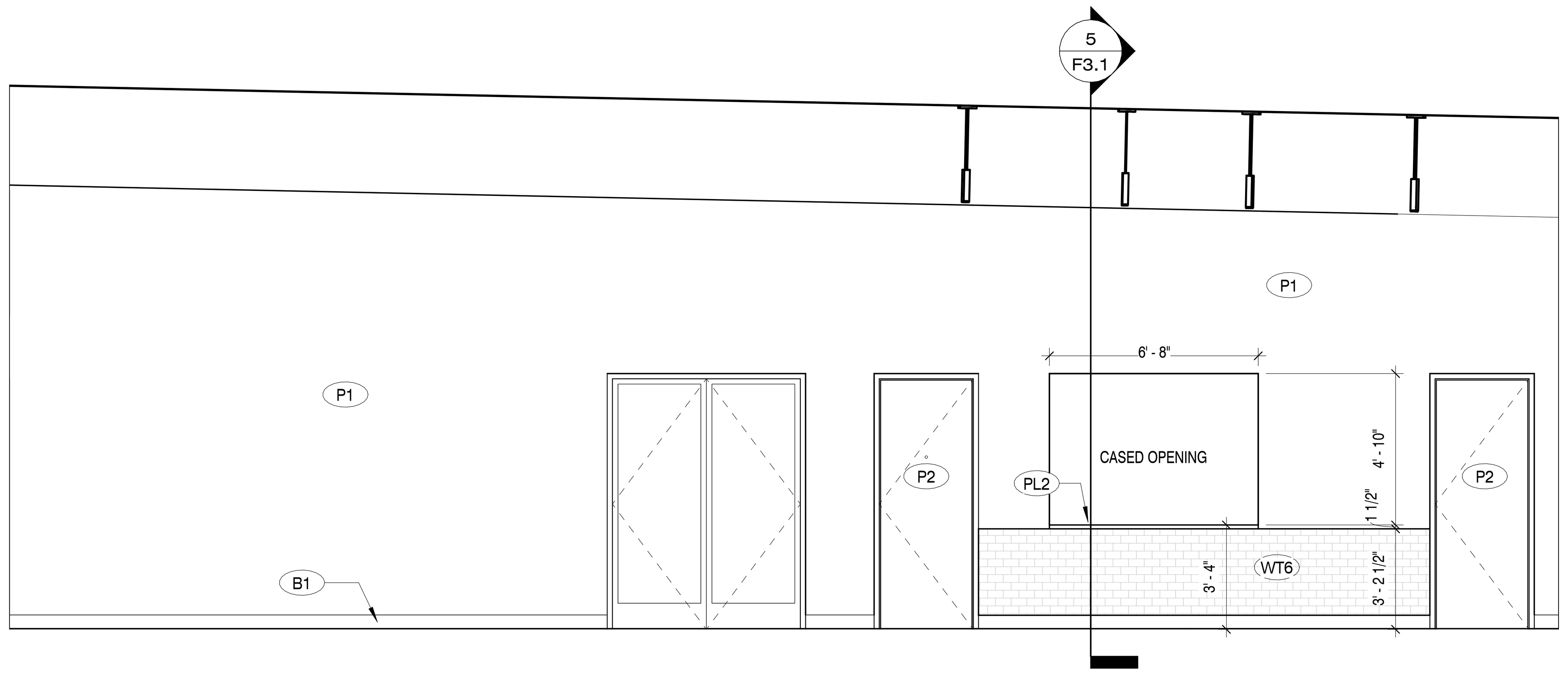
11 MENS RESTROOM RIGHT ELEVATION
1/4" = 1'-0"



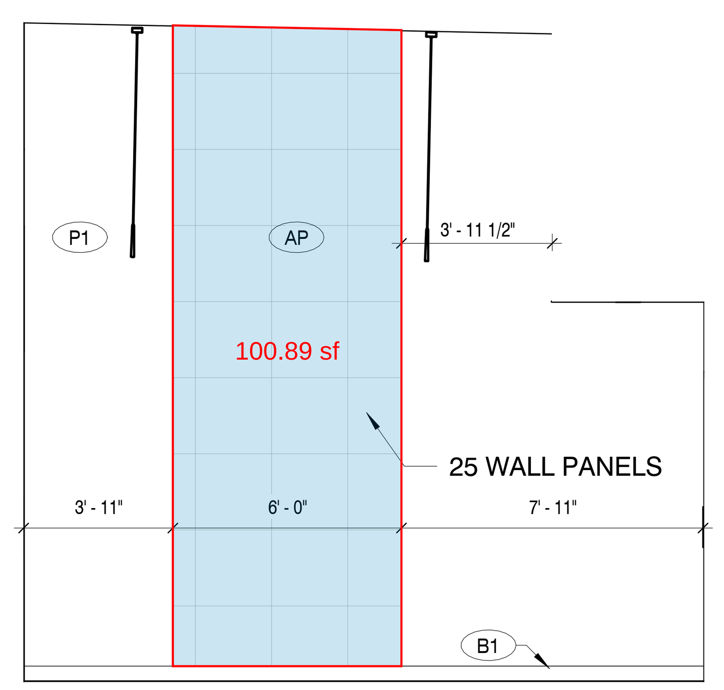
10 WOMENS RESTROOM LEFT ELEVATION
1/4" = 1'-0"



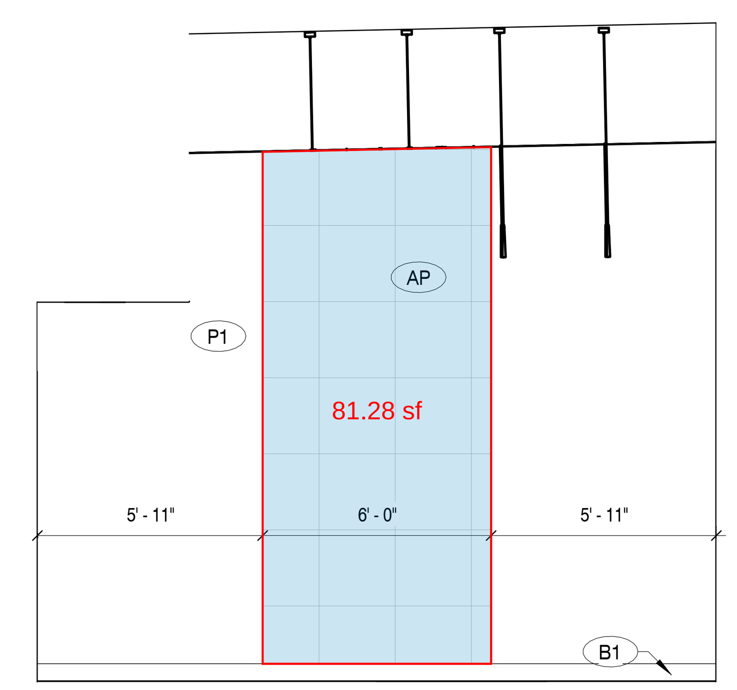
9 WOMENS RESTROOM RIGHT ELEVATION
1/4" = 1'-0"



15 LODGE RIGHT ELEVATION
1/4" = 1'-0"



14 COMPUTER LAB RIGHT ELEVATION
1/4" = 1'-0"



13 COMPUTER LAB LEFT ELEVATION
1/4" = 1'-0"

FINISH SCHEDULE

CODE	ITEM	MANUFACTURER	DESCRIPTION
	FLOORING		NOTE: LOCATE THE FLOOR FINISH CHANGE AT DOOR OPENINGS AT THE CENTERLINE OF THE DOOR LEAF
LVT1	LUXURY VINYL TILE	COBALT SURFACES	KATANGA COLLECTION/ COLOR: MUSHROOM/ SIZE: 3MM 20 MIL / SKU: K320-831
LVT2	LUXURY VINYL TILE	COBALT SURFACES	KATANGA COLLECTION/ COLOR: MAGNOLIA/ SIZE: 3MM 20 MIL / SKU: K320-833
RF	RESILIENT FLOORING	NORA FLOORING	COLLECTION: NORAMENT SATURA / COLOR: VULPECULA 5106
RF2	VINYL COMP. TILE	FORBO FLOORING	COLLECTION: SPHERA ELITE/ COLOR: 50480 SELENTINE
RF3	RESILIENT FLOORING	NORA FLOORING	COLLECTION: NORAPLAN ULTRA GRIP / COLOR: 6016
FT1	FLOOR TILE	CROSSVILLE TILE	TRAJECTORY COLLECTION/ COLOR: DIRECTION / SIZE: 12" X 24" / SKU: TRY02/ GROUT: TEC 908 DOVE GRAY
SC	SEALED CONCRETE	————	————
TR1	TRANSITION	SCHLUTER	STYLE: RENO-U EDGE-PROTECTION PROFILE FOR ABUTTING TILE TO LOWER TRANSITIONS COLOR: SATIN ANODIZED, HEIGHT: 10 MM, TILE TO LVT
TR2	TRANSITION	SCHLUTER	STYLE: VINPRO-S RESILIENT SURFACE EDGE PROTECTION MINIMAL REVEAL COLOR: BRUSHED ANTIQUE BRONZE ANODIZED. HEIGHT: 4MM
TR3	TRANSITION	SCHLUTER	STYLE: VINPRO-S RESILIENT SURFACE EDGE PROTECTION MINIMAL REVEAL COLOR: BRUSHED ANTIQUE BRONZE ANODIZED. HEIGHT: 3MM
TR4	TRANSITION	SCHLUTER	STYLE: RONDEC-DB DECORATIVE PROFILE PRONOUNCED VISIBLE SURFACE. COLOR: SATIN ANODIZED. HEIGHT 14MM
B1	BASE	SHAW CONTRACT	RUBBER COVE BASE / COLOR: EBONY
B2	BASE	CROSSVILLE TILE	COLLECTION: TRAJECTORY / COLOR: DIRECTION / SIZE: 6" X 12"
DECK	DECK	TREX	COLLECTION: ENHANCE / 1" SQUARE EDGE PROFILE / 1" x 12" x 12' FASCIA / COLOR: TBD
	WALLS		
P1	PAINT	SHERWIN WILLIAMS	COLOR: PURE WHITE / SKU: SW 7005
P2	PAINT	SHERWIN WILLIAMS	COLOR: CLOVE (DOOR PAINT) / SKU: SW 9605
P3	PAINT	SHERWIN WILLIAMS	COLOR: EVERGREEN FOG (ACCENT) / SKU: SW 9130
P4	PAINT	SHERWIN WILLIAMS	COLOR: BALANCED BEIGE (DOOR PAINT) / SKU: SW 7037
WT1	WALL TILE	PORCELANOSA	COLLECTION: NEBRASKA / COLOR: ICE NEBRASKA TEA / SIZE: 18" X 47" / INSTALLATION: VERTICAL / SKU: 100300114 / GROUT: TEC 961 SANDSTONE BEIGE
WT2	WALL TILE	PORTOBELLO	COLLECTION: ARTFACT / COLOR: STRAW / SIZE: 2.5" X 0.9" GLOSSY PRESSED/ INSTALLATION: HORIZONTAL / SKU: 900728E / GROUT: TEC 988 PEARL
WT3	WALL TILE	PORTOBELLO	COLLECTION: LIGHTHOUSE / COLOR: SEA / SIZE: 2" X 10" GLOSSY PRESSED/ INSTALLATION: VERTICAL / SKU: 204628E / GROUT: TEC 953 STARRY NIGHT
WT4	WALL TILE	CROSSVILLE TILE	COLLECTION: ARGENT 2.0 / COLOR: WHITE TIGER UPS / SIZE: 12" 24" / INSTALLATION: HORIZONTAL / SKU: ARG31.11224UPS / GROUT: TBA
WT5	WALL TILE	MSI	COLLECTION: STELLA / COLOR: OCEANO / SIZE: 2 X 10 / INSTALLATION: VERTICAL / SKU: NSTEOCE2X10G / GROUT: TEC 909 STERLING
WT6	WALL TILE	PORCELANOSA	COLLECTION: MOSAICO / COLOR: NANTES CALIZA / SIZE: 18X47 / SKU: 100239861/ INSTALLATION: HORIZONTAL / GROUT: TEC 909 STERLING
WT7	WALL TILE	FLOOR & DECOR	COLLECTION: CANVAS / COLOR: MERINGUE II / SIZE: 3X12 / SKU:100655299/ GROUT: TEC 949 SILVERADO
	PLASTIC LAMINATE	————	
PL1	P. LAM	WILSON ART	COLOR: KING LAKE / SKU: 8263
PL2	P. LAM	WILSON ART	COLOR: QUARTZ FROST / SKU: 5031
	CEILING		
ACT1	ACOUSTICAL CEILING TILE	ARMSTRONG	STYLE: FINE FISSURED SECOND LOOK II-1761, SIZE: 24" x 48" x 3/4", COLOR: WHITE, GRID: 15/16"
ACT2	ACOUSTICAL CEILING TILE	ARMSTRONG	STYLE: ULTIMA-1912 BEVELED TEGULAR, SIZE: 24" x 24" x 3/4", COLOR: WHITE FINE TEXTURE, GRID: 9/16"
GB1	GYP. BD. BULKHEAD	SHERWIN WILLIAMS	PAINT (P2) UNLESS NOTED OTHERWISE
	MISC.		
SS1	SOLID SURFACE	WILSON ART	COLOR: FROSTY WHITE / SKU: 1573SL
CB1	CABINETS	WILSON ART	SPECTRUM COLOR ALTERATION / COLOR: SW9130 EVERGREEN FOG
AP	ACOUSTIC PANELING	FELT RIGHT	STYLE: SUBWAY TILE /COLOR: INDIGO / SIZE : 2' X 2'
WS	WOOD SLATS	TERRAMAI	WHITE OAK SQUARE SLAT MODULAR PANEL/ SKU: 2864-S
FB1	CUSHION FABRIC	MOMENTUM	PATTERN: MIXER EPU / COLOR: ICE BLUE / SKU: 09554906
AS	ACOUSTIC SLATS	SOELBERG	COLLECTION: MUTO BOX / COLOR: NATURAL WALNUT
AG	ACOUSTIC GRID	SOELBERG	COLLECTION: MUTO GRID / COLOR: NATURAL ASH

NOTES:
1. ALL DOOR TRIM COLOR TO BE THE SAME AS DOOR PAINT
2. PAINT FINISHES:
   - WALLS: SATIN
   - CEILINGS: EGGSHELL
   - TRIM: SEMI-GLOSS
   - RESTROOMS: SEMI GLOSS

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THOMAS DANIEL BRITT
REGISTERED ARCHITECT
COMMERCE
No. 102274
STATE OF TENNESSEE
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

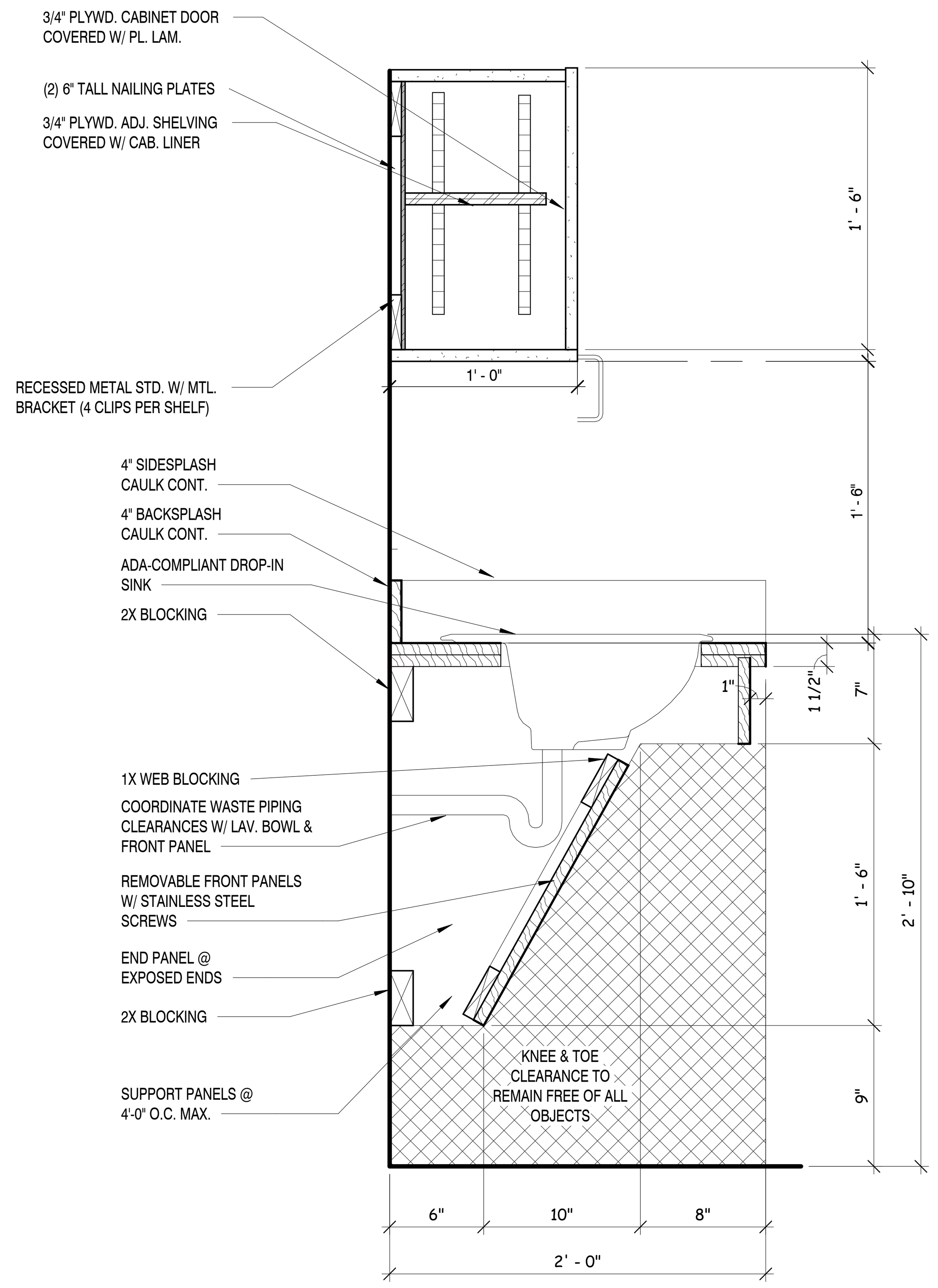
No.	Date

Drawing Title:
FINISH INTERIOR ELEVATIONS

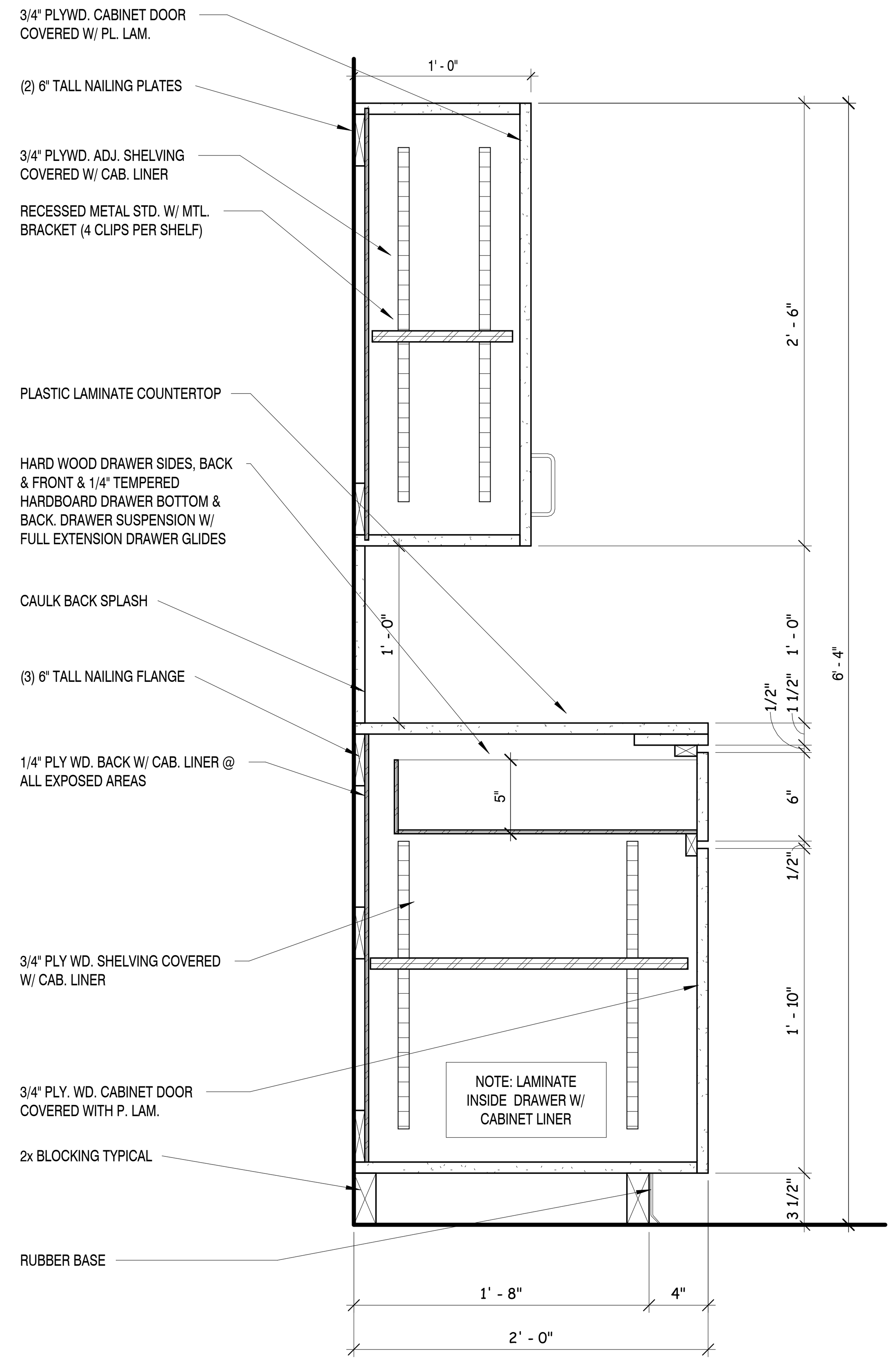
Date: 11/26/2025

Project No. 25026

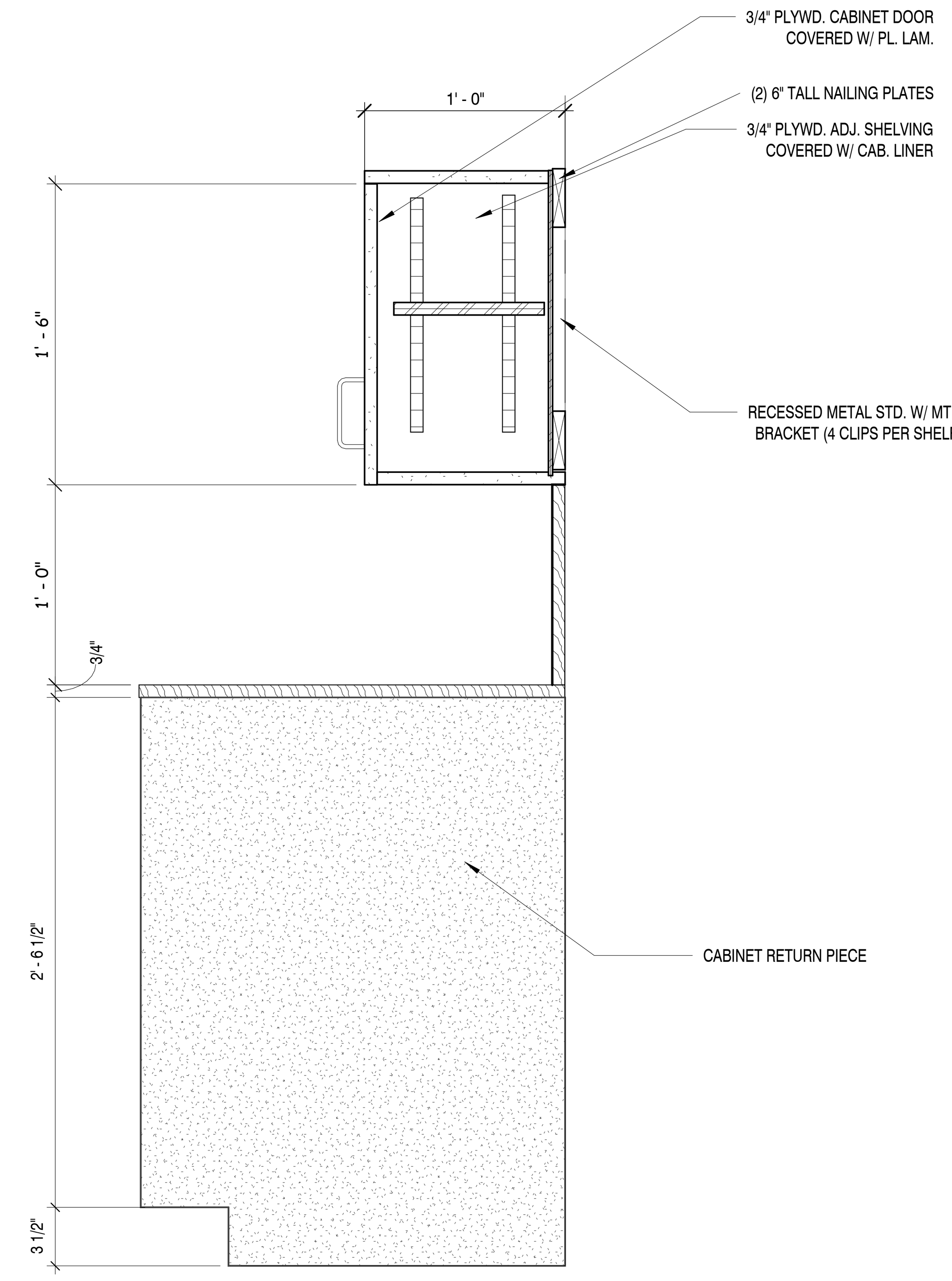
Sheet No. F2.1



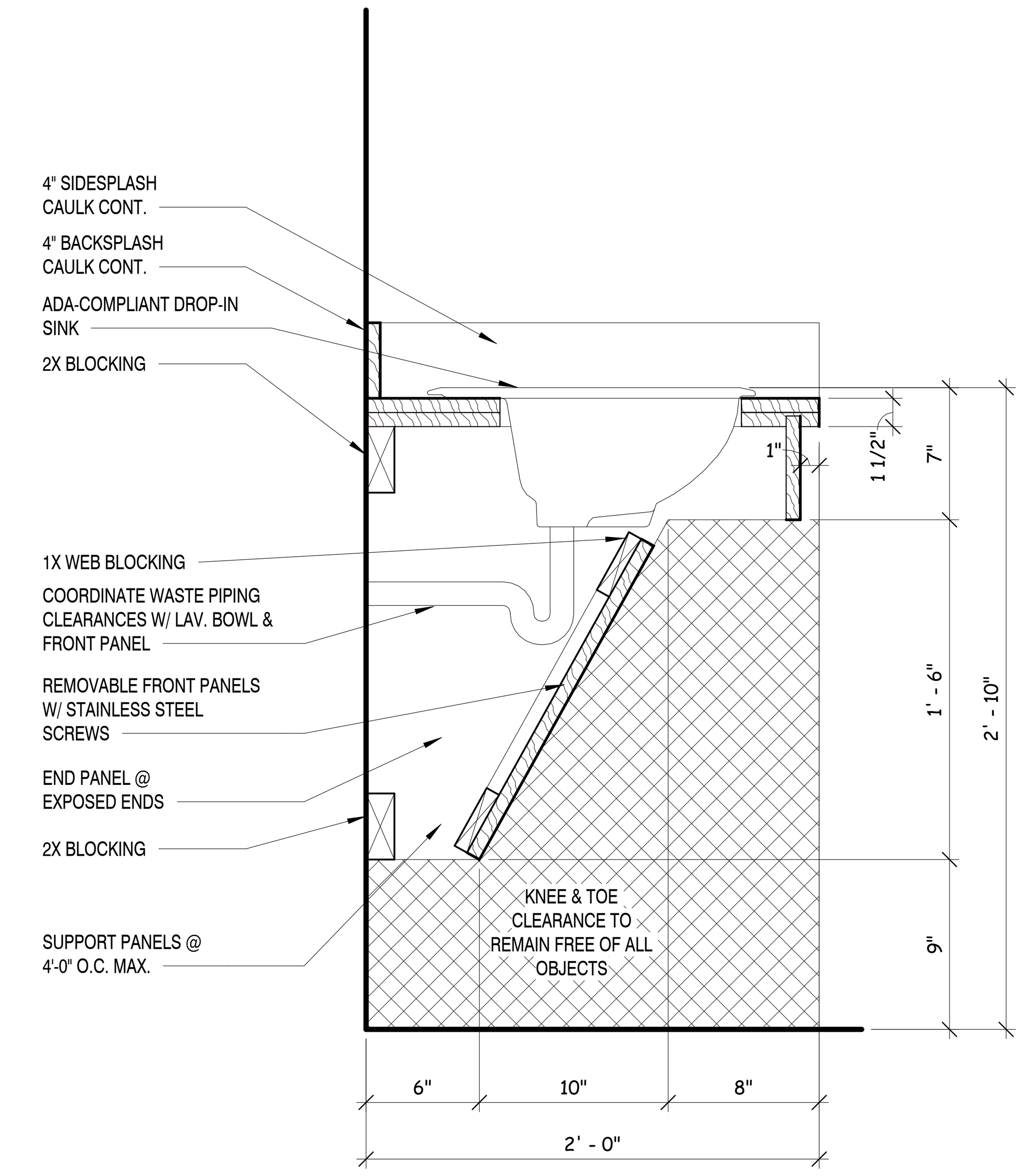
④ EXAM ROOM MILLWORK SECTION
1 1/2" = 1'-0"



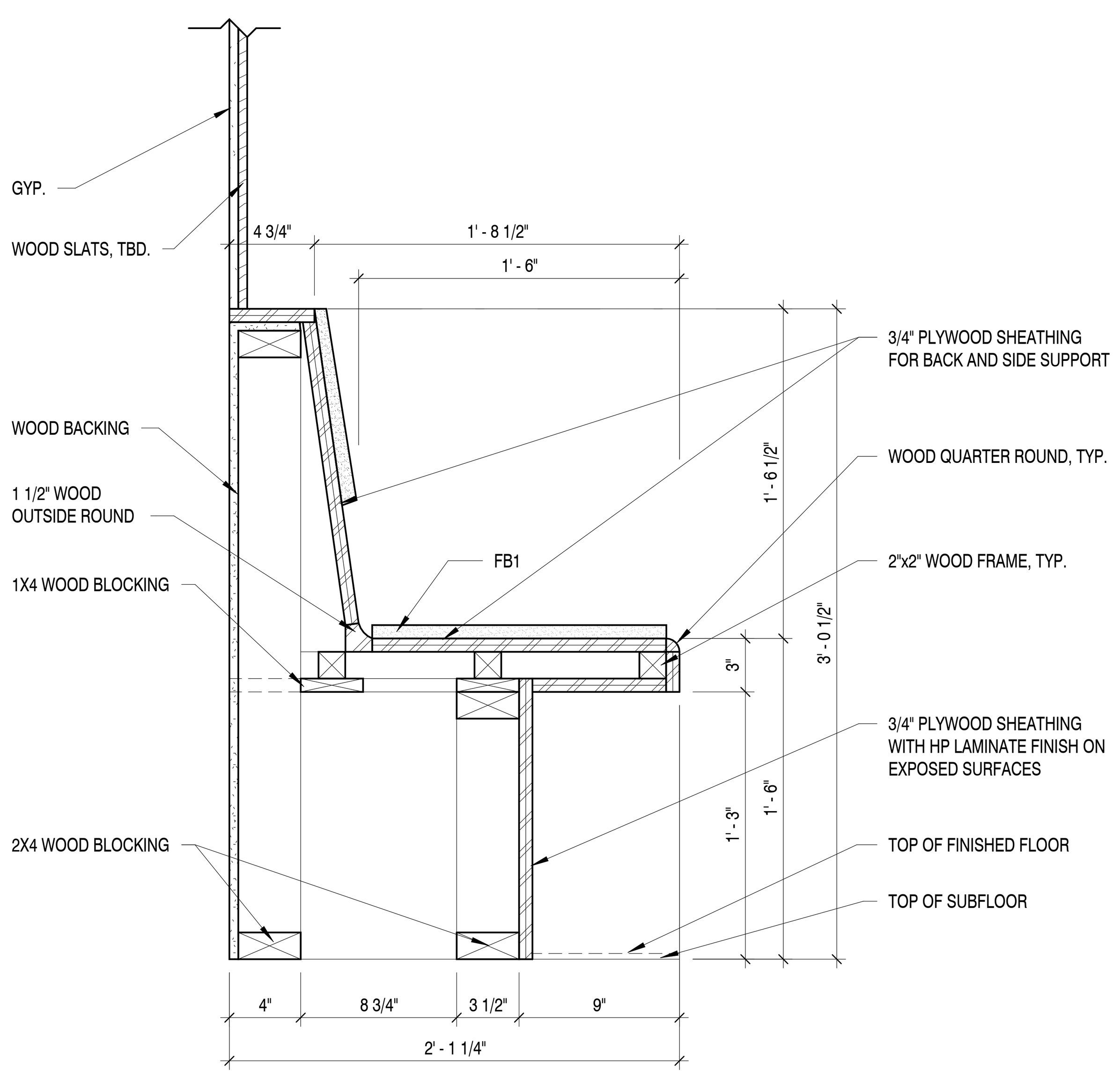
③ EXAM ROOM MILLWORK SECTION
1 1/2" = 1'-0"



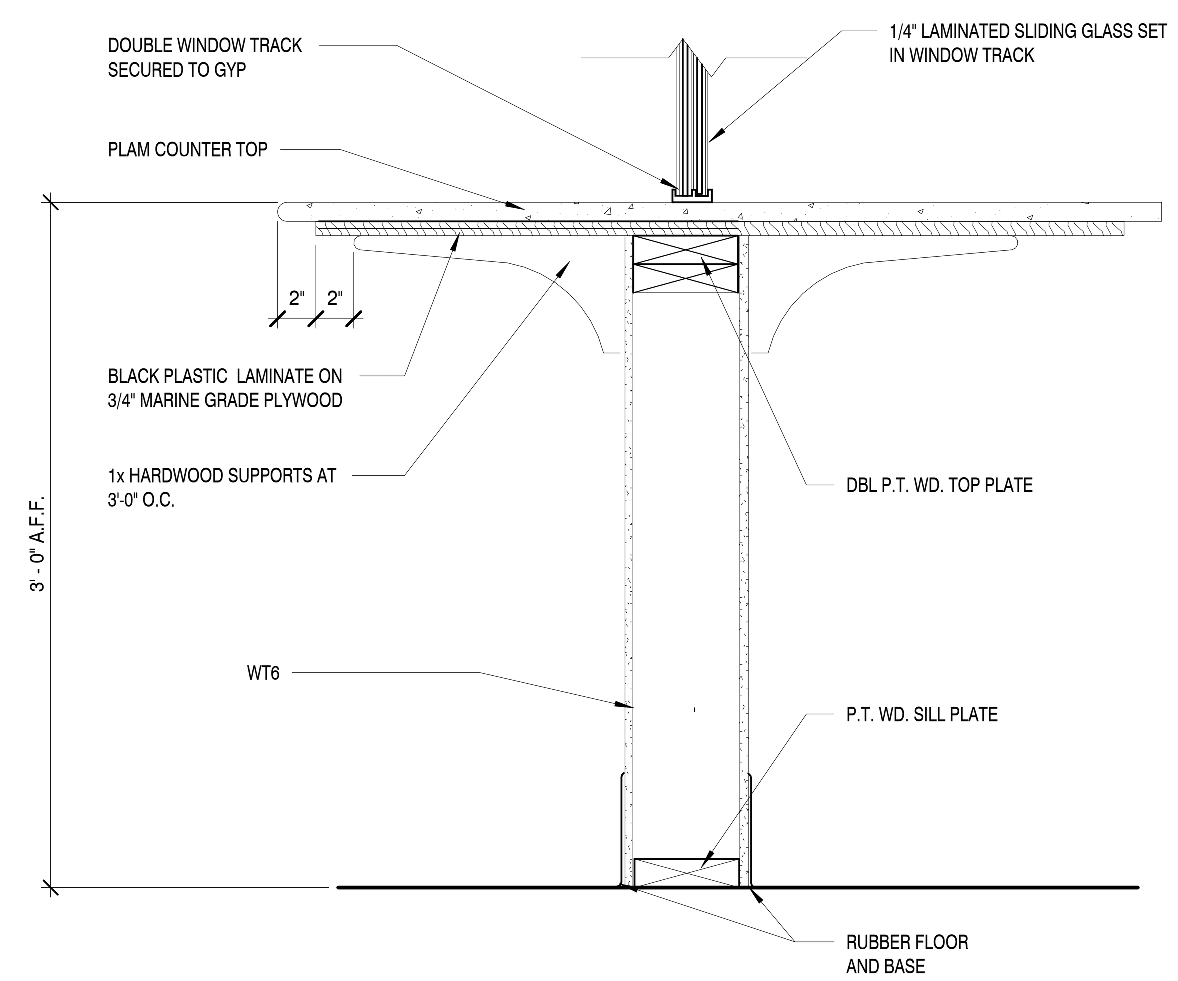
② EXAM ROOM 2 MILLWORK SECTION
1 1/2" = 1'-0"



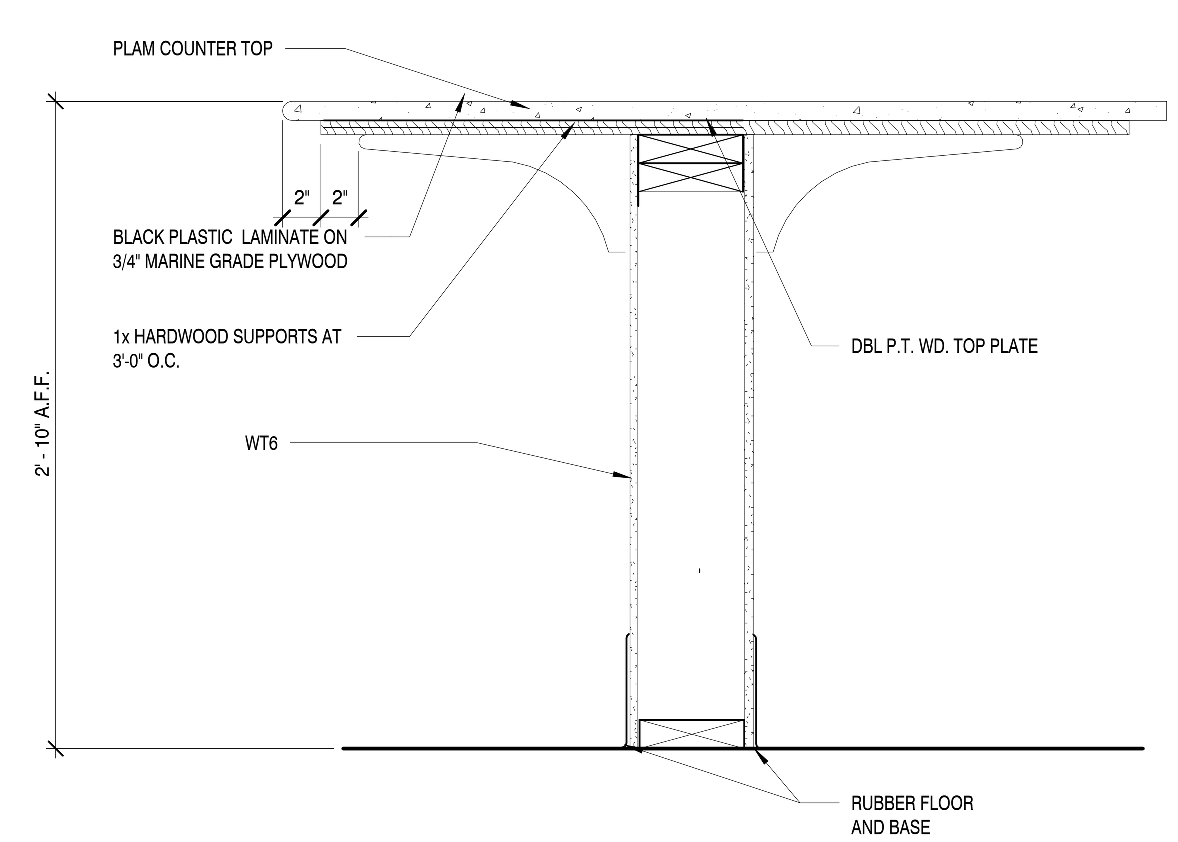
① EXAM ROOM 2 MILWORK SECTION
1 1/2" = 1'-0"



⑦ BUILT-IN BENCH MILWORK SECTION
1 1/2" = 1'-0"



⑥ CLINIC LOBBY RECEPTION DESK SECTION
1 1/2" = 1'-0"



⑤ LODGE FOOD COUNTER SECTION
1 1/2" = 1'-0"

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THOMAS DANIEL BRITT
REGISTERED ARCHITECT
COMMERCE
No. 102271
STATE OF TENNESSEE
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

Drawing Title:
MILLWORK SECTIONS

Date: 11/26/2025

Project No.
25026

Sheet No.
F3.1

GENERAL NOTES :

1.01 ALL CONSTRUCTION SHALL CONFORM TO THE 2018 INTERNATIONAL BUILDING CODE. REFERENCE TO OTHER STANDARD SPECIFICATIONS OR CODES SHALL MEAN THE LATEST STANDARD OR CODE ADOPTED AND PUBLISHED UNLESS SPECIFIED OTHERWISE.

1.02 DRAWINGS SHOW TYPICAL AND CERTAIN SPECIFIC CONDITIONS ONLY. FOR DETAILS NOT SPECIFICALLY SHOWN, PROVIDE DETAILS SIMILAR TO THOSE SHOWN.

1.03 VERIFY ALL EXISTING CONDITIONS, DIMENSIONS, AND ELEVATIONS BEFORE STARTING WORK. NOTIFY THE STRUCTURAL ENGINEER OF ANY DISCREPANCY. NOTIFY THE STRUCTURAL ENGINEER IN WRITING OF CONDITIONS ENCOUNTERED IN THE FIELD CONTRADICTORY TO THOSE SHOWN ON THE STRUCTURAL CONTRACT DOCUMENTS.

1.04 THE STRUCTURE IS DESIGNED FOR A COMPLETED CONDITION ONLY, AND THEREFORE REQUIRE TEMPORARY SUPPORT BRACING DURING CONSTRUCTION. THE STRUCTURE SHALL BE CONSIDERED STABLE WHEN : ALL THE FRAMING HAS BEEN ERECTED AND CONNECTED AS SHOWN ON THE DESIGN AND SHOP FABRICATION DRAWINGS, SLAB, FLOOR, AND ROOF DIAPHRAGMS ARE COMPLETELY ATTACHED AND CURED AND THE FOOTINGS HAVE BEEN COMPLETELY BACKFILLED. THE DESIGN, ADEQUACY, AND SAFETY OF ERECTION BRACING, SHORING, TEMPORARY SUPPORTS, ETC. IS THE SOLE RESPONSIBILITY OF THE CONTRACTOR.

1.05 FOR DIMENSIONS NOT SHOWN, SEE ARCHITECTURAL DRAWINGS.

1.06 REVIEW OF SUBMITTALS AND/OR SHOP DRAWINGS BY THE STRUCTURAL ENGINEER DOES NOT RELIEVE THE CONTRACTOR OF THE RESPONSIBILITY TO REVIEW AND CHECK SHOP DRAWINGS BEFORE SUBMITTAL TO THE STRUCTURAL ENGINEER. THE CONTRACTOR REMAINS SOLELY RESPONSIBLE FOR ERRORS AND OMISSIONS ASSOCIATED WITH THE PREPARATION OF SHOP DRAWINGS AS THEY PERTAIN TO MEMBER SIZES, DETAILS, ASSEMBLY REQUIREMENTS, AND DIMENSIONS SPECIFIED IN THE CONTRACT DOCUMENTS. CONTRACTOR IS ALSO RESPONSIBLE FOR MEANS, METHODS, TECHNIQUES, SEQUENCES, PROCEDURES AND SAFETY OF CONSTRUCTION.

1.07 CONTRACTOR TO REFER TO DRAWINGS OF OTHER TRADES AND VENDOR DRAWINGS FOR EMBEDDED ITEMS AND RECESSES NOT SHOWN ON THE STRUCTURAL DRAWINGS.

1.08 CONTRACTOR SHALL VERIFY SIZE AND LOCATION OF ALL MECHANICAL AND ELECTRICAL OPENINGS AND EQUIPMENT PADS WITH THE MECHANICAL AND ELECTRICAL EQUIPMENT DETAILS AND APPROVED SHOP DRAWINGS. IT SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR TO PROVIDE ALL OPENINGS AND SLEEVES FOR PROPER DISTRIBUTION FOR ALL UTILITY LINES THROUGHOUT THE BUILDING.

1.09 CONTRACTOR SHALL BE RESPONSIBLE FOR THE DESIGN, INSTALLATION AND FINAL REMOVAL/CLEARANCE OF ANY REQUIRED SHORING OR BRACING OF STRUCTURES.

DESIGN LOADS: 2018 IBC (ASCE 7-16)

DEAD LOADS:

ROOF:......SELF WEIGHT OF MATERIALS AS SHOWN
FLOOR......SELF WEIGHT OF MATERIALS AS SHOWN

LIVE LOADS:

ROOF:......20 psf
FLOOR SLAB:......100 psf

WIND LOADS:

ULTIMATE DESIGN WIND SPEED:......104 MPH
EXPOSURE CATEGORY:......C
OCCUPANCY CATEGORY:......II
IMPORTANCE FACTOR:......1.0 (MAIN WIND FORCE RESISTING SYSTEM)
1.15 (COMPONENTS AND CLADDING)
ENCLOSURE CLASSIFICATION:......ENCLOSED
INTERNAL PRESSURE COEFFICIENT:......+/-0.18

WALL COMPONENTS AND CLADDING:......25 sf +/- 38 psf
50 sf +/- 33 psf
75 sf +/- 30 psf
100 sf +/- 28 psf

SEISMIC LOADS

MAPPED SPECTRAL RESPONSE ACCELERATION:......S_S = 52.0%
S_1 = 12.3%
OCCUPANCY CATEGORY:......II
SEISMIC IMPORTANCE FACTOR:......1.0
SITE CLASS:......D
SPECTRAL RESPONSE COEFFICIENT:......S_{DS} = 0.480
S_{D1} = 0.194
SEISMIC DESIGN CATEGORY:......C
BASIC SEISMIC FORCE RESISTING SYSTEM:......WOOD-FRAMED WALLS SHEATHED WITH RATED SHEAR PANELS
RESPONSE MODIFICATION FACTOR, R:......6.5
ANALYSIS PROCEDURE USED:......EQUIVALENT LATERAL FORCE
SEISMIC RESPONSE COEFFICIENTS, Cs:......0.074
DESIGN BASE SHEAR, V:......0.074W

SNOW LOADS

GROUND SNOW LOAD, Pg:......10 psf
FLAT ROOF SNOW LOAD, Pf:......7 psf
SNOW EXPOSURE FACTOR, Ce:......1.0
SNOW LOAD IMPORTANCE FACTOR, I:......1.0
THERMAL FACTOR, Ct:......1.0
RAIN ON SNOW SURCHARGE:......5 psf (BALANCED LOAD)

ALLOWABLE SOIL BEARING PRESSURE:......2,000 psf (DESIGN BASIS)

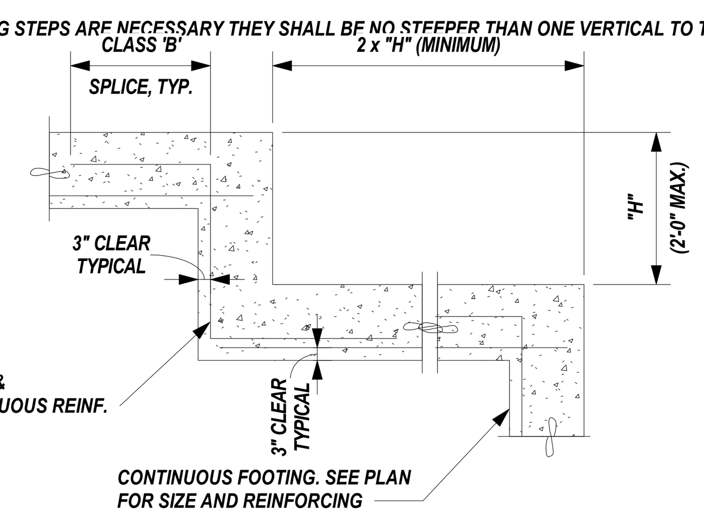
FOUNDATION NOTES :

2.01 OWNER OR CONTRACTOR'S GEOTECHNICAL ENGINEER SHALL VERIFY CONDITION AND/OR ADEQUACY OF ALL SUBGRADE, FILLS, AND BACKFILLS BEFORE PLACEMENT OF FOUNDATIONS, FOOTING, SLABS, WALLS, FILLS, BACKFILLS, ETC. ALL FOOTINGS SHALL REST EITHER ON UNDISTURBED SOIL OR NEWLY PLACED STRUCTURAL FILL. OWNER OR CONTRACTOR'S GEOTECHNICAL ENGINEER SHALL VERIFY ALLOWABLE SOIL BEARING CAPACITY PREPARATION REQUIREMENTS INCLUDING SUBGRADE IMPROVEMENT AND STRUCTURAL FILL PLACEMENT REQUIREMENTS. A MANUALLY OPERATED VIBRATOR SLED OR TAMPER SHOULD BE USED TO DENSIFY ANY SOILS IN THE BOTTOM OF THE FOOTING TRENCHES LOOSENED DURING THE EXCAVATION PROCESS.

2.02 SIDES OF THE FOUNDATIONS SHALL BE FORMED UNLESS CONDITIONS PERMIT EARTH FORMING. FOUNDATIONS POURED AGAINST THE EARTH REQUIRE THE FOLLOWING PRECAUTIONS : SLOPE SIDES OF EXCAVATIONS AS APPROVED BY THE GEOTECHNICAL ENGINEER AND CLEAN UP SLOUGHING BEFORE AND DURING CONCRETE PLACEMENT.

2.03 CONTRACTOR IS RESPONSIBLE FOR ADEQUATELY PROTECTING ALL EXCAVATION SLOPES.

2.04 WHERE FOOTING STEPS ARE NECESSARY THEY SHALL BE NO STEEPER THAN ONE VERTICAL TO TWO HORIZONTAL.



TYPICAL FOOTING STEP DETAIL

NOT TO SCALE

2.05 THE BOTTOM OF FOOTINGS SHALL BEAR BELOW THE FROST DEPTH AS SPECIFIED IN THE GEOTECHNICAL REPORT OR BY THE LOCAL MUNICIPALITY.

2.06 FOOTINGS MUST BE BACKFILLED BEFORE THE STRUCTURE IS CONSIDERED STABLE. CONCRETE SLAB SHALL REACH 28 DAY COMPRESSIVE STRENGTH BEFORE THE STRUCTURE IS CONSIDERED STABLE.

2.07 COORDINATE FOOTING STEPS WITH MECHANICAL, PLUMBING, AND ELECTRICAL DRAWINGS.

2.08 SOIL SUPPORTED SLABS ON GRADE ARE DESIGNED TO BEAR ON A SUBGRADE WITH A MINIMUM MODULUS OF SUBGRADE REACTION OF 150 PCI. WHERE POSSIBLE. SEE GEOTECHNICAL REPORT FOR AREAS WHERE MINIMUM MODULUS OF SUBGRADE REACTION WILL DIFFER.

2.09 WHERE FOUNDATION EXCAVATIONS MUST REMAIN OPEN AND ARE SUBJECT TO RAINFALL, THE EXCAVATIONS SHALL BE UNDERCUT AND A 3" THICK MUD MAT OF 2,000 PSI CONCRETE SHALL BE PLACED IN THE BOTTOM TO PROTECT THE BEARING SOILS.

REINFORCED CONCRETE NOTES:

3.01 ALL CONCRETE WORK SHALL CONFORM TO ACI 301, SPECIFICATIONS FOR STRUCTURAL CONCRETE FOR BUILDINGS. DESIGN IS BASED ON ACI 318, BUILDING CODE REQUIREMENTS FOR REINFORCED CONCRETE.

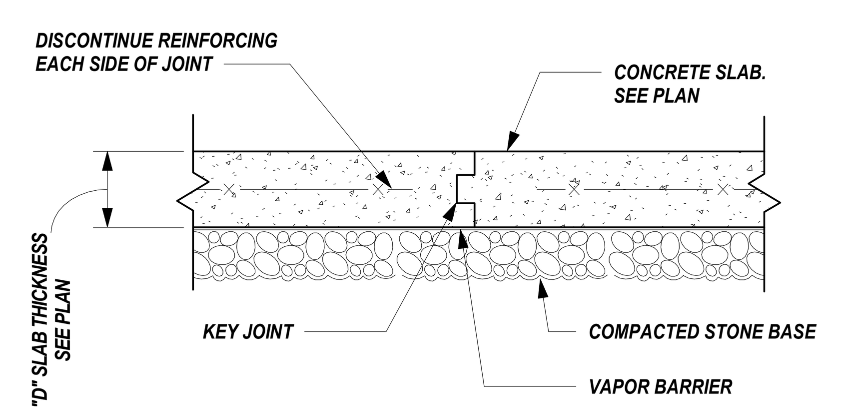
3.02 UNLESS NOTED OTHERWISE, ALL CONCRETE SHALL BE NORMAL WEIGHT AND HAVE THE FOLLOWING MINIMUM 28 DAY COMPRESSIVE STRENGTHS.
SPREAD FOOTINGS......3000 psi
SLAB ON GRADE AND ELEVATED FLOORS......4000 psi U.N.O.
BEAMS, WALLS, FOUNDATION WALLS, AND PILE CAPS......4000 psi
ALL OTHER......3000 psi

ALL EXTERIOR CONCRETE SHALL HAVE 5% - 7% ENTRAINED AIR, U.N.O.

3.03 THE PROPOSED MATERIALS AND MIX DESIGN SHALL BE FULLY DOCUMENTED AND REVIEWED BY A TESTING LABORATORY. RESPONSIBILITY FOR OBTAINING THE REQUIRED DESIGN STRENGTH IS THE CONTRACTOR'S. CONCRETE PROPORTIONS SHALL BE ESTABLISHED ON THE BASIS OF FIELD EXPERIENCE AND/OR TRIAL MIXTURES WITH MATERIALS TO BE EMPLOYED IN ACCORDANCE WITH ACI 318 AND 301.

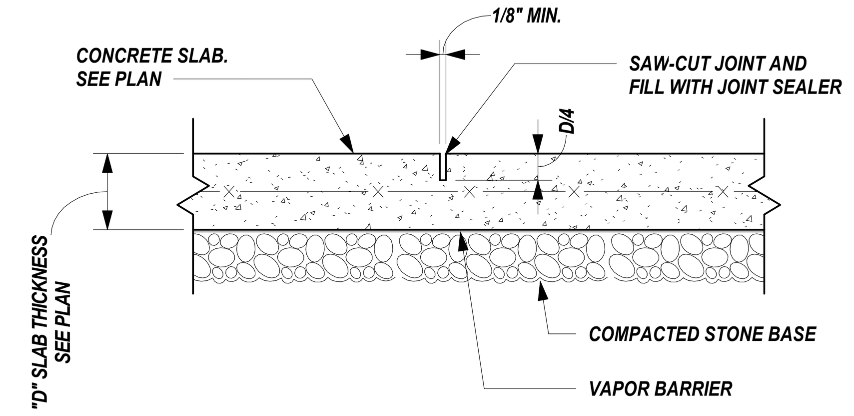
3.04 USE OF CALCIUM CHLORIDE, CHLORIDE IONS, OR OTHER SALTS IN CONCRETE IS NOT PERMITTED.

3.05 DETAIL CONSTRUCTION AND CONTRACTION JOINTS AS SHOWN IN DETAILS BELOW WITH MAXIMUM JOINT SPACING OF 100 SQUARE FEET



CONSTRUCTION JOINT DETAIL

NOT TO SCALE



CONTRACTION JOINT DETAIL

NOT TO SCALE

3.06 CHAMFER OR ROUND ALL EXPOSED CORNERS MINIMUM 3/4".

3.07 DETAIL CONCRETE REINFORCEMENT AND ACCESSORIES IN ACCORDANCE WITH ACI 315 DETAILING MANUAL. SUBMIT SHOP DRAWINGS FOR APPROVAL, SHOWING ALL FABRICATION DIMENSIONS AND LOCATIONS FOR PLACING REINFORCING STEEL AND ACCESSORIES. DO NOT BEGIN FABRICATION UNTIL SHOP DRAWINGS ARE COMPLETED, REVIEWED AND APPROVED. WRITTEN DESCRIPTION OF REINFORCEMENT WITHOUT ADEQUATE SECTIONS, ELEVATIONS, AND DETAILS IS NOT ACCEPTABLE.

3.08 REINFORCING STEEL SHALL CONFORM TO ASTM A615, GRADE 60 DEFORMED BARS UNLESS NOTED OTHERWISE.

3.09 TIE ALL REINFORCING STEEL AND EMBEDMENTS SECURELY IN PLACE PRIOR TO PLACING CONCRETE. PROVIDE SUFFICIENT SUPPORTS TO MAINTAIN THE POSITION OF REINFORCEMENT WITHIN SPECIFIED TOLERANCE DURING ALL CONSTRUCTION ACTIVITIES. "STICKING" DOWELS INTO WET CONCRETE IS NOT PERMITTED.

3.10 PROVIDE CONTINUOUS REINFORCEMENT WHEREVER POSSIBLE. SPLICE ONLY AS SHOWN OR APPROVED. STAGGER SPLICE WHERE POSSIBLE. USE FULL TENSION SPLICE (CLASS "B") UNLESS NOTED OTHERWISE. DOWELS SHALL MATCH THE SIZE AND SPACING OF THE SPECIFIED REINFORCEMENT AND SHALL BE LAPPED WITH FULL TENSION SPLICES (CLASS "B") UNLESS NOTED OTHERWISE. TERMINATE BARS WITH STANDARD HOOKS.

3.11 REINFORCING STEEL SHALL HAVE THE FOLLOWING CONCRETE COVER UNLESS NOTED OTHERWISE.

CONCRETE AGAINST EARTH (NOT FORMED)......3"

FORMED CONCRETE EXPOSED TO EARTH OR WEATHER
#6 THROUGH #18 BARS......2"
#5 BARS AND SMALLER......1 1/2"
COVER FOR TOP BARS IN CONCRETE FOOTINGS SHALL BE 2".

CONCRETE NOT EXPOSED TO EARTH OR WEATHER
SLABS AND WALLS......1"

3.12 DO NOT WELD OR TACK WELD REINFORCING STEEL UNLESS APPROVED OR DIRECTED BY THE STRUCTURAL ENGINEER.

3.13 THE DESIGN AND CONSTRUCTION OF FORMS SHALL CONFORM TO THE FOLLOWING REQUIREMENTS :

A. FORMS SHALL CONFORM TO SHAPE, FORM, AND LINES ON DRAWINGS.
B. ADEQUATE BRACING SHALL BE USED.
C. FORMS SUPPORTED ON GROUND SHALL HAVE ADEQUATE MUD SILLS.
D. QUALIFIED WORKMEN SHALL CONSTANTLY OBSERVE AND ADJUST, AS REQUIRED, ALL SHORES DURING CONCRETE PLACING.
E. THE CONTRACTOR SHALL BE SOLELY RESPONSIBLE FOR THE ADEQUATE DESIGN AND CONSTRUCTION OF ALL FORMS.

3.14 SHORING SHALL REMAIN IN PLACE UNTIL CONCRETE HAS ATTAINED 75% OF ITS 28 DAY STRENGTH.

3.15 ALL REINFORCING STEEL PLACEMENT SHALL BE REVIEWED BY THE GENERAL CONTRACTOR FOR COMPLIANCE WITH APPROVED SHOP DRAWINGS AND THE REQUIREMENTS OF THE SPECIFICATIONS.

3.16 THE FOLLOWING REINFORCING IS TO BE PROVIDED UNLESS NOTED OR DETAILED OTHERWISE.

A. PROVIDE CORNER BARS WITH CLASS 'B' SPLICE IN CORNERS OF ALL FOOTINGS, AND REINFORCED WALLS. PROVIDE SAME BAR SIZE, NUMBER OF BARS, AND SPACING AS CONTINUOUS HORIZONTAL REINFORCEMENT.
B. PROVIDE "Z" BARS IN ALL FOOTING STEPS FOR EACH CONTINUOUS BAR.

3.17 FOR MISC. CONCRETE PADS NOT SHOWN ON THE STRUCTURAL DRAWINGS, SEE ARCHITECTURAL DRAWINGS.

3.18 DO NOT PLACE PIPES OR CONDUIT IN THE PLANE OF SLABS ON GRADE. DO NOT PLACE PIPES OR DUCTS WITH DIAMETER EXCEEDING ONE HALF OF THE PENETRATED WALL THICKNESS THROUGH THE WALL UNLESS SPECIFICALLY SHOWN OR DETAILED ON THE STRUCTURAL DRAWINGS.

3.19 SEE ARCHITECTURAL DRAWINGS FOR CONCRETE FILL AND REINFORCING REQUIRED FOR CONCRETE ITEMS NOT SHOWN ON THE STRUCTURAL DRAWINGS.

3.20 SEE ARCHITECTURAL DRAWINGS FOR EXTERIOR SIDEWALKS OR CONCRETE PAVING.

3.21 REFER TO DRAWING OF OTHER TRADES AND VENDOR DRAWINGS FOR EMBEDDED ITEMS AND RECESSES NOT SHOWN ON THE STRUCTURAL DRAWINGS.

3.22 PROVIDE BONDING AGENT ON CONCRETE SURFACES THAT WILL BE JOINED WITH FRESH CONCRETE.

3.23 WELDED WIRE FABRIC (WWF) SHALL LAP TWO FULL MESHES AND BE SECURELY WIRED AT EACH SIDE AND END. WWF SHALL CONFORM TO ASTM A185 AND HAVE A MINIMUM ULTIMATE STRENGTH OF 75,000 PSI.

3.24 EMBEDDED STRUCTURAL STEEL SHALL BE ASTM A36. ANCHOR BOLTS SHALL BE A36 THREADED RODS WITH CUT THREADS AND NUTS CONFORMING TO ASTM A563. GALVANIZE ALL ANCHOR BOLTS AND NUTS EXPOSED TO WEATHER AND WHERE INDICATED.

3.25 PROVIDE TWO #4 BARS IN TOP OF WALL FOOTINGS SUPPORTING MASONRY WALLS WHERE OPENINGS OF DOORS OCCUR. EXTEND BARS 2'-0" BEYOND EDGE OF OPENINGS.

3.26 SEE SCHEDULE BELOW FOR REINFORCING EMBEDMENT/SPLICE LENGTHS :

REINFORCING EMBEDMENT/SPLICE LENGTHS	
SIZE	MINIMUM SPLICE LENGTH (inches)
3	19
4	25
5	31
6	37

3.27 CONTRACTOR TO PROVIDE AN ADDITIONAL 1 YARD OF CONCRETE AND 0.25 TONS OF REINFORCING STEEL TO THEIR BID TO BE USED AT ARCHITECT OR ENGINEER'S DISCRETION (IN PLACE ALLOWANCE)

WOOD TRUSS NOTES:

6.01 WOOD TRUSSES SHALL BE DESIGNED AND CONSTRUCTED IN ACCORDANCE WITH THE TRUSS PLATE INSTITUTE'S TPI 1-2014 "NATIONAL DESIGN STANDARDS FOR METAL-PLATE-CONNECTED WOOD TRUSS CONSTRUCTION" AND THE 2018 INTERNATIONAL BUILDING CODE..

6.02 WOOD TRUSSES TO BE DESIGNED BY A REGISTERED ENGINEER IN THE STATE OF TENNESSEE.

6.03 WOOD TRUSS DESIGNER TO COORDINATE LAYOUT AND SPACING WITH MECHANICAL, ELECTRICAL, AND PLUMBING. DESIGNER TO PROVIDE MAXIMUM ALLOWABLE SHIFTS BASED ON MECHANICAL, ELECTRICAL, AND PLUMBING LAYOUT.

6.04 ALL RECOMMENDATIONS FOR BRACING, QUALITY CONTROL, AND DESIGN OF THE TRUSSES SHALL BE PER THE TRUSS PLATE INSTITUTE SPECIFICATIONS.

6.05 PROVIDE SHOP DRAWINGS SHOWING STRESS DIAGRAMS STAMPED WITH SEAL OF DESIGNING ENGINEER. THE FOLLOWING ITEMS TO BE INCLUDED WITH SHOP DRAWINGS:
A. LAYOUT PLAN INDICATING LOCATIONS AND TRUSS TYPE.
B. TRUSS DETAILS AND TRUSS TO TRUSS CONNECTIONS.
C. BRACING REQUIREMENTS.
D. ANY OTHER INFORMATION REQUIRED TO COMPLETE TRUSS PORTION OF CONTRACT.

6.06 PROVIDE SPECIAL DESIGNED TRUSSES FOR CORNERS, GIRDERS, HEADERS, ETC.

6.07 ROOF TRUSSES TO BE DESIGNED FOR FOLLOWING LOADS (UNLESS NOTED OTHERWISE):
A. LIVE LOAD TOP CHORD......20 psf
B. DEAD LOAD TOP CHORD......10 psf
C. DEAD LOAD BOTTOM CHORD......10 psf
D. WIND LOADS, SEISMIC LOADS, AND SNOW LOADS PER DESIGN LOADS IN GENERAL NOTES

6.08 FLOOR TRUSSES TO BE DESIGNED FOR FOLLOWING LOADS (UNLESS NOTED OTHERWISE):
A. LIVE LOAD TOP CHORD......40 psf
B. DEAD LOAD TOP CHORD......15 psf
C. DEAD LOAD BOTTOM CHORD......10 psf
D. MAXIMUM LIVE LOAD DEFLECTION......L/480
E. MAXIMUM TOTAL LOAD DEFLECTION......L/360

6.09 ALL LUMBER TO BE A MINIMUM SIZE OF 2 X 4 #2 SOUTHERN PINE, KILN DRIED, FOR TOP AND BOTTOM CHORDS AND #3 SOUTHERN PINE, KILN DRIED, FOR WEBS.

6.10 ALTHOUGH WEB LAYOUT MAY BE SHOWN ON PLANS, IT IS RESPONSIBILITY OF TRUSS DESIGNER TO MODIFY AS REQUIRED FOR DESIGN PURPOSES.

6.11 TRUSSES TO BE CONNECTED TO NAILERS WITH SIMPSON STRONG TIE OR "H2.5A" TRUSS CONNECTORS, OR EQUIVALENT, WITH A MINIMUM G185 GALVANIZED COATING AT EACH BEARING LOCATION, UNLESS NOTED OTHERWISE.

6.12 MAXIMUM SPACING OF ROOF TRUSSES TO BE 24" ON CENTER.

6.13 THE VERTICALS OF GABLE END TRUSSES SHALL BE AT 16" O.C. AND DESIGNED FOR OUT-OF-PLANE WIND LOADS PER DESIGN LOADS IN GENERAL NOTES.

6.14 CONTRACTOR AND SUPPLIER TO PROVIDE ALL LABOR AND MATERIALS TO COMPLETE TRUSS PORTION OF PROJECT.

6.15 ALL ROOF TRUSSES SHALL BE BRACED DURING AND AFTER ERECTION IN ACCORDANCE WITH THE TRUSS PLATE INSTITUTE STANDARDS BCSI-B1, B2 AND B3, 2012 EDITION. THE CONTRACTOR IS SOLELY RESPONSIBLE FOR PROVIDING TEMPORARY BRACING OF THE TRUSS SYSTEM DURING INSTALLATION. PERMANENT BRACING AS REQUIRED BY THE TRUSS DESIGN DRAWINGS SHALL BE PROVIDED AS PART OF THE TRUSS PORTION OF THE CONTRACT BY A DESIGN SERVICE SUCH AS ALPINE STRUCTURAL CONSULTANTS. THE PLANS FOR THE PERMANENT BRACING SYSTEM SHALL BEAR THE STAMP OF AN ENGINEER REGISTERED IN TENNESSEE AND BE SUBMITTED AS PART OF THE TRUSS SHOP DRAWING SUBMITTAL FOR REVIEW BY THE STRUCTURAL ENGINEER. THE LOADS IMPARTED INTO THE STRUCTURE LATERAL BRACING SYSTEM BY THE TRUSS BRACING SYSTEM SHALL BE SHOWN ON THE SHOP DRAWING SUBMITTALS.

6.16 AS A MINIMUM, PERMANENT 2X4 DIAGONAL BRACING SHALL BE NAILED TO THE TOP SIDE OF THE ROOF TRUSS BOTTOM CHORDS AND BOTTOM SIDE OF THE TRUSS TOP CHORDS AT EACH END OF THE BUILDING AND AT 20 FOOT INTERVALS. ATTACH BRACING TO CHORDS USING (2) 16d x 3 1/2" COMMON NAILS.

6.17 WOOD TRUSSES SHALL NOT BE CUT, NOTCHED, OR BORED.

WOOD NOTES :

4.01 ALL LUMBER TO BE #2 SOUTHERN PINE, OR BETTER KILN DRIED, UNLESS NOTED OTHERWISE. 2x4 NON-BEARING STUDS CAN BE SPF STUD GRADE. ALL LUMBER IN CONTACT WITH CONCRETE OR MASONRY, OR EXPOSED TO WEATHER SHALL BE PRESSURE TREATED TO A MINIMUM RETENTION LEVEL OR 0.25. PRESSURE TREATED LUMBER USED AS A BEARING PLATE SHALL BE KILN DRIED AFTER TREATMENT. OTHER LUMBER SHALL BE EQUAL TO OR GREATER THAN THE FOLLOWING:

MEMBER	SPECIES	GRADE	Fb	Fb
2x4	SYP	NO. 2	1,100 PSI	1,400,000 PSI
2x4	SPF	NO. 2	775 PSI	1,100,000 PSI
2x6	SYP	NO. 2	1,000 PSI	1,400,000 PSI
2x6	SPF	NO. 2	775 PSI	1,100,000 PSI
2x8	SYP	NO. 2	925 PSI	1,400,000 PSI
2x10	SYP	NO. 2	800 PSI	1,400,000 PSI
2x12	SYP	NO. 1	1,000 PSI	1,600,000 PSI
LVL	N/A	2.0E	2,900 PSI	2,000,000 PSI
LSL RIM BOARD	N/A	1.3E	1,700 PSI	1,300,000 PSI
PSL COLUMN	N/A	1.8E	2,400 PSI	1,800,000 PSI

4.02 CONTRACTOR SHALL USE 'SIMPSON STRONG TIE' (OR APPROVED EQUAL) WOOD FRAMING ANCHORS, CONNECTORS, HANGERS, ETC. FOR ALL WOOD TO WOOD CONNECTIONS. ALL ANCHORS TO BE INSTALLED IN ACCORDANCE WITH MANUFACTURER'S SPECIFICATIONS IN ORDER TO ACHIEVE MAXIMUM CONNECTOR CAPACITY. ALL CONNECTORS SHALL BE GALVANIZED CONNECTORS IN CONTACT WITH PRESSURE TREATED LUMBER, AND CONNECTORS SHALL HAVE A MINIMUM G185 COATING IN ACCORDANCE WITH ASTM A153.

4.03 FLOOR SHEATHING TO BE 3/4" TONGUE AND GROOVE RATED PLYWOOD FLOORING. BLOCK AS NEEDED SO THAT ALL PANEL EDGES CAN BE ATTACHED. ALL DIAPHRAGM BOUNDARIES AND PANEL EDGES TO BE ATTACHED WITH 10d NAILS AT 6" O.C. AND ATTACHED TO INTERMEDIATE FRAMING AT 12" O.C.

4.04 ROOF SHEATHING TO BE 5/8" (OR 19/32") EXTERIOR GRADE PLYWOOD WITH AN APA SPAN RATING OF 32/16, U.N.O. ALL EDGES SHALL BE BLOCKED WITH LUMBER OR PROVIDE PLYWOOD CLIPS (1 CLIP PER SPAN). ROOF SHEATHING SHALL BE FASTENED TO JOISTS AND BLOCKING WITH 10d COMMON NAILS AT 6" O.C. EDGES AND 12" O.C. INTERMEDIATE, U.N.O. SHEATHING SHALL BE INSTALLED IN ACCORDANCE WITH LAYOUT CASE 1 OR 2018 IBC TABLE 2306.3.1. WALL SHEATHING TO BE 7/16" WITH PANEL SPAN RATING OF 24/16.

4.05 CUTTING, NOTCHING, BORED HOLES IN STUD WALLS, RAFTER, ETC., SHALL BE DONE IN ACCORDANCE WITH THE 2018 INTERNATIONAL BUILDING CODE SECTION 2308.

4.06 ALL WOOD CONNECTIONS NOT SHOWN SHALL BE DETAILED PER THE INTERNATIONAL BUILDING CODE "FASTENING SCHEDULE" TABLE 2304.9.1.

4.07 ALL STEEL HARDWARE INCLUDING PLATES, NAILS, NUTS AND BOLTS SHALL BE HOT DIPPED GALVANIZED.

4.08 ALL STEEL IN CONTACT WITH PRESSURE TREATED WOOD SHALL BE SEPARATED WITH 15# FELT.

4.09 ATTACH DECK TO JOISTS WITH (2) #10 x 3" SCREWS AT EACH JOIST.

4.10 WOOD DECKING SHALL BE #1 GRADE PRESSURE TREATED SOUTHERN YELLOW PINE OR DOUGLAS FIR.

4.11 ALL GLULAM MEMBERS TO BE PARALLAM PLUS RATED FOR EXTERIOR EXPOSURE, SERVICE LEVEL 2 AND HAVE THE FOLLOWING DESIGN PROPERTIES :

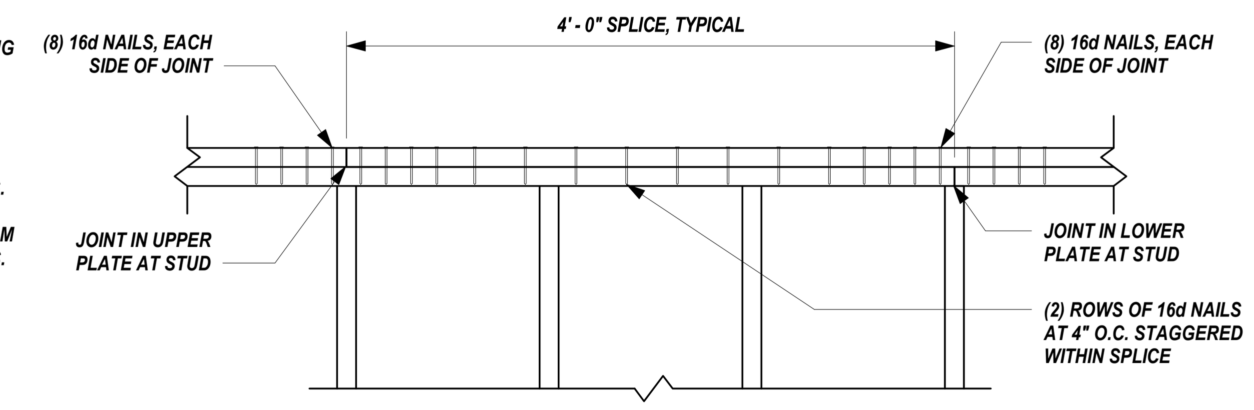
SERVICE LEVEL	G SHEAR MODULUS OF ELASTICITY (PSI)	E MODULUS OF ELASTICITY (PSI)	E min. ADJUSTED MODULUS OF ELASTICITY (PSI)	Fb FLEXURAL STRESS (1) (PSI)	Ft⊥ TENSION STRESS (2) (PSI)	Fc ∥ COMPRESSION PERPENDICULAR TO GRAIN (3) (PSI)	Fc COMPRESSION PARALLEL TO GRAIN (PSI)	Fv HORIZONTAL SHEAR PARALLEL TO GRAIN (PSI)	Crp CREEP FACTOR	SG EQUIVALENT SPECIFIC GRAVITY
BEAM APPLICATION										
1	103,750	1.660x10	843,725	2,117	1,519	533	2,030	241	0.20	0.50
2	91,250	1.460x10	742,070	1,827	1,397	368	1,508	197	0.60	0.50
3	83,750	1.340x10	681,080	1,624	1,337	263	1,189	171	0.85	0.50
COLUMN APPLICATION										
1	93,375	1.494x10	759,350	1,752	1,316	215	1,750	160	0.20	0.50
2	82,125	1.314x10	667,865	1,512	1,211	150	1,300	120	0.60	0.50
3	75,375	1.206x10	612,970	1,344	1,158	110	1,025	100	0.85	0.50

4.12 SEE HEADER SCHEDULE BELOW FOR NON-LOAD BEARING HEADERS NOT SHOWN ON PLAN.

NON-LOAD BEARING HEADER SCHEDULE
HEADER SIZE AND POSTS
EQUAL TO OR LESS THAN 6'-0" (2) 2x8 HEADER (2) JACK, (10 KING
FROM 6'-0" TO 10'-0" (2) 2x12 HEADER (2) JACK, (1) KING

4.13 SEE BELOW FOR TYPICAL CONTINUOUS TOP PLATE SPLICE

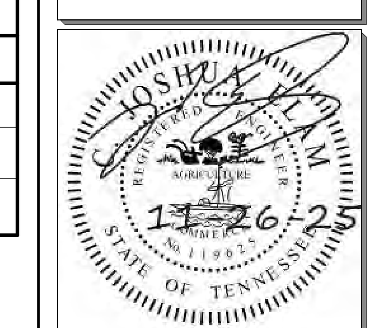
NOTE:
PROVIDE CONTINUOUS TOP PLATES WITHOUT JOINTS WHERE WALLS ARE 12'-0" OR LESS IN LENGTH



ELAM STRUCTURAL ENGINEERING

TAC
The Architecture Collaborative
6700 Baum Drive, Suite 23 | Knoxville, TN 37919
www.tac-45.com | 865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830



THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

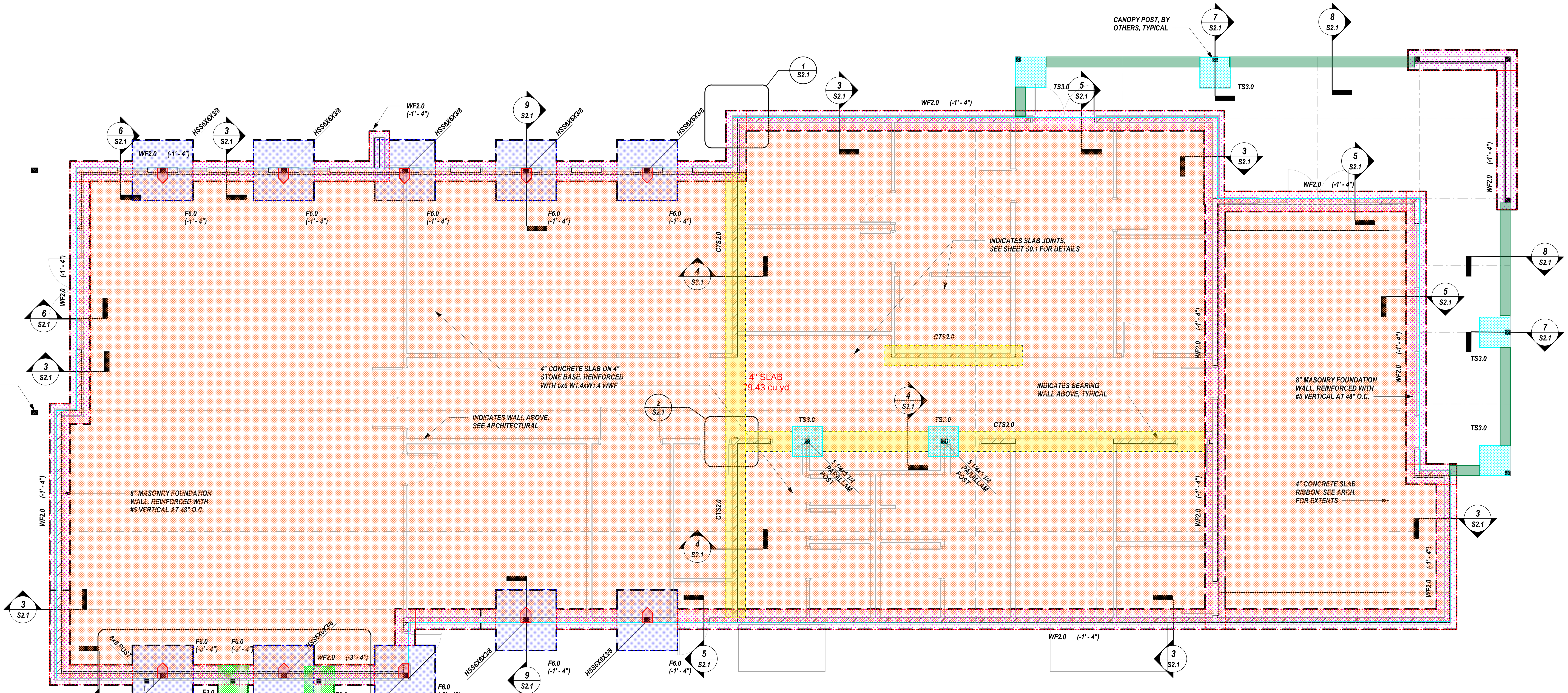
No.	Date

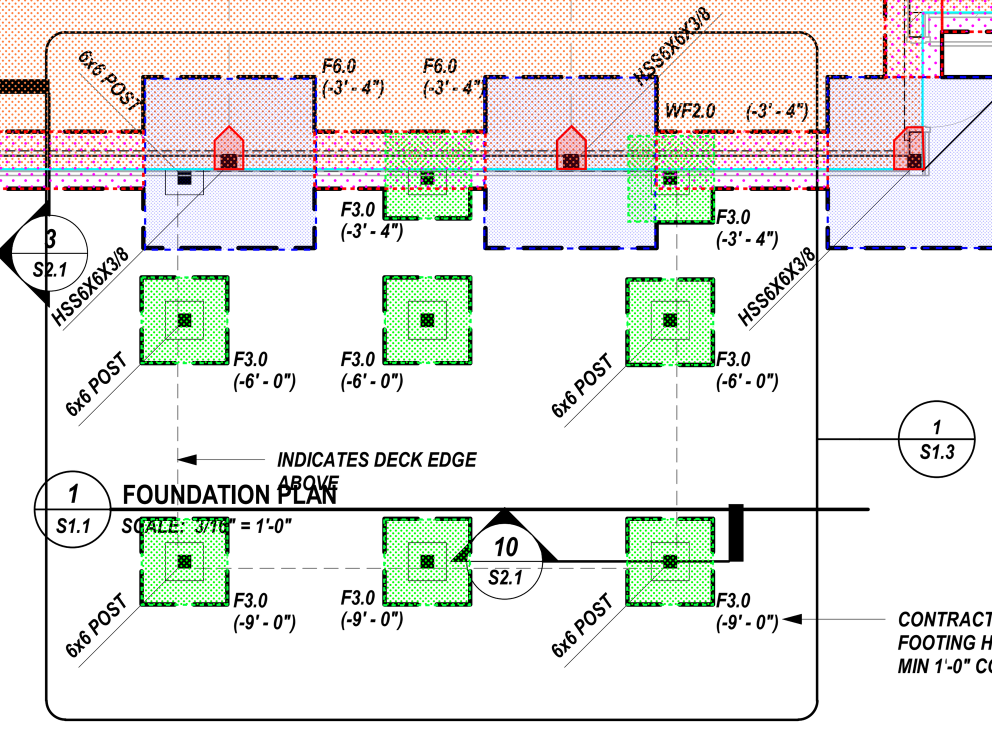
Drawing Title:
STRUCTURAL NOTES

Date: 11-26-2025

Project No.
25026

Sheet No.
S0.1





1 S1.1 FOUNDATION PLAN
SCALE: 3/16" = 1'-0"

CONTRACTOR TO VERIFY TOP OF FOOTING HEIGHTS IN FIELD. MAINTAIN MIN 1'-0" COVER

FOOTING SCHEDULE		
MARK	SIZE	REINFORCING
CTS2.0	2'-0"x1'-0"xCONT.	(3) #5 CONTINUOUS WITH #4 BARS AT 24" O.C., TRANSVERSE
WF2.0	2'-0"x1'-0"xCONT.	(3) #5 CONTINUOUS WITH #4 BARS AT 24" O.C., TRANSVERSE
TS3.0	3'-0"x3'-0"x1'-0"	(4) #5 EACH WAY, BOTTOM
F3.0	3'-0"x3'-0"x1'-0"	(4) #5 EACH WAY, BOTTOM
F4.0	4'-0"x4'-0"x1'-0"	(5) #5 EACH WAY, BOTTOM
F6.0	6'-0"x6'-0"x1'-0"	(9) #5 EACH WAY, TOP AND BOTTOM

FOUNDATION PLAN NOTES

1. ELEVATIONS ARE BASED ON A REFERENCE FLOOR ELEVATION OF 0'-0", U.N.O. TOP OF SLAB ON GRADE IS AT THE REFERENCE ELEVATION U.N.O.
2. T.O. EXTERIOR FOOTING ELEVATION = -1'-4", TYP., U.N.O.
3. (#'-##") INDICATES TOP OF FOOTING ELEVATION AT NON-TYPICAL LOCATIONS
4. WF#.# INDICATES WALL OR CONTINUOUS FOOTING MARK. SEE SCHEDULE FOR SIZE AND REINFORCEMENT
5. TS#.# INDICATES THICKENED SLAB AREAS TO BE POURED MONOLITHICALLY WITH SLAB ON GRADE. SEE SCHEDULE FOR SIZE AND REINFORCEMENT
6. F.S. INDICATES FOOTING STEP. STEP FOOTING SO AS TO MAINTAIN MINIMUM 1'-4" COVER ABOVE FOOTING. COORDINATE FINAL LOCATIONS OF FOOTING STEPS IN FIELD.
7. SEE ARCHITECTURAL DRAWINGS FOR ELEVATIONS, WALL SECTIONS, AND DIMENSIONS NOT SHOWN
8. SEE ARCHITECTURAL AND CIVIL DRAWINGS FOR SIDEWALKS, PAVING, AND SITE DETAILS AT THE BUILDING'S EXTERIOR
9. SEE SHEETS S0.1 AND S0.2 FOR STRUCTURAL NOTES

CONCRETE				
	Description	Label	Quantity	Unit
	CONCRETE	4" SLAB	79.43	cu yd
	CONCRETE	CTS2.0	10.34	cu yd
	CONCRETE	F3	2.67	cu yd
	CONCRETE	F6	13.29	cu yd
	CONCRETE	F6'	0.63	cu yd
	CONCRETE	TS3.0	2.01	cu yd
	CONCRETE	WF2.0	29.11	cu yd
	CONCRETE	WTS3.0	4.26	cu yd

ELAM STRUCTURAL ENGINEERING

TAC The Architecture Collaborative
6700 Baum Drive, Suite 23 Knoxville, TN 37919
www.tac-45.com | 865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

11-26-25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

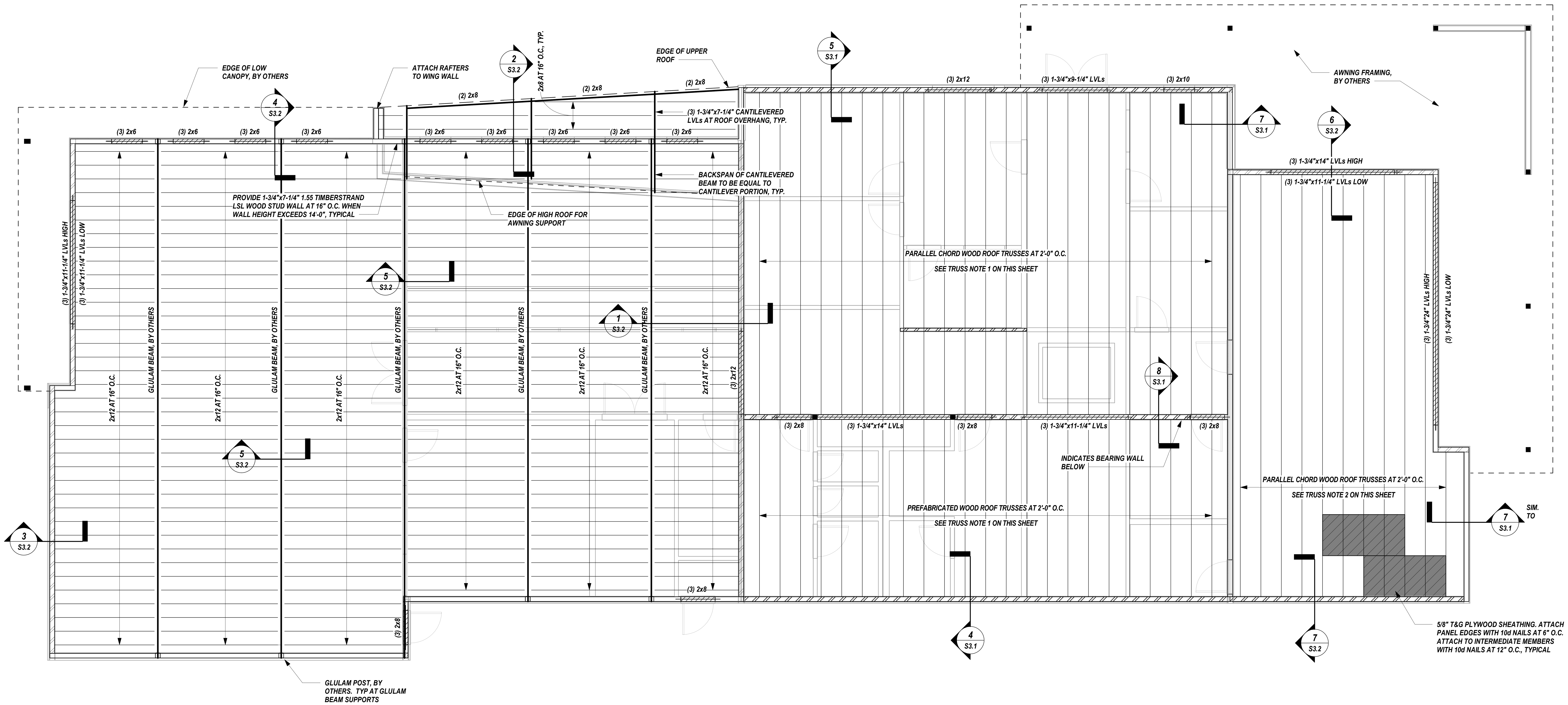
No.	Date

Drawing Title:
FOUNDATION PLANS

Date: 11-26-2025

Project No.
25026

Sheet No.
S1.1



1 S1.2 ROOF FRAMING PLAN
SCALE: 3/16" = 1'-0"

ROOF FRAMING PLAN NOTES

1. ALL TRUSSES TO BE AT 24" O.C.

2. ALL TRUSSES SHALL BE ORIENTED IN THE DIRECTION SHOWN. TRUSS AND BEAM LAYOUT SHALL NOT BE SIGNIFICANTLY ALTERED FROM THE CONSTRUCTION DOCUMENTS. IF THERE IS A CONFLICT, OR THE LAYOUT AS SHOWN IN THE CONSTRUCTION DOCUMENTS CANNOT PHYSICALLY BE CONSTRUCTED, CONTACT STRUCTURAL ENGINEER.

3. PROVIDE 5/8" EXTERIOR GRADE PLYWOOD ROOF SHEATHING.

4. ALL BEAM SOLID BEARING POINTS TO BE CONTINUOUS TO FOUNDATION BELOW, U.N.O.

5. SEE DETAIL 1/S3.1 FOR TYPICAL WALL ELEVATIONS.

6. SEE SECTION 2 AND 3/S3.1 FOR TYPICAL HEADER FRAMING.

7. ALL EXTERIOR LUMBER SHALL BE PRESSURE TREATED.

8. SEE ARCHITECTURAL DRAWINGS FOR ELEVATIONS, WALL SECTIONS, AND DIMENSIONS NOT SHOWN

9. IF THERE IS A DISCREPANCY BETWEEN ARCHITECTURAL PLANS/SECTIONS AND STRUCTURAL PLAN/SECTIONS, STRUCTURAL DRAWINGS SHALL CONTROL FOR ALL BEAM LOCATIONS AND BEAM BEARING POINTS.

10. SIMILAR CONDITIONS CALL FOR SIMILAR DETAILS. REFER TO GENERAL NOTES AND DETAILS IF SPECIFIC CONDITIONS ARE NOT SHOWN EXPLICITLY ON PLAN.

11. SEE SHEETS S0.1 AND S0.2 FOR STRUCTURAL NOTES.

TRUSS DESIGNER NOTES

1. ASSUMED TRUSS DEPTH AT REAR OF BUILDING = 24"

2. ASSUMED TRUSS DEPTH AT REAR OF BUILDING = 36"

2. PROVIDE 2x6 MINIMUM TOP AND BOTTOM CHORDS IF NEEDED TO MAINTAIN ASSUMED TRUSS DEPTH

3. NOTIFY ARCHITECT/STRUCTURAL ENGINEER IF DEPTH CANNOT BE ACHIEVED

4. TOP CHORD SLOPING TRUSSES

5. ROOF SLOPE = SEE ARCH.

ELAM STRUCTURAL ENGINEERING

TAC
The Architecture Collaborative
6700 Baum Drive, Suite 23 | Knoxville, TN 37919
www.tac-45.com | 865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

JOSHUA ELAM
REGISTERED ENGINEER
AGRICULTURE COMMERCE
No. 119625
STATE OF TENNESSEE
11-26-25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

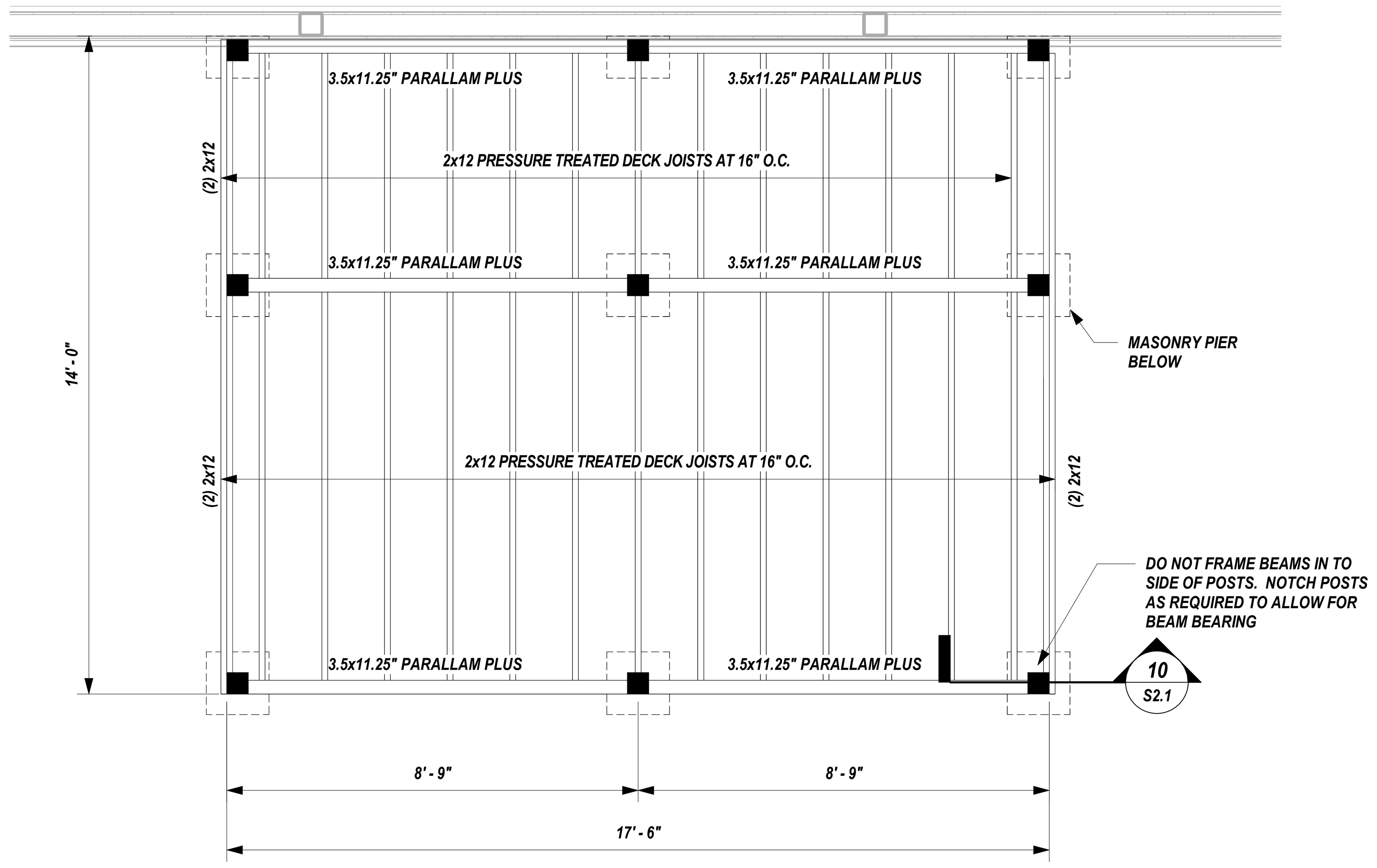
No.	Date

Drawing Title:
ROOF FRAMING PLAN

Date: 11-26-2025

Project No.
25026

Sheet No.
S1.2



1
S1.3
DECK FRAMING PLAN
SCALE: 3/8" = 1'-0"

TAC
The Architecture Collaborative
6700 Baum Drive, Suite 23 | Knoxville, TN 37919
www.tac-45.com | 865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

11-26-25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

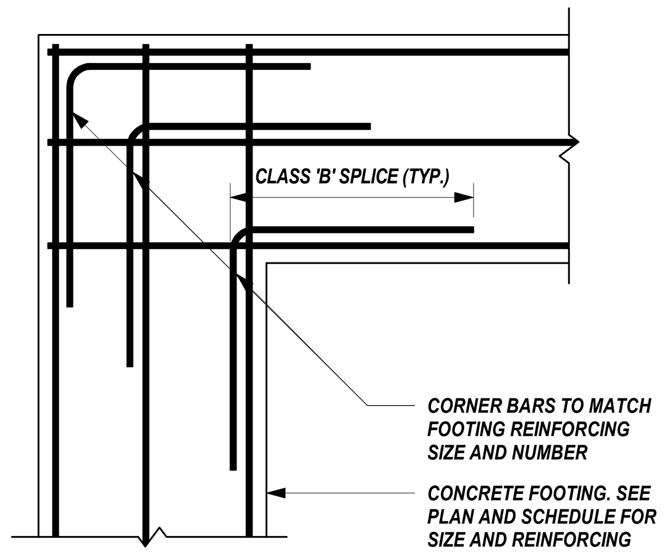
Drawing Title:
DECK FRAMING PLAN

Date: 11-26-2025

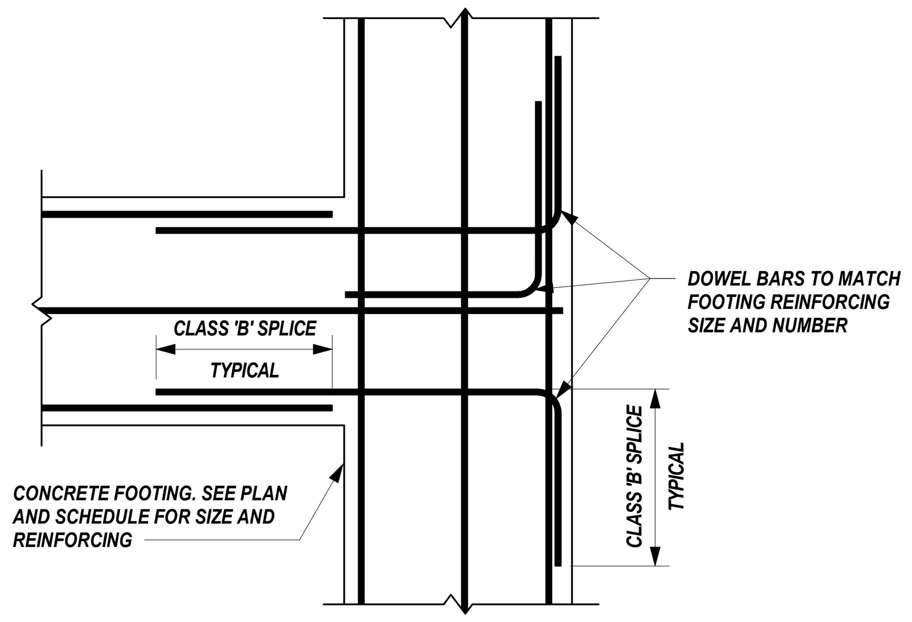
Project No. 25026

Sheet No. S1.3

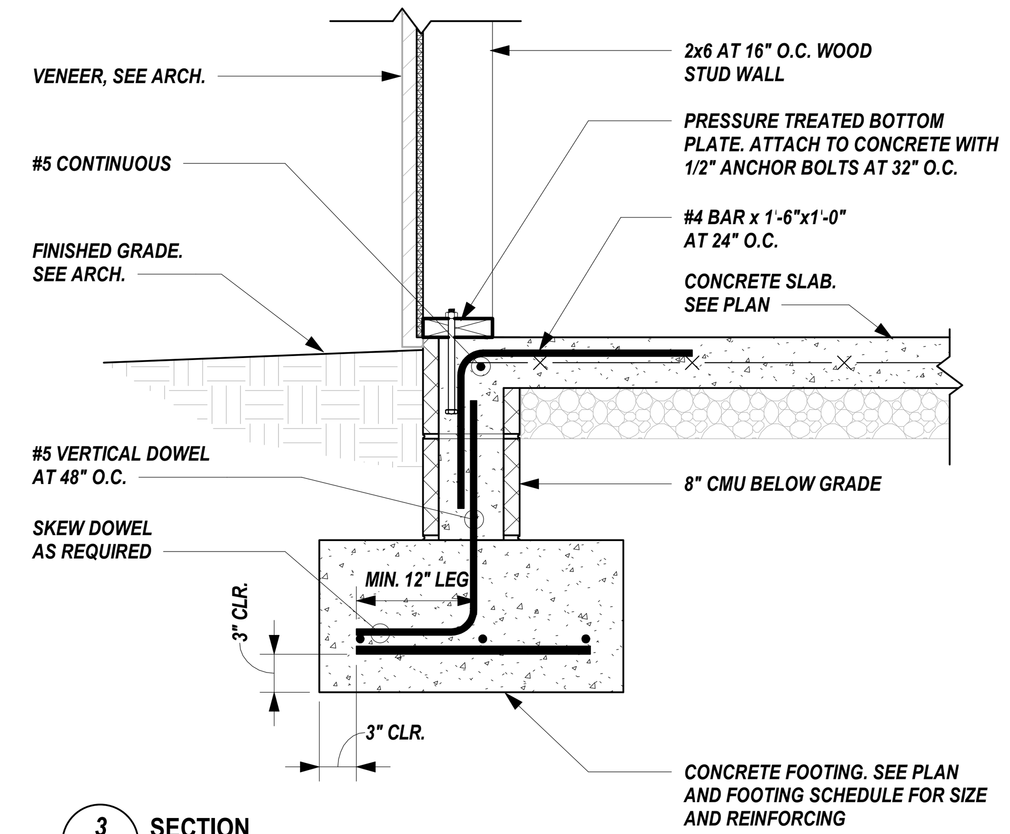
ELAM
STRUCTURAL ENGINEERING



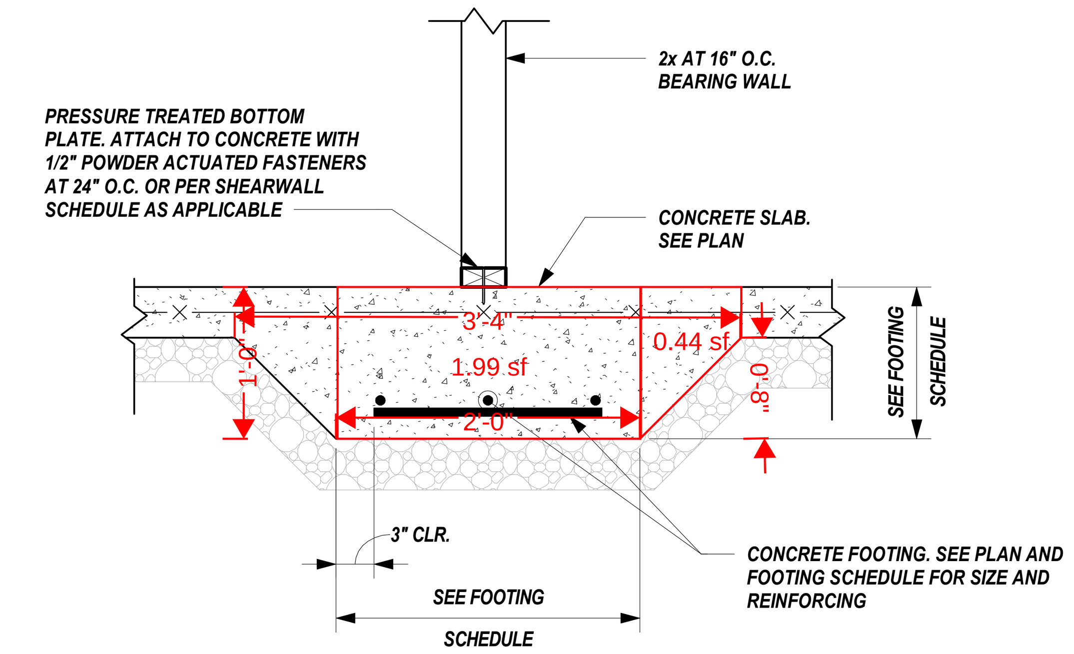
1 FOOTING CORNER DETAIL
S2.1 SCALE: 3/4" = 1'-0"



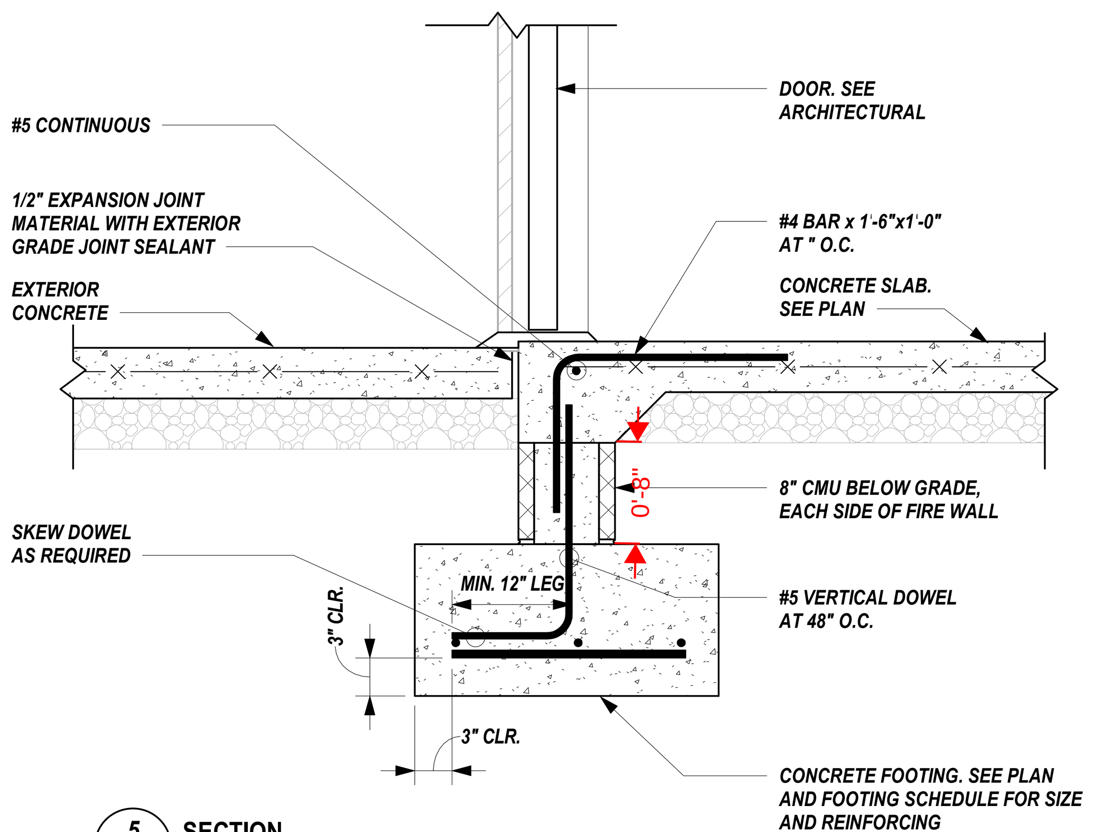
2 FOOTING INTERSECTION DETAIL
S2.1 SCALE: 3/4" = 1'-0"



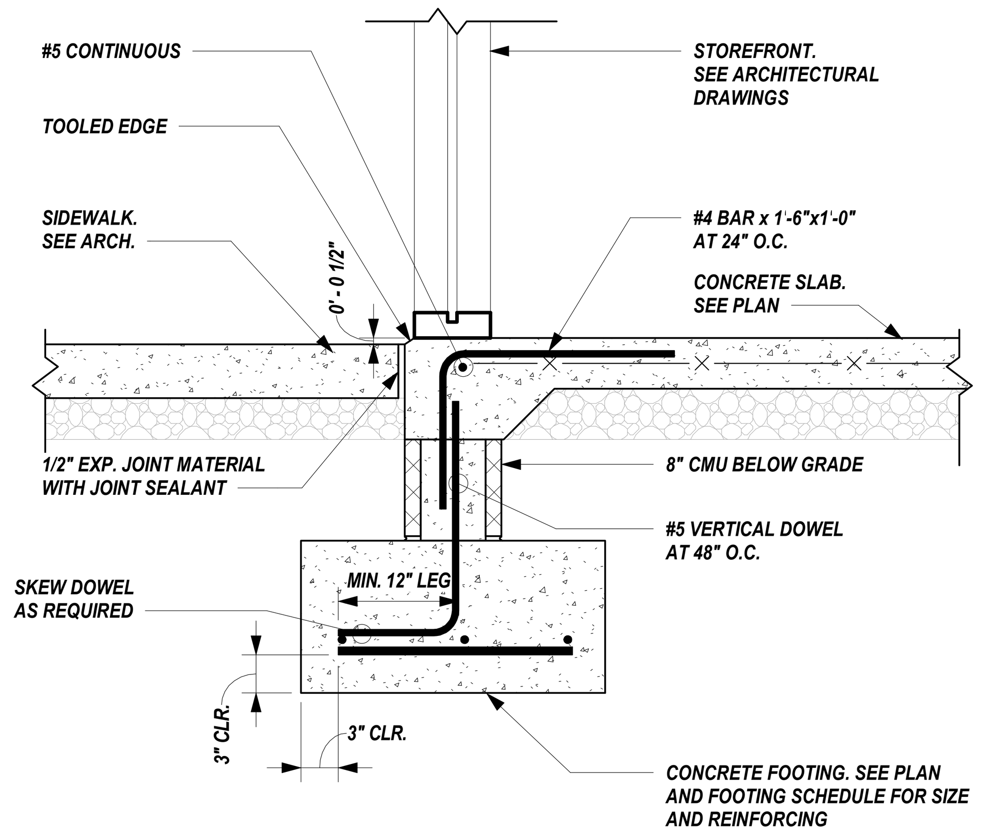
3 SECTION
S2.1 SCALE: 1" = 1'-0"



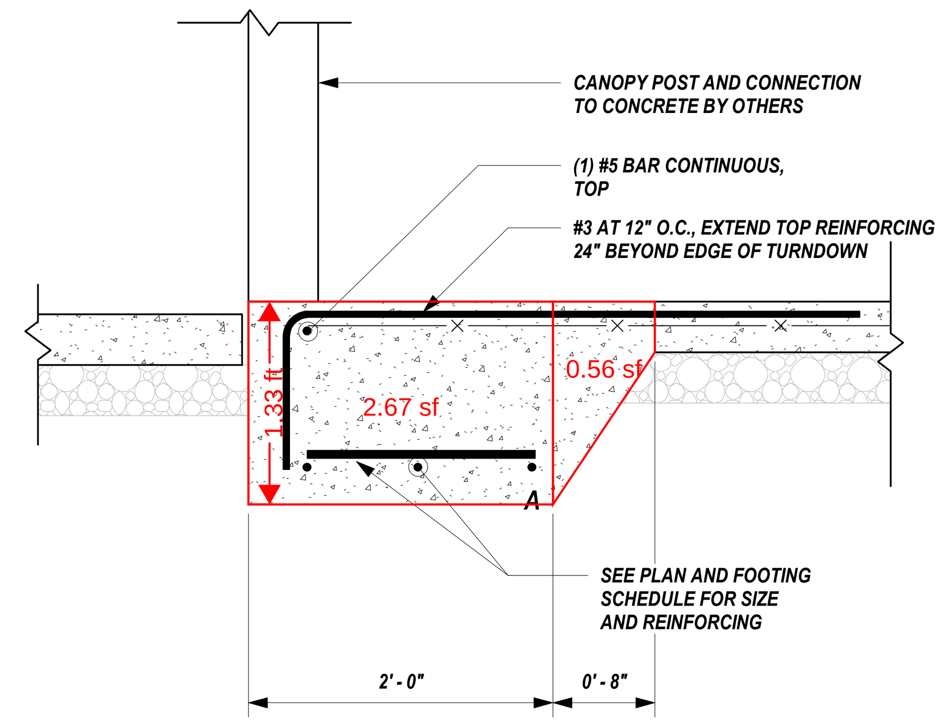
4 SECTION
S2.1 SCALE: 1" = 1'-0"



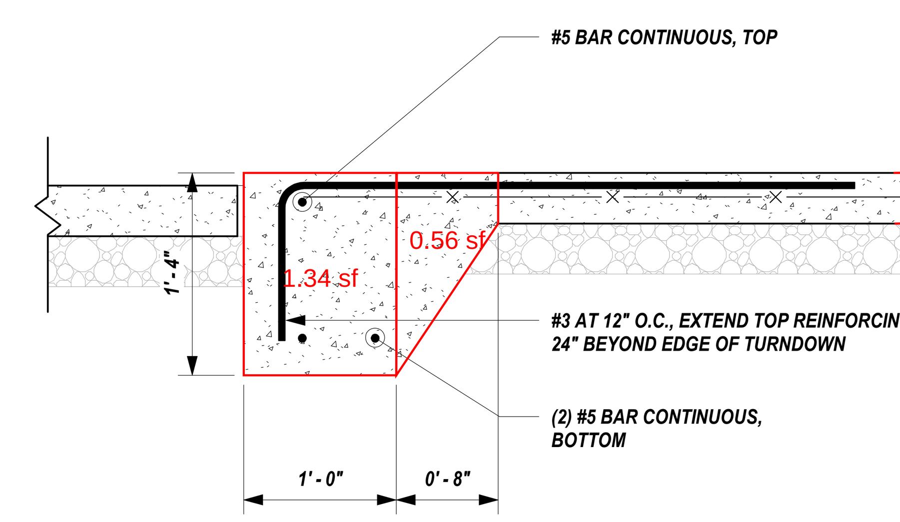
5 SECTION
S2.1 SCALE: 1" = 1'-0"



6 SECTION
S2.1 SCALE: 1" = 1'-0"

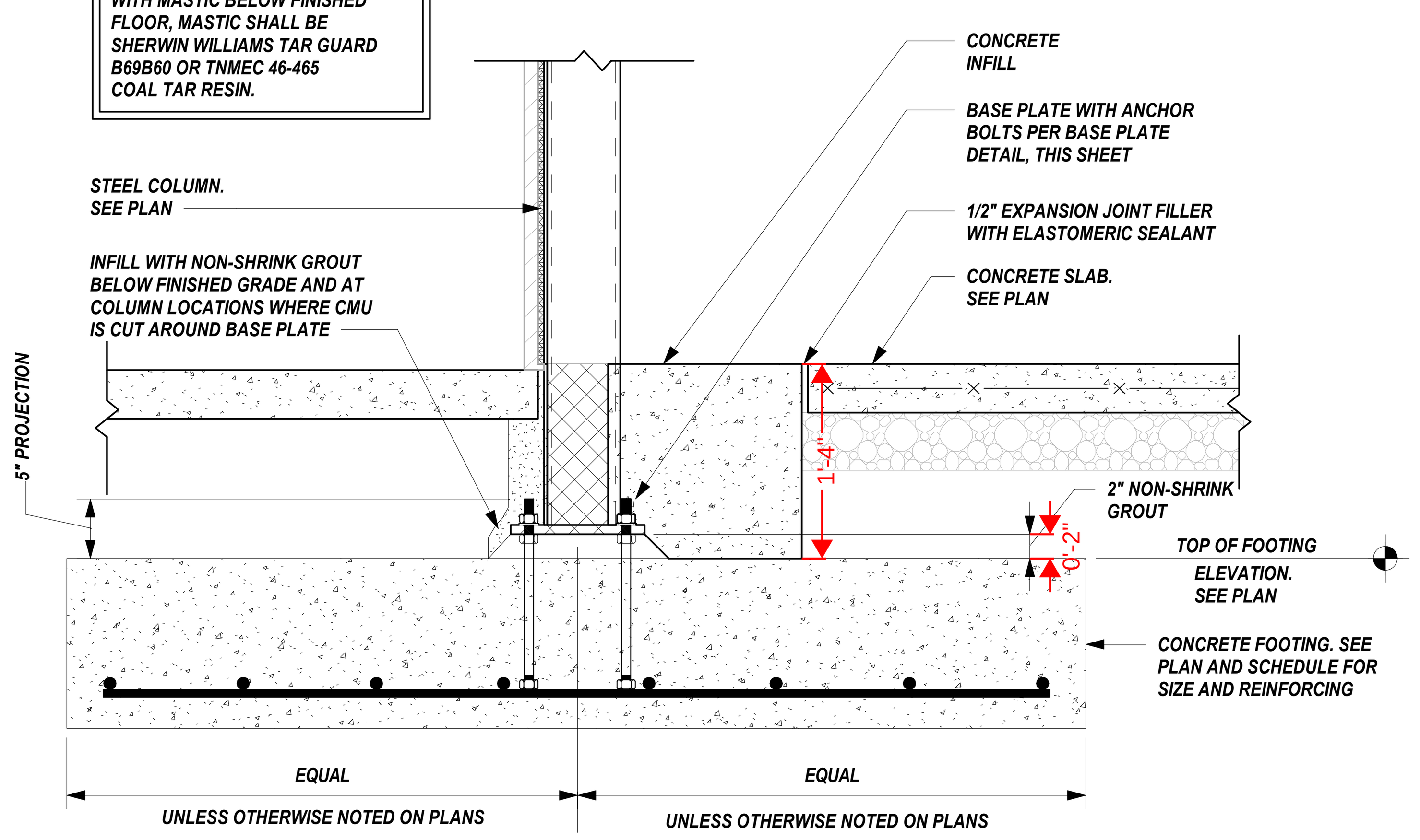


7 SECTION
S2.1 SCALE: 1" = 1'-0"

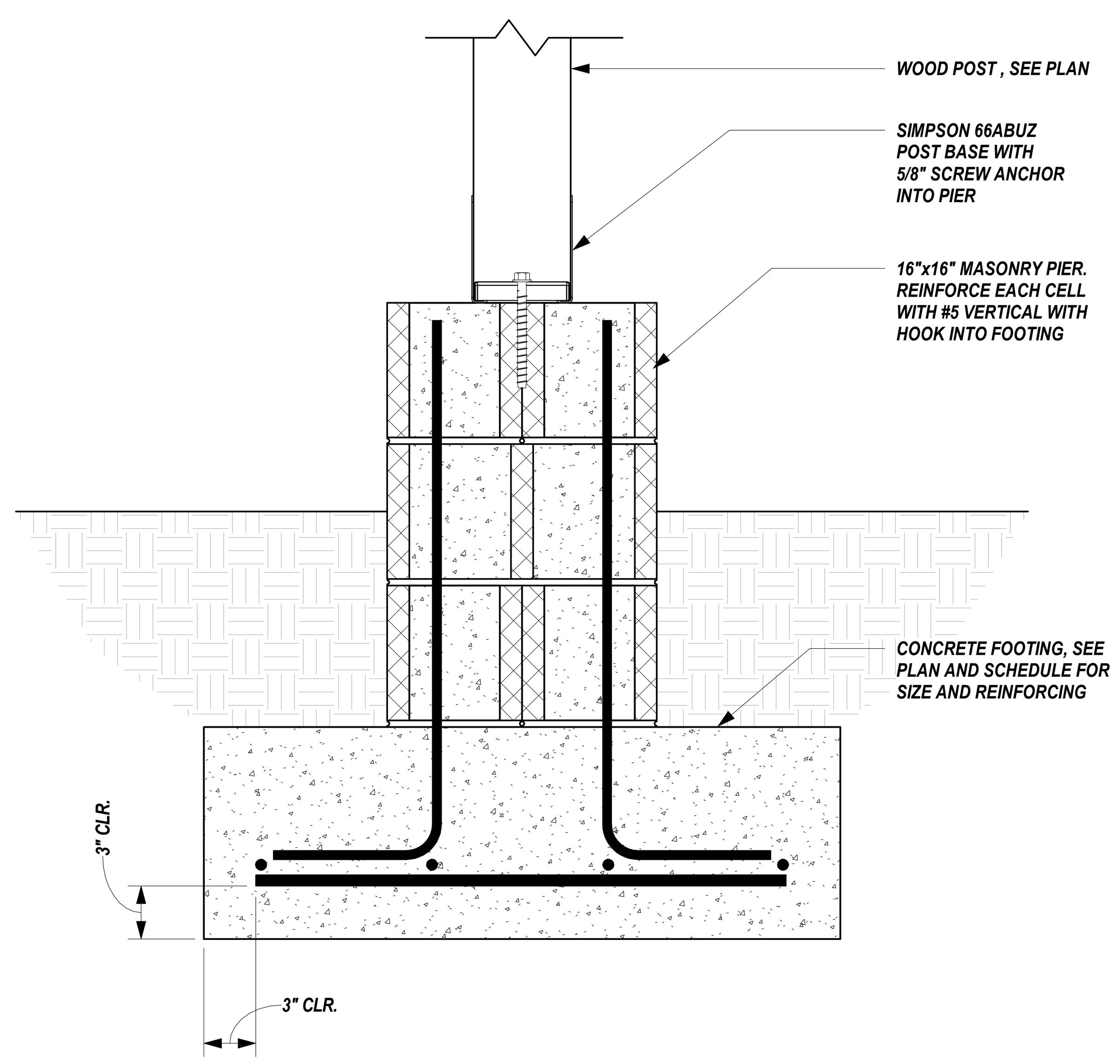


8 SECTION
S2.1 SCALE: 1" = 1'-0"

NOTE:
COAT COLUMN AND BASEPLATE WITH MASTIC BELOW FINISHED FLOOR, MASTIC SHALL BE SHERWIN WILLIAMS TAR GUARD B69B60 OR TNMEC 46-465 COAL TAR RESIN.



9 SECTION
S2.1 SCALE: 1" = 1'-0"



10 SECTION
S2.1 SCALE: 1 1/2" = 1'-0"

ELAM STRUCTURAL ENGINEERING

TAC The Architecture Collaborative
6700 Baum Drive, Suite 23 | Knoxville, TN 37919
www.tac-45.com | 865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

STATE OF TENNESSEE
No. 119625
11-26-25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

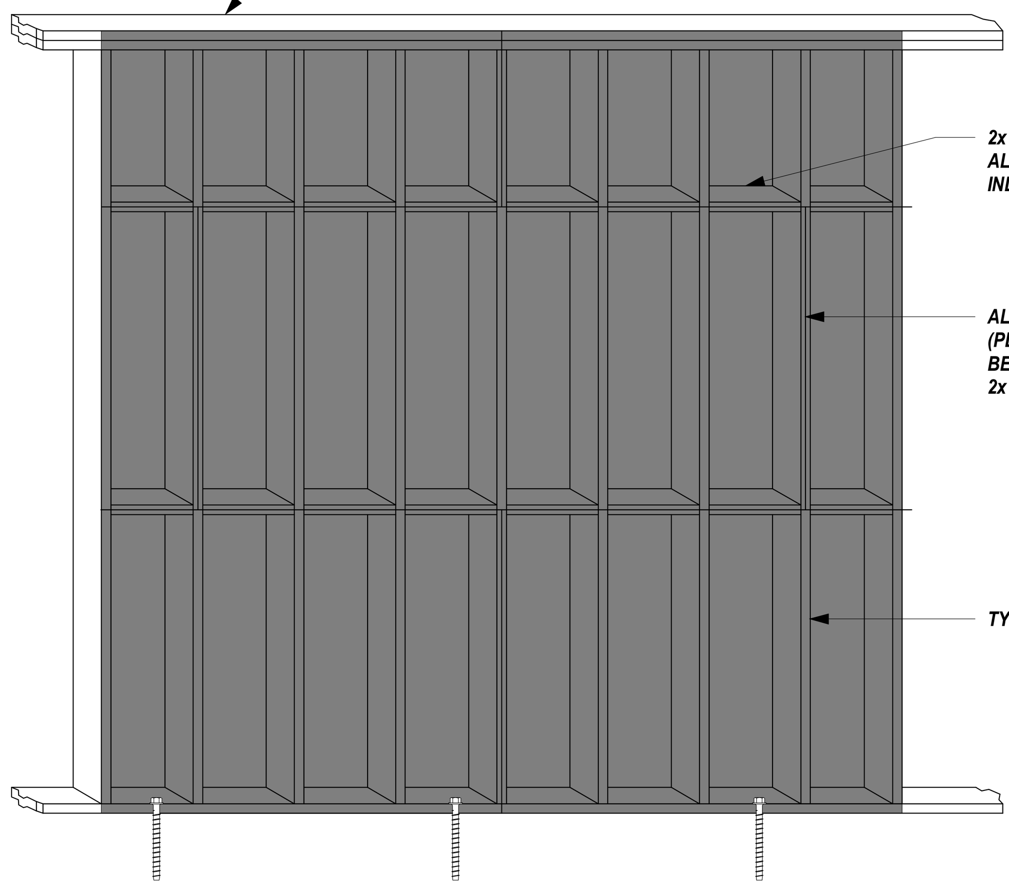
No.	Date

Drawing Title:
FOUNDATION SECTIONS AND DETAILS

Date: 11-26-2025

Project No. 25026

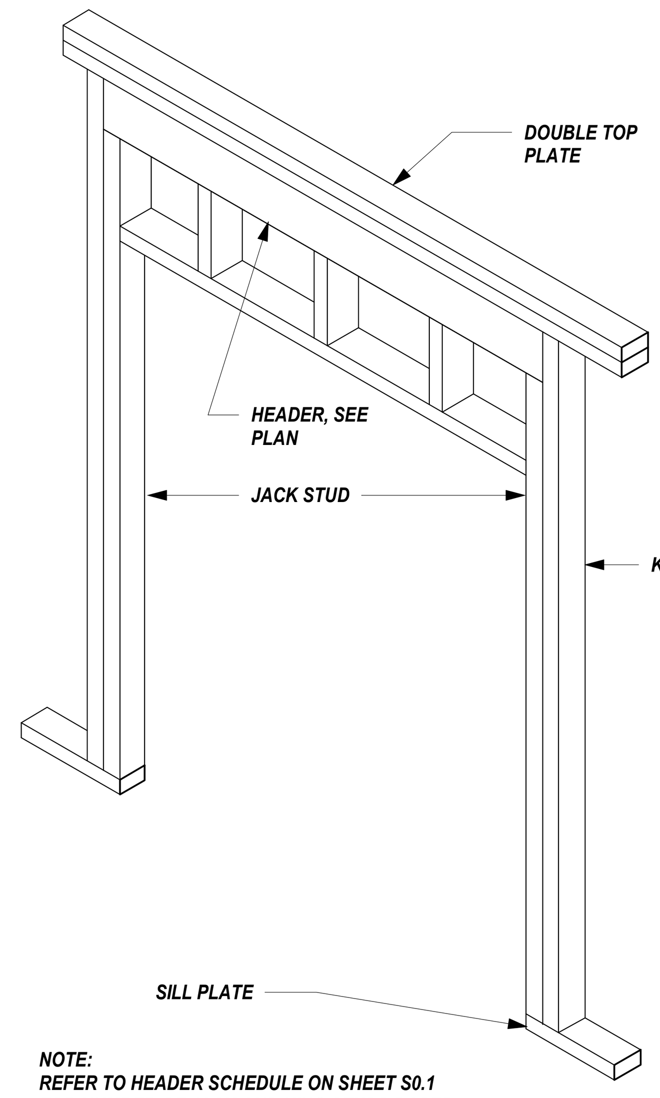
Sheet No. S2.1



NOTES :

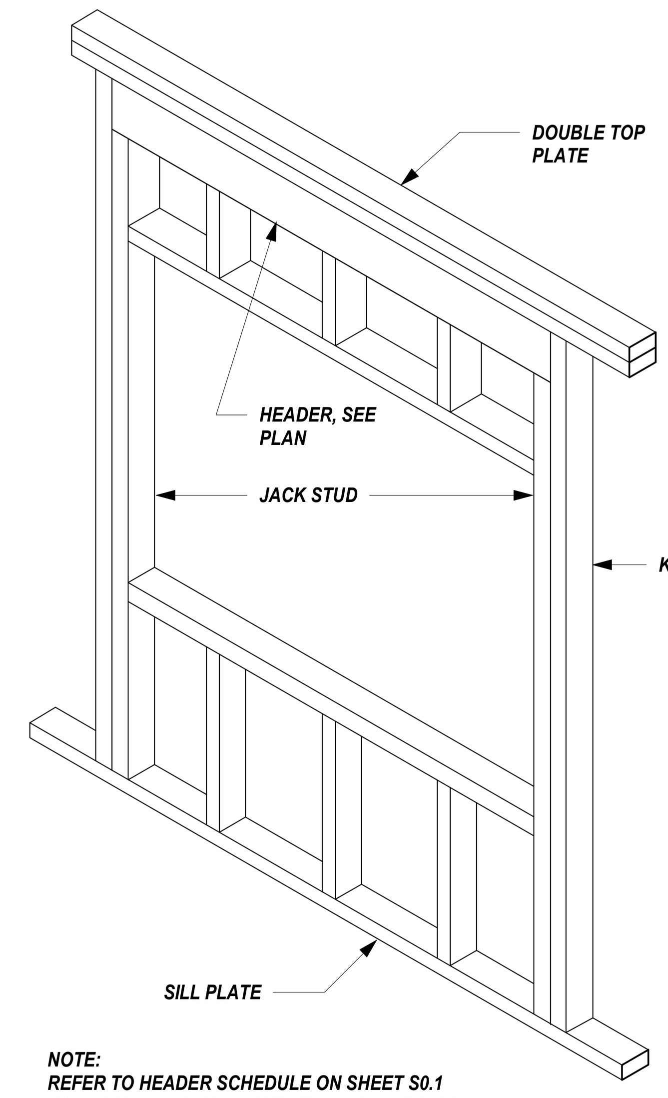
1. EXTERIOR WALLS SHALL BE SHEATHED WITH 7/16" (MINIMUM) APA RATED SHEATHING WITH EXPOSURE 1 DURABILITY CLASSIFICATION. SHEATHING SHALL BE LAID EITHER HORIZONTALLY (AS SHOWN) OR VERTICALLY. ATTACH SHEATHING WITH 10d NAILS AT 6" O.C. ALONG PANEL EDGE FRAMING AND AT 12" O.C. ALONG INTERMEDIATE STUDS UNLESS NOTED OTHERWISE IN SHEAR WALL ELEVATIONS
2. STUDS TO BE 2x SPACED AT 16" O.C.
3. IF USING SYP STUDS, THE MAXIMUM SPACING OF THE BLOCKING IS 4'-0" O.C.
4. IF USING SPF STUDS, BLOCKING IS REQUIRED AT 3 PLACES, BUT CANNOT EXCEED 3'-4" O.C., MAXIMUM.
5. ANCHORS TO BE 1/2" DIAMETER x 6" EMBEDMENT SPACED AT 48" O.C., MAX. ANCHORS ALSO LOCATED AT 4" MAX. FROM WALL CORNERS AND SILL PLATE TERMINATION, SUCH AS OPENINGS. IN LIEU OF HEADED STUD ANCHORS, 1/2" DIAMETER x 6" TITEN HD SCREW ANCHORS CAN BE USED.

1 S3.1 TYPICAL EXTERIOR WALLS AND INTERIOR BEARING WALLS
SCALE: 1" = 1'-0"



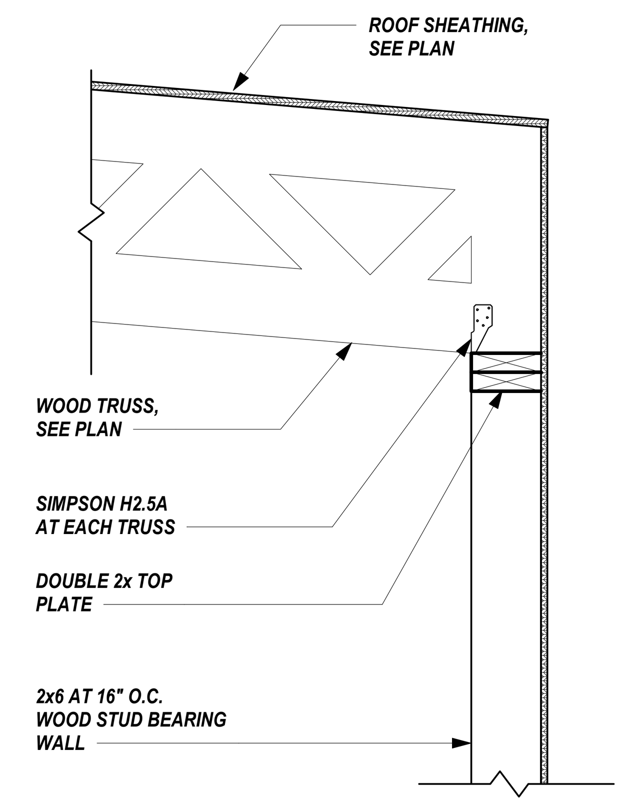
NOTE:
REFER TO HEADER SCHEDULE ON SHEET S0.1 FOR NUMBER OF JACK AND KING STUDS BASED ON BEAM SPAN.

2 S3.1 HEADER DETAIL AT DOOR
SCALE: 1" = 1'-0"



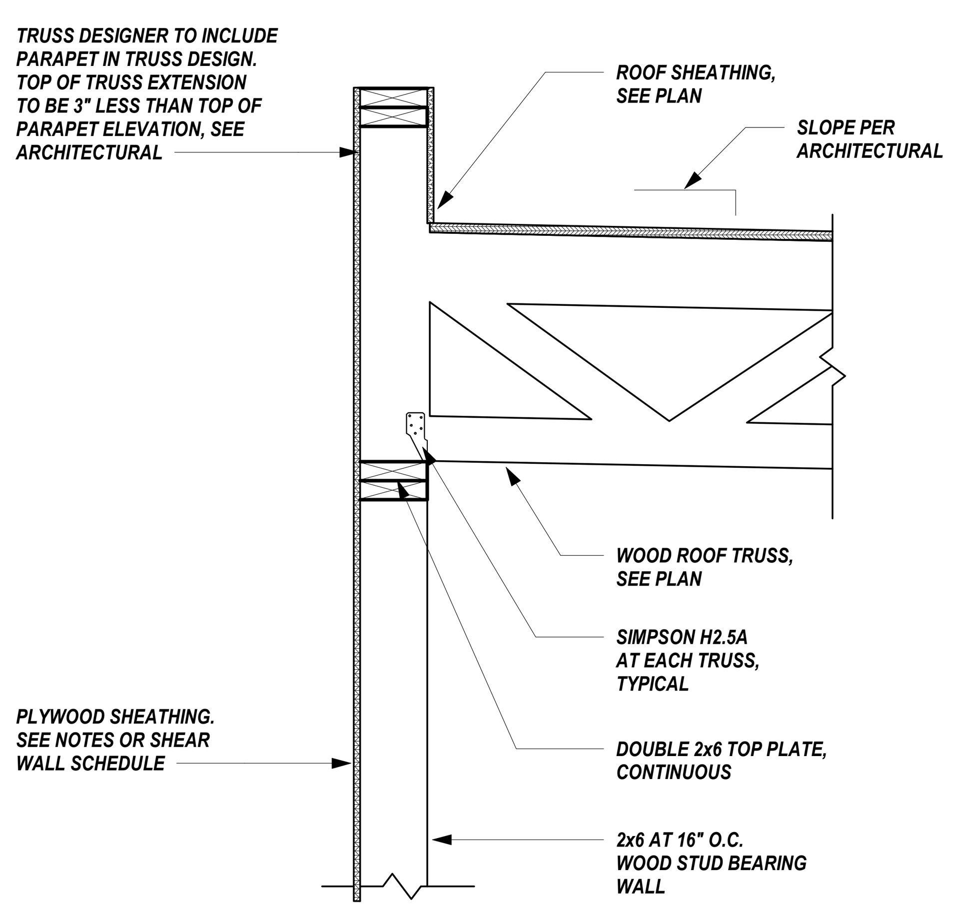
NOTE:
REFER TO HEADER SCHEDULE ON SHEET S0.1 FOR NUMBER OF JACK AND KING STUDS BASED ON BEAM SPAN.

3 S3.1 HEADER DETAIL AT WINDOW
SCALE: 1" = 1'-0"

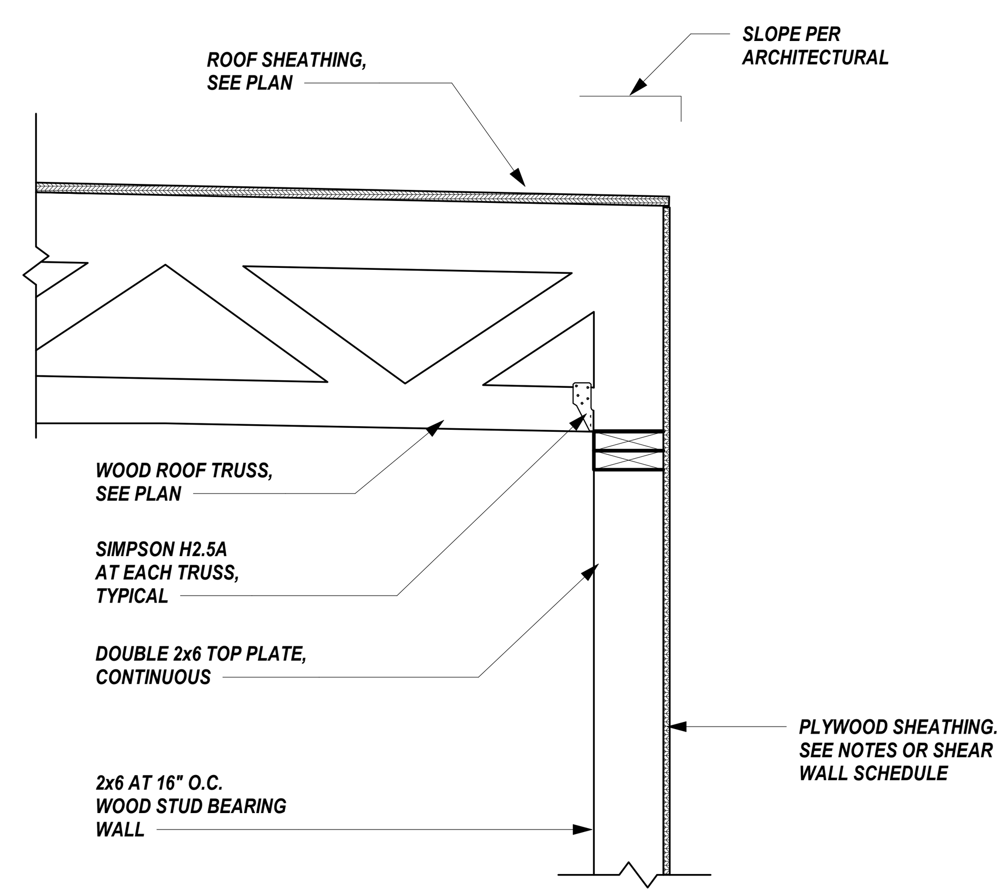


NOTE:
SEE ARCHITECTURAL FOR EXTERIOR FINISHES

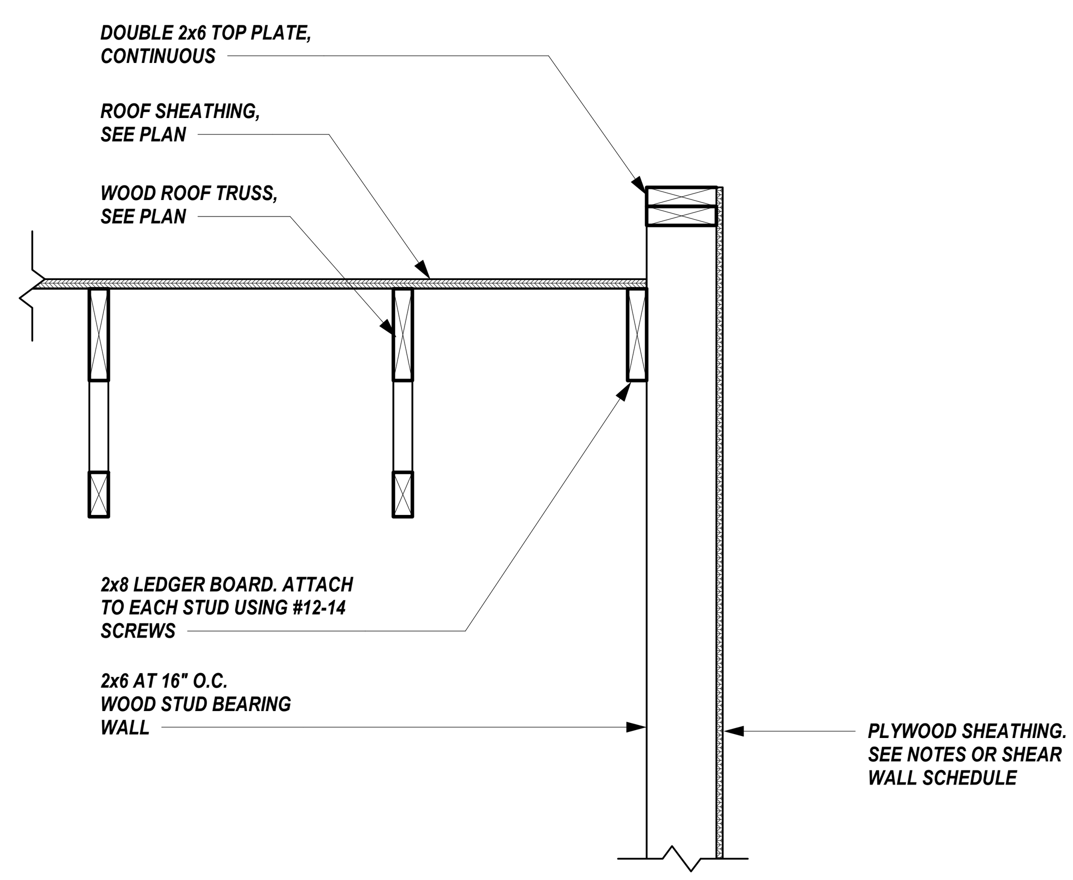
4 S3.1 SECTION
SCALE: 1" = 1'-0"



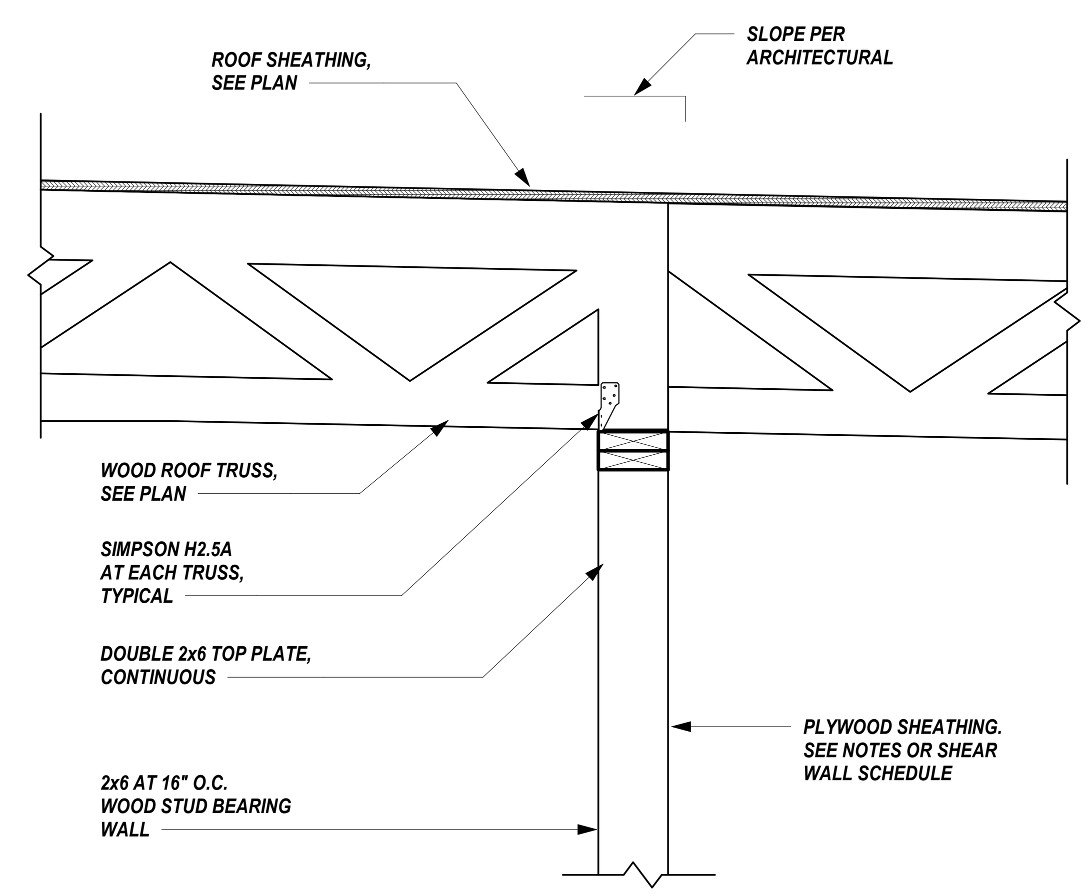
5 S3.1 SECTION
SCALE: 1" = 1'-0"



6 S3.1 SECTION
SCALE: 1" = 1'-0"



7 S3.1 SECTION
SCALE: 1" = 1'-0"



8 S3.1 SECTION
SCALE: 1" = 1'-0"

ELAM
STRUCTURAL ENGINEERING

TAC
The Architecture Collaborative
6700 Baum Drive, Suite 23 Knoxville, TN 37919
www.tac-45.com | 865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

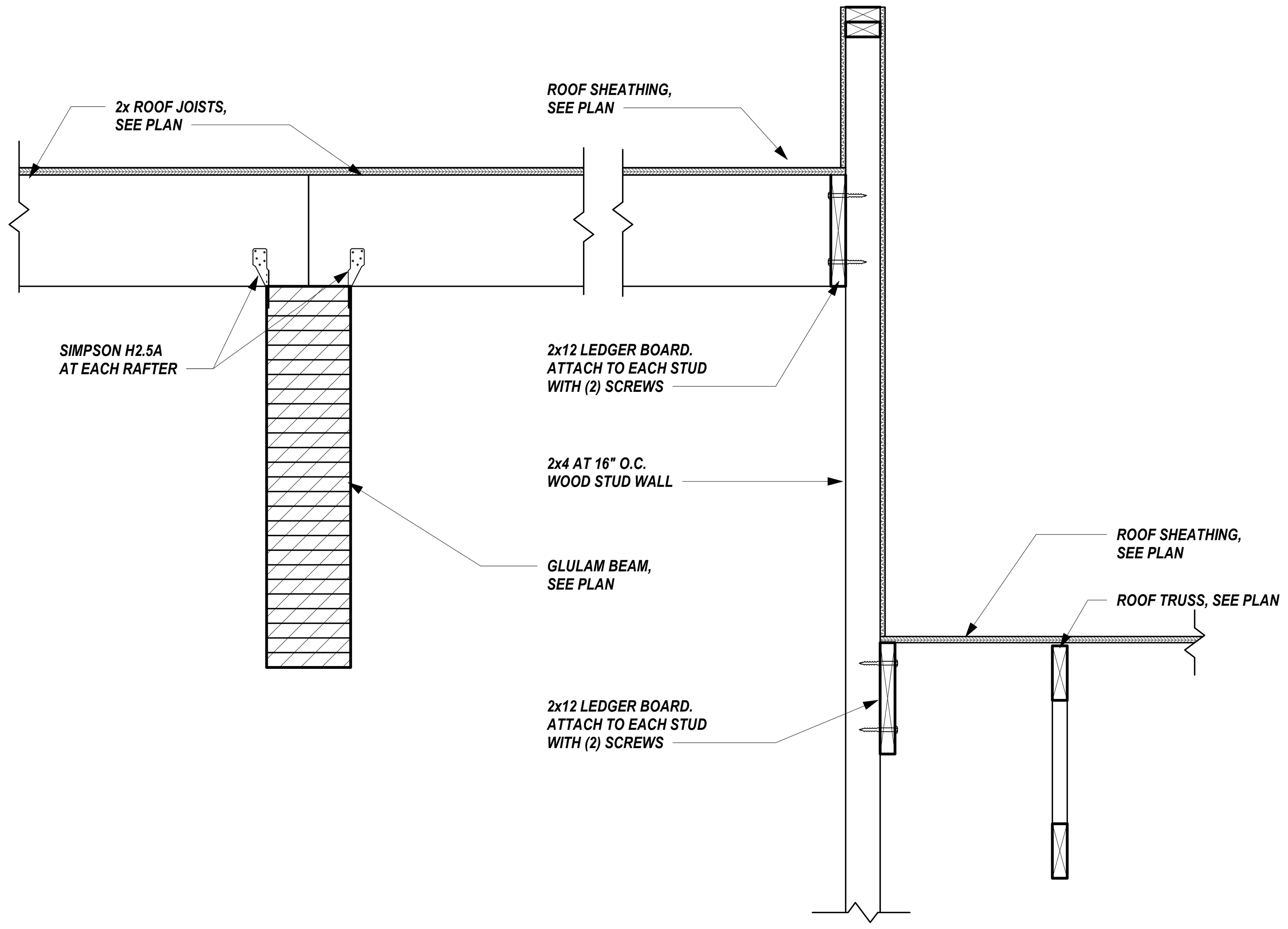
No.	Date

Drawing Title:
FRAMING SECTIONS AND DETAILS

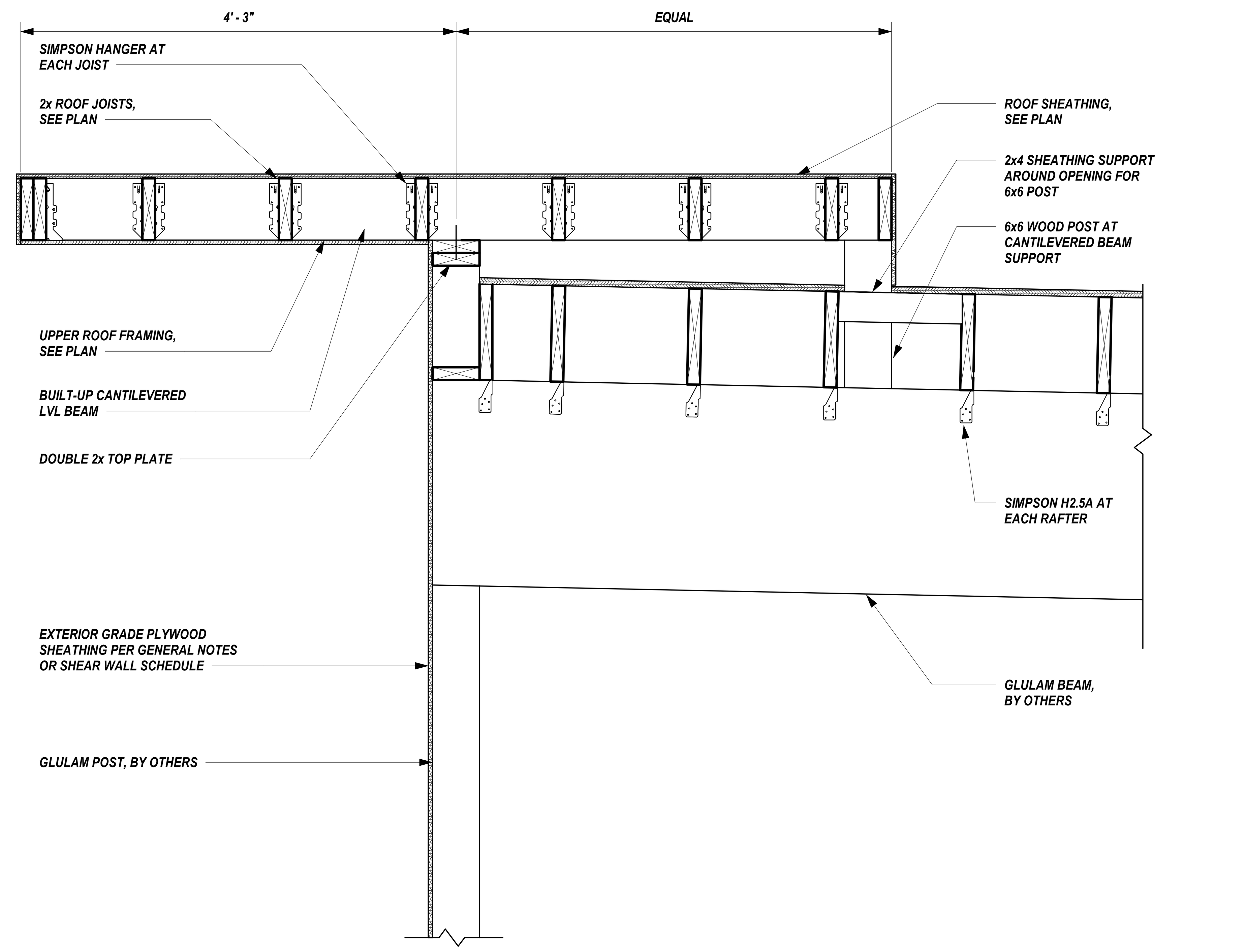
Date: 11-26-2025

Project No.
25026

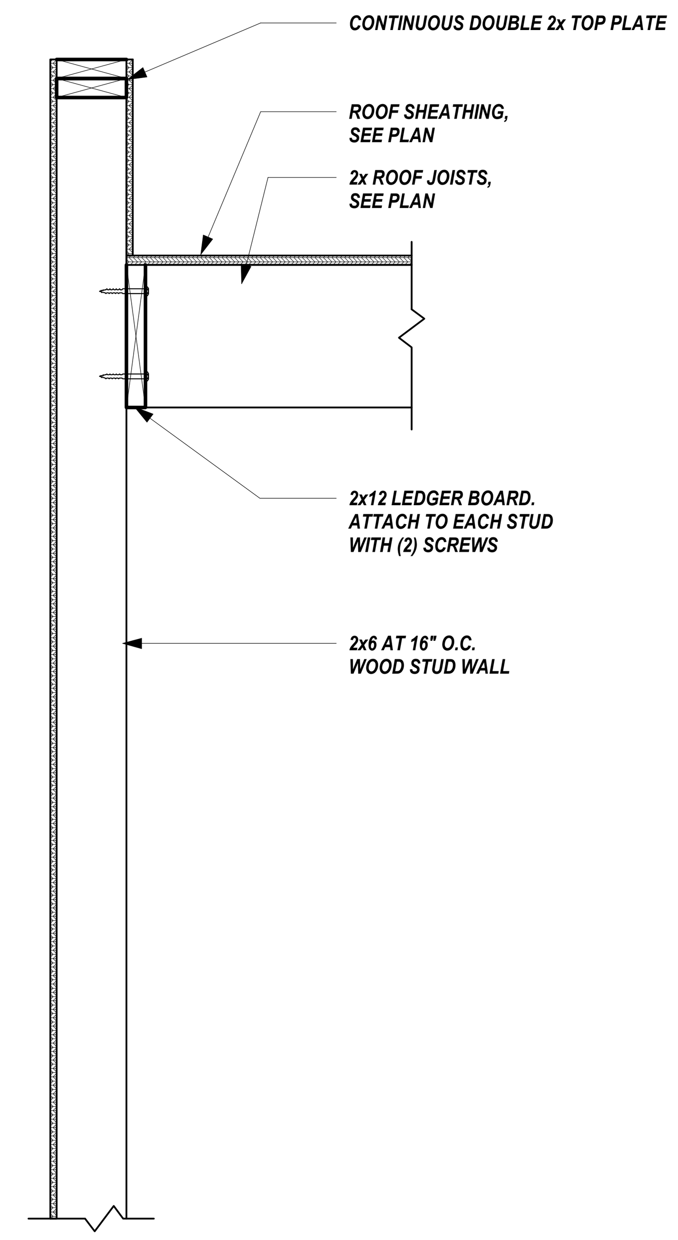
Sheet No.
S3.1



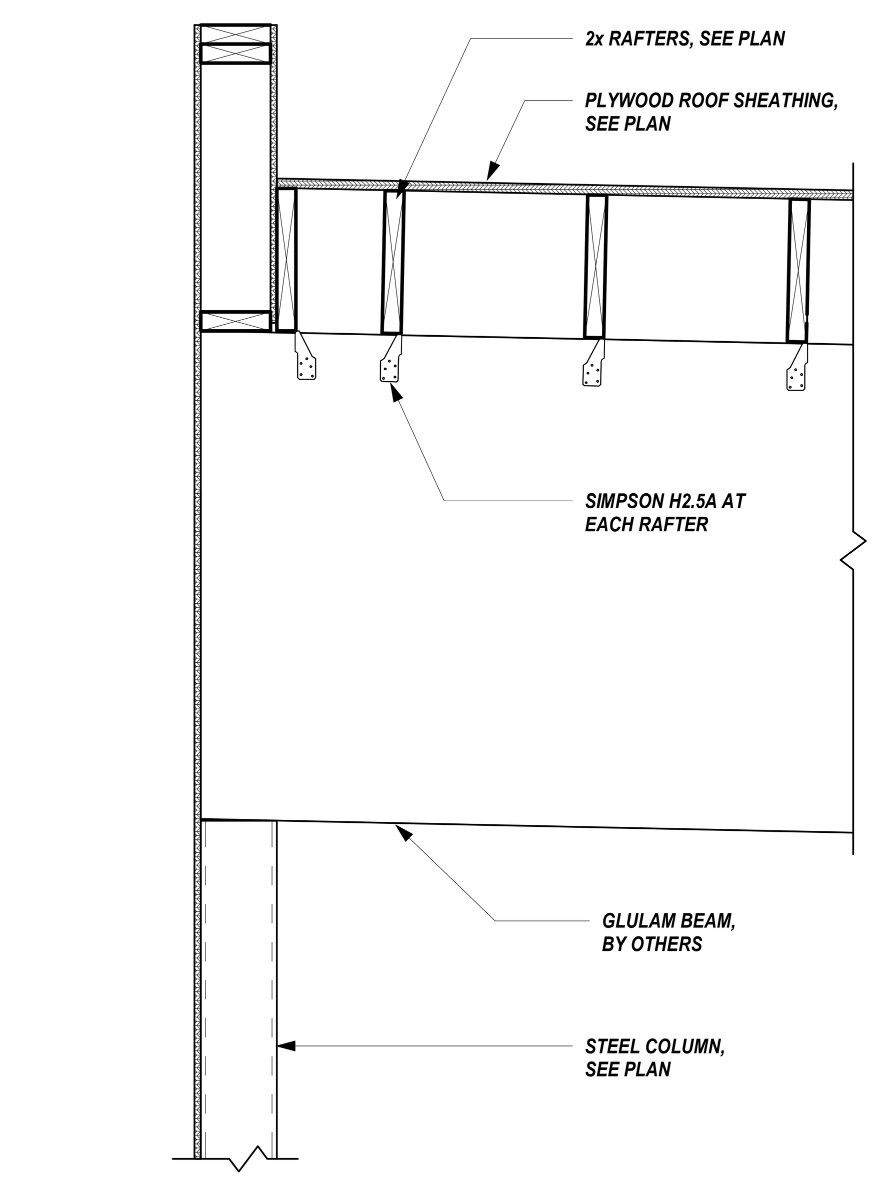
1 S3.2 SECTION
SCALE: 1" = 1'-0"



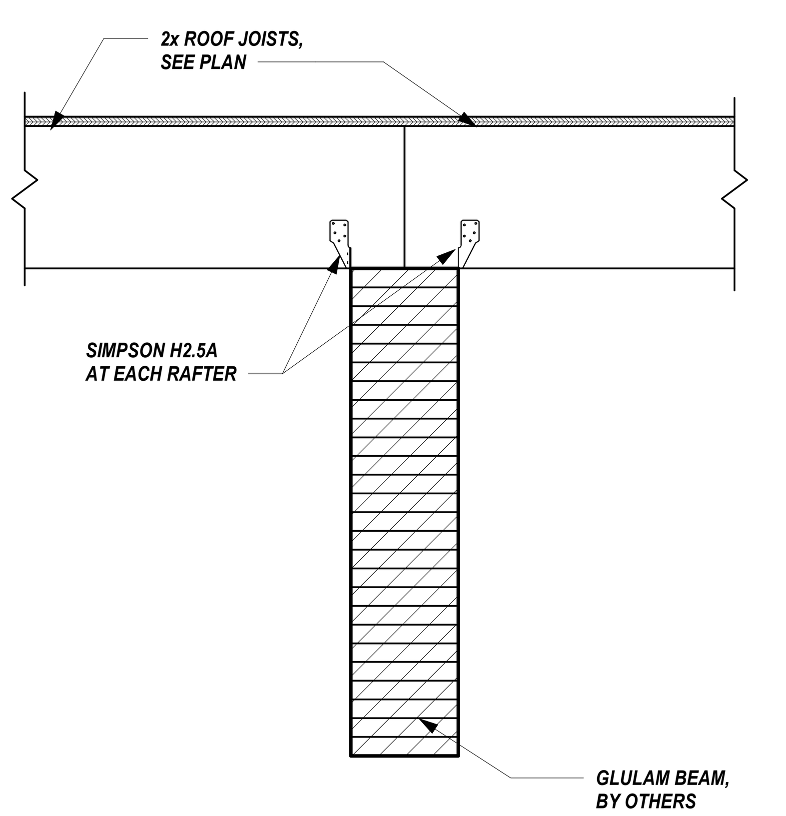
2 S3.2 SECTION
SCALE: 1" = 1'-0"



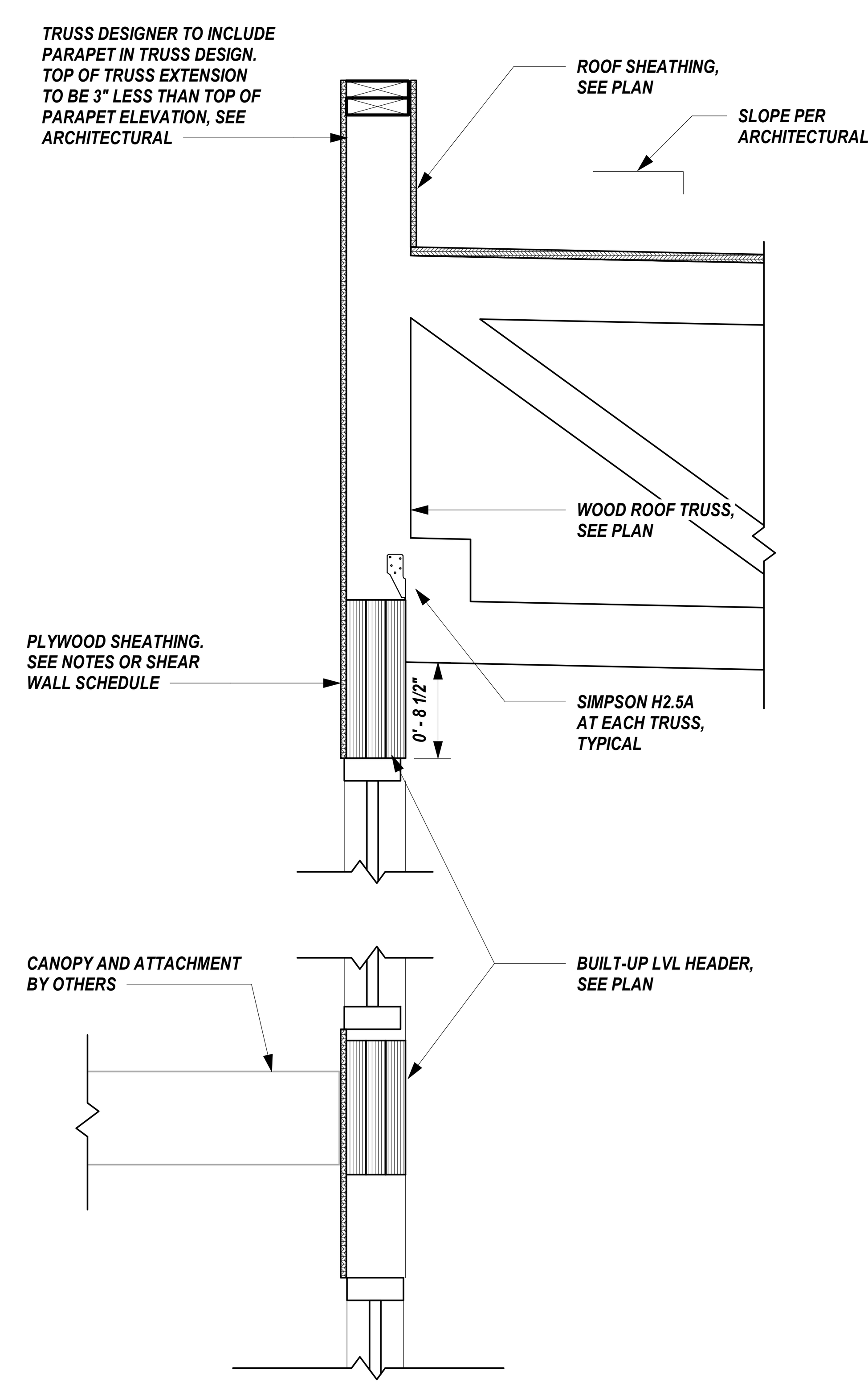
3 S3.2 SECTION
SCALE: 1" = 1'-0"



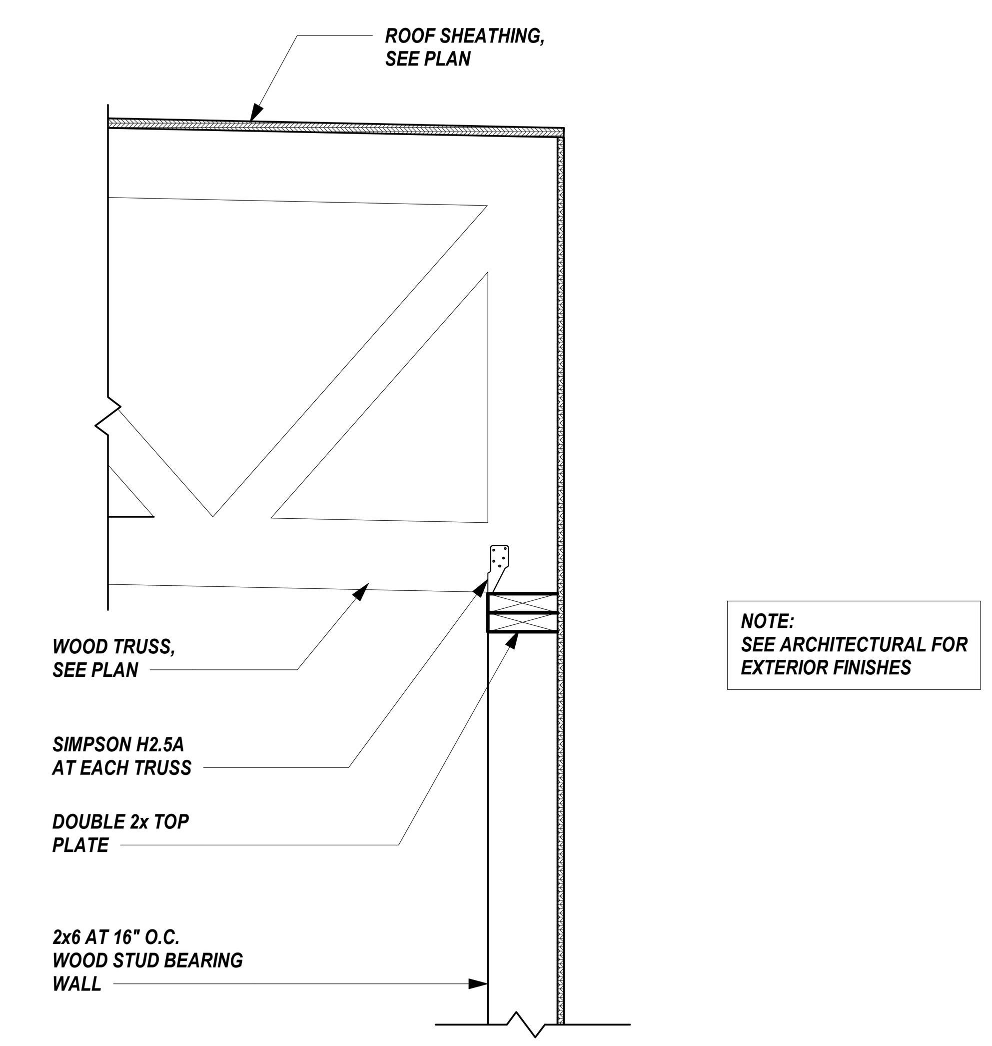
4 S3.2 SECTION
SCALE: 1" = 1'-0"



5 S3.2 SECTION
SCALE: 1" = 1'-0"



6 S3.2 SECTION
SCALE: 1" = 1'-0"



NOTE: SEE ARCHITECTURAL FOR EXTERIOR FINISHES

7 S3.2 SECTION
SCALE: 1" = 1'-0"

TAC
The Architecture Collaborative
6700 Baum Drive, Suite 23 | Knoxville, TN 37919
www.tac-45.com | 865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

11-26-25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

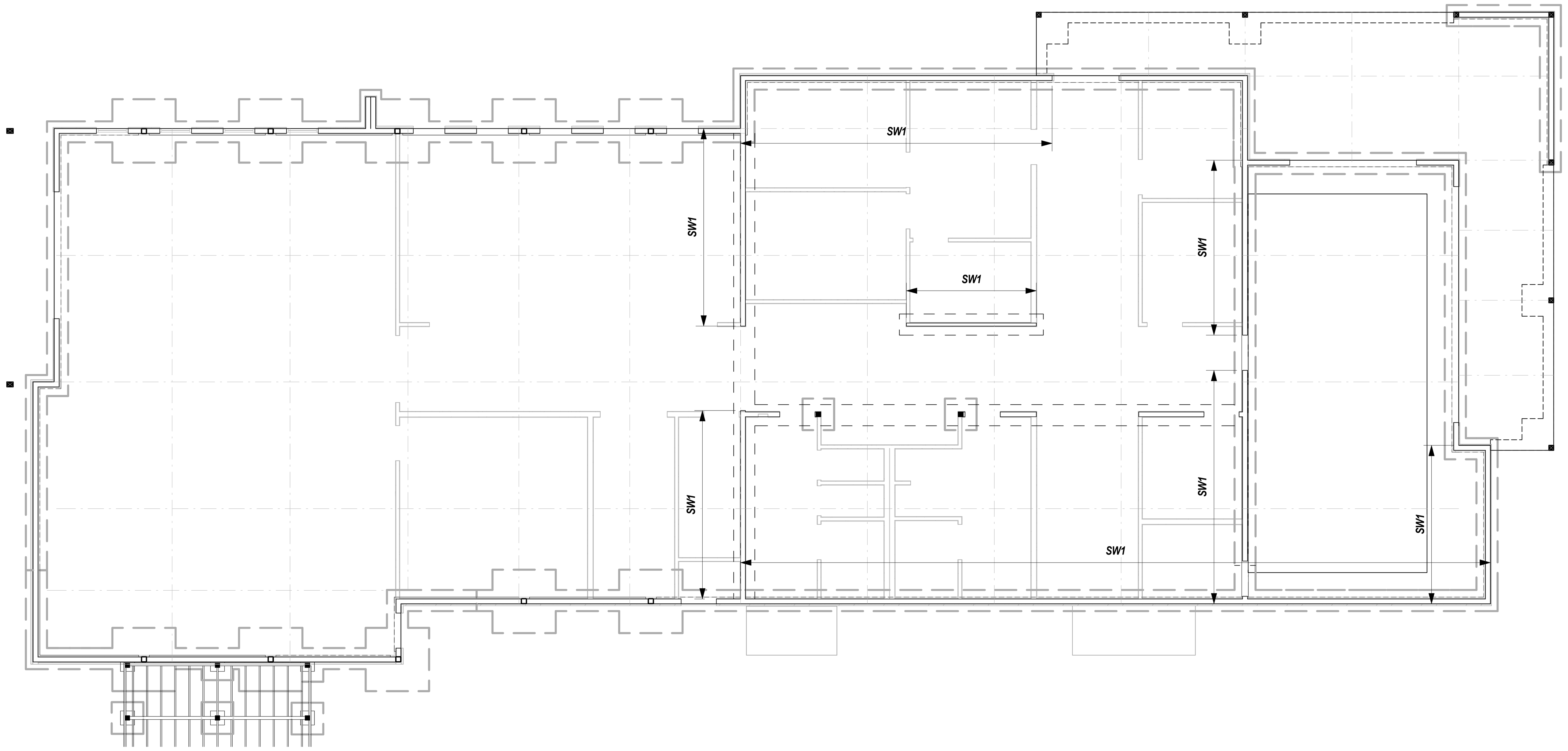
Drawing Title:
FRAMING SECTIONS AND DETAILS

Date: 11-26-2025

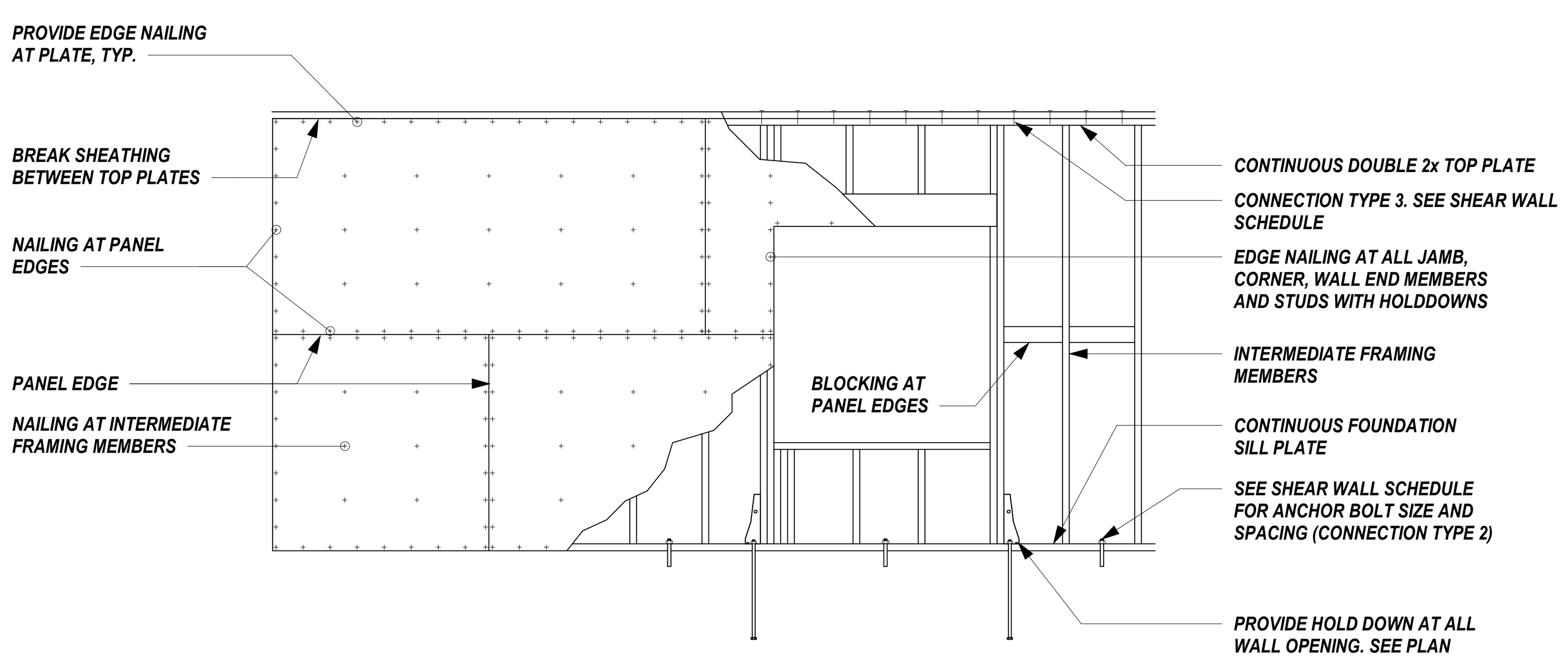
Project No. 25026

Sheet No. S3.2

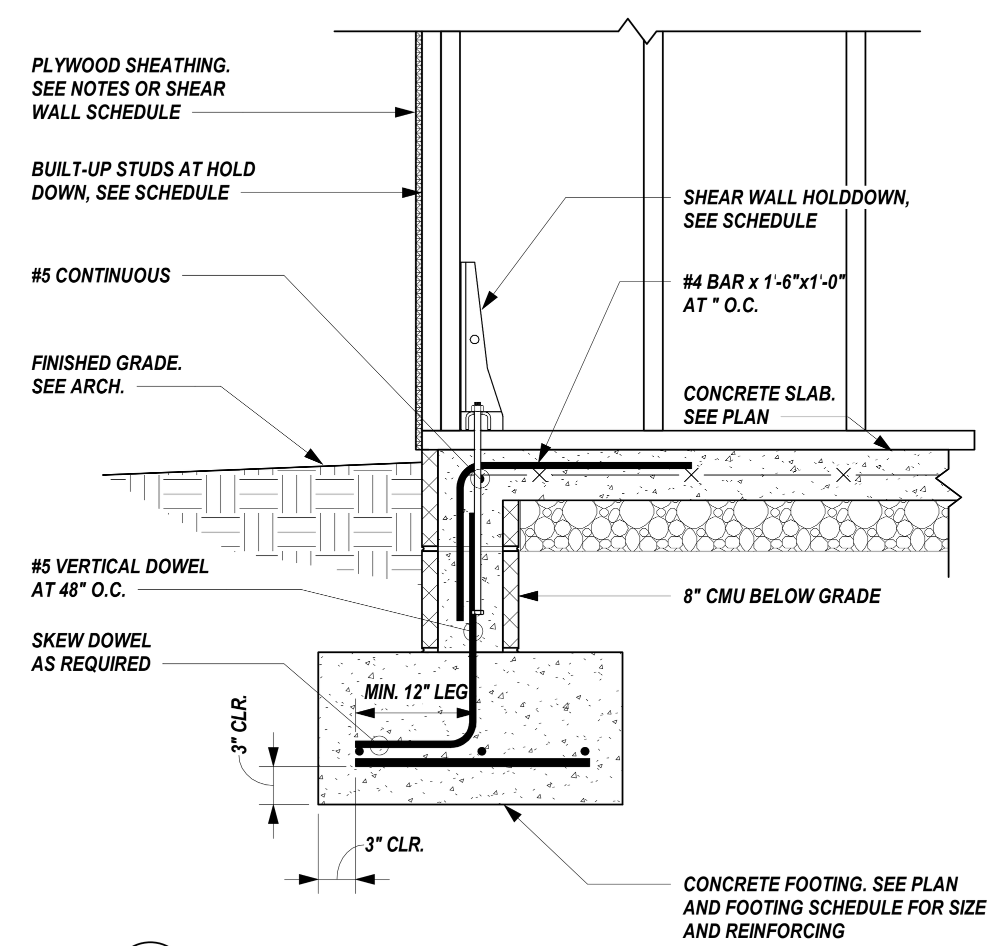
ELAM STRUCTURAL ENGINEERING



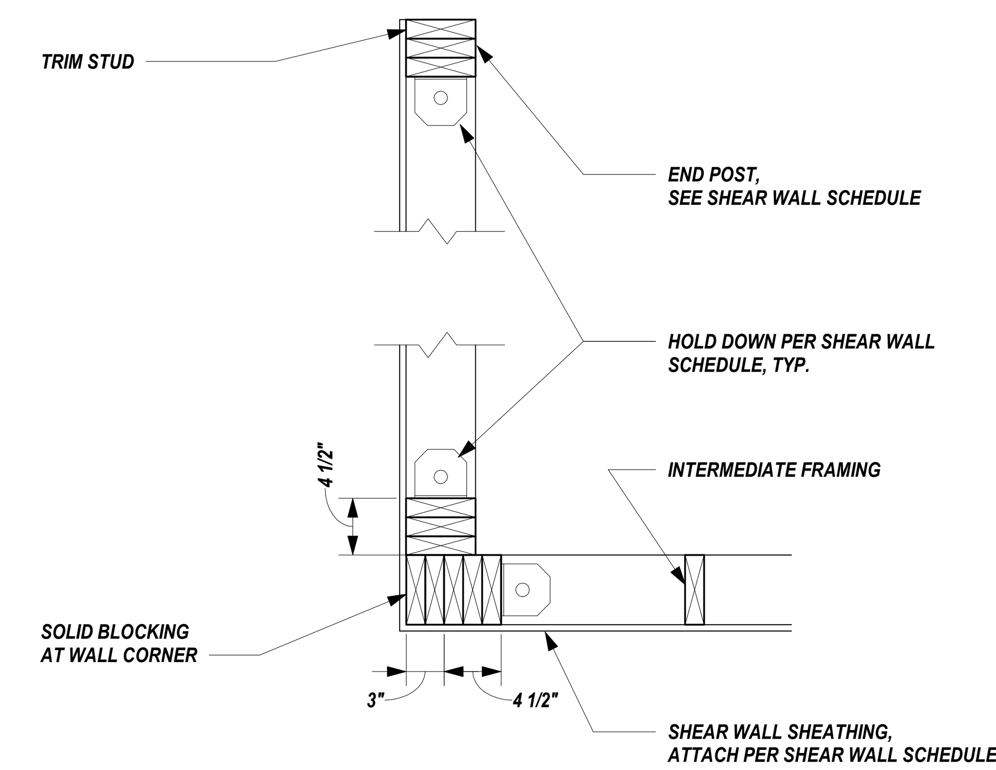
1 SHEAR WALL PLAN
S4.1 SCALE: 1/8" = 1'-0"



2 TYPICAL SHEAR WALL ELEVATION
S4.1 SCALE: 3/8" = 1'-0"



3 SECTION AT SHEAR WALL HOLDDOWN
S4.1 SCALE: 1" = 1'-0"



4 PLAN AT SHEAR WALL CORNER
S4.1 SCALE: 1" = 1'-0"

SHEAR WALL SCHEDULE

	FRAMING REQUIREMENTS	NAILING REQUIREMENTS		WALL BOTTOM PLATE CONNECTION		TOP PLATE CONNECTION
SHEAR WALL TYPE	SHEATHING	PANEL EDGE ATTACHMENTS	INTERMEDIATE FRAMING MEMBERS	SILL TO WOOD FRAMING BELOW (CONNECTION TYPE 1)	SILL TO CONCRETE BELOW (CONNECTION TYPE 2)	WALL TOP PLATE TO FRAMING ABOVE (CONNECTION TYPE 3)
SW1	7/16" SHEATHING, 1 SIDE	8d @ 4" O.C.	8d @ 12" O.C.	16d @ 6" O.C.	5/8" DIA. BOLTS AT 32" O.C., 8" EMBED.	10d TOENAILS @ 6" O.C.

TYPICAL SHEAR WALL NOTES :
1. SHEATHING TO BE APPLIED WITH THE LONG PANEL DIMENSION PERPENDICULAR TO WALL STUDS.
2. USE 1/2" SHEATHING IF PANELS ARE INSTALLED WITH LONG DIMENSION PARALLEL TO WALL STUDS.
3. PANEL JOINTS ON OPPOSITE SIDES OF THE WALL SHALL NOT OCCUR AT THE SAME FRAMING MEMBER.
4. IN LIEU OF 3x MEMBERS, DOUBLE 2x MEMBERS MAY BE USED WHEN NAILED TOGETHER PER WALL BOTTOM PLATE TO WOOD FRAMING BELOW.
5. WALL STUDS NOT OCCURRING AT ABUTTING PANEL EDGES MAY BE 2x.
6. WALL BOTTOM PLATES OCCURRING ABOVE WOOD FRAMING MAY BE 2x.
7. STAGGER NAILS AT ABUTTING PANEL EDGES.
8. INSTALL BLOCKING AT PANEL EDGES NOT OCCURRING AT STUDS OR PLATES.
9. PROVIDE SIMPSON BPS TYPE BEARING PLATES OR EQUIVALENT ON ALL ANCHOR BOLTS TO CONCRETE. PROVIDE BPS5/8-3 AT 2x4 WALLS AND BPS5/8-6 AT 2x6 WALLS. PROVIDE STANDARD CUT WASHER BETWEEN NUT AND BPS.
10. ANCHOR BOLTS SHALL BE CAST-IN-PLACE, SCREW ANCHORS, OR ADHESIVE ANCHORS AND EMBEDMENT SHALL BE AS NOTED. LOCATE ANCHOR BOLTS WITHIN 8" OF ENDS OF WALLS.
11. DO NOT OVERDRIVE NAILS.
12. (2) FLOOR-TO-FLOOR HOLD DOWNS SHALL BE USED AT EACH END POST AT ALL LEVELS ABOVE AND BELOW FLOOR PLATE.
13. END POST STUDS SHALL BE ALIGNED FOR THE FULL HEIGHT OF THE SHEAR WALL. IF END POSTS DO NOT ALIGN, UPPER END POSTS MUST BE CARRIED DOWN TO THE FOUNDATION AND ANCHORED PER HOLDDOWN SCHEDULE.

HOLD DOWN SCHEDULE, ANCHOR BOLT TYPES, AND INSTALLATION CRITERIA

HOLD DOWN MARK		WOOD POST AND CONNECTION		ANCHOR SIZE	ADHESIVE ANCHOR TO FOUNDATION	
PLAN MARK	HOLD DOWN TYPE	HOLD DOWN WOOD POST	WOOD FASTENERS	DIAMETER (IN)	EMBED (IN)	EDGE DISTANCE (IN)
SW1	HDU5-SDS2.5	(2) 2x	14-SDS 1/4x2.5	5/8	8	2-3/4"

TAC
The Architecture Collaborative
6700 Baum Drive, Suite 23 | Knoxville, TN 37919
www.tac-45.com | 865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

Drawing Title:
SHEAR WALL DETAILS

Date: 11-26-2025

Project No.
25026

ELAM STRUCTURAL ENGINEERING

Sheet No.
S4.1

Preliminary Sprinkler Calculation Sheet

Flow test Data

Static Pressure:	98 psi
Residual Pressure:	48 psi
Flow (GPM) :	920 gpm
Date taken:	9/18/2025
Time:	3:50 PM
Test taken by:	ASSOCIATED FIRE
Elevation of Hydrant:	HARPER AVE & N. COURT...

GPM Demand of BLDG.

Most remote area or highest demand (Room Name)	LIGHT HAZARD
Design Density (NFPA 13 or supplied by Insurance Co.)	0.1
Design Area (Square footage)	1500 sq. ft.
Overage Factor (1.20 typ.)	1.2
Remote area GPM demand(Density x Area x Overage)	180 gpm
Standpipe GPM demand (If required)(500 gpm for the first, 250 after)	0 gpm
Hose GPM demand (100 Light, 250 ordinary, 500 extra hazard)	100 gpm
Total GPM (Remote Area + Standpipe + Hose)	280 gpm

Available Pressure

Density	0.1
Max Sprinkler Head coverage (As per NPFA 13 table 8.6.2.2(a-d))	225 sq. ft.
Square footage spacing x Density = GPM sprinkler head (Q)	22.5 gpm
K-Factor of Sprinkler head (K)	5.6 (K)
Equation: Pressure required at head=(Q / K)2	16.14317602 psi
Elevation difference from test hydrant to base of riser x .433	0 ft
Elevation difference from base of riser to remote area x .433	12 ft
Backflow Preventer pressure drop	12 psi
Safety Factor (5 psi min.) (SF)	5 psi
Fixed Pressure drop =	38.33917602 psi

Estimated Friction Drop Thru Fire Line

Length of run from test hydrant to riser (HR)	500 Lin. Ft.
Pipe C Factor (Ductile Iron C-100)	100
Nominal Pipe Inside Diameter (8", 6", 4")	6
Friction loss in pipe (psi/ft) (Based on Hazen William Equation)(HW1)	0.004929288 psi/ft
HR x 1.30 x HW1 =	3.204037276 Est. psi/sqft
Length of run from riser to last sprinkler head (estimated.) (RS)	150 Lin. Ft.
Base of Riser to farthest sprinkler	12
Pipe C Factor (Black Steel C-120)	120
Nominal Pipe Inside Diameter (6", 4")	4
Friction loss in pipe (psi/ft) (Based on Hazen William Equation)(HW2)	0.01119116 psi/ft
RS x 1.30 x HW2 =	2.182276207 Est. psi/sqft

Estimated Required Flow Data for Building

Required GPM	280 gpm
Required psi	42 psi

Preliminary Sprinkler Calculation Sheet

Flow test Data

Static Pressure:	98 psi
Residual Pressure:	48 psi
Flow (GPM) :	920 gpm
Date taken:	9/18/2025
Time:	3:50 PM
Test taken by:	ASSOCIATED FIRE
Elevation of Hydrant:	HARPER AVE & N. COURT...

GPM Demand of BLDG.

Most remote area or highest demand (Room Name)	RETAIL - OH2
Design Density (NFPA 13 or supplied by Insurance Co.)	0.2
Design Area (Square footage)	1500 sq. ft.
Overage Factor (1.20 typ.)	1.2
Remote area GPM demand(Density x Area x Overage)	360 gpm
Standpipe GPM demand (If required)(500 gpm for the first, 250 after)	0 gpm
Hose GPM demand (100 Light, 250 ordinary, 500 extra hazard)	250 gpm
Total GPM (Remote Area + Standpipe + Hose)	610 gpm

Available Pressure

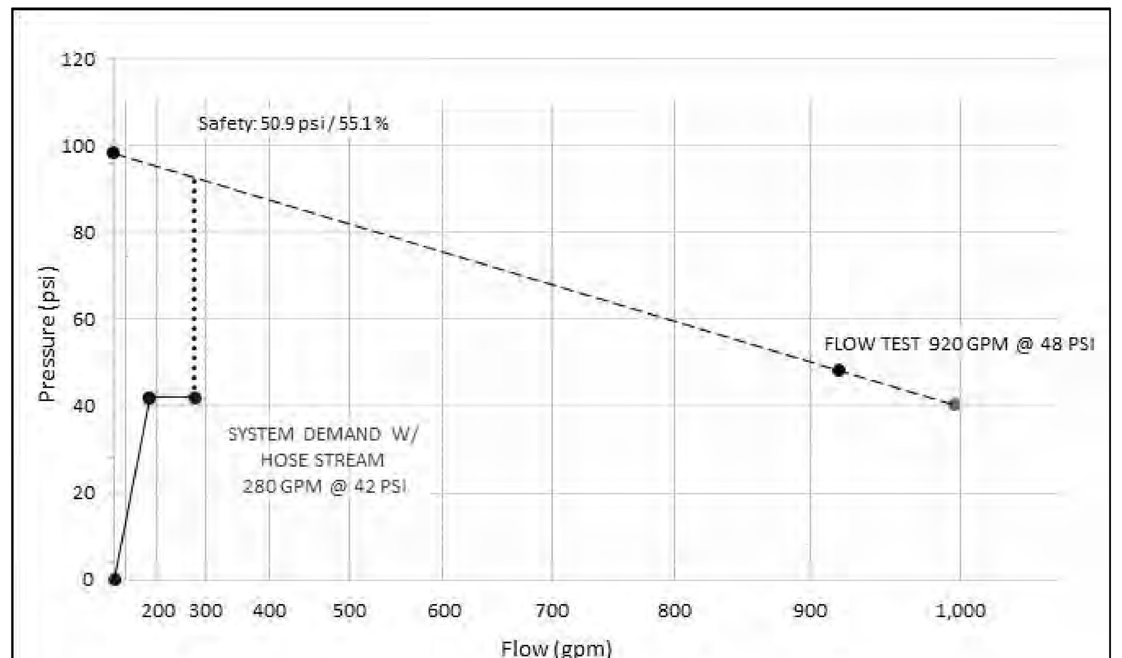
Density	0.2
Max Sprinkler Head coverage (As per NPFA 13 table 8.6.2.2(a-d))	130 sq. ft.
Square footage spacing x Density = GPM sprinkler head (Q)	26 gpm
K-Factor of Sprinkler head (K)	5.6 (K)
Equation: Pressure required at head=(Q / K)2	21.55612245 psi
Elevation difference from test hydrant to base of riser x .433	0 ft
Elevation difference from base of riser to remote area x .433	12 ft
Backflow Preventer pressure drop	12 psi
Safety Factor (5 psi min.) (SF)	5 psi
Fixed Pressure drop =	43.75212245 psi

Estimated Friction Drop Thru Fire Line

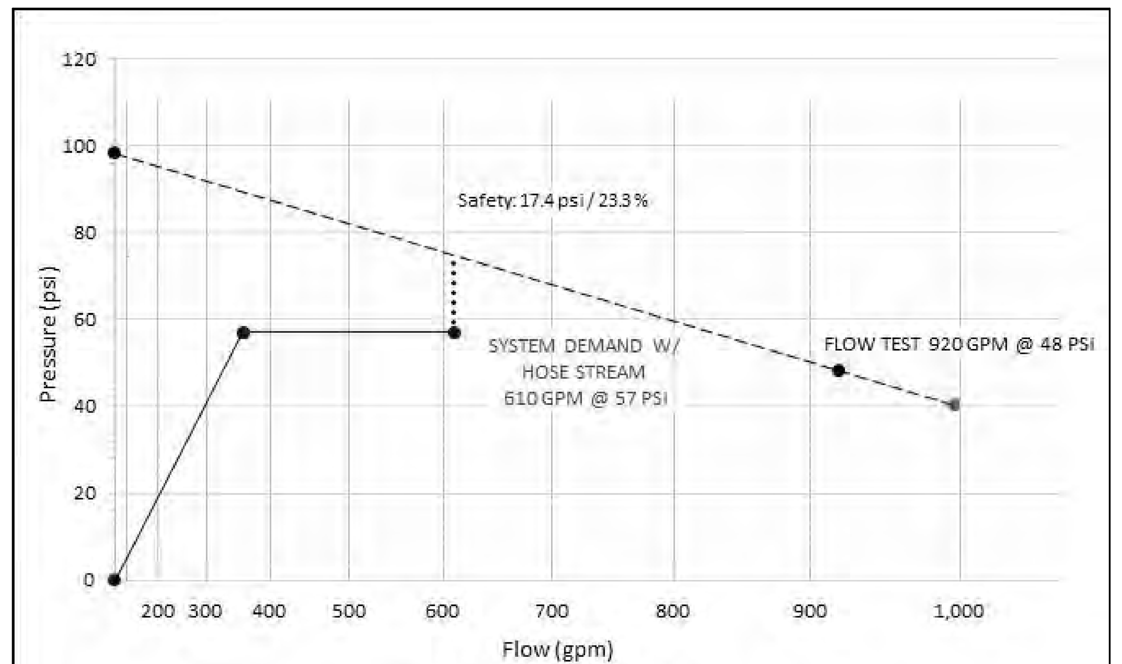
Length of run from test hydrant to riser (HR)	500 Lin. Ft.
Pipe C Factor (Ductile Iron C-100)	100
Nominal Pipe Inside Diameter (8", 6", 4")	6
Friction loss in pipe (psi/ft) (Based on Hazen William Equation)(HW1)	0.020816228 psi/ft
HR x 1.30 x HW1 =	13.53054833 Est. psi/sqft
Length of run from riser to last sprinkler head (estimated.) (RS)	150 Lin. Ft.
Base of Riser to farthest sprinkler	12
Pipe C Factor (Black Steel C-120)	120
Nominal Pipe Inside Diameter (6", 4")	4
Friction loss in pipe (psi/ft) (Based on Hazen William Equation)(HW2)	0.040344153 psi/ft
RS x 1.30 x HW2 =	7.867109765 Est. psi/sqft

Estimated Required Flow Data for Building

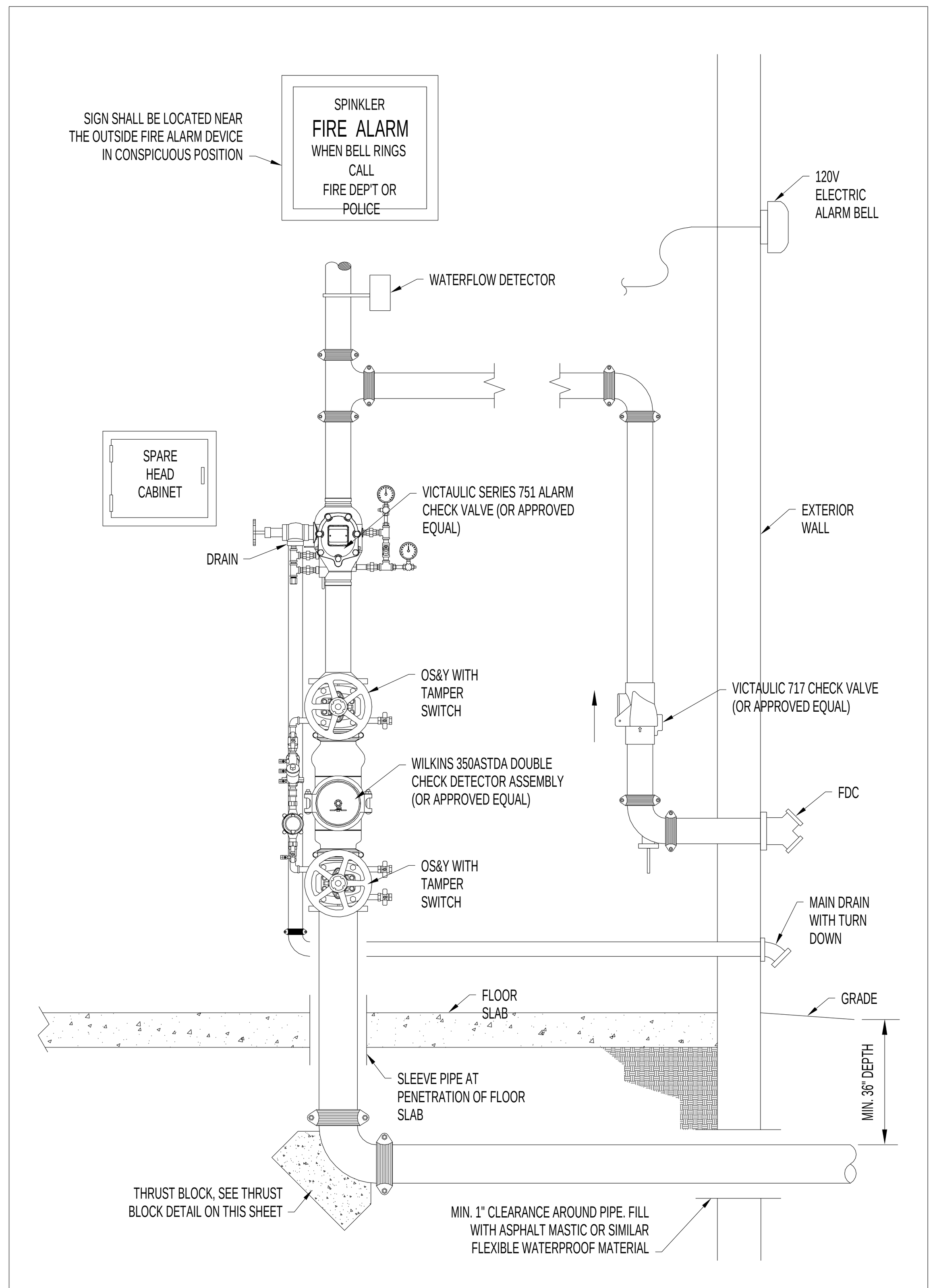
Required GPM	610 gpm
Required psi	57 psi



HYDRAULIC GRAPH
LIGHT HAZARD



HYDRAULIC GRAPH
RETAIL - OH2



TYPICAL WET SYSTEM FIRE RISER DETAIL
NO SCALE

SPRINKLER HEAD LEGEND								
SYMBOL	TYPE	TEMP	K	MAX PRESSURE	MANUFACTURE	MODEL	SERIES	COMMENTS
	UPRIGHT	165°	5.6	175	TYCO	TY3121	TY-FRL	QUICK RESPONSE
	PENDENT	165°	5.6	175	TYCO	TY3221	TY-FRL	QUICK RESPONSE
	WALL	155°	5.6	175	TYCO	TY3335	DS-1	QUICK RESPONSE, DRY

* PROVIDE ALL BRACING, SUPPORTS AND HANGERS PER NFPA 13 - 2019 EDITION
* "WG" DENOTES WIRE GUARD
* COORDINATE SPRINKLER HEAD FINISH WITH ARCHITECT

FIRE SPRINKLER SYSTEM NOTES:

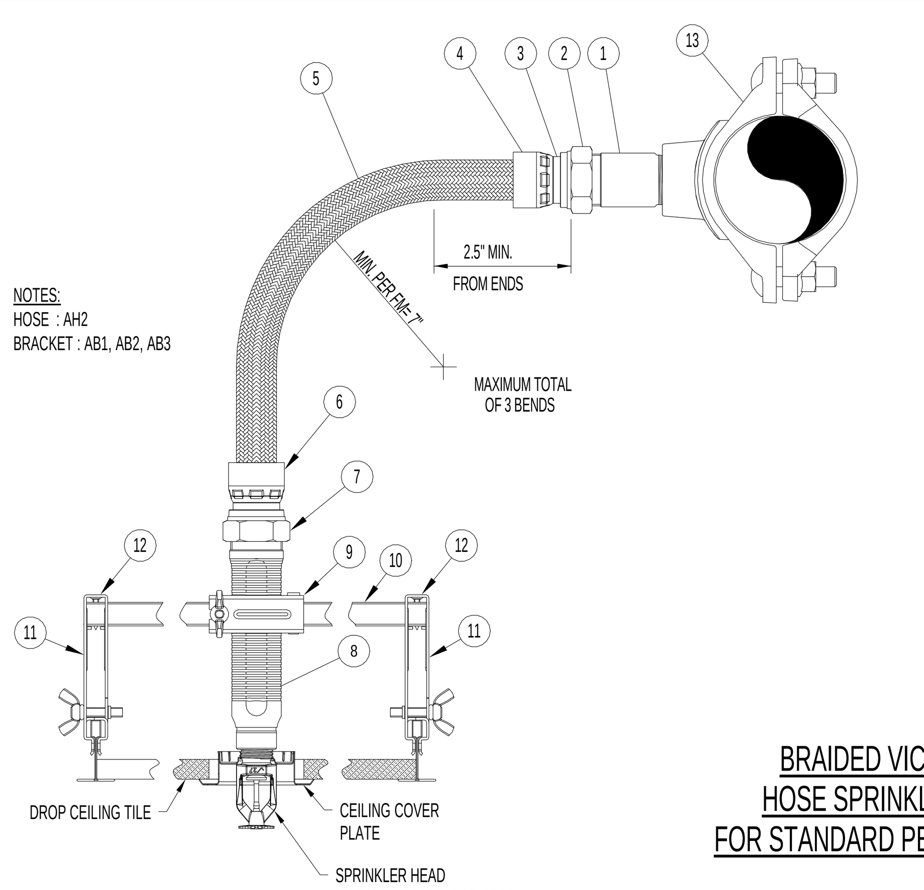
1. FIRE SPRINKLER CONTRACTOR SHALL PROVIDE A FIRE SPRINKLER SYSTEM DESIGN COMPLIANT WITH ALL APPLICABLE PROVISIONS OF NFPA 13 2019 EDITION. FURNISH AND INSTALL A WET SPRINKLER SYSTEM THROUGHOUT THE BUILDING.
2. THESE DRAWINGS ARE SCHEMATIC FOR DESIGN INTENT ONLY AND THE DESIGN-BUILD CONTRACTOR IS RESPONSIBLE FOR A COMPLETE AND FUNCTIONAL SYSTEM WITH ANY NECESSARY APPURTENANCES.
3. FIRE SPRINKLER SHOP DRAWINGS (2 SETS OF WORKING PLANS, PRODUCT DATA AND HYDRAULIC CALCULATIONS) ARE TO BE SUBMITTED FOR REVIEW AFTER THE ENGINEER OF RECORD IS SATISFIED THAT THE SHOP DRAWINGS SATISFY THE REQUIREMENTS OF THE NFPA 13 AND THE PROJECT DOCUMENTS. THE ENGINEER OF RECORD SHALL CITE SUCH APPROVAL ON THE SHOP DRAWINGS
4. ALL DETAIL DESIGN DRAWINGS AND CALCULATIONS SHALL BE SEALED BY A SPRINKLER SYSTEM ENGINEER OR R.M.E. LICENSED IN THE STATE OF TENNESSEE.
5. THE SPACES ARE CLASSIFIED AS "LIGHT HAZARD" (EXCEPT WHERE NOTED BELOW) THROUGHOUT SYSTEM, DESIGN CALCULATIONS SHALL INCLUDE SPRINKLERS TO PROVIDE A DESIGN DENSITY OF 0.10 GPM/FT^2 FOR THESE OCCUPANCIES. JANITOR CLOSETS, MECHANICAL ROOMS, AND CHEMICAL STORAGE ROOMS SHALL BE CLASSIFIED AS "ORDINARY HAZARD", USING A DESIGN DENSITY OF 0.15 GPM/FT^2.
6. PIPE AND FITTINGS
   A. DUCTILE-IRON PIPE: AWWA C151, PUSH-ON-JOINT TYPE, WITH CEMENT-MORTAR LINING AND SEAL COAT ACCORDING TO AWWA C104. INCLUDE RUBBER GASKET ACCORDING TO AWWA C111.
   B. DUCTILE-IRON PIPE: AWWA C151, MECHANICAL-JOINT TYPE; WITH CEMENT-MORTAR LINING AND SEAL COAT ACCORDING TO AWWA C104. INCLUDE GLAD, RUBBER ACCORDING TO NFPA 1963 AND MATCHING LOCAL FIRE DEPARTMENT SIZES AND THREADS, AND BOTTOM OUTLET WITH PIPE THREADS. INCLUDE BRASS, LUGGED CAPS, GASKETS, AND BRASS CHAINS; BRASS, LUGGED SWIVEL CONNECTION AND DROP CLAPPER FOR EACH HOSE-CONNECTION INLET; EIGHTEEN (18") INCH (460-MM) HIGH BRASS SLEEVE; AND ROUND, FLOOR, BRASS, ESCUTCHEON PLATE WITH MARKING "AUTO SPKR."
      1. FINISH INCLUDING SLEEVE: POLISHED CHROME-PLATED.
      2. FINISH INCLUDING SLEEVE: ROUGH CHROME-PLATED.
      3. FINISH INCLUDING SLEEVE: POLISHED BRASS.
   C. STEEL PIPE: ERW OR CW SCHEDULE 10 OR 40. ALL FITTINGS SHALL COMPLY WITH NFPA 13.
   D. (WHERE APPLICABLE) BLAZEMASTER (FM APPROVED) CPVC SDR 13.5 CPVC PIPE AND FITTINGS (OR APPROVED EQUAL)
7. CONTRACTOR SHALL COORDINATE WITH ARCHITECT TO ENSURE INSULATION IN ATTIC COMPLETELY COVERS PIPING FOR MAIN FLOOR SPRINKLER SYSTEM. PIPING MAY REQUIRE ROUTING THROUGH JOIST/TRUSSES IN ACCORDANCE WITH NFPA 13R. CONFIRM WITH STRUCTURAL ENGINEER.
8. QUICK RESPONSE SIDEWALL SPRINKLERS SHALL BE INSTALLED AT EACH STAIRWAY LANDING.
9. A SIDEWALL SPRINKLER SHALL BE PROVIDED AT THE BOTTOM OF ELEVATOR SHAFT AND LOCATED PER NFPA CODE. AN UPRIGHT STYLE HEAD SHALL BE LOCATED AT THE TOP OF THE ELEVATOR SHAFT AND SHALL BE OF ORDINARY OR INTERMEDIATE TEMPERATURE RATING PER NFPA 13 STANDARDS.
10. ALL SYSTEM VALVES AND GAUGES SHALL BE ACCESSIBLE FOR OPERATION, INSPECTION, TEST, AND MAINTENANCE.
11. AIR COMPRESSOR SHALL BE PROVIDED FOR DRY PIPE SYSTEMS. PROVIDE AIR PIPING VALVES, GAUGES AND PRESSURE SENSORS AS PER NFPA.
12. COORDINATE LOCATION OF SPRINKLER WITH ALL OTHER DISCIPLINES. SPRINKLER HEADS SHALL BE CENTER OR QUARTERED IN CEILING TILE UNLESS NOTED OTHERWISE. ALL SPRINKLERS IN GRID CEILING TO BE ON RETURN BENDS OR UTILIZE FLEXIBLE SPRINKLER DROP (VICTAULIC AH2CC OR APPROVED EQUAL).
13. COORDINATE WITH THE ELECTRICAL CONTRACTOR FOR INSTALLATION AND MONITORING OF ALL TAMPER AND FLOW SWITCHES. FLOW SWITCHES AND ALARM CHECK VALVES SHALL BE CONNECTED TO GENERAL FIRE ALARM AND SHALL SOUND WITHIN 90 SECONDS OF FLOW.
14. CONTRACTOR SHALL SUPPLY FLEXIBLE PIPE COUPLINGS ON ALL PIPES 2" OR LARGER AT ALL FLEXIBLE JOINTS PER NFPA 13. FLEXIBLE COUPLINGS SHALL ALSO BE PROVIDED WITHIN 1' OF BOTH SIDES OF STRUCTURAL ELEMENTS THAT PIPING PASSES THROUGH.
15. ALL PIPING SHALL HAVE HANGERS INSTALLED PER NFPA 13.
16. SPARE HEAD CABINET SHALL BE LOCATED AS CLOSE TO RISER AS POSSIBLE AND MUST CONTAIN A MINIMUM OF SIX HEADS. THIS SHALL INCLUDE TWO SPRINKLER HEADS OF EACH TYPE AND TEMPERATURE RATING.
17. A 100 GPM HOSE STREAM SHALL BE ADDED TO THE SPRINKLER REQUIREMENTS FOR COMBINED INSIDE AND OUTSIDE HOSE, AT THE POINT OF CONNECTION TO THE SYSTEM.
18. ALLOW 10 PSI PRESSURE LOSS SAFETY FACTOR FOR REDUCTION OR FLUCTUATION OF WATER PRESSURE.
19. PENETRATION OF FIRE AND SMOKE BARRIERS/PARTITIONS SHALL BE ADEQUATELY SEALED/PROTECTED. SEE ALSO FIRESTOP DETAILS ON DWG.
20. THE PRESENT WATER SUPPLY SYSTEM SERVING THE FACILITY HAS NOT BEEN DETERMINED AS BEING ADEQUATE FOR THE FIRE PROTECTION SYSTEM. DETERMINATION WILL BE MADE UPON RECEIVING FLOW TEST DATA.

PIPING LEGEND

WET FIRE LINE
DRY FIRE LINE

LEGEND

FLEXIBLE COUPLING
LATERAL BRACE
LONGITUDINAL BRACE
4-WAY BRACE
PIPE UP
PIPE DOWN
HYDRAULICALLY MOST REMOTE AREA



PRODUCT LEGEND	
1	1" NIPPLE (INLET)
2	NUT
3	ADAPTER RING
4	COLLAR
5	304 STAINLESS STEEL FLEXIBLE BRAIDED HOSE (1" NOMINAL DIAMETER)
	(LENGTHS AVAILABLE 31", 48", 60", & 72")
6	COLLAR
7	NUT
8	REDUCING OUTLET (AVAILABLE 1/2" OR 3/4" FMPT)
9	CENTER BRACKET ASSEMBLY
10	SQUARE BAR (AVAILABLE IN 24" AND 48" LENGTHS)
11	END BRACKET ASSEMBLY
12	SHEET METAL SCREW
13	VIC MECHANICAL-T OUTLET STYLE 922 FIRELOCK (1" FMPT)
SEE VICTAULIC SUBMITTAL NUMBER 10.85	

NOTES:

a. All hoses shall be factory-pressure tested to 400 psi. (2760 kPa).
b. Approvals:
a. FM-1637 (Braided)
b. UL 2443 (Unbraided / Corrugated)
c. NYC MEA 60-05-E
d. City of Los Angeles 5/1/09 - 5/1/10.

BRAIDED VICFLEX FLEXIBLE
HOSE SPRINKLER ASSEMBLIES
FOR STANDARD PENDENT SPRINKLERS

FACILITY SYSTEMS CONSULTANTS, LLC

TAC
The Architecture Collaborative
6215 Kingston Pike Knoxville, TN 37919
www.tac-45.com 865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

PRESTON C. HALE REGISTERED ENGINEER AGRICULTURE COMMERCE NO. 114592 STATE OF TENNESSEE
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

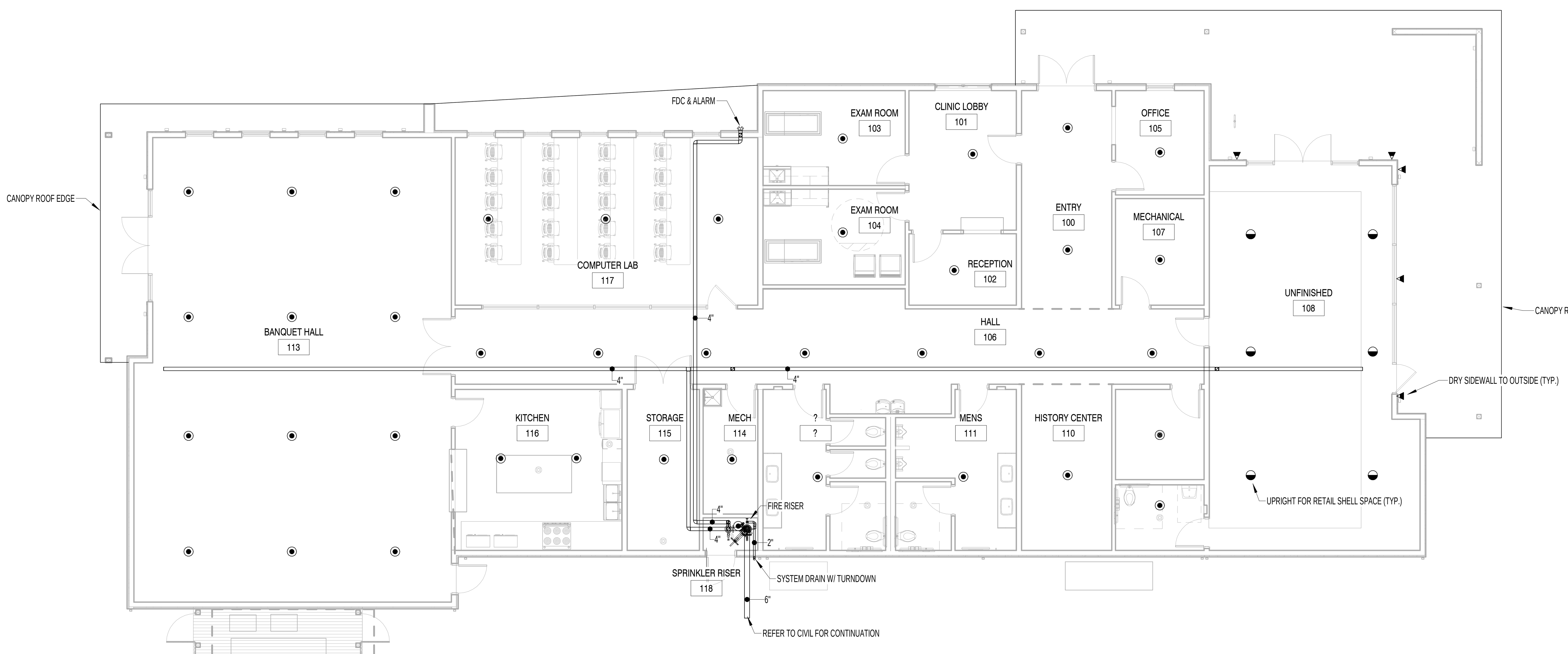
No.	Date

Drawing Title:
FIRE PROTECTION SCHEDULES, NOTES & DETAILS

Date: 2025

Project No. 25026

Sheet No. FP0.1



1 FIRE PROTECTION PLAN
1/8" = 1'-0"

FACILITY SYSTEMS CONSULTANTS, LLC

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

PRESTON C. HALL
REGISTERED ENGINEER
AGRICULTURE
COMMERCE
NO. 114592
STATE OF TENNESSEE
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

Drawing Title:
FIRE PROTECTION PLAN

Date: 2025

Project No.
25026

Sheet No.
FP1.1

ELECTRIC SPLIT SYSTEM HEAT PUMP UNIT SCHEDULE

DESIGNATION		CFM	O.A. CFM	EXT. S.P.	EVAP. FAN HP	CAPACITY (BTUH)			INDOOR UNIT					OUTDOOR UNIT				SEER	COP @ 47F	HSPF (BTUH/WATT)	MANUFACTURER AND MODEL NO.	
AHU	HP					TOTAL COOLING	SENSIBLE COOLING	HEATING	WEIGHT (LBS)	AUX. HEAT (KW)	MCA	MOP	VOLTAGE	WEIGHT (LBS)	MCA	MOP	VOLTAGE				INDOOR	OUTDOOR
1	1	1600	200	.5	.75	47,438	35,420	43,500	174.0	10.8	44.0	45.0	208/3/60	251.0	18.1	30.0	208/3/60	15.20	4.30	8.10	TRANE 5TEM4D07AC51	TRANE 5TWA4048A3
2	2	800	90	.5	.50	23,100	17,100	22,400	117.0	7.2	29.0	30.0	208/3/60	174.0	13.0	20.0	208/1/60	14.30	4.20	7.50	TRANE 5TEM4B03AC31	TRANE 5TWR4024A1
3	3	600	65	.5	.50	18,600	13,300	17,900	117.0	7.2	29.0	30.0	208/3/60	174.0	12.0	20.0	208/1/60	14.30	3.80	7.50	TRANE 5TEM4B03AC31	TRANE 5TWR4018A1
4	4	1000	90	.5	.50	27,200	20,600	25,600	138.0	7.2	29.0	30.0	208/3/60	174.0	16.0	25.0	208/1/60	14.30	4.10	7.50	TRANE 5TEM4D04AC31	TRANE 5TWR4030A1
5	5	1400	270	.5	.75	41,691	31,297	38,500	174.0	10.8	44.0	45.0	208/3/60	251.0	18.1	30.0	208/3/60	16.00	4.30	7.80	TRANE 5TEM4D07AC51	TRANE 5TWA4042A3

ACCESSORIES AND FEATURES:

* ESP DOES NOT INCLUDE FILTER SECTION.
* PROVIDE FILTER RACK AND 2" THICK THROW-AWAY FILTER.
* PROVIDE 7-DAY PROGRAMMABLE THERMOSTAT.
* PROVIDE ACR COPPER REFRIGERANT TUBING WITH FILTER-DRYER, SIGHT GLASS, AND SERVICE VALVES.
* PROVIDE A MINIMUM 5 YEAR COMPRESSOR WARRANTY.
* EQUIPMENT TO BE ARI CERTIFIED AND UL APPROVED.
* ALTERNATE MANUFACTURERS: LENNOX, CARRIER.
* COOLING CAPACITY AT 80/67 EAT, 95 ODT.
* HEATING CAPACITY AT 47 ODT.
* PROVIDE CONCRETE PAD UNDER CONDENSING UNIT.
* SAFETY DISCONNECT BY ELECTRICAL.
* PROVIDE MANUAL O.A. BACKDRAFT DAMPER.

PACKAGED HEAT PUMP UNIT WITH ELECTRIC HEAT SCHEDULE

PHP	CFM	O.A. CFM	EXT. S.P.	EVAP. FAN HP	CAPACITY (BTUH)			AUX. HEAT (KW)	WEIGHT (LBS)	ELECTRICAL			SEER/EER	COP	HSPF	MANUF. AND MODEL NO.
					TOTAL	SENSIBLE	HEATING			MCA	MOP	VOLTAGE				
1	5000	1250	.5	2.9	152,620	112,090	133,410	36.0	2565.0	172.0	175.0	208/3/60	-/10.60	3.30		TRANE WSK150A3S0N

ACCESSORIES AND FEATURES:

* PROVIDE A MINIMUM 5 YEAR COMPRESSOR WARRANTY.
* EQUIPMENT TO BE ARI CERTIFIED AND UL APPROVED.
* PROVIDE 7-DAY PROGRAMMABLE THERMOSTAT / HUMIDISTAT.
* COOLING CAPACITY AT 80/67 EAT AND 95° F OUTDOOR AIR TEMPERATURE.
* ESP DOES NOT INCLUDE FILTER SECTION AND ANY ACCESSORIES.
* PROVIDE FILTER RACK AND 2" THICK THROW-AWAY FILTER WITH UNIT.
* PROVIDE EACH UNIT WITH SAFETY DISCONNECT, AND LOW AMBIENT CONTROLS AND ANTI-SHORT CYCLE TIMER.
* PROVIDE BAROMETRIC RELIEF DAMPER.
* PROVIDE CONCRETE PAD UNDER UNIT.
* PROVIDE HOT GAS REHEAT.
* PROVIDE ECONOMIZER WITH 100% O.A. DAMPER.
* PROVIDE SMOKE DETECTORS WITH AUTOMATIC SHUT DOWN IN THE SUPPLY AND RETURN DUCTS OF UNIT PER 2018 IMC FOR RETURN DUCT AND NFPA 90A FOR SUPPLY DUCT. PROVIDED BY ELECTRICAL, INSTALLED BY MECHANICAL.
* ALTERNATE MANUFACTURERS: LENNOX, CARRIER.

EXHAUST FAN SCHEDULE

DESIGNATION EF	CFM	EXHAUST S.P.	RPM	WATTS	TYPE	CONTROL	VOLTAGE	SONES	WEIGHT	MANUF. AND MODEL NO.	NOTES
1	50	.25	836	12.0	CEILING	LIGHT SWITCH	120/1/60	<0.3	13.0	GREENHECK SP-A70	1,2
2	300	.35	967	105.0	INLINE	LIGHT SWITCH	120/1/60	1.5	38.0	GREENHECK CSP-A410	1,2
3	300	.35	967	105.0	INLINE	LIGHT SWITCH	120/1/60	1.5	38.0	GREENHECK CSP-A410	1,2
4	75	.25	884	14.0	CEILING	LIGHT SWITCH	120/1/60	0.4	13.0	GREENHECK SP-A90	1,2

ACCESSORIES AND FEATURES:

1. PROVIDE UNIT MOUNTED SOLID STATE SPEED CONTROL.
2. PROVIDE ROOF CAP WITH BIRD SCREEN.

NOTE:

* PROVIDE RADIATION DAMPER FOR ALL EXHAUST FANS LOCATED IN FIRE RATED CEILINGS. CONTRACTOR TO VERIFY CEILING TYPES PRIOR TO FAN PURCHASE.

ROOF RELIEF HOOD SCHEDULE

DESIGNATION		CFM THRU HOOD	FREE AIR VELOCITY	HOOD FREE AREA (FT)2	P.D. THRU HOOD (INCHES W.G.)	HOOD DIMENSIONS (DIA.)	HOOD CONSTRUCTION / FINISH	MANUFACTURER AND MODEL NO.
TYPE	NO.							
RH	1	600	732.0	0.82	0.066	29"	SPUN ALUMINUM	GREENHECK GRSR-12

ACCESSORIES AND FEATURES:

* PROVIDE ROOF CURB AND BIRD SCREEN.

ELECTRIC HEATER SCHEDULE

DESIGNATION EH	KW	CONTROL	AMPS	VOLTAGE	WEIGHT (LBS)	MANUF. AND MODEL NO.
1	3.3	UNIT MTD. TSTAT	11.9	208/1/60	25.0	MARKEL F2F5103N

ACCESSORIES AND FEATURES:

* UNITS MUST BE UL-LISTED.
* PROVIDE INTERNAL DISCONNECT SWITCH.
* PROVIDE MANUFACTURER'S HANGING KITS AND BRACKETS.
* ALTERNATE MANUFACTURERS: INDEECO

HVAC LEGEND & SYMBOLS

	A. RECTANGULAR DUCT: * MANUAL OPPOSED BLADE DAMPER WITH LOCKING QUADRANT. LEVER OPERATOR, OF STEEL CONSTRUCTION. * LOUVERS & DAMPERS MODEL CD-400; KRUEGER MODEL 0BD-DM TYPE 2 OPERATION FOR LESS THAN 10" WIDE. A. ROUND DUCT: * ROUND BLADE CONTROL DAMPER OF STEEL CONSTRUCTION WITH MANUAL OPERATOR * LOUVERS AND DAMPERS MODEL CD-600.
	RECTANGULAR ELBOW OF STEEL CONSTRUCTION.
	ROUND 90° ELBOW OF STEEL CONSTRUCTION.
45° L W	CONSTRUCT ALL BRANCH CONNECTIONS WITH 45° FITTING PER SMACNA STANDARDS. DIMENSION: L = 1/4W, 4"MINIMUM
D	SHEET METAL CONNECTORS, INC. HIGH EFFICIENCY TAKEOFF WITH DAMPER AND LOCKING QUADRANT FOR ROUND DUCT TAKEOFFS FROM RECTANGULAR DUCTWORK. INCREASE MAIN DUCT SIZE AT FITTING TO ACCOMMODATE ITS INSTALLATION IF REQUIRED TO MEET SMACNA AND MANUFACTURER'S INSTRUCTIONS. DIMENSION: D = RUNOUT DIAMETER (6"-14")
	DUCT RISER
	DUCT DROP
	ROUND DUCT RISER
	ROUND DUCT DROP
	FLEXIBLE DUCT EQUAL TO FLEXMASTER TYPE 3 INSULATED FLEXIBLE DUCT WITH ALUMINUM FOIL JACKET AND ALUMINUM FOIL FIBERGLASS POLYESTER LAMINATE LINER. INSTALL IN ACCORDANCE WITH SECTION III OF SMACNA'S "HVAC DUCT CONSTRUCTION STANDARDS, METAL AND FLEXIBLE".
	SUPPLY DUCT
	RETURN DUCT
	EXHAUST DUCT
XX-X	INDICATES NEW HVAC EQUIPMENT
	DUCT TRANSITION
	BALANCING DAMPER
U/C	INDICATES 3/4" UNDERCUT DOOR
T	WALL MOUNTED THERMOSTAT
T/H	WALL MOUNTED THERMOSTAT / HUMIDISTAT
	SUPPLY DIFFUSER W/ FLEX DUCT (CD)
	EGGCRATE CEILING RETURN (CR)
	CEILING EXHAUST GRILLE (CE)
	EXHAUST FAN (EF)
SWS - #	SIDEWALL SUPPLY GRILLE (SWS)
SWR - #	SIDEWALL RETURN GRILLE (SWR)

AIR DISTRIBUTION SCHEDULE

DESIGNATION	SERVICE	DESCRIPTION	MATERIAL	MANUF. AND MODEL NO.	ACCESSORIES AND FEATURES
CD-1	LOUVERED SUPPLY REGISTER	* FOUR-WAY DEFLECTION * FACE AIR VOLUME CONTROL	STEEL CONSTRUCTION WITH WHITE BAKED ENAMEL FINISH	PRICE SCD	* OPPOSED BLADE DAMPER * NC<35
CE-1	EGGCRATE EXHAUST GRILLE	* FREE PATTERN	ALUMINUM CONSTRUCTION WITH WHITE BAKED ENAMEL FINISH	PRICE 80	* NC<35
CR-1	EGGCRATE RETURN GRILLE	*FREE PATTERN	ALUMINUM CONSTRUCTION WITH WHITE BAKED ENAMEL FINISH	PRICE 80	*NC<35
DG-1	DOOR GRILLE	SIGHT PROOF LOUVERS	STEEL CONSTRUCTION WITH MILL FINISH	PRICE STG	*NC<35
SWR-1	SIDEWALL RETURN GRILLE	*45 DEGREE DEFLECTION *FIXED BARS	STEEL CONSTRUCTION WITH WHITE BAKED ENAMEL FINISH	PRICE 530	*NC<35
SWS-1	SPIRAL DUCT SUPPLY GRILLE	*DOUBLE DEFLECTION *FOR MOUNTING ON ROUND OR SPIRAL DUCT	ALUMINUM WITH PAINTABLE OR CLEAR FINISH	PRICE SDG	*NC<35
SWS-2	SIDEWALL SUPPLY GRILLE	*DOUBLE DEFLECTION	STEEL CONSTRUCTION WITH WHITE BAKED ENAMEL FINISH	PRICE 520	*3/4" BLADE SPACING *NC<35

* SEE ARCHITECTURAL REFLECTED CEILING PLANS FOR MOUNTING TYPE REQUIRED. PROVIDE PLASTER FRAME FOR ALL AIR DEVICES LOCATED IN GYP. BOARD CEILINGS.
* VERIFY THE MOUNTING OF ALL CEILING AIR DISTRIBUTION DEVICES COMPLY WITH BUILDING STRUCTURE PRIOR TO PURCHASE.

GENERAL NOTES:

1. CONTRACTOR SHALL FIELD VERIFY ALL DIMENSIONS BEFORE FABRICATION OF H.V.A.C. COMPONENTS OR PURCHASE OF EQUIPMENT.
2. CONTRACTOR SHALL COORDINATE ALL OTHER TRADES WITH THE INSTALLATION OF H.V.A.C. SYSTEM.
3. H.V.A.C. LEGEND MAY CONTAIN SYMBOLS AND ABBREVIATIONS NOT USED ON THIS SPECIFIC PROJECT, LEGEND SHALL BE USED FOR REFERENCE PURPOSES.
4. CONTRACTOR IS RESPONSIBLE FOR THE COMPLETE H.V.A.C. SYSTEM AS IT RELATES TO DRAWINGS AND SPECIFICATIONS.
5. CONTRACTOR IS REQUIRED TO REVIEW ARCHITECTURAL PLANS FOR RATED ASSEMBLIES. CONTRACTOR IS RESPONSIBLE FOR INSTALLATION OF FIRE AND/OR SMOKE DAMPERS IN ACCORDANCE WITH THE SPECIFICATIONS AND APPLICABLE BUILDING CODES.
6. "STANDARD" ACCESSORIES OR CONTROLS ON H.V.A.C. EQUIPMENT SHALL BE THOSE WHICH MANUFACTURER PROVIDES ON THE MAJORITY OF STOCK MERCHANDISE.
7. BRAND NAMES AND MODEL NUMBERS ARE PROVIDED TO ESTABLISH A LEVEL OF QUALITY AND PERFORMANCE. "EQUAL TO" ITEMS MAY BE SUBMITTED FOR CONSIDERATION BY THE ENGINEER AND OWNER.
8. THE DRAWINGS ARE GENERALLY DIAGRAMMATIC AND INDICATE THE APPROXIMATE ROUTING OF PIPING AND DUCTWORK. CONTRACTOR SHALL COORDINATE HIS WORK WITH OTHER TRADES. MINOR OFFSETS AND ADJUSTMENTS SHALL BE PROVIDED WHERE REQUIRED AT NO ADDITIONAL COST TO THE OWNER.
9. COORDINATE CEILING DIFFUSERS AND REGISTER LOCATIONS WITH ARCHITECT'S REFLECTED CEILING PLAN.

HVAC NOTES:

1. THE CONTRACTOR IS RESPONSIBLE TO PROVIDE SUPPLY, RETURN AND EXHAUST DUCT AS FOLLOWS: DUCTWORK TO BE DESIGNED, BRACED, AND SUPPORTED IN ACCORDANCE WITH SMACNA FOR LOW PRESSURE APPLICATIONS, SEAL CLASS C PER SMACNA HVAC DUCT CONSTRUCTION STANDARDS - METAL AND FLEXIBLE DUCTWORK (FLEXIBLE DUCT MAX. 5') SMACNA-06). SINGLE WIRE HANGERS SHALL NOT BE ALLOWED FOR FLEXIBLE DUCTWORK SUPPORT. FLEXIBLE DUCTWORK SHALL BE SUPPORTED IN A MANNER THAT PREVENTS CONSTRICTION OR DIPS. INSULATION SHALL BE AS NOTED BELOW.
2. ALL DUCT ELBOWS SHALL BE 1.5 R/D, UNLESS NOTED OTHERWISE.
3. MANUAL, OPPOSED BLADE DAMPERS SHALL BE PLACED IN EACH BRANCH OF SUPPLY DUCTWORK FOR FINAL BALANCING PURPOSES. BALANCING DEVICES SHALL BE IN ACCORDANCE WITH IMC (2018) 603.18.
4. CONTRACTOR SHALL FIELD VERIFY ALL DUCT ROUTING DIMENSIONS AND TERMINAL DEVICES TO AVOID INTERFERENCES. CONTRACTOR IS RESPONSIBLE FOR PROVIDING ALL SUPPORTS FOR PIPING AND DUCTWORK.
5. CONDENSATE DRAIN PIPING SHALL BE FULL SIZE PER EQUIPMENT CONNECTION WITH PVC ROUTED TO INDIRECT CONNECTION WITHOUT CREATING AN OBSTRUCTION. ALL SUPPORTS FOR THE CONDENSATE DRAIN PIPING IS BY THE MECHANICAL / HVAC CONTRACTOR.
6. THE MECHANICAL SYSTEMS SHALL HAVE TESTING AND BALANCING PERFORMED BY THE CONTRACTOR RESPONSIBLE FOR THE INSTALLATION OF THE SYSTEM(S). THE CONTRACTOR SHALL PREPARE AND SUBMIT A COMPLETE REPORT IDENTIFYING ALL MAJOR PIECES OF HVAC EQUIPMENT AND AIR DISTRIBUTION DEVICES WITH PERFORMANCES AND FINAL AIR BALANCE OF EACH. SUBMITTAL SHALL BE PRESENTED TO THE ENGINEER AND BUILDING OWNER OR TO THE OWNER'S REPRESENTATIVE FOR REVIEW AND APPROVAL. KITCHEN HOOD AND FANS TO BE BALANCED BY IT'S SUPPLIER. BOTH PROCEDURES ARE TO BE DONE AT THE SAME TIME AND TO BE COORDINATED TO ATTAIN DESIGN RESULTS.
7. PROVIDE MINIMUM 10 FEET SEPARATION BETWEEN OUTSIDE AIR INTAKES AND EXHAUST VENTS, PLUMBING VENTS, ETC.
8. PROVIDE SMOKE DETECTORS IN THE SUPPLY AND RETURN AIR DUCTS OF ALL UNITS 2000 CFM AND OVER. LOCATE DETECTOR IN THE RETURN AIR DUCT UPSTREAM OF FRESH AIR MIXING, AND UPSTREAM OF EXHAUST IN ACCORDANCE WITH IMC 606.1. LOCATE DETECTOR IN THE SUPPLY AIR DUCT DOWNSTREAM OF THE FILTERS AND UPSTREAM OF THE FIRST BRANCH CONNECTION IN ACCORDANCE WITH NFPA 90A. DETECTORS SHALL BE WIRED INTO BUILDING FIRE ALARM SYSTEM AND SHALL BE WIRED TO SHUT UNIT/S DOWN UPON DETECTION OF SMOKE.
9. THERMOSTATS SHALL BE MOUNTED 48" ABOVE FINISHED FLOOR UNLESS INDICATED OTHERWISE.
10. PROVIDE ACCESS DOOR (12"x12" MIN) AS REQUIRED FOR DAMPER AND CONTROL ACCESS IN WALLS AND CEILINGS.

DUCT SEALING:

1. PRESSURE SENSITIVE TAPE USED AS THE PRIMARY SEALANT IS CERTIFIED AND SHALL COMPLY WITH UL-181A OR UL-181B.
2. PROVIDE LONGITUDINAL SEAMS ON RIGID DUCT AND TRANSVERSE SEAMS ON ALL DUCTS.
3. PROVIDE MECHANICAL FASTENERS AND SEALANTS SHALL BE USED TO CONNECT DUCTS AND AIR DISTRIBUTION DEVICES.

INSULATION:

1. DUCTWORK SHALL BE INSULATED IN ACCORDANCE WITH THE FOLLOWING:
   A. NO INSULATION FOR EXHAUST OR EXPOSED SUPPLY AND RETURN DUCTWORK.
   B. ALL OTHER DUCTWORK PROVIDE 2" FIBERGLASS BLANKET TYPE INSULATION WITH FOIL VAPOR BARRIER COVER IN ACCORDANCE WITH SMACNA HVAC DUCT CONSTRUCTION STANDARDS.

PENETRATIONS:

1. SLEEVES SHALL BE INSTALLED WHERE DUCTS, LOUVERS, OR PIPING PENETRATE NON-RATED EXTERIOR WALLS, PARTITIONS, FLOORS, OR ROOF. PACK AROUND SLEEVES AND SEAL WEATHER TIGHT. INSTALL FLASHING AS REQUIRED. SLEEVES SHALL BE MINIMUM 16 GAUGE GALVANIZED STEEL AND SHALL BE FIRMLY SET IN BUILDING STRUCTURE.

SUBMITTALS AND ACCEPTANCE:

1. UNLESS OTHERWISE INSTRUCTED, THE CONTRACTOR SHALL SUBMIT ONE (1) DIGITIAL SET (PDF) OF HVAC SHOP DRAWINGS TO THE PROJECT MANAGER WHO SHALL THEN RELAY THEM TO THE DESIGN ENGINEER FOR REVIEW AND APPROVAL PRIOR TO THE PURCHASE OF EQUIPMENT.
2. OPERATION AND MAINTENANCE MANUALS FOR ALL MECHANICAL EQUIPMENT SHALL BE COMPILED INTO A THREE RING BINDER AND TURNED OVER TO BUILDING OWNER UPON PROJECT COMPLETION.

FACILITY SYSTEMS CONSULTANTS, LLC

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

STATE OF TENNESSEE
AGRICULTURE
COMMERCE
NO. 114592
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

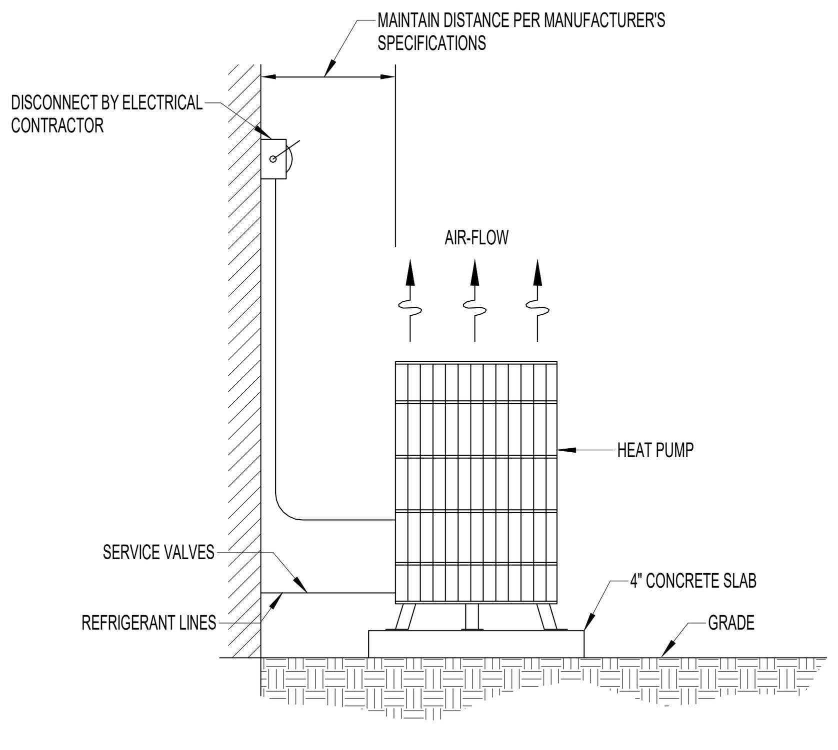
No.	Date

Drawing Title:
HVAC SCHEDULES, NOTES, LEGENDS

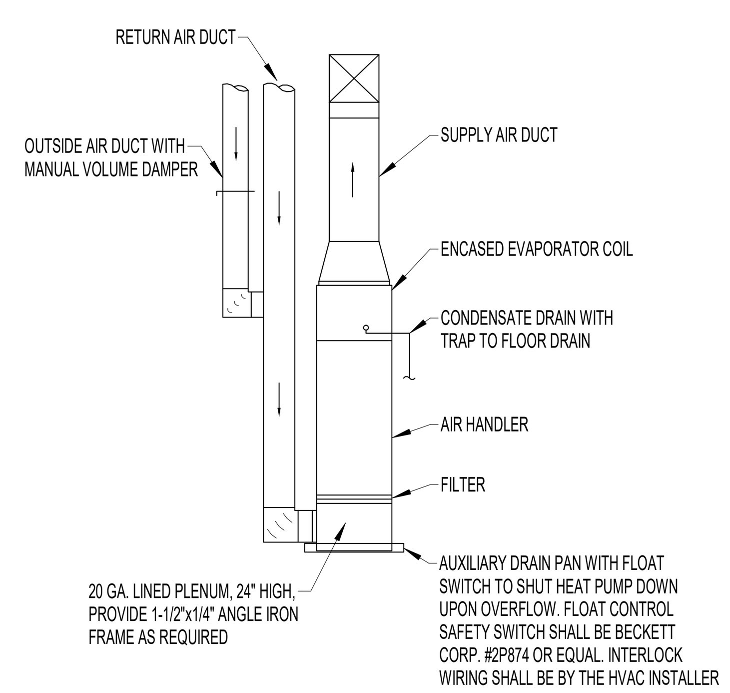
Date: 11.26.2025

Project No. 25026

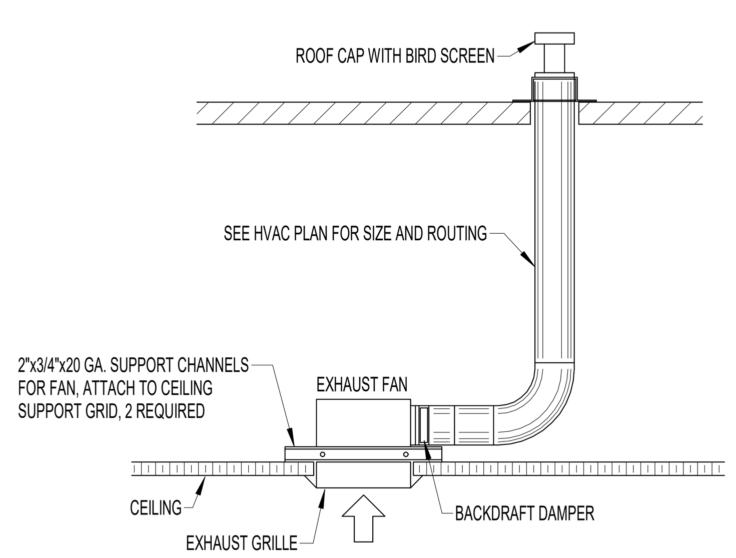
Sheet No. M0.1



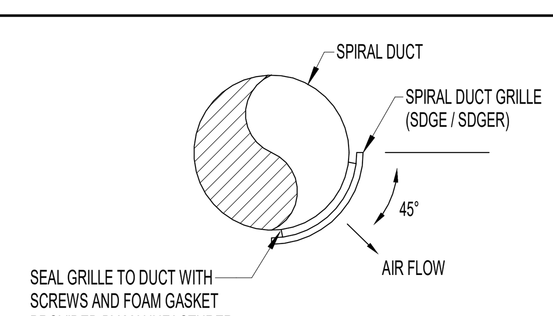
OUTDOOR AIR COOLED HEAT PUMP UNIT DETAIL
NO SCALE



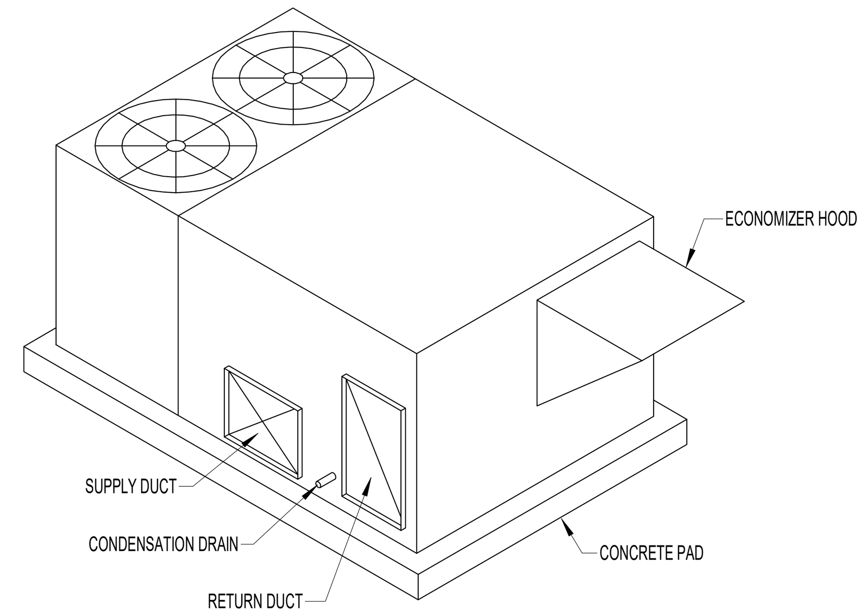
VERTICAL AIR HANDLER (W/ MANUAL DAMPER) INSTALLATION DETAIL
NO SCALE



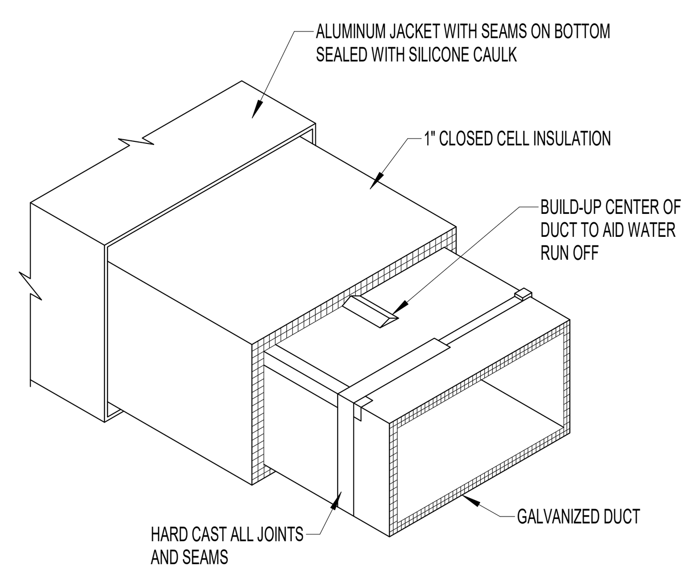
CEILING EXHAUST FAN DETAIL
NO SCALE



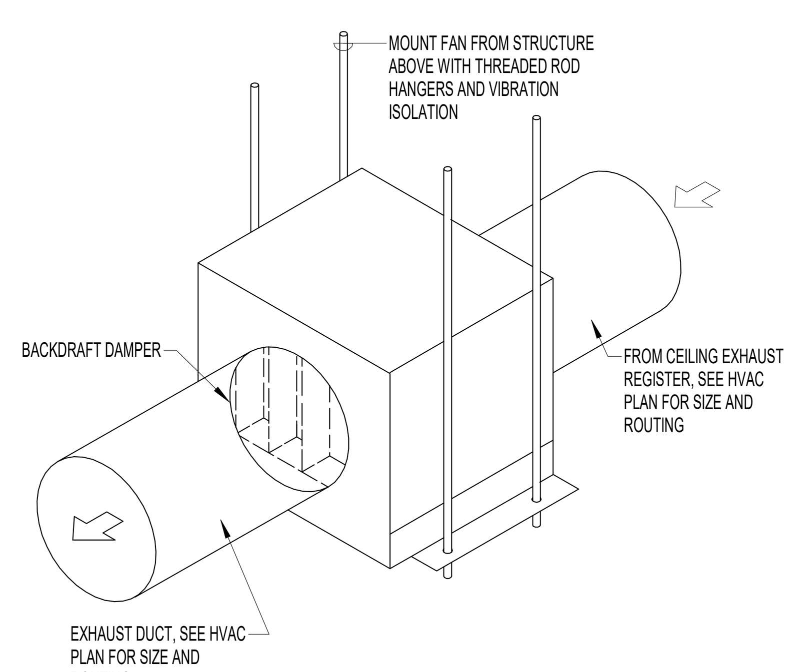
SPIRAL DUCT SUPPLY GRILLE DETAIL
NO SCALE



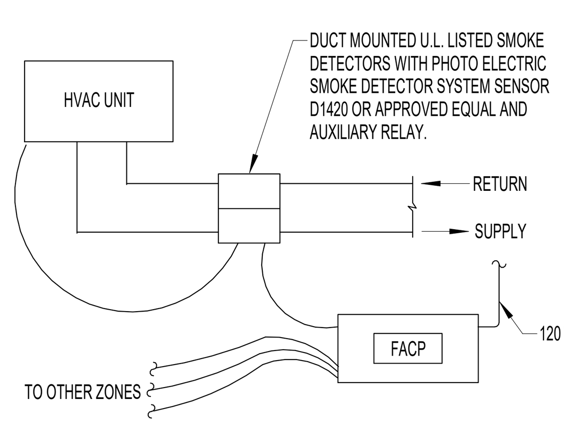
PACKAGED HVAC UNIT DETAIL
NO SCALE



DUCT EXPOSED TO WEATHER DETAIL
NO SCALE

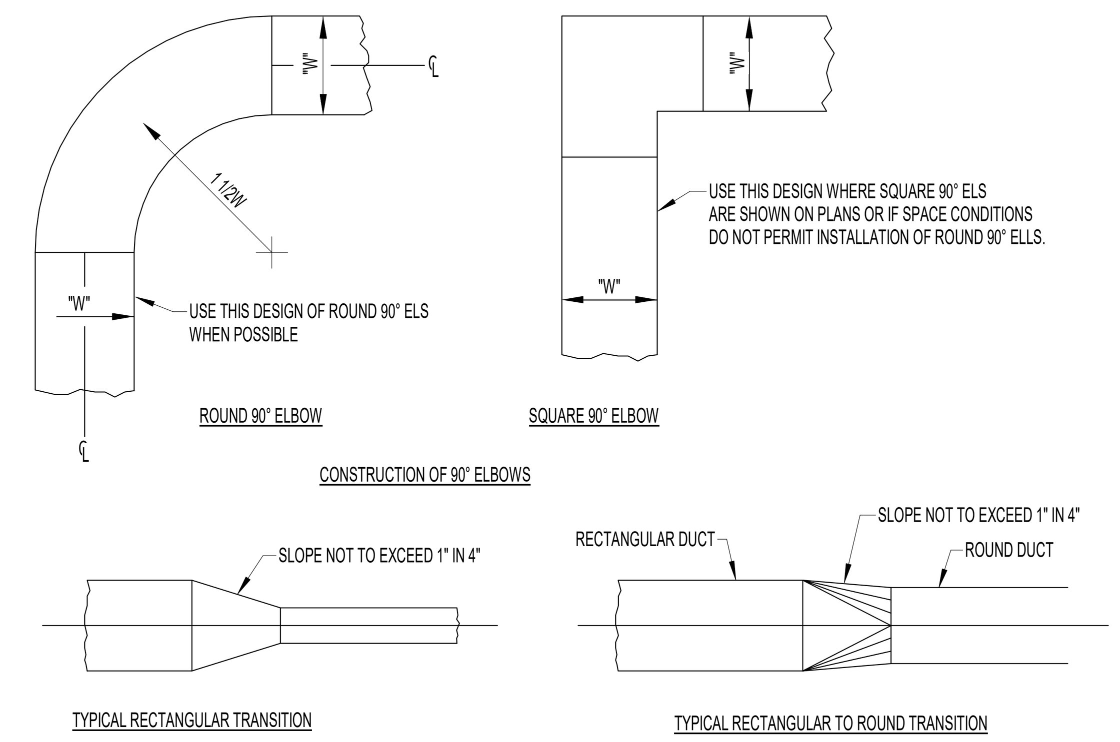


INLINE CEILING EXHAUST FAN DETAIL
NO SCALE

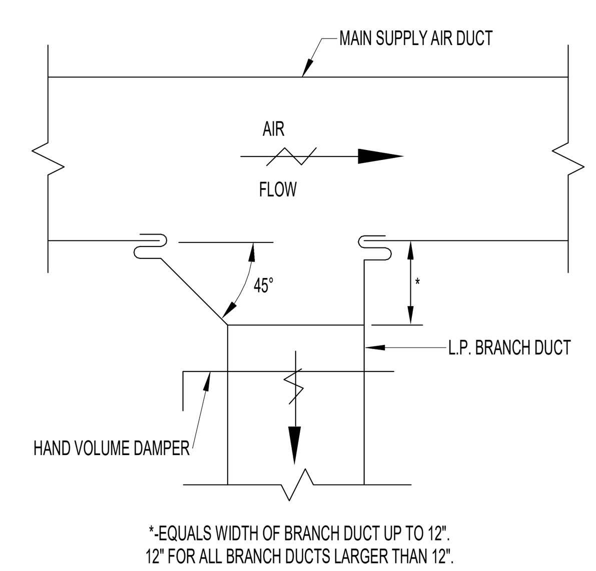


NOTES:
1. INSTALL DUCT SMOKE DETECTORS ON SUPPLY AND/OR RETURN ON UNITS AS INDICATED ON SCHEDULE. INSTALL PER APPLICABLE CODES.
2. INTERLOCK HVAC UNIT TO SHUT DOWN UPON AN ALARM CONDITION.
3. ALL INSTALLATION AND UNIT CONTROL WIRING TO EQUIPMENT SHALL BE BY MECHANICAL INSTALLER, DETECTORS AND FA WIRING BY ELECTRICAL.

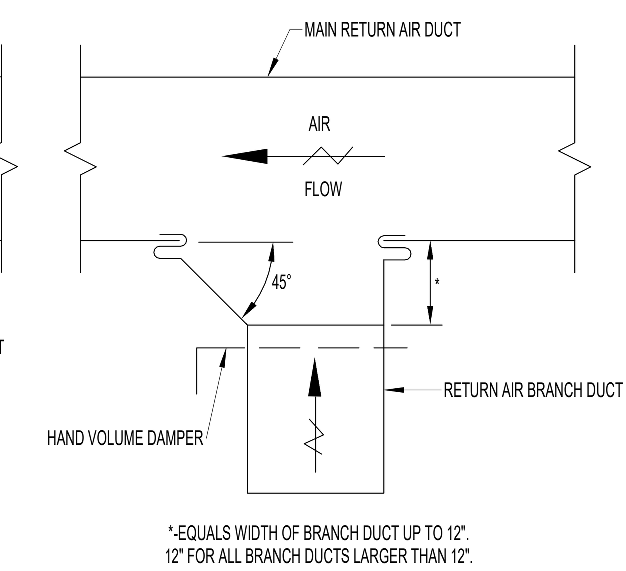
FIRE ALARM CONTROL PANEL DETAIL
NO SCALE



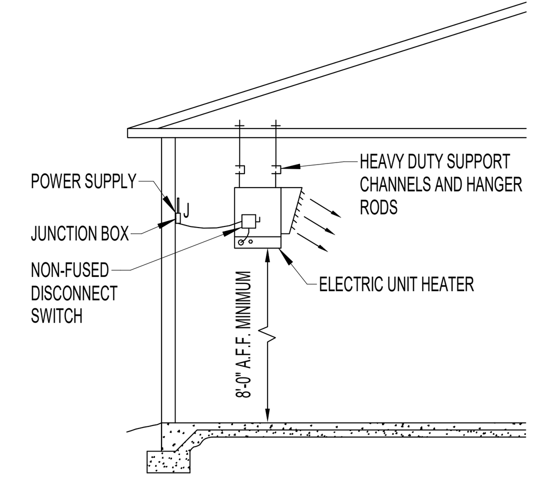
DETAILS OF THE LOW VELOCITY DUCT LAYOUT
NO SCALE



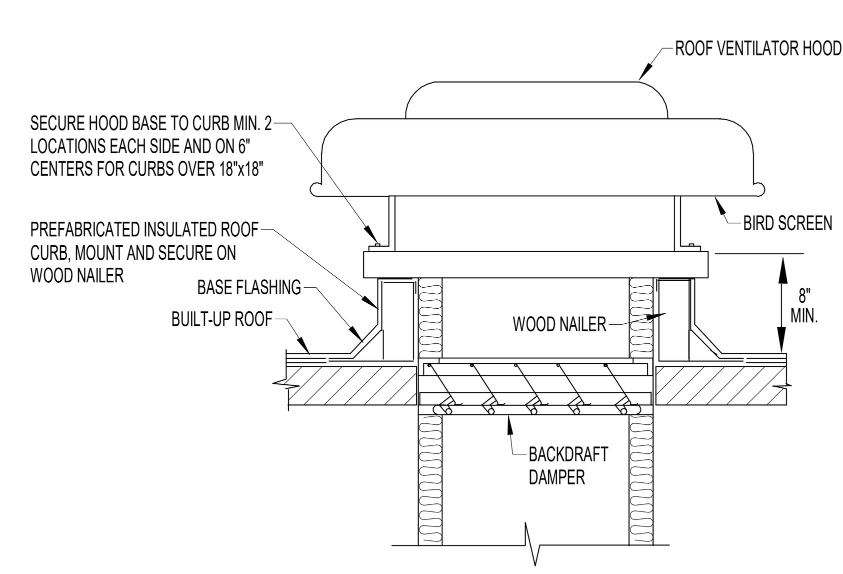
TYPICAL SUPPLY AIR BRANCH DUCT TAKE-OFF
NO SCALE



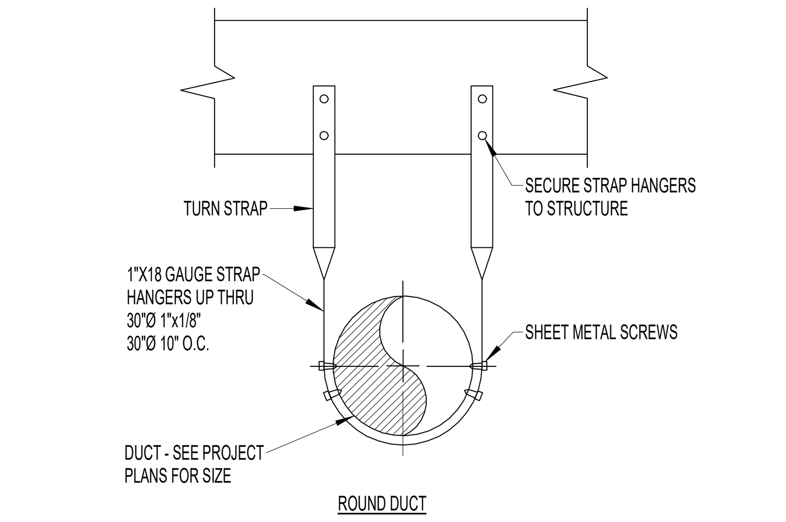
TYPICAL RETURN AIR BRANCH DUCT TAKE-OFF
NO SCALE



ELECTRIC UNIT HEATER MOUNTING DETAIL
NO SCALE

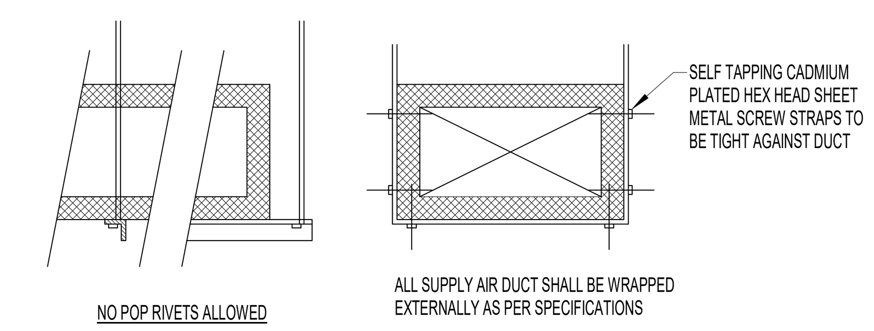


ROOF INTAKE / RELIEF HOOD MOUNTING DETAIL
NO SCALE

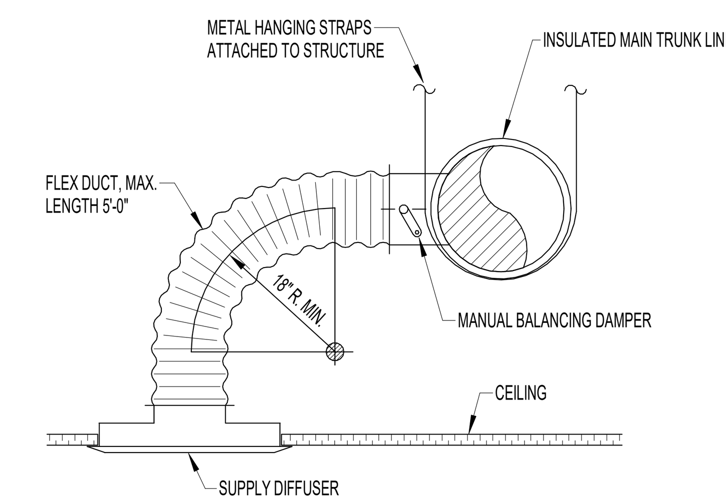


HANGER SIZES FOR RECTANGULAR DUCT			
MAX. SIDE	HANGER	HORIZONTAL SUPPORT ANGLE	MAXIMUM SPACING
30"	1"x18" GAGE STRAP	NONE REQUIRED	10'-0"
36"	1/4" ROUND ROD	1 1/2"x1 1/2"x1/8"	8'-0"
48"	1/4" ROUND ROD	2"x2"x1/8"	8'-0"
60"	5/16" ROUND ROD	2"x2"x1/8"	8'-0"
84"	3/8" ROUND ROD	2"x2"x1/8"	8'-0"

ALL METAL SUPPLY AIR DUCT, EXCEPT EXPOSED, SHALL BE WRAPPED EXTERNALLY AS PER SPECIFICATIONS

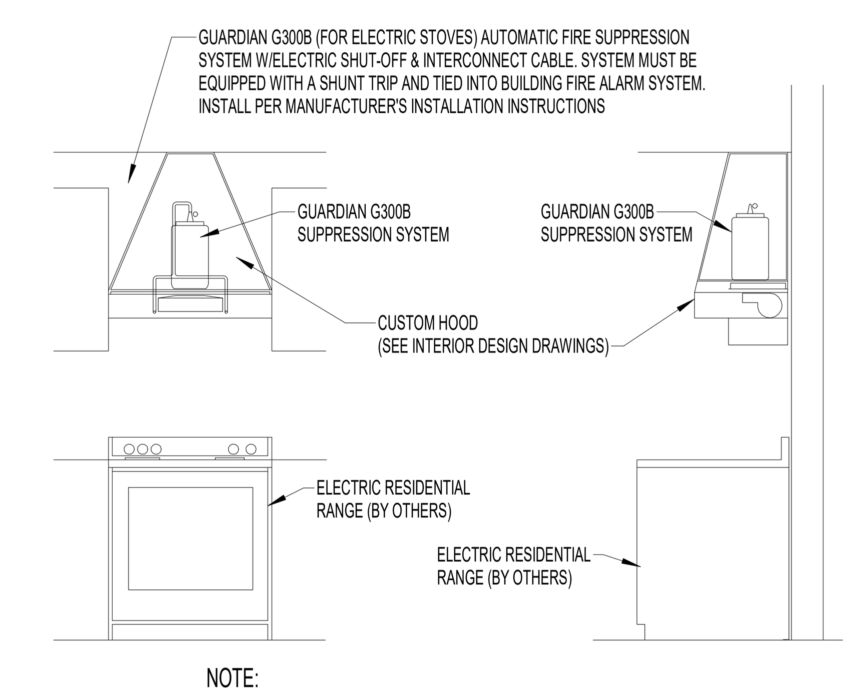


DUCT STRAP HANGER DETAIL
NO SCALE



NOTE:
THE MECHANICAL CONTRACTOR MUST PROVIDE ADAPTERS FOR DIFFUSERS WITH SQUARE AND RECTANGULAR NECKS AS REQUIRED.

SIDE DISCHARGE FLEX DUCT DETAIL
NO SCALE



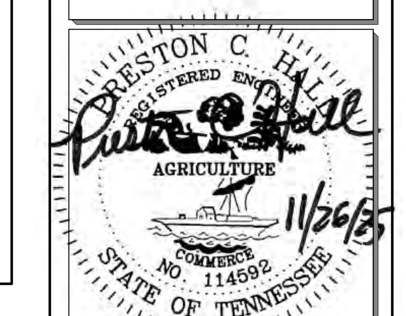
NOTE:
A CLASS K FIRE EXTINGUISHER SHALL BE LOCATED WITHIN THE KITCHEN.

SECTION / DETAIL AT RESIDENTIAL RANGE HOOD
NO SCALE

FACILITY SYSTEMS CONSULTANTS, LLC

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830



THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

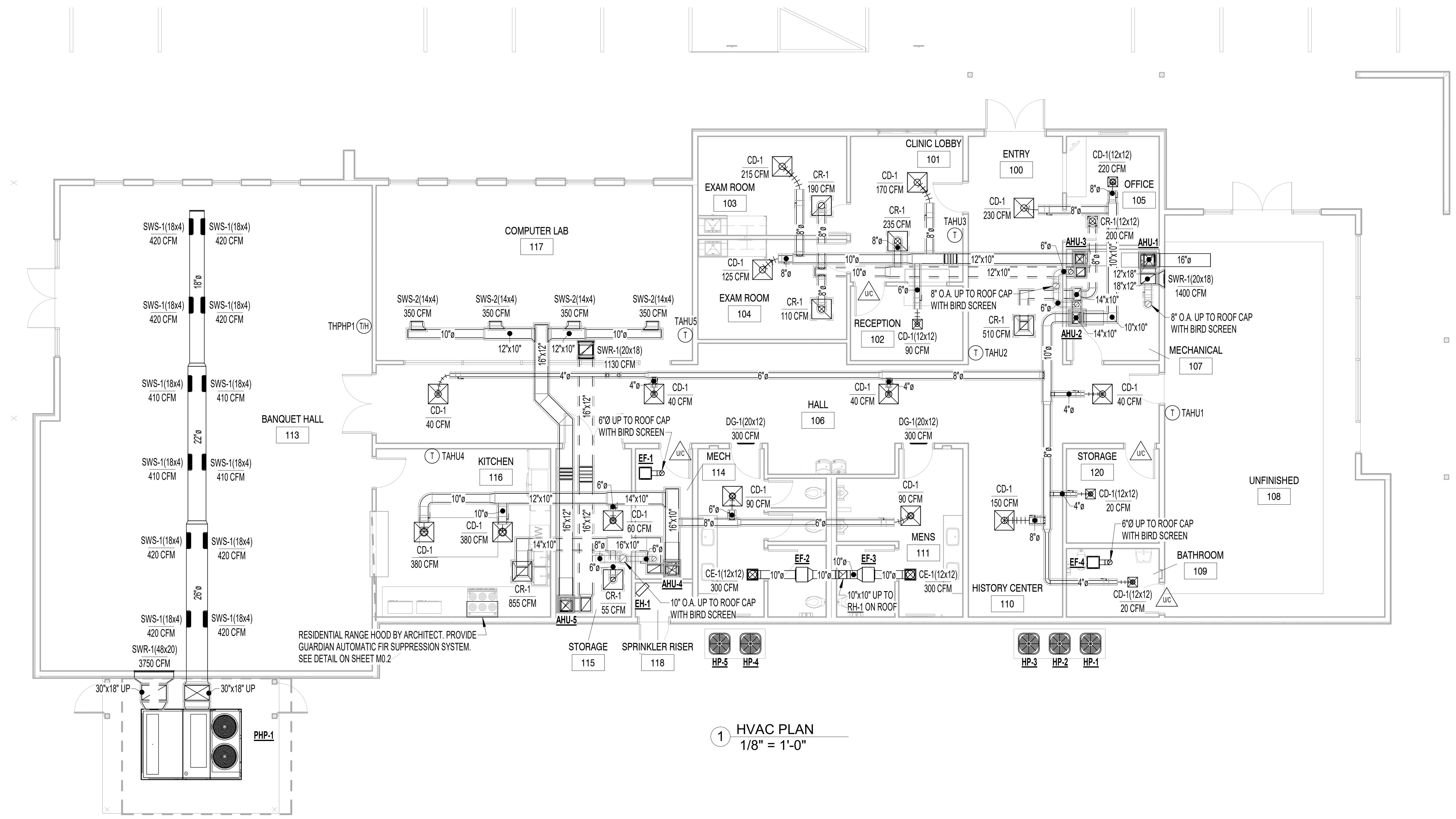
No.	Date

Drawing Title:
HVAC DETAILS

Date: 11.26.2025

Project No.
25026

Sheet No.
M0.2



1 HVAC PLAN
1/8" = 1'-0"

FACILITY SYSTEMS CONSULTANTS, LLC

TAC
The Architecture Collaborative

6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:

ELKS COMMUNITY ENRICHMENT CENTER

262 WILBERFORCE AVE
OAK RIDGE, TN 37830

PRESTON C. HALL
REGISTERED ENGINEER
AGRICULTURE
COMMERCE
NO. 114592
STATE OF TENNESSEE
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

Drawing Title:
HVAC PLAN

Date: 11.26.2025

Project No.
25026

Sheet No.
M1.1

PLUMBING FIXTURE SCHEDULE

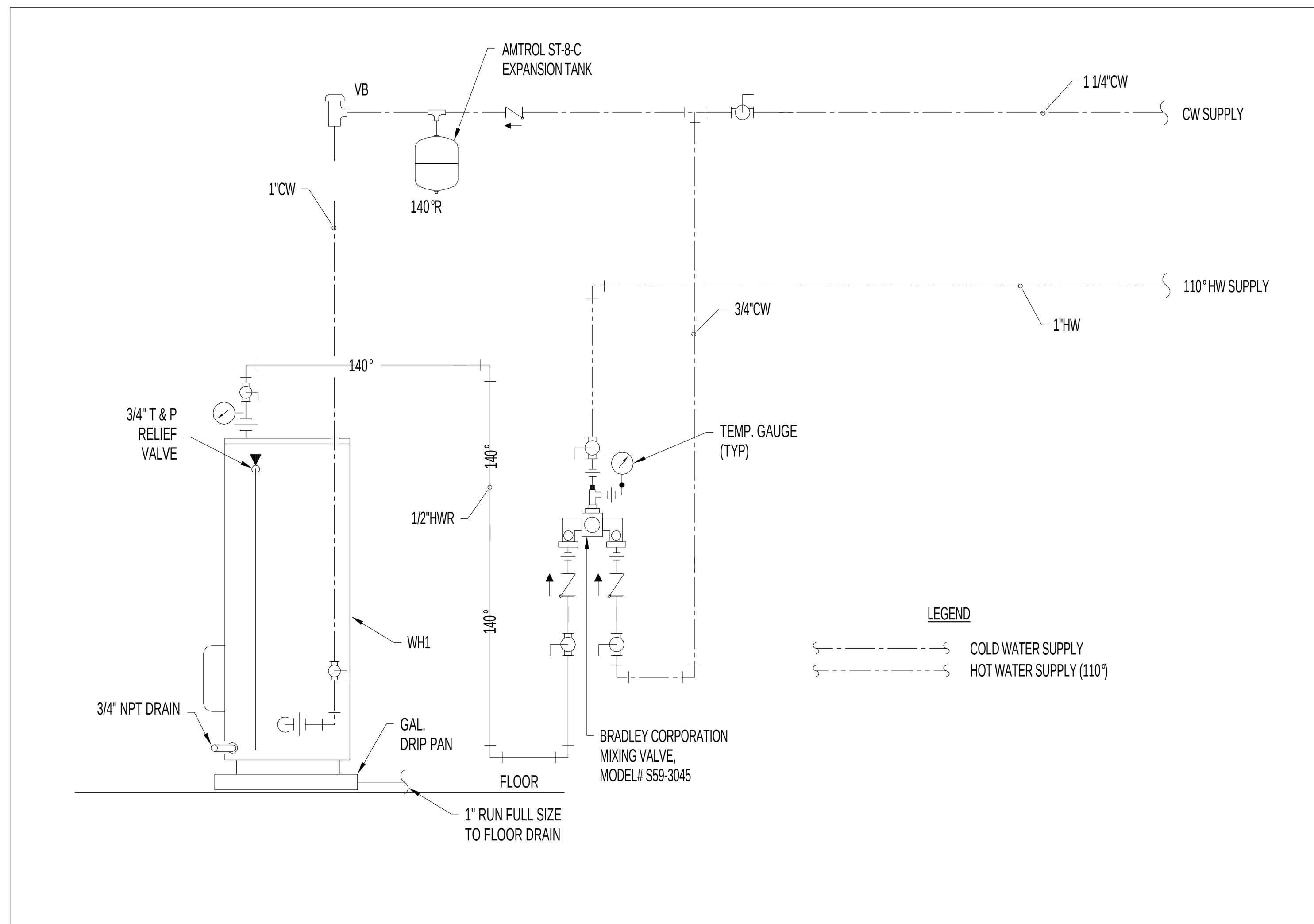
DESIGNATION	FIXTURE TYPE	MANUFACTURER	MODEL#	DESCRIPTION	NOTES
FCO	CLEANOUT	ZURN	ZN1400-BZ1	DURA-COAT CAST IRON ADJUSTABLE CLEANOUT WITH NICKEL BRONZE TOP	
GCO	CLEANOUT	ZURN	Z1400	DURA-COAT CAST IRON ADJUSTABLE CLEANOUT WITH HEAVY DUTY CAST IRON TOP	
WCO	CLEANOUT ACCESS COVER	ZURN ZURN	Z1445 Z1469	CLEANOUT TEE, DURA-COATED CAST IRON BODY W/ A GAS AND WATERTIGHT ABS TAPERED THREAD PLUG ROUND STAINLESS STEEL ACCESS OVER W/ SECURING SCREW	
FD1	FLOOR DRAIN	ZURN ZURN	ZN415B-3NH-5-R-Z1-BZ1 Z1072	DURA-COATED CAST IRON BODY FLOOR DRAIN WITH 5" POLISHED NICKEL BRONZE STRAINER BARRIER TRAP SEAL DEVICE	
WC1	WATER CLOSET VALVE SEAT CONNECTION SIZES	ZURN ZURN ZURN	Z5655-BWL1 Z6000AV-WS1 Z5955SS-EL-STS	1.6GPF SIPHON JET FLUSH ACTION FLOOR MOUNTED STANDARD HEIGHT WATER CLOSET WITH 2-1/8" FULLY GLAZED TRAPWAY AQUAVANTAGE MANUAL OPERATED FLUSH VALVE 1.6 GPF CLOG RESISTANT TRIPLE FILTERED BY-PASS, DUAL SEAL AND CHLORAMINE RESISTANT INTERNAL PARTS. ELONGATED WHITE OPEN FRONT TOILET SEAT LESS COVER WITH SELF SUSTAINING STAINLESS STEEL CHECK HINGE CW 1",WASTE 4",VENT 2"	
WC2 ADA	WATER CLOSET VALVE SEAT CONNECTION SIZES	ZURN ZURN ZURN	Z5665-BWL1 Z6000AV-WS1 Z5955SS-EL-STS	1.6GPF ADA SIPHON JET FLUSH ACTION FLOOR MOUNTED ADA HEIGHT WATER CLOSET WITH 2-1/8" FULLY GLAZED TRAPWAY AQUAVANTAGE MANUAL OPERATED FLUSH VALVE 1.6 GPF CLOG RESISTANT TRIPLE FILTERED BY-PASS, DUAL SEAL AND CHLORAMINE RESISTANT INTERNAL PARTS. ELONGATED WHITE OPEN FRONT TOILET SEAT LESS COVER WITH SELF SUSTAINING STAINLESS STEEL CHECK HINGE CW 1",WASTE 4",VENT 2"	
U1	URINAL VALVE CONNECTION SIZES	ZURN ZURN	Z5755-U Z6003AV-ULF	OMNI-FLOW .125 TO 1GPF WALL MOUNTED TOP SPUD ASYMMETRIC BACK WALL URINAL WITH INTEGRAL P-TRAP AND VANDAL RESISTANT OUTLET STRAINER AQUAVANTAGE MANUAL OPERATED FLUSH VALVE .125GPF CLOG RESISTANT TRIPLE FILTERED BY-PASS, DUAL SEAL AND CHLORAMINE RESISTANT INTERNAL PARTS CW 3/4",WASTE 2",VENT 1-1/2"	
L1	SINK FAUCET DRAIN TRAP SUPPLIES INSTALLATION KIT CONNECTION SIZES	SLOAN T&S BRASS TRUEBRO	SS-3021 B-2701-VF05 102	21 5/8"x 15 3/4"x 8 1/4" VITREOUS CHINA, RECTANGLE UNDERMOUNT COUNTERTOP LAVATORY, FRONT OVERFLOW - SINGLE HOLE SINGLE LEVER CHROME PLATED FAUCET - STANDARD, .5 GPM FLOW GRID STRAINER W/ 1-1/2" TAILPIECE 1-1/2" CAST BRASS P-TRAP W/C.O. PLUG ANGLE SUPPLIES W/LOOSE KEY STOPS HANDI-GUARD, MOLDED CLOSED CELL VINYL INSULATION WASTE 1-1/2" VENT 1-1/2" H & CW 1/2"	
L2	SINK FAUCET DRAIN TRAP SUPPLIES CARRIER INSTALLATION KIT CONNECTION SIZES	SLOAN T&S BRASS TRUEBRO	SS-3003 B-2701-VF05 102	20"x18" VITREOUS CHINA WALL HUNG W/ BACKSPLASH - 4" CENTERSET SINGLE LEVER CHROME PLATED FAUCET - STANDARD, .5 GPM FLOW GRID STRAINER W/ 1-1/2" TAILPIECE 1-1/2" CAST BRASS P-TRAP W/ C.O. PLUG ANGLE SUPPLIES W/ LOOSE KEYS CONCEALED ARM SUPPORT HANDI-GUARD, MOLDED CLOSED CELL VINYL INSULATION WASTE 1-1/2", VENT 1 1/2", H&CW 1/2"	
KS1	SINK FAUCET DRAIN P-TRAP CONTINUOUS WASTE SUPPLIES CONNECTION SIZES	ELKAY SYMMONS	LRAD332255 S-23-2-0.5	LUSTERTONE 2 BOWL 18GA STAINLESS STEEL 33" X 22" X 5-1/2" DROP IN SINK ORIGINS 8"CC FAUCET WITH INTEGRAL 8-3/4" CAST SWING SPOUT, CERAMIC DISK CARTRIDGE, METAL SINGLE LEVER HANDLE AND MATCHING SIDE SPRAY HEAVY DUTY BASKET STRAINER WITH CAST BRASS LOCK AND COUPLING NUT 1-1/2" CAST BRASS 17 GAUGE P-TRAP WITH CLEANOUT 1-1/2" 20 GAUGE CONTINUOUS WASTE END OUTLET WITH CAST BRASS TEE 1/2" X 3/8" COMP X COMP LAVATORY SUPPLY KIT WITH ESCUTCHEONS, 1/4 TURN CHROME PLATED STOPS AND CHROME PLATED COPPER TUBE SUPPLY LINES CW 1/2",HW 1/2",WASTE 1-1/2",VENT 1-1/2"	
S1	SINK FAUCET CONNECTION SIZES	ELKAY T&S BRASS	DLR191910 B-2866-05-LF15	LUSTERTONE 18GA STAINLESS STEEL 19-1/2" X 19" X 10" DROP IN SINK MEDICAL FAUCET, 8" O.C., 4" WRIST HANDLES, SWIVEL GOOSENECK W/ LAMINAR FLOW CONTROL INCLUDE 2" CAST BRASS P-TRAP W/ C.O. PLUG & ANGLE SUPPLIES W/ LOOSE KEY STOPS CW 1/2",HW 1/2",WASTE 1-1/2",VENT 1-1/2"	
MS1	MOP SINK FAUCET CONNECTION SIZES	STERN WILLIAMS T&S BRASS	MTB2424 B-0665-BSTR	24" X 24" TERRAZZO SINK FAUCET W/VACUUM BREAKER AND PALE HOOK PROVIDE W/ MOP HANGER & HOSE BRACKET WASTE 3", VENT 1 1/2", H & CW 3/4"	
HB1	HOSE BIBB	WOODFORD	MODEL B65	WALL HYDRANT, BOX ENCLOSED, NON-FREEZE, ANTI-SIPHON, AUTOMATIC DRAINING, BRONZE CASING, ALL BRONZE INTERIOR PARTS - 3/4" FEMALE FITTING	
EWC1	WATER COOLER P-TRAP SUPPLIES CONNECTION SIZES	ELKAY	EZSTLDDLC	BI-LEVEL ADA WALL MOUNTED STAINLESS STEEL NON-PRESSURIZED WATER COOLER NON-FILTERED, NON-REFRIGERATED, LIGHT GREY GRANITE WITH FLEX GUARD BUBBLER AND 115V/60HZ 1-1/4" CAST BRASS 17GA P-TRAP WITH CLEANOUT 1/2" X 3/8" COMP X COMP LAVATORY SUPPLY KIT WITH ESCUTCHEON, 1/4 TURN CHROME PLATED STOP AND CHROME PLATED COPPER TUBE SUPPLY LINE CW 1/2",WASTE 1-1/4",VENT 1-1/4"	
IB1	ICE MAKER BOX CONNECTION SIZE	OATEY	39156	6"x6" ICE MAKER BOX, HIGH IMPACT POLYSTYRENE CW 1/2"	
HA1	HAMMER ARRESTOR	ZURN	Z-1700	STAINLESS STEEL BELLOW TYPE PLUMBING DRAINAGE INSTITUTE RATINGS: "100" (1-11 FU), "200" (12-32 FU), "300"(33-60 FU), "400"(61-113)	

ELECTRIC WATER HEATER SCHEDULE

MARK	WH1
DESIGN MANUFACTURER	BRADFORD WHITE
MODEL NUMBER	LE340S3-3
GALLON CAPACITY	40 GAL.
RECOVERY @ 100 DEG. F. RISE	18 GPH
HEATNG. CAP. (KW) - NON-SIMULTANEOUS	4.5 KW
ELECTRICAL (VOLTS/HZ/PH)	208/60/1

NOTES:
1. ALTERNATIVE MANUFACTURERS: LOCHINVAR, A.O. SMITH, STATE IND.
2. PROVIDE RECIRULATING PUMP SEE SCHEDULE.
3. RECIRCULATING PUMP SHALL BE EQUIPPED W/ AUTOMATIC TIMER
4. UNITS EXCEEDING 120 GAL. OR 58.6 kW SHALL BE ASME LISTED.

RECIRCULATING PUMP		
	MARK	P-1
	DESIGN MANUFACTURER	BELL AND GOSSET
	MODEL NUMBER	PL-36B
	HORSE POWER	1/6
	RPM	3300
ELECTRICAL		
	VOLTS/HZ/PH	115/60/1
	F.L.A.	2.1



WH1 - ELECTRIC WATER HEATER DETAIL
NOT TO SCALE

PLUMBING NOTES:

1. SANITARY WASTE AND VENT PIPING BOTH ABOVE AND BELOW GRADE SHALL BE SCHEDULE 40 PVC-DWV PLASTIC PIPE AND FITTINGS WITH SOLVENT WELD JOINTS. PLASTIC PIPING AND PIPING COMPONENTS SHALL BE LISTED AS CONFORMING WITH ANSI/NSF STD. 14 AND ASTM D-2665.
2. UNLESS INDICATED OTHERWISE ON DRAWINGS, INTERNAL WATER PIPING IS TO BE ROUTED IN CEILING SPACES, ATTICS, CRAWL SPACES AND IN AND BETWEEN WALL STUDS, ETC. (AS AND WHERE APPLICABLE) AND ON INSIDE OF INSULATED BUILDING ENVELOPE. THIS PIPING SHALL BE TYPE "L" COPPER AND INSTALLED IN ACCORDANCE WITH 2024 INTERNATIONAL PLUMBING CODE. TYPE "A" AND/OR "B" PEX IS PERMISSIBLE UPON OWNERS APPROVAL ONLY ON WATER LINES 2" AND SMALLER. WHERE PEX IS USED COLD AND HOT WATER MAINS SHALL BE TYPE L COPPER WITH BRANCH PIPING BEING PEX. WHERE PEX IS USED AS MAINS IN LIEU OF COPPER, PIPING SHALL BE A PIPE SIZE LARGER THAN WHAT IS SHOWN ON PLANS.
3. ALL VENT PIPING TO PENETRATE ROOF A MINIMUM OF 12" ABOVE ROOF. FLASH AND SEAL TO ROOF WEATHERTIGHT. PAINT VENT PIPING ABOVE ROOF AND WITH 2 COATS EPOXY BASED PAINT. COLOR TO MATCH ROOF.
4. CONTRACTOR SHALL INSPECT SITE AND BECOME FAMILIAR WITH EXISTING CONDITIONS WHICH MAY AFFECT WORK, INCLUDING VERIFICATION OF LOCATIONS AND RELATIONSHIP BETWEEN FIXTURES AND CONNECTIONS.
5. PIPING PLACED IN TRENCHES SHALL BE EMBEDDED IN 6" OF LOOSE AGGREGATE FILL, TAMP FILL MATERIAL ON EACH SIDE IN 6" LAYERS. ALL PIPING UNDER SLAB SHALL HAVE A MINIMUM 1" COVER FROM BOTTOM OF SLAB TO TOP OF PIPE AT HIGH POINT. PROTECT PIPING FROM BEING CRUSHED OR OTHERWISE CONSTRICTED.
6. EACH SINK, WATER CLOSET, ETC. SHALL HAVE SHUT-OFF VALVES LOCATED AT THE FIXTURE.
7. THE PLUMBING SYSTEM IN ITS ENTIRETY SHALL NOT BE COVERED UNTIL IT HAS BEEN INSPECTED, TESTED, AND APPROVED BY THE OWNER.
8. PRIOR TO COVERING THE WATER SUPPLY SYSTEM, IT SHALL BE PRESSURE TESTED AND PROVED TIGHT UNDER A WATER PRESSURE NOT LESS THAN 25 P.S.I. ABOVE THE WORKING PRESSURE UNDER WHICH IT IS TO BE OPERATED. THIS TEST SHALL BE COMPLETED AND APPROVED IN THE PRESENCE OF THE OWNER.
9. ALL SOLDERED JOINTS SHALL BE CLEANED BRIGHT AND ALL BURRS SHALL BE REMOVED AND THE SHALL BE RETURNED TO FULL BORE.
10. ALL SOLDER AND FLUX USED IN THE INSTALLATION OR REPAIR OF THE WATER SUPPLY OR DISTRIBUTION SYSTEM SHALL BE LEAD FREE.
11. ALL SOLDERED JOINT MATERIAL SUCH AS FITTINGS, SOLDER, TUBING SHALL BE APPROVED BY THE OWNER PRIOR TO INSTALLATION.
12. ALL MATERIALS, METHODS, AND PRACTICES SHALL BE IN ACCORDANCE WITH THE 2024 INTERNATIONAL PLUMBING CODE.
13. CONTRACTOR IS RESPONSIBLE FOR ALL REQUIRED FITTINGS TO CREATE A COMPLETE AND FUNCTIONAL PLUMBING SYSTEM. CONTRACTOR SHALL DETERMINE ANY FITTINGS REQUIRED FOR CONNECTION TO FIXTURES SPECIFIED.
14. PROVIDE REMOVABLE PVC COVERS ON ALL EXPOSED SUPPLY AND WASTE FITTINGS TO COMPLY WITH ANSI STD. A117.1 REQUIREMENTS.
15. CLEANOUTS:
    A. INTERIOR FINISHED FLOOR AREAS (FCO) - LACQUERED CAST IRON BODY WITH ANCHOR FLANGE, REVERSIBLE CLAMPING COLLAR, THREADED TOP ASSEMBLY AND ROUND GASKETED DEPRESSED COVER TO ACCEPT FLOOR FINISH.
    B. INTERIOR FINISHED WALL AREAS (WCO) - LINE TYPE WITH LACQUERED CAST IRON BODY AND ROUND EPOXY COATED GASKET COVER, AND ROUND STAINLESS STEEL ACCESS COVER SECURED WITH MACHINE SCREW.
    C. EXTERIOR SURFACED AREAS - ROUND CAST NICKEL BRONZE ACCESS FRAME AND NON-SKID COVER.
    D. EXTERIOR UN-SURFACED AREAS - LINE TYPE WITH LACQUERED CAST IRON BODY AND ROUND EPOXY COATED GASKET COVER.
16. ALL HOT WATER PIPE ABOVE GRADE SHALL BE INSULATED WITH 1 ½" FIBERGLASS, LOW PRESSURE INSULATION WITH WHITE UNIVERSAL JACKET. ALL COLD WATER PIPE ABOVE GRADE SHALL BE INSULATED WITH 1/2" FIBERGLASS, LOW PRESSURE INSULATION WITH WHITE UNIVERSAL JACKET. ALL INSULATION SHALL BE INSTALLED IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS.
17. ALL BALL/CONTROL/BALANCING VALVES WHICH ARE NOT READILY ACCESSIBLE VIA LAY-IN CEILING OR OPEN TO SPACE SHALL BE PROVIDED WITH AN ACCESSIBLE LOCKING PANEL EQUAL TO MIFAB TYPE CAD-FL - ACCESS PANEL SHALL BE PAINTED TO MATCH CEILING OR WALL FINISH.
18. ALL CONDENSATE PIPING SHALL BE INSULATED WITH 1" THICK ARMAFLEX INSULATION WITH GLUED JOINTS, OR 1 ½" THICK FIBERGLASS INSULATION WITH VAPOR BARRIER MASTIC WRAP.
19. PROVIDE VACUUM BREAKERS WHERE ANY THREADED CONNECTIONS ARE PRESENT ON WATER SUPPLY LINE.
20. WATER HAMMER ARRESTORS TO BE INSTALLED ON EQUIPMENT PER MANUFACTURER RECOMMENDATIONS.

PLUMBING LEGEND

COLD WATER LINE
HOT WATER SUPPLY
HOT WATER RETURN
140° — HOT WATER - 140°F
140°R — HOT WATER RETURN - 140°F
SANITARY SEWER LINE
GW — GREASE WASTE
G — NATURAL GAS
LP — LIQUID PROPANE
VENT LINE
C — CONDENSATE
RW — RAINWATER PIPING
N — NITROGEN
NO — NITROUS OXIDE
O — OXYGEN
V — VACUUM
A — AIR
FW — FILTERED WATER
POINT OF CONNECTION TO EXISTING PLUMBING
FLOOR DRAIN
FLOOR SINK
ROOF DRAIN
WATER CLOSET CONNECTION
FLOOR/GRADE CLEAN-OUT
WALL CLEAN-OUT
FIXTURE CONNECTION
HOSE BIBB
A — WATER HAMMER ARRESTOR

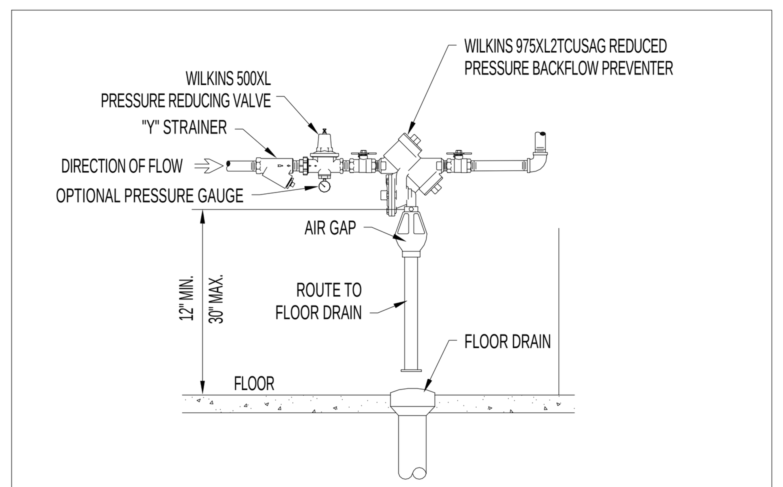
NOTE:
1. FOR CONNECTION SIZES AT FIXTURES, SEE PLUMBING FIXTURE SCHEDULE.

PIPING SYMBOLS

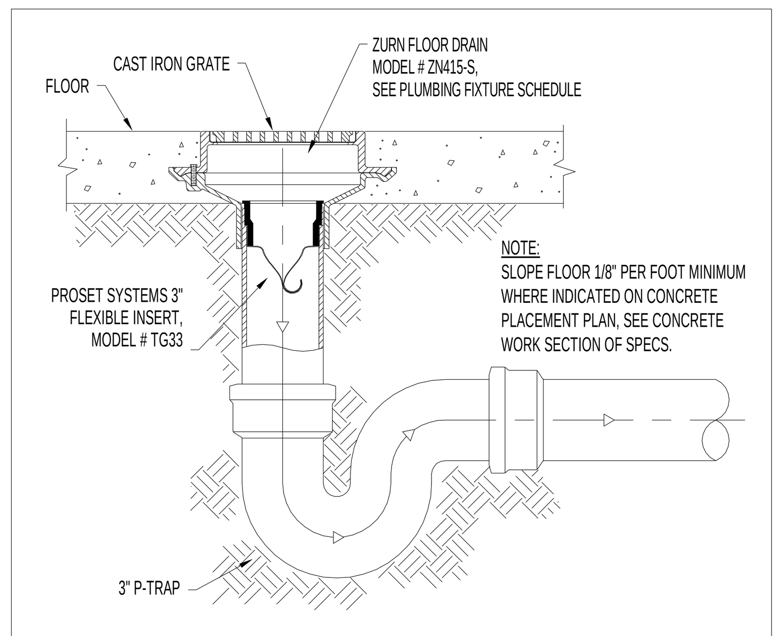
THREE WAY VALVE
BALL VALVE
BUTTERFLY VALVE
GATE VALVE
CHECK VALVE
GAS COCK / PLUG VALVE
GLOBE VALVE
UNION
CIRCUIT SETTER
PRESSURE REGULATING VALVE
PRESSURE RELIEF VALVE
PIPE TURN DOWN
PIPE TURN UP
PIPE TEE DOWN
PIPE TEE UP
PIPE TRANSITION
STRAINER
CLEAN OUT

ABBREVIATIONS

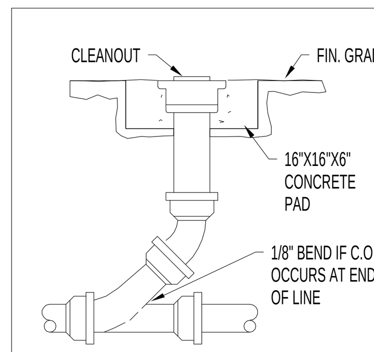
BV BALL VALVE
CI CAST IRON
CO CLEAN OUT
CON CONDENSATE
CW COLD WATER
CHK. V CHECK VALVE
EX EXISTING
FD FLOOR DRAIN
FS FLOOR SINK
GW GREASY WASTE
HB HOSE BIB/WALL HYDRANT
HW HOT WATER
HWS HOT WATER SUPPLY
HWR HOT WATER RETURN
P1 FIXTURE NUMBER (SEE SCHEDULE)
SS SANITARY SEWER
VS VENT STACK
VT VENT LINE
VTR VENT THRU ROOF
VB VACUUM BREAKER
WH WATER HEATER
WS WASTE STACK



TYPICAL BACKFLOW PREVENTER DETAIL
NOT TO SCALE

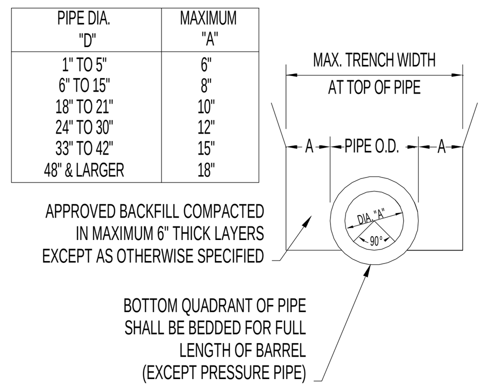


FLOOR DRAIN DETAIL
NOT TO SCALE

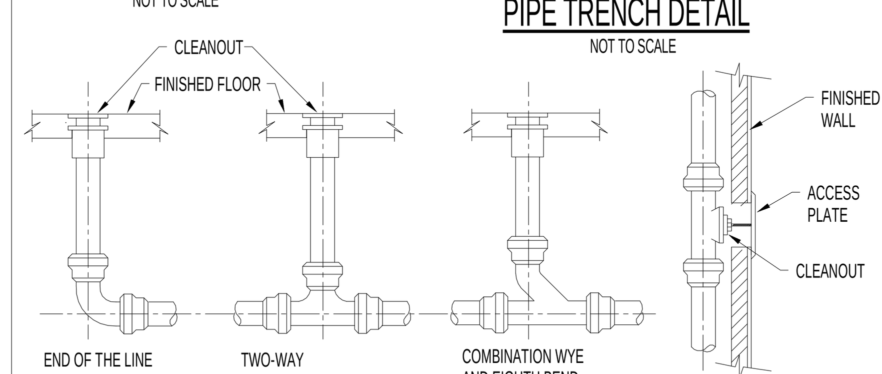


CLEANOUT TO GRADE
NOT TO SCALE

PIPE DIA. "D"	MAXIMUM "A"
1" TO 5"	6"
6" TO 15"	8"
18" TO 21"	10"
24" TO 30"	12"
33" TO 42"	15"
48" & LARGER	18"



PIPE TRENCH DETAIL
NOT TO SCALE



FLOOR & WALL CLEANOUTS
NO TO SCALE

FACILITY SYSTEMS CONSULTANTS, LLC
713 South Central Street, Suite 101
Knoxville, Tennessee 37902
Phone: (865) 246-0164
Fax: (865) 246-1084

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

AGRICULTURE
COMMERCE
NO. 114592
STATE OF TENNESSEE
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

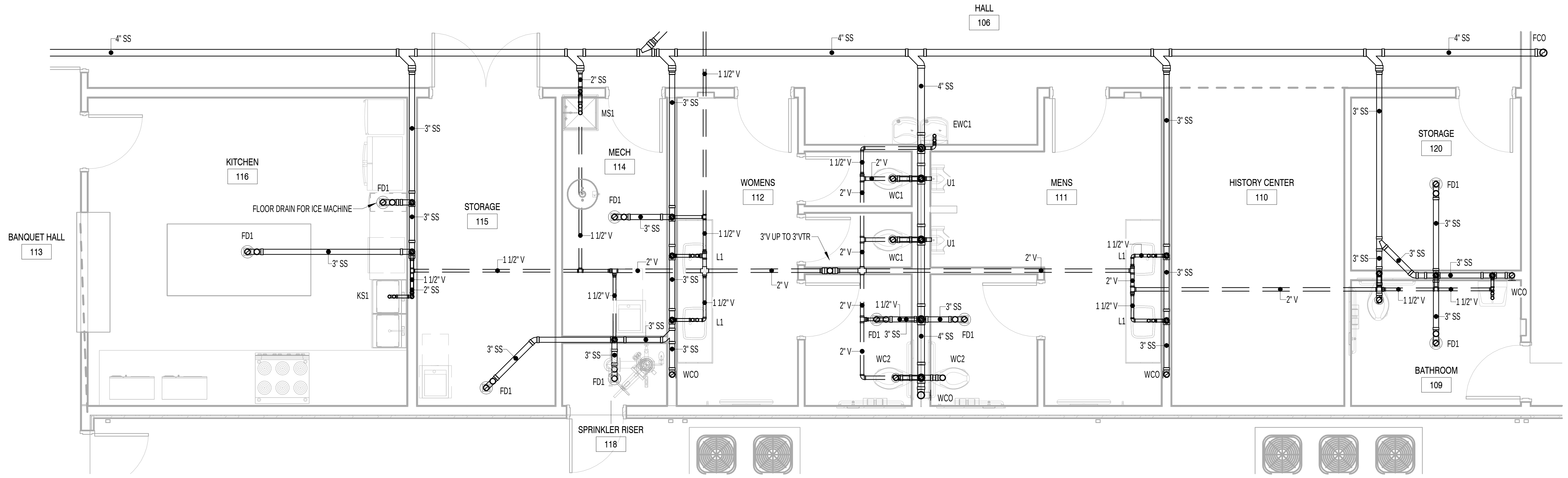
No.	Date

Drawing Title:
PLUMBING SCHEDULES, NOTES & DETAILS

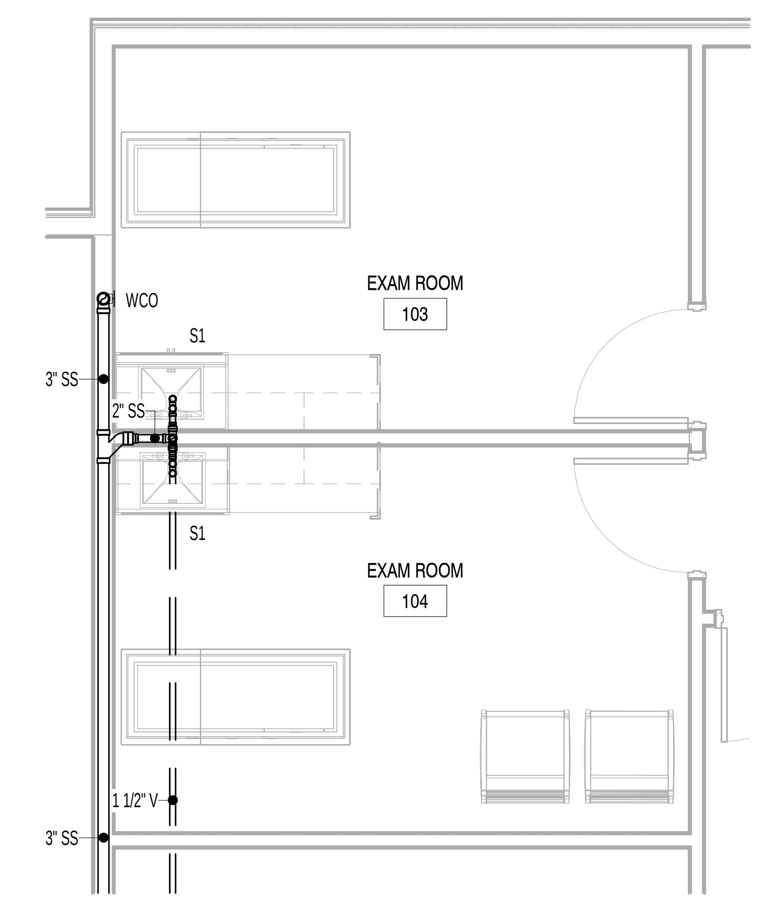
Date: 2025

Project No. 25026

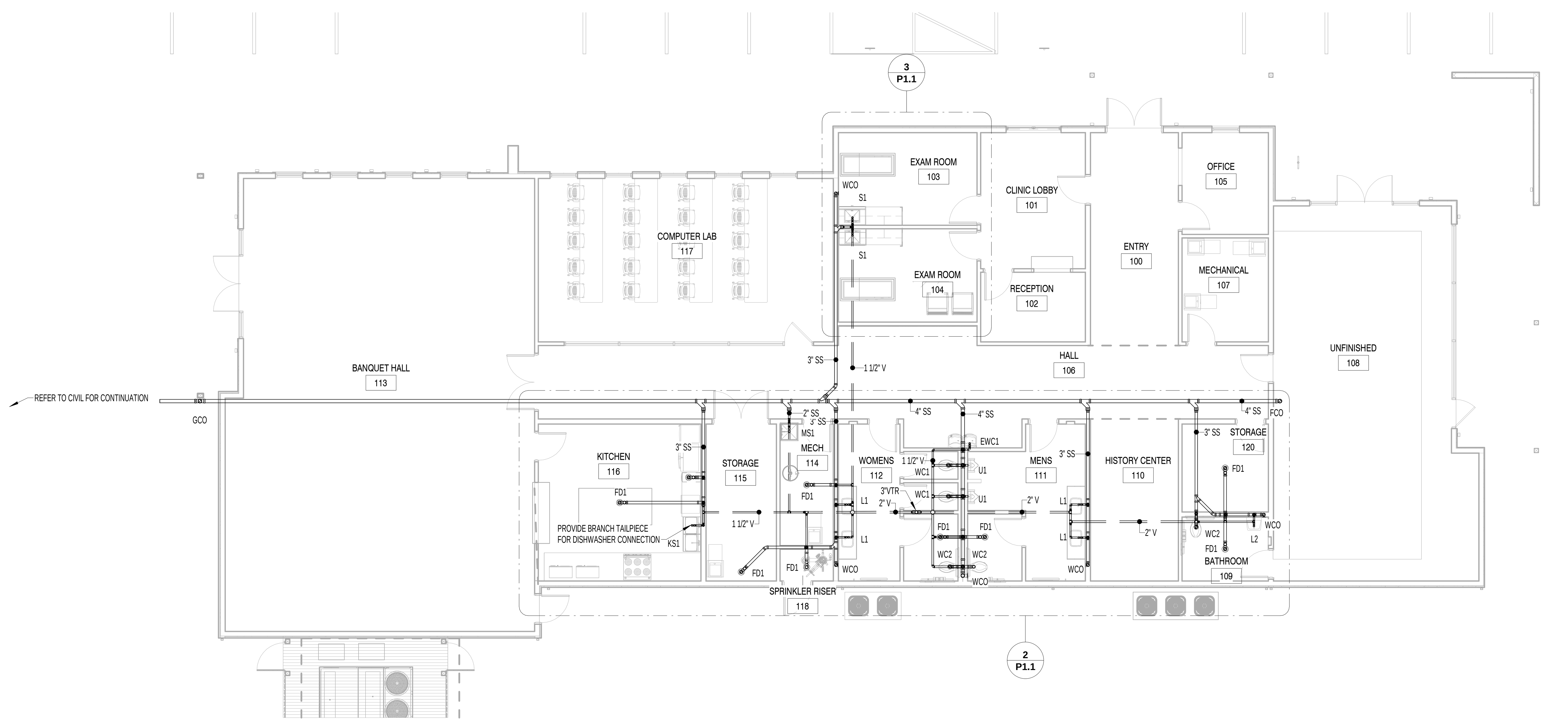
Sheet No. P0.1



2 SANITARY PLAN - ENLARGED
1/4" = 1'-0"



3 SANITARY PLAN - ENLARGED
1/4" = 1'-0"



1 SANITARY PLAN
1/8" = 1'-0"

FACILITY SYSTEMS CONSULTANTS, LLC
713 South Central Street, Suite 101
Knoxville, Tennessee 37902
Phone: (865) 246-0164
Fax: (865) 246-1084

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

PRESTON C. HALL
REGISTERED ENGINEER
AGRICULTURE
COMMERCE
NO. 114592
STATE OF TENNESSEE
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

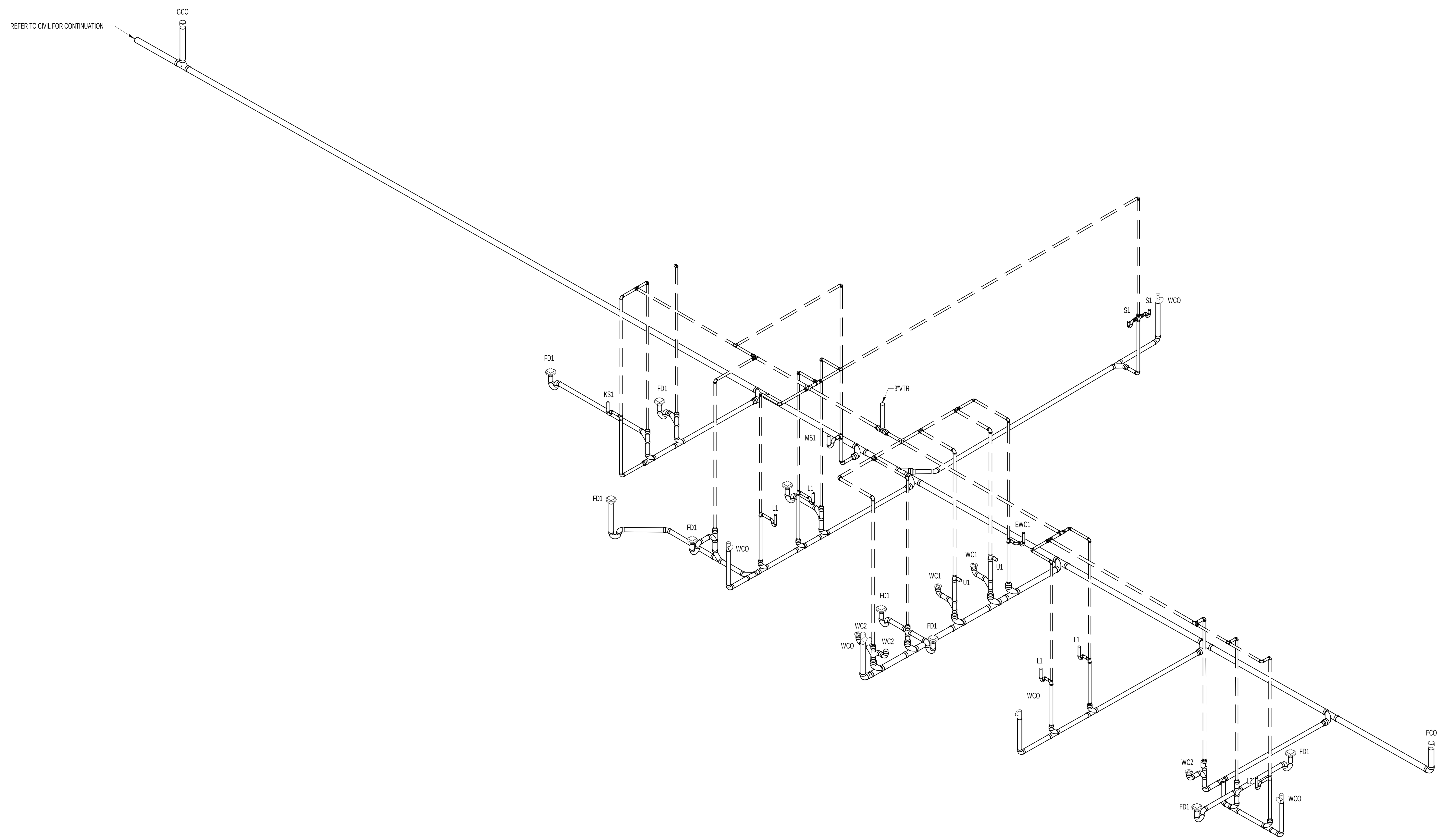
No.	Date

Drawing Title:
SANITARY PLAN

Date: 2025

Project No.
25026

Sheet No.
P1.1



1 SANITARY ISOMETRIC

TAC
The Architecture Collaborative

6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:

ELKS COMMUNITY ENRICHMENT CENTER

262 WILBERFORCE AVE
OAK RIDGE, TN 37830

PRESTON C. HALL
REGISTERED ENGINEER
AGRICULTURE
COMMERCE
NO. 114592
STATE OF TENNESSEE
11/26/25

THIS DRAWING IS ISSUED FOR:

CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

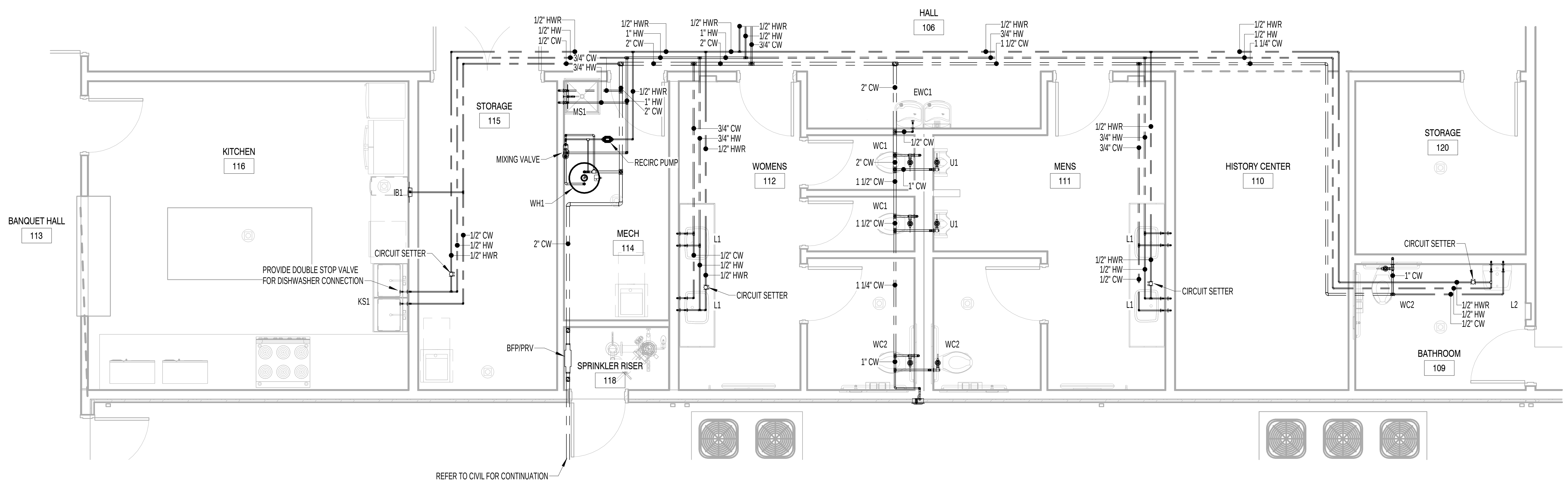
Drawing Title:
SANITARY ISOMETRIC

Date: 2025

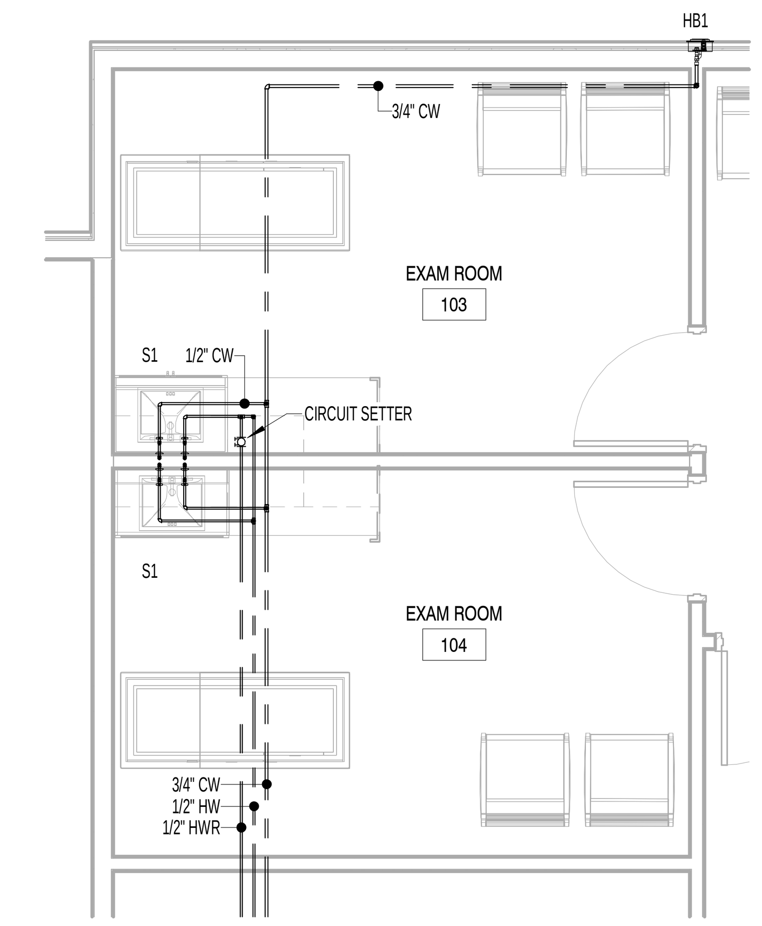
Project No. 25026

Sheet No. P1.2

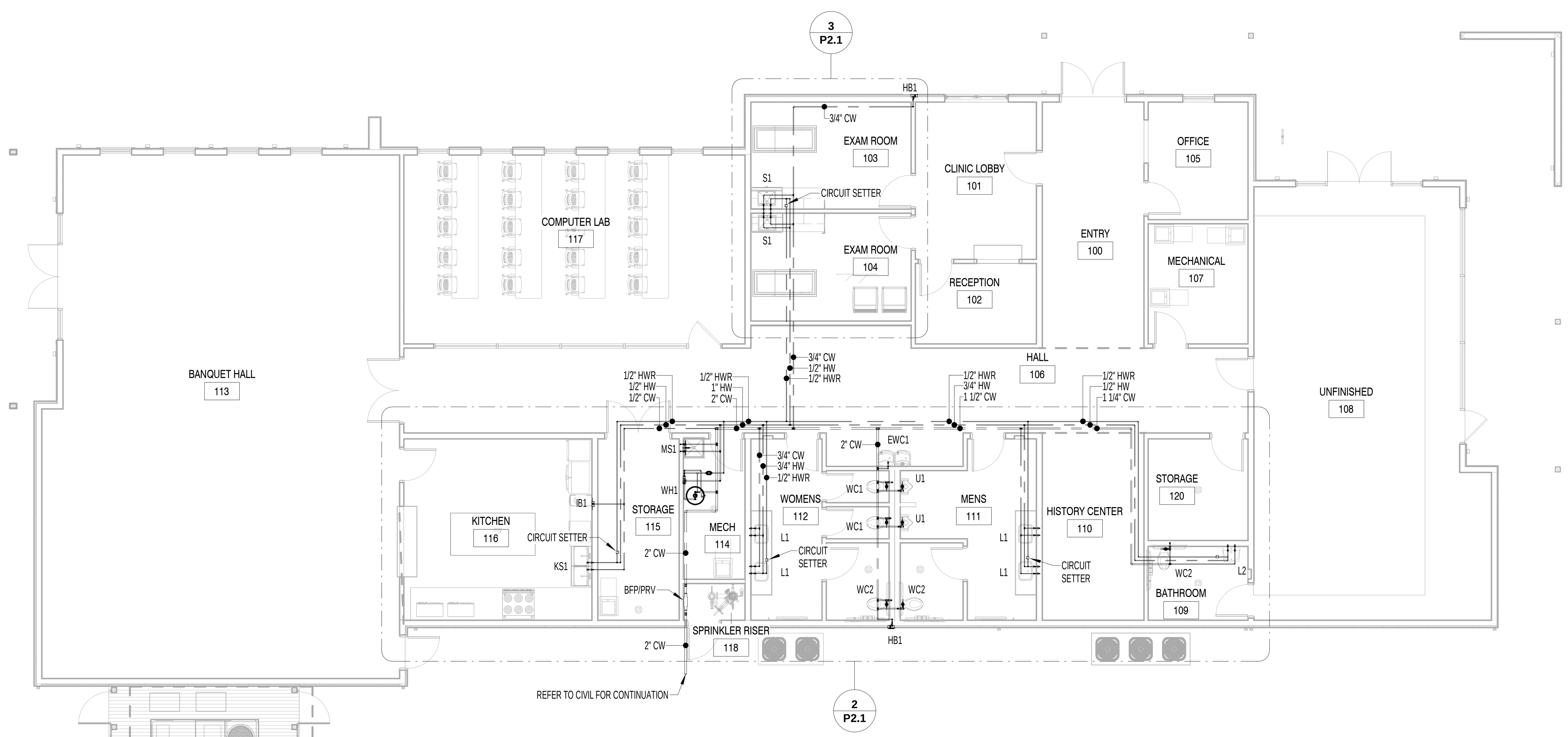
FACILITY SYSTEMS CONSULTANTS, LLC



2 DOMESTIC WATER PLAN - ENLARGED
1/4" = 1'-0"



3 DOMESTIC WATER PLAN - ENLARGED
1/4" = 1'-0"



1 DOMESTIC WATER PLAN
1/8" = 1'-0"

FACILITY SYSTEMS CONSULTANTS, LLC

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:

ELKS COMMUNITY ENRICHMENT CENTER

262 WILBERFORCE AVE
OAK RIDGE, TN 37830

PRESTON C. HALL
REGISTERED ENGINEER
AGRICULTURE
COMMERCE
NO. 114592
STATE OF TENNESSEE
11/26/25

THIS DRAWING IS ISSUED FOR:

CONSTRUCTION DOCUMENTS

Revisions:

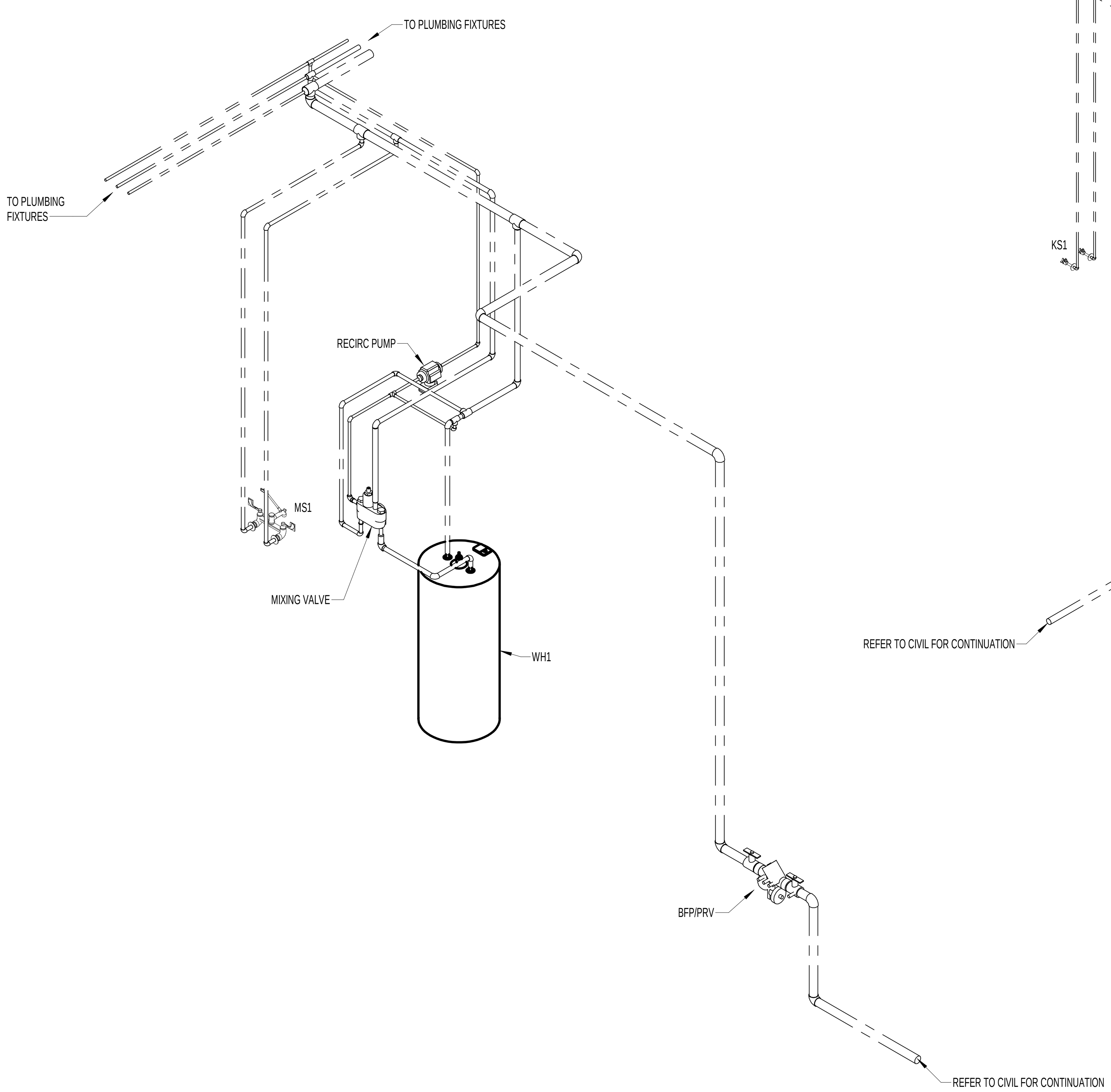
No.	Date

Drawing Title:
DOMESTIC WATER PLAN

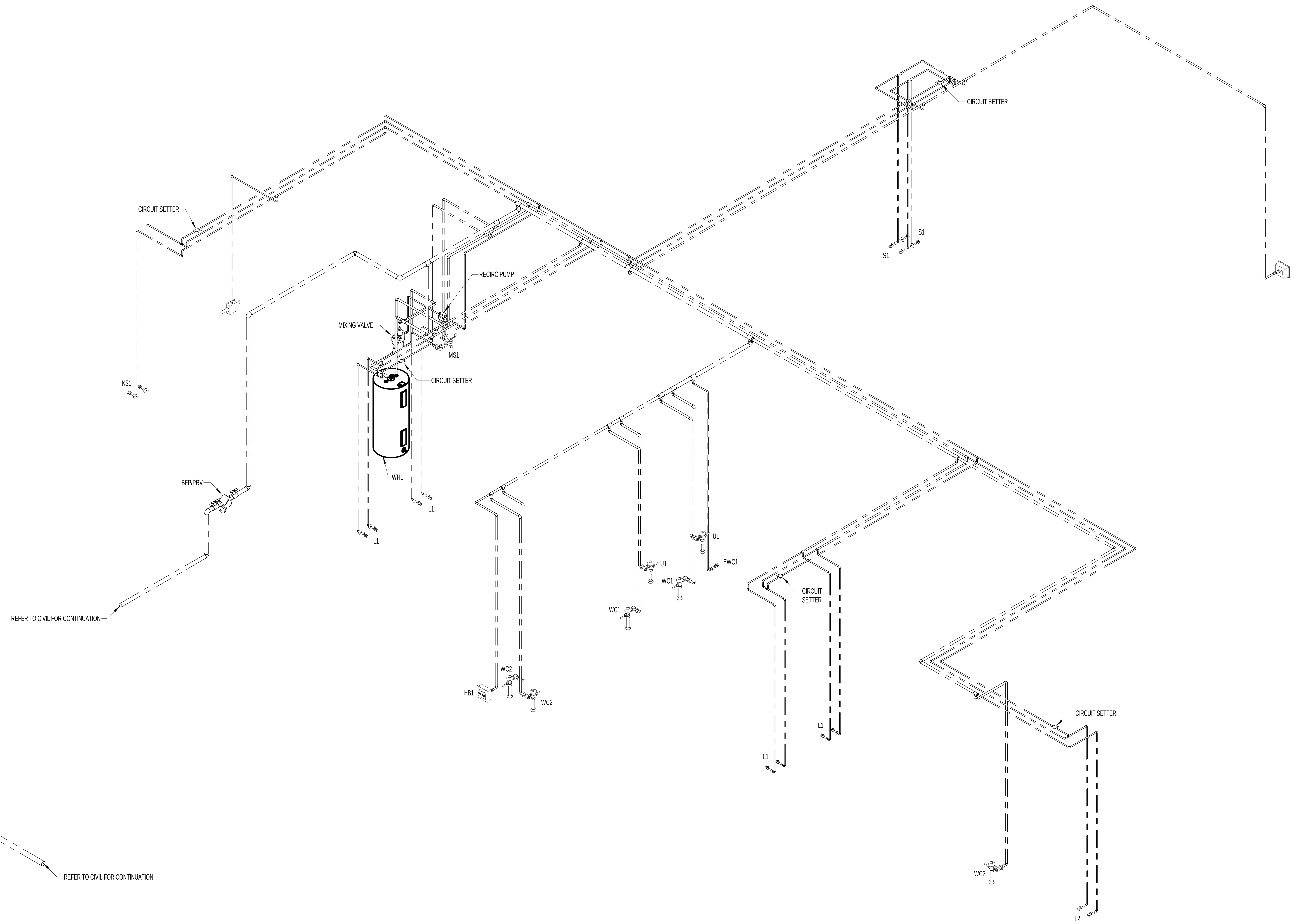
Date: 2025

Project No. 25026

Sheet No. P2.1



2 WATER ENTRANCE / WATER HEATER ISOMETRIC



1 DOMESTIC WATER ISOMETRIC

FACILITY
SYSTEMS
CONSULTANTS, LLC

TAC
The Architecture Collaborative

6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:

ELKS COMMUNITY ENRICHMENT CENTER

262 WILBERFORCE AVE
OAK RIDGE, TN 37830

PRESTON C. HALL
REGISTERED ENGINEER
AGRICULTURE
COMMERCE
NO. 114592
STATE OF TENNESSEE
11/26/25

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

Drawing Title:
DOMESTIC WATER
ISOMETRIC

Date: 2025

Project No.
25026

Sheet No.
P2.2

ELECTRICAL LEGEND

SYMBOL:	DESCRIPTION:
	METER, PROVIDE AND INSTALL IN ACCORDANCE WITH LOCAL UTILITY COMPANY REQUIREMENTS.
	PANELBOARD, MOUNT TOP 6'-0" AFF, REFER TO PANELBOARD SCHEDULES FOR REQUIREMENTS. SIDE WITH DESIGNATION IS FRONT UNLESS OTHERWISE NOTED.
TTB	TELEPHONE/COMMUNICATIONS TERMINAL BOARD. PROVIDE 3/4"x 4'x 4' PLYWOOD BACKBOARD PAINTED WITH GRAY FIRE RETARDANT PAINT. MOUNT TOP 8'-0" AFF. PROVIDE GROUND BAR SIMILAR TO HUBBELLCOMPANY MODEL NUMBER HBBB14416G, 4" BY 16". EXTEND #4 AWG BARE COPPER FROM GROUND BAR TO SERVICE ENTRANCE PANEL/DISCONNECT AND BOND TO GROUNDING SYSTEM.
LCP	LIGHTING CONTROL PANEL
A-1	HOME RUN TO PANEL WITH CIRCUIT SHOWN, CROSS MARKS INDICATE NUMBER OF CURRENT CARRYING CONDUCTORS WHERE MORE THAN TWO, NOT INCLUDING GROUND, MINIMUM #12 AWG.
	OVERHEAD WIRING
	UNDERGROUND WIRING
b A, b A	LED LIGHTING FIXTURE; "A" INDICATES TYPE, REFER TO LIGHTING FIXTURE SCHEDULE. "b" CORRESPONDS TO CONTROLLING SWITCH.
b A	LED LIGHTING FIXTURE; "A" INDICATES TYPE, REFER TO LIGHTING FIXTURE SCHEDULE. "b" CORRESPONDS TO CONTROLLING SWITCH.
	EXIT SIGN; SHADED QUADRANTS INDICATE NUMBER OF FACES, PROVIDE DIRECTIONAL ARROWS AS SHOWN. WHERE WALL MOUNTED, INSTALL 96" AFF. UNIT SHALL BE EQUIPPED WITH INTEGRAL BATTERY BACK-UP. REFER TO TYPE "EX" IN LIGHTING FIXTURE SCHEDULE.
	COMBINATION EMERGENCY LIGHT/EXIT SIGN; SHADED QUADRANTS INDICATE NUMBER OF FACES, PROVIDE DIRECTIONAL ARROWS AS SHOWN. WHERE WALL MOUNTED, INSTALL 96" AFF. UNIT SHALL BE EQUIPPED WITH INTEGRAL BATTERY BACK-UP. REFER TO TYPE CEX IN LIGHTING FIXTURE SCHEDULE.
	EMERGENCY LIGHT; WHERE WALL MOUNTED, INSTALL 96" AFF. UNIT SHALL BE EQUIPPED WITH INTEGRAL BATTERY BACK-UP. REFER TO TYPE EM IN LIGHTING FIXTURE SCHEDULE.
ED	EXIT DISCHARGE EMERGENCY LIGHT; UNIT SHALL BE WET LOCATION LISTED AND EQUIPPED WITH INTEGRAL BATTERY BACK-UP. REFER TO TYPE ED IN LIGHTING FIXTURE SCHEDULE.
EF	EMERGENCY FIXTURE, REMOTE WEATHERPROOF UNIT POWERED FROM COMBINATION UNIT, INSTALL 96" AFF. REFER TO TYPE "EF" IN LIGHTING FIXTURE SCHEDULE.
S_a	LIGHT SWITCH, TOGGLE, 20 AMP. 277V, MTD 48" AFF, UNO. "a" CORRESPONDS TO FIXTURES CONTROLLED BY SWITCH.
S_3	3-WAY LIGHT SWITCH, 20 AMP, 277V, MTD 48" AFF, UNO.
S_4	4-WAY LIGHT SWITCH, 20 AMP, 277V, MTD 48" AFF, UNO.
D	LIGHT SWITCH, DIMMING, 120VOLT, MTD 48" AFF, UNO. COMPATIBLE WITH FIXTURES, VERIFY WITH MANUFACTURER.
D_3	DIMMER AND SWITCH, 20AMP, 120VOLT, MTD 48" AFF, UNO. COMPATIBLE WITH 3-WAY SWITCH OR DIMMER.
PS	4-BUTTON SCENE CONTROLLER, ON/OFF AND RAISE/LOWER DIMMING CONTROL nLIGHT, GENERATION 2 MODEL # RPOD BA-4S-DX-?-G2; EQUAL BY WATTSTOPPER OR LUTRON
OS	OCCUPANCY SENSOR, DUAL TECHNOLOGY, CEILING MOUNTED OR 9' AFF WALL MOUNTED IN ROOMS WITH CEILINGS 10' OR ABOVE, WATTSTOPPER DT-200 OR EQUAL, PROVIDE BZ-50 POWER PACK
PS	PHOTOCELL SENSOR
J	JUNCTION BOX, SIZE AND USE AS REQUIRED
U	20A, 125V, 2 POLE, 3 WIRE, GROUNDING DUPLEX RECEPTACLE MTD. 42" AFF, UNO. "U" INDICATES DEVICE WITH USB CHARGING PORT.
U	20A, 125V, 2 POLE, 3 WIRE, GROUNDING DUPLEX RECEPTACLE, MTD 18" AFF, UNO. "U" INDICATES DEVICE WITH USB CHARGING PORT.
	20A, 125V, 2 POLE, 3 WIRE, GROUNDING QUAD RECEPTACLE, TWO GANG BOX, MTD 42" AFF, UNO.
	20A, 125V, 2 POLE, 3 WIRE, GROUNDING QUAD RECEPTACLE, TWO GANG BOX, MTD 18" AFF, UNO.
	60A, 250V, 3 POLE, 4-WIRE GROUNDING, SINGLE RECEPTACLE.
	20A, 250V, 2 POLE, 3-WIRE GROUNDING, SINGLE RECEPTACLE.
	30A, 250V, 2 POLE, 3-WIRE GROUNDING, SINGLE RECEPTACLE.
	EXHAUST FAN, PROVIDE 120 VOLT CONNECTION PER NEC.
	HEAVY DUTY FUSED DISCONNECT SWITCH, PROVIDE FUSES AS RECOMMENDED BY EQUIPMENT MFGR. USE NEMA CONFIGURATION AS REQUIRED
M	ELECTRIC MOTOR
TV	TELEVISION OUTLET. PROVIDE 1" CONDUIT FROM BOX TO LOCATION ABOVE ACCESSIBLE CEILING. PROVIDE BUSHING ON OPEN END OF CONDUIT. MTD AT 7'-6" AFF
	COMBINATION PHONE AND DATA OUTLET. PROVIDE 4" SQUARE BOX WITH SINGLE GANG DEVICE RING. STUB 1" CONDUIT WITH PULLSTRING FROM BOX TO LOCATION ABOVE ACCESSIBLE CEILING AS INDICATED ON DRAWINGS. PROVIDE BUSHING ON OPEN END OF CONDUIT. MTD 18" AFF, UNO. STUB UP 12" ABOVE CEILING
	COMBINATION PHONE AND DATA OUTLET. PROVIDE 4" SQUARE BOX WITH SINGLE GANG DEVICE RING. STUB 1" CONDUIT WITH PULLSTRING FROM BOX TO LOCATION ABOVE ACCESSIBLE CEILING AS INDICATED ON DRAWINGS. PROVIDE BUSHING ON OPEN END OF CONDUIT. MTD 42" AFF, UNO. STUB UP 12" ABOVE CEILING
FB	FLOOR BOX, USE WALKER CO. MODEL NO. RFB4SS WITH COVER, PROVIDE DUPLEX RECEPTACLE AND APPROPRIATE BRACKETS FOR DATA JACKS. SINGLE BOX IS SHOWN ON BOTH PLANS FOR WIRING REQUIREMENTS.

ELECTRICAL ABBREVIATIONS

ABBREVIATIONS	DESCRIPTION:
A	AMPERE
AFF	ABOVE FINISHED FLOOR - MEASURED FROM FLOOR TO CENTER OF DEVICE, EXCEPT AS OTHERWISE SPECIFICALLY NOTED.
ADA	AMERICANS WITH DISABILITIES ACT OF 1990
AFG	ABOVE FINAL GRADE
AGTD	AUTOMATIC GENERATOR TRANSFER DEVICE
C	CONDUIT
CM	INDICATES DEVICE TO BE CEILING MOUNTED
EUH	ELECTRIC UNIT HEATER
F	FUSE
FAR	FUSE AS REQUIRED
FPN	FUSE PER NAMEPLATE REQUIREMENTS
G	GROUND
GF	INDICATES RECEPTACLE OR CIRCUIT BREAKER, AS APPLICABLE, TO HAVE GROUND FAULT PROTECTION
IG	INDICATES RECEPTACLE TO BE ISOLATED GROUND TYPE
MCM	Kcmil (THOUSAND CIRCULAR MILS)
NEC	NATIONAL ELECTRICAL CODE
N	NEUTRAL
NL	INDICATES FIXTURE TO BE CONNECTED UNSWITCHED TO SERVE AS A "NIGHT" LIGHT.
PH	PHASE
QR	INDICATES FIXTURE TO BE EQUIPPED WITH QUARTZ RESTRIKE, PROVIDE TIME DELAY MODULE.
SCIR	SHORT CIRCUIT INTERRUPTING RATING
S.O.	SPACE ONLY
TVSS	TRANSIENT VOLTAGE SURGE SUPPRESSOR, LEVITON 57??? SERIES OR EQUAL.
UNO	UNLESS NOTED OTHERWISE
VIF	VERIFY IN FIELD
WG	INDICATES ITEM IS SUBJECT TO PHYSICAL DAMAGE: PROVIDE WIRE GUARD PROTECTION.
WP	INDICATES DEVICE TO HAVE WEATHERPROOF COVER, TAYMAC MODEL NO. MX3200 OR EQUAL.

GENERAL ELECTRICAL NOTES:

1. THE CONTRACTOR SHALL VISIT THE JOB SITE AND CAREFULLY EXAMINE THOSE PORTIONS OF THE SITE AFFECTED BY THIS WORK SO AS TO BECOME FAMILIAR WITH EXISTING CONDITIONS THAT WILL AFFECT EXECUTION OF THE WORK.
2. THE CONTRACTOR IS RESPONSIBLE FOR OBTAINING ALL NECESSARY PERMITS AND PAYING ALL UTILITY CO. AID TO CONSTRUCTION FEES.
3. ALL WORK SHALL BE IN ACCORDANCE WITH THE CURRENT/APPLICABLE NATIONAL ELECTRICAL CODE, NFPA 70, LOCAL CODES/ORDINANCES AND THE APPLICABLE ACCESSIBILITY CODE. SHOULD PLANS AND CODES CONFLICT, THE CODE TAKES PRECEDENCE. MAKE NO CHANGES, EVEN IN THE CASE OF CONFLICT, WITHOUT FIRST OBTAINING APPROVAL OF THE ARCHITECT/ENGINEER.
4. "PROVIDE" AS USED HERE AND ON THE DRAWINGS, IS AN ALL-INCLUSIVE TERM REQUIRING CONTRACTOR TO FURNISH, INSTALL, WIRE, AND CONNECT ALL SPECIFIED EQUIPMENT AS WELL AS COMPONENTS, ACCESSORIES, AND MOUNTING HARDWARE TO ENSURE THAT SPECIFIED EQUIPMENT FUNCTIONS TO MEET SYSTEM REQUIREMENTS.
5. IT SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR TO PROTECT OTHER FACILITIES AND EQUIPMENT FROM DAMAGE. THE CONTRACTOR SHALL BEAR ALL EXPENSE FOR REPAIR OR REPLACEMENT OF FACILITIES, EQUIPMENT, OR OTHER PROPERTY DAMAGED BY OPERATIONS IN CONJUNCTION WITH THE COMPLETION OF THIS WORK. ELECTRICAL CONTRACTOR SHALL GIVE ADEQUATE NOTIFICATION TO TENNESSEE ONE CALL, (800) 351-1111, PRIOR TO COMMENCEMENT OF ANY EXCAVATION.
6. PROVIDE SPECIFIED EQUIPMENT, AS NOTED ON DRAWINGS, OR APPROVED EQUAL. ADDITIONAL EQUIPMENT AND MATERIAL MAY BE REQUIRED OTHER THAN THAT SHOWN ON DRAWINGS TO INSTALL THE SPECIFIED EQUIPMENT SUCH AS HANGERS, SUPPORTS, ETC. ELECTRICAL CONTRACTOR SHALL PROVIDE ALL MATERIAL, LABOR, AND EQUIPMENT REQUIRED.
7. THE CONTRACTOR SHALL VERIFY THAT THE ACTUAL EQUIPMENT SUPPLIED HAS THE SAME ELECTRICAL SPECIFICATIONS AS THE EQUIPMENT USED AS THE BASIS OF DESIGN. IF THE EQUIPMENT IS DIFFERENT, THE CONTRACTOR SHALL MAKE ADJUSTMENTS TO THE PANELS AND CIRCUITS AND INCLUDE THEM IN SUBMITTALS.
8. ALL ITEMS SHALL BE NEW. USED EQUIPMENT AND MATERIALS WILL NOT BE ALLOWED UNLESS SPECIFICALLY NOTED TO BE EXISTING OR RELOCATED ON RESPECTIVE PROJECT SITE.
9. ALL MATERIALS SHALL BE LISTED AND LABELED BY UNDERWRITERS LABORATORY, INC.
10. THE CONTRACTOR IS DIRECTED TO VERIFY THE SERVICE ARRANGEMENT WITH THE LOCAL ELECTRIC UTILITY BEFORE ORDERING EQUIPMENT. DESIGN IS BASED ON A 208Y/120 VOLT, THREE PHASE, FOUR WIRE, SOLIDLY GROUNDED WYE SERVICE. THIS DESIGN ASSUMES AN UNDERGROUND SERVICE WITH SERVICE CONDUITS AND CONDUCTORS BY ELECTRICAL CONTRACTOR. ELECTRICAL CONTRACTOR SHALL INSTALL PRIMARY CONDUITS AS SPECIFIED BY THE LOCAL ELECTRIC UTILITY.
11. UNLESS OTHERWISE INSTRUCTED, THE CONTRACTOR SHALL SUBMIT A DIGITAL (PDF) COPY OF ELECTRICAL SHOP DRAWINGS TO THE PROJECT MANAGER WHO SHALL RELAY THEM TO THE DESIGN ENGINEER FOR REVIEW AND APPROVAL PRIOR TO THE PURCHASE OF EQUIPMENT. THE SUBMITTAL SHALL INCLUDE LIGHTING FIXTURES, SWITCHGEAR, GENERATOR AND FIRE ALARM EQUIPMENT, WHEN INCLUDED IN THE PROJECT. OPERATION AND MAINTENANCE MANUALS FOR ALL ELECTRICAL EQUIPMENT SHALL BE COMPILED AND SUBMITTED IN DIGITAL (PDF) TO THE BUILDING OWNER UPON PROJECT COMPLETION.
12. ALL WIRES SHALL BE TERMINATED AND LABELED. ALL JUNCTION BOXES SHALL BE LABELED TO INDICATED THE CIRCUITS CONTAINED IN THE BOX.
13. PANELBOARD LEGENDS SHALL BE TYPED. LABEL ALL PANELBOARDS/SWITCHGEAR INDICATING LOCATION OF BREAKER SERVING PANEL IN ACCORDANCE WITH N.E.C.
14. ALL FEEDERS #4 AND LARGER SHALL BE MEGGER TESTED AFTER INSTALLATION.
15. UNLESS OTHERWISE NOTED, ALL CONDUCTORS SHALL BE COPPER AND #12 AWG MINIMUM WITH THHN/THWN, 600 VOLT INSULATION.
16. PROVIDE A DEDICATED NEUTRAL, COLOR CODED, FOR EACH UNGROUNDED CONDUCTOR. SHARING OF NEUTRALS IS PROHIBITED.
17. DO NOT INSTALL MORE THAN THREE CIRCUITS (SIX CURRENT CARRYING CONDUCTORS) IN A CONDUIT.
18. THE MINIMUM CONDUIT SIZE SHALL BE 1/2". INTERIOR CONDUITS SHALL BE EMT; UNDERGROUND CONDUIT AND CONCRETE ENCASED CONDUIT SHALL BE SCHEDULE 40 PVC. EXTERIOR EXPOSED CONDUIT SHALL BE SCHEDULE 40 PVC., UNLESS NOTED OTHERWISE.
19. MC CABLE MAY BE USED FOR CONCEALED BRANCH CIRCUIT WIRING IN INTERIOR DRY LOCATIONS.
20. A GREEN, COPPER GROUND WIRE SHALL BE INSTALLED IN ALL CONDUIT SYSTEMS AND SHALL BE BONDED TO ALL ENCLOSURES, BOXES, AND EQUIPMENT.
21. BONDING JUMPERS SHALL BE USED TO BOND CONDUIT TO ENCLOSURES, BOXES, AND EQUIPMENT WHERE KNOCKOUTS ARE USED.
22. ALL DIMENSIONS ARE MEASURED TO THE CENTER OF THE DEVICE.
23. THE CONTRACTOR SHALL PROVIDE FIRESTOPPING OF ALL RATED PENETRATIONS PER DETAILS. ELECTRICAL BOXES INSTALLED ON OPPOSITE SIDES OF A FIRE RATED WALL SHALL HAVE A TWO FOOT MINIMUM HORIZONTAL SEPARATION.
24. THE CONTRACTOR SHALL PROVIDE ADEQUATE TEMPORARY POWER AND LIGHT. EQUIVALENT TO ONE 150-WATT INCANDESCENT LAMP PER 200 SQ. FT.
25. THE CONTRACTOR SHALL GUARANTY ALL WORK TO BE FREE OF DEFECTS IN WORKMANSHIP AND MATERIALS FOR ONE YEAR AFTER SUBSTANTIAL COMPLETION.

LIGHTING NOTES:

1. CONTRACTOR SHALL FURNISH AND INSTALL LIGHT SWITCHES/CONTROLS FOR ALL LIGHTING AT LOCATIONS AS SHOWN ON THE DRAWINGS.
2. CONFIRM EXACT LIGHT FIXTURE LOCATIONS WITH ARCHITECTURAL REFLECTED CEILING PLAN.
3. CONNECT ALL EXIT AND EMERGENCY LIGHTS TO UNSWITCHED LIGHTING CIRCUITS. UNITS SHALL OPERATE AUTOMATICALLY UPON LOSS OF POWER.
4. PRIOR TO ORDERING THE SPECIFIED LIGHT FIXTURES, THE CONTRACTOR SHALL VERIFY THE FIXTURE IS SUITABLE FOR THE CEILING TYPE. FOR EXAMPLE, A FIRE RATED FIXTURE SHALL BE INSTALLED IN A FIRE RATED ASSEMBLY. IF DISCREPANCIES ARE FOUND, THE CONTRACTOR SHALL NOTIFY THE ARCHITECT AND ENGINEER PRIOR TO PROCEEDING.

POWER NOTES:

1. CONTRACTOR SHALL FURNISH AND INSTALL FUSED DISCONNECTS FOR ALL HVAC EQUIPMENT WITH FUSES AS PER MANUFACTURER RECOMMENDATIONS. AMPACITY, POLES, AND TYPE NEMA ENCLOSURE OF DISCONNECT SWITCHES AS REQUIRED. FURNISH AND INSTALL A WEATHERPROOF, GFCI DUPLEX RECEPTACLE OUTLET WITHIN 25 FEET OF EACH HVAC PIECE OF EQUIPMENT.
2. MOUNT ALL SWITCHES AND OTHER ELECTRICAL EQUIPMENT IN COMPLIANCE WITH APPLICABLE PROVISIONS OF THE APPLICABLE ACCESSIBILITY CODE.
3. ALL RESTROOM, EXTERIOR, COUNTER TOP, AND ROOF TOP HVAC SERVICE RECEPTACLES SHALL BE GFCI.
4. ALL EXTERIOR RECEPTACLES SHALL HAVE APPROVED WEATHERPROOF COVERS AS PER NEC 406.8 (B).
5. ALL DAMP AND WET LOCATION DEVICES SHALL BE WEATHER RESISTANT.
6. ALL 125V AND 250V, 15A AND 20A RECEPTACLES SHALL BE TAMPER-RTESISTANT TYPE IN ACCORDANCE WITH NEC 406.12 WHERE REQUIRED. THIS INCLUDES (BUT NOT LIMITED TO) IN OR OUTSIDE OF DWELLING UNITS; COMMON AREAS OF MULTI-FAMILY DWELLING UNITS; DORM ROOMS, GUEST ROOMS, AND COMMON AREAS OF HOTELS; CHILDCARE, DAYCARE, AND K-12 EDUCATION FACILITIES; PATIENT CARE, COMMON AREAS, AND CORRIDORS OF MEDICAL FACILITIES; ASSEMBLY OCCUPANIES CORRIDORS. SEE NEC 406.12 FOR MORE INFORMATION.
7. ALL EXTERIOR 120V, 15A AND 20A RECEPTACLES INSTALLED IN DAMP OR WET LOCATUIONS SHALL BE WET LOCATION RATED WITH APPROPRIATE WEATHER-PROOF COVERS IN ACCORDANCE WITH NEC 406.9.

COMMUNICATIONS NOTES:

1. CONTRACTOR SHALL FURNISH AND INSTALL ALL COMBINATION TELEPHONE AND DATA CONDUITS, BOXES, PLYWOOD TERMINAL BOARD, ETC.
2. ACTIVE EQUIPMENT TO BE PROVIDED/INSTALLED BY OTHERS.
3. PROVIDE APPROPRIATE NYLON PULLSTRING/ROPE IN ALL EMPTY CONDUITS.

DEMOLITION NOTES:

1. REMOVE ALL EXISTING DEVICES IN WALL AND CEILINGS BEING REMOVED AND PROPERLY ABANDON CONDUIT SYSTEM. REMOVE ALL EXISTING UNUSED OR ABANDONED CONDUIT, WIRING, JUNCTION BOXES, ETC. ABOVE CEILING. REMOVE ALL LIGHT FIXTURES IN AREAS WHERE NEW FIXTURES ARE ILLUSTRATED, PROPERLY DISPOSE OF OR TURN OVER TO OWNER AS DIRECTED.

SPRINKLER MONITORING DEVICE LEGEND

SYMBOL:	DESCRIPTION OF DEVICE	DEVICE MODEL NUMBER(S)	HEIGHT
S "SMP"	SPRINKLER MONITORING CONTROL PANEL	4010-9101	TOP @ 6'-0"
S	SMOKE DETECTOR	4098-9792 BASE WITH 4098-9714 SENSOR	CLG
HS WP	WEATHER PROOF HORN/STROBE UNIT	4906-9131	7'-6"
TS	TAMPER SWITCH	F.B.O.-SUPERVISED BY I.A.M. MODEL 4090-9001	AS REQUIRED
FS	FLOW SWITCH	F.B.O.-SUPERVISED BY I.A.M. MODEL 4090-9001	AS REQUIRED
H 135°	HEAT DETECTOR 135° FIXED TEMP	4098-9733	CLG
DD	DUCT MOUNTED SMOKE DETECTOR	4098-9756	DUCT MOUNTED
R	REMOTE TEST STATION FOR DD	2098-9806	5'-6"

NOTES:
MODEL NUMBERS ARE SIMPLEX GRINNELL UNLESS OTHERWISE NOTED

SPRINKLER MONITORING WIRING SCHEDULE

DEVICE TYPE	WIRE QUANTITY, SIZE, AND TYPE
MAPNET/IDNET CIRCUIT, COMMUNICATION/DATA LINE	2 #18 TWISTED/SHIELDED
STROBE CIRCUIT, HORN/STROBE CIRCUIT	2 #14
24 VDC MAPNET/IDNET POWER	2 #14

NOTES :
SPRINKLER MONITORING SYSTEM WIRING SHALL BE INSTALLED IN CONDUIT.

SPRINKLER MONITORING SYSTEM NOTES:

1. FURNISH AND INSTALL A COMPLETE SPRINKLER MONITORING SYSTEM AS DESCRIBED HEREIN AND AS SHOWN ON THE PLANS; TO BE WIRED, CONNECTED, TESTED, AND LEFT IN FIRST-CLASS OPERATING CONDITION. THE SYSTEM SHALL USE ADDRESSABLE TRUEALARM INITIATING DEVICE CIRCUITS WITH REQUIRED SUPERVISION. ALL EQUIPMENT SHALL BE UL LISTED AND LABLED. HEREIN SPECIFIED IS THAT OF SIMPLEX AND DEPICTS THE TYPE AND QUALITY OF EQUIPMENT TO BE FURNISHED. THE ENTIRE INSTALLATION SHALL CONFORM TO THE APPLICABLE SECTIONS OF NFPA 72, NATIONAL FIRE ALARM CODE; NFPA 101, LIFE SAFETY CODE; NFPA 70, NATIONAL ELECTRICAL CODE; THE AMERICANS WITH DISABILITIES ACT; AND LOCAL AUTHORITIES HAVING JURISDICTION TO MEET THE COMPLETE FUNCTIONALITY REQUIREMENTS AS SET FORTH IN THESE SPECIFICATIONS.
2. COORDINATE AND INSTALL REQUIRED ANCILLARY ALARM FUNCTIONS WHERE SHOWN ON THE PLANS. EXAMPLES ARE MONITORING OF SPRINKLER SYSTEMS AND HVAC SHUTDOWN.
3. SPRINKLER MONITORING CONTROL PANEL - SIMPLEX MODEL 4004 OR EQUAL WITH A MINIMUM OF 8 ZONES, CAPABLE OF SUPPORTING ALARM INITIATING APPLIANCES AND ALARM INDICATING APPLIANCES AS REQUIRED ON THE PLANS. THE CONTROL PANEL SHALL BE PROGRAMMABLE VIA THE FRONT PANEL FOR INPUT/OUTPUT FUNCTIONS AND SHALL CONTAIN A HISTORICAL EVENT LOG, 80 CHARACTER LCD DISPLAY, AND A SYSTEM WALK TEST OPTION.
4. CONTROL PANEL REMOTE REPORTING - THE SYSTEM SHALL HAVE AN INTERNALLY MOUNTED DIGITAL ALARM COMMUNICATING TRANSMITTER (DACT), CAPABLE OF BEING PROGRAMMED FOR COMMUNICATIONS WITH A MINIMUM OF TWO TELEPHONE CIRCUITS AND WITH PROGRAMMABLE LOCAL AC POWER FAIL TIME DELAY REPORTING. 24 VDC OPERATING VOLTAGE AND BATTERY BACKUP SHALL BE PROVIDED BY THE FIRE ALARM CONTROL PANEL. DACT SHALL REPORT TO MANUFACTURER'S, UL LISTED, CENTRAL STATION MONITORING FACILITY.
5. PROVIDE BATTERY CALCULATIONS TO SHOW THAT THE PROPER QUANTITY OF BATTERIES ARE SUPPLIED THAT UPON LOSS OF 120 VAC POWER WILL PROVIDE A MINIMUM OF 24 HOURS OF NORMAL SUPERVISORY MODE FOLLOWED BY 5 MINUTES OF ALARM INDICATION.
6. PROVIDE SUBMITTALS CONTAINING COMPLETE DESCRIPTIVE DATA AND CADD DRAWING SHOWING CONDUIT LAYOUT, WIRE COUNT, AND DEVICE LOCATIONS.
7. SYSTEM SHALL BE FULLY TESTED BY A NICET CERTIFIED TECHNICIAN IN THE PRESENCE OF THE OWNER'S REPRESENTATIVE AND BE WARRANTED FOR ONE YEAR.
8. ELECTRICAL CONTRACTOR IS RESPONSIBLE FOR ENSURING THAT THE FIRE ALARM SYSTEM IS ACCEPTABLE TO THE LOCAL FIRE OFFICIAL HAVING JURISDICTION. ALL LABOR, MATERIALS, AND EQUIPMENT NECESSARY FOR AN OPERATING AND FULLY FUNCTIONAL SYSTEM IS INCLUDED IN THIS CONTRACT.

LIGHTING FIXTURE SCHEDULE

SYM	CATALOG NUMBER		PERFORMANCE				MOUNTING	DESCRIPTION
	COMPANY	MODEL NUMBER	WATTS	TEMP IN K	MINCRI	DELIVERED LUMENS		
CEX	LITHONIA	ECR LED HO	3.8	N/A	N/A	44.6	UNIVERSAL	COMBINATION EMERGENCY LIGHT/EXIT SIGN WITH LED ILLUMINATION, INTEGRAL BATTERY BACK-UP
EX	LITHONIA	EXR LED EL	1	N/A	N/A	N/A	UNIVERSAL	THERMOPLASTIC EXIT SIGN WITH LED ILLUMINATION AND INTEGRAL BATTERY BACK-UP
EM	LITHONIA	ELM4L	2	N/A	N/A	640	UNIVERSAL	TWIN HEAD EMERGENCY LIGHT WITH INTEGRAL BATTERY BACK-UP
EF	LITHONIA	ERE ?? T WP SQ	2	N/A	N/A	73.5	WALL	WET LOCATION REMOTE EMERGENCY HEAD FOR USE WITH TYPE CEX FIXTURE
A	LITHONIA	2BLT2 33L ADP GZ10 LP835	26.5	3000	82	3300	RECESSED	2'X2' LED FLAT PANEL WITH LUMEN OUTPUT 3300 AND COLOR TEMPERATURE 35K
B	LITHONIA	2BLT2 40L ADP GZ10 LP835	31	3000	82	4034	RECESSED	2'X2' LED FLAT PANEL WITH LUMEN OUTPUT 4000 AND COLOR TEMPERATURE 35K
C	LITHONIA	CLX L48 4000LM SEF WDL MVOLT GZ10 35K 80CRI	31.8	3000	80	4000	SURFACE	1' X 4' LINEAR STRIP LIGHT, MVOLT
D	VAKKER LIGHTING	VKC-230907211202	6	2700	80	460	PENDANT	4 LAMP CHANDELIER, 10W G4 LED EQUIVALENT LAMP
F	ACCESS LIGHTING	23941-BS/OPL	8	2700	90	800	PENDANT	10" GLOBE PENDANT
G	ACCESS LIGHTING	29001LEDDLP-MBL	10	3000	90	800	PENDANT	CYLINDER PENDANT
H	KUZCO	PD43216-BK/OP-5CCT-UNV	20	3000	90	1750	PENDANT	16" GLOBE PENDANT, 120-277V
K	AFX LIGHTING	ELLA LED VANITY LIGHT	9	3000	90	1400	WALL, 6" ABOVE MIRROR	LED VANITY LIGHT
L	VAKKER LIGHTING	VKC-230907211206	9	2700	80	690	PENDANT	6 LAMP CHANDELIER, 10W G4 LED EQUIVALENT LAMP
M	WVH	WVH SLAT-PANEL GLOW LED LIGHT STRIPS	3.4/FT	3000	80		WALL AND CEILING, SEE ARCH ELEV.	SLAT PANEL GLOW LED LIGHT STRIPS; MOUNTED BETWEEN WOOD SLATS ON WALL AND CEILING; SEE ARCHITECTURAL DETAILS, RCP, AND ELEVATIONS. CONNECT REMOTE DRIVERS, SYNC, AND WALL MOUNT REMOTE CONTROL.
N	LITHONIA	LDN6 35/15 L06 ? ? ? MVOLT GZ10	17.5	3000	80	1514	RECESSED	6" LED RECESSED DOWNLIGHT, MVOLT (120-277), 0-10V DIMMING DRIVER
P	WAC LIGHTING	DS-WD0534-930S-BK	50	3000	90	3940	WALL, SEE ARCH ELEV.	EXTERIOR TUBE WALL SCONCE, MVOLT (120-277)
Q	WAC LIGHTING	WS-W17310-30-BK	13	3000	90	765	WALL, SEE ARCH ELEV.	EXTERIOR WALL SCONCE
R	LITHONIA	WDGE2 LED P3 40K 80CRI VW MVOLT-??	23	4000	80	3213	WALL, 11'-0" AFG	FULL CUT-OFF WALL PACK WITH VISUAL COMFORT LENS, TYPE 3 WIDE DISTRIBUTION
T	T-BAR LED	TBSL-HW-4-24-D-U-W	32	3000	82	2336	CEILING GRID REPLACEMENT, RECESSED	T-BAR LED CEILING GRID LIGHTING MODULES, 4FT LONG, TAKES PLACE OF GRID PIECES. COORDINATE INSTALLATION WITH CEILING GRID INSTALLER, CANNOT TAKE PLACE OF MAIN RUNS. INSTALL REMOTE DRIVERS AS NECESSARY, ONE FOR EVERY 12FT OF LIGHTING (3) MODULES.
U	HINKLEY	DAX 32377BK	8.5	3000	90	600	PENDANT	EXTRA SMALL LED PENDANT
V	VISUAL COMFORT	6018EN3-112	9	3000	90	800	PENDANT	8" GLOBE PENDANT, 120-277V; LED 60W E26 A19 EQUIVALENT LAMP; FINISH BY ARCHITECT
SA	LITHONIA	HEAD - RSX1 LED P4 40K R4 MVOLT SPA ?? POLE - SSS 18 4C DM19AS UL ??	133	4000	70	16500	POLE, 18FT	AREA LIGHT, 1@90 DEGREES, TYPE 4 DISTRIBUTION, FULL CUT-OFF, 18 FT POLE, 2FT BASE (AFG)
SB	LITHONIA	HEAD - RSX1 LED P4 40K R2 MVOLT SPA ?? POLE - SSS 18 4C DM19AS UL ??	133	4000	70	16500	POLE, 18FT	AREA LIGHT, 1@90 DEGREES, TYPE 2 DISTRIBUTION, FULL CUT-OFF, 18 FT POLE, 2FT BASE (AFG)

NOTES:
1. THE FINISH OF ALL FIXTURES (NOTED BY ? IN THE MODEL NUMBER) SHALL BE VERIFIED WITH AND APPROVED BY THE ARCHITECT.
2. REFER TO THE ARCHITECT'S REFLECTED CEILING PLAN FOR THE EXACT LOCATION OF FIXTURES.
3. ALL FIXTURES SHALL BE FURNISHED COMPLETE WITH ELECTRONIC DRIVERS WITH MAXIMUM 10% THD.
4. ALL FIXTURES IN KITCHEN OR FOOD PREP AREAS SHALL BE LENSED OR HAVE SHATTER PROOF LAMPS.
5. PROVIDE TENMAT FIRE RATED COVERS FOR ALL RECESSED FIXTURES IN FIRE RATED CEILINGS.

FACILITY SYSTEMS CONSULTANTS, LLC
713 South Central Street, Suite 101
Knoxville, Tennessee 37902
Phone: (865) 246-0164
Fax: (865) 246-1084

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

NATHAN A. BROWN
REGISTERED ENGINEER
AGRICULTURE
COMMERCE
No. 104449
STATE OF TENNESSEE
11/26/2025

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

Drawing Title:
ELECTRICAL LEGEND & NOTES

Date: 11.26.2025

Project No.
25026

Sheet No.
E0.1

ERRC SYSTEM SPECIFICATIONS:

EMERGENCY RADIO COMMUNICATION AND CELLULAR SIGNAL ENHANCEMENT SYSTEM TESTING, DESIGN, & ROUGH-IN

1. CONTRACTOR SHALL INCLUDE IN BASE BID PRICE TESTING OF EMERGENCY RESPONDER RADIO COMMUNICATIONS SIGNALS AND CELLULAR SIGNALS. TEST RESULTS SHALL BE DOCUMENTED FOR BOTH THE OUTSIDE AND INTERIOR OF THE BUILDING FOR DETERMINATION OF NEED FOR A BI-DIRECTIONAL AMPLIFIER SYSTEM TO BOOST THESE SIGNALS WITHIN THE FACILITY. PRICE SHALL ALSO INCLUDE PRE-CABLING THE BUILDING FOR ALL REQUIRED ANTENNAS AND BACKBONE TRUNK CABLES.
2. ALL TESTS SHALL BE CONDUCTED, DOCUMENTED, AND SIGNED BY A PERSON IN POSSESSION OF AN FCC GENERAL RADIO TELEPHONE OPERATORS LICENSE. ALL TESTING PERSONNEL SHALL BE CERTIFIED AND AUTHORIZED BY THE SIGNAL BOOSTER MANUFACTURER IN THE INSTALLATION AND OPERATION OF THEIR EQUIPMENT. PERSONNEL QUALIFICATIONS MUST BE ACCEPTABLE TO THE AHJ.
3. THE TESTING SHALL BE IN ACCORDANCE WITH UL2524 1ST EDITION, NFPA-72 2010 EDITION, NFPA 1221 2016 EDITION, IFC 2018 EDITION AND FCC COMPLIANT TO ESTABLISH STANDARDS OF QUALITY FOR MATERIALS AND PERFORMANCE.
4. EC SHALL SUB-CONTRACT AN APPROVED MANUFACTURER OR A QUALIFIED AND APPROVED VENDOR TO PERFORM THE TESTS AND SHOULD THE TEST RESULTS SHOW THE NEED FOR A SIGNAL ENHANCEMENT SYSTEM(S) THAT VENDOR SHALL TEST AND DETERMINE RECOMMENDED LOCATIONS OF COMPONENTS AND EQUIPMENT WHICH ARE REQUIRED FOR PROPER COVERAGE TO MEET THE AFOREMENTIONED CODE REQUIREMENTS. A PROPOSED COST FOR THE COMPLETE SYSTEM SHALL ALSO BE FURNISHED AT THAT TIME.
5. CRITICAL AREAS SUCH AS EXIT STAIRS, EXIT PASSAGEWAYS, ELEVATOR LOBBIES, SPRINKLER SECTIONAL VALVE LOCATIONS AND SIMILAR CRITICAL AREAS SHALL BE PROVIDED WITH 100% FLOOR AREA RADIO COVERAGE. GENERAL BUILDING AREAS SHALL BE PROVIDED WITH 95% RADIO COVERAGE, OR AS SPECIFIED BY AHJ.
6. THE IN-BUILDING EMERGENCY RADIO COMMUNICATION ENHANCEMENT SYSTEMS MUST PROVIDE THE FOLLOWING SIGNAL STRENGTHS:
   a. DOWNLINK - MINIMUM SIGNAL STRENGTH OF -95 DBM THROUGHOUT THE COVERAGE AREA.
   b. UPLINK - MINIMUM SIGNAL STRENGTH OF -95 DBM RECEIVED AT THE AHJ RADIO SYSTEM.
7. CONDUIT SLEEVES SHALL BE INSTALLED AS REQUIRED AND ONE 2 "C SHALL BE INCLUDED FROM DAS LOCATION TO ROOF FOR DONOR ANTENNA. WHERE INACCESSIBLE CEILINGS (GYP, ETC) ARE ENCOUNTERED, CONTRACTOR SHALL INCLUDE IN BASE BID PRICE DESIGN OF EMERGENCY RADIO RESPONDER COVERAGE SYSTEM. INDIVIDUAL UNIT ANTENNA LOCATIONS SHALL BE IDENTIFIED AND ROUGH-IN BOX INSTALLED AT EACH ANTENNA LOCATION."3/4 SHALL BE INSTALLED FROM EACH ANTENNA BACK BOX TO ACCESSIBLE LOCATION, TTB SERVING SAME AREA, OR DAS LOCATION AS REQUIRED FOR UNINHIBITED PATH OF DAS SYSTEM CABLING IF COMPLETE DAS SYSTEM IS REQUIRED UPON TESTING COMPLETION.

ADD ALTERNATE FOR INSTALLED SYSTEM (IF REQUIRED)

1. IN ORDER TO MEET INTERNATIONAL BUILDING CODE (IBC) AND INTERNATIONAL FIRE CODE (IFC) REQUIREMENTS ADOPTED BY THE LOCAL AHJ AND FIRE DEPARTMENT, THE BUILDING SHALL HAVE APPROVED RADIO COVERAGE FOR EMERGENCY RESPONDERS WITHIN 95% OF ALL AREAS OF THE BUILDING.
2. TO MEET THESE REQUIREMENTS, THE BUILDING SHALL HAVE INSTALLED A HYBRID DISTRIBUTED ANTENNA SYSTEM (DAS). THE DAS WILL BE DESIGNED TO ENHANCE THE RADIO SIGNAL COVERAGE OF THE EXISTING LOCAL 911 PUBLIC SAFETY COMMUNICATIONS SYSTEM TO MEET OR EXCEED THE RADIO SIGNAL REQUIREMENTS MANDATED IN THE IFC, SECTION 510, ET AL. CONFIRM EXISTING AHJ RADIO SYSTEM TYPE AND FREQUENCY BAND. THE DAS SHALL SUPPORT TWO-WAY COMMUNICATIONS ON ALL 911 FREQUENCIES.
3. SYSTEM DESIGN AND SOLUTION IS TO INCLUDE BI-DIRECTIONAL AMPLIFIER(S) (BDA) AND A HYBRID DAS. THE HYBRID DAS IS TO BE A BROADBAND ARRANGEMENT OF BDA BASE STATION EQUIPMENT, ANNUNCIATOR, COAX CABLE, SINGLE MODE FIBER OPTIC CABLE, OPTICAL DISTRIBUTION UNITS, REMOTE UNITS, DIRECTIONAL COUPLERS, POWER SPLITTERS, AND INDOOR AND OUTDOOR ANTENNAS TACTICALLY POSITIONED THROUGHOUT THE FACB TO ACHIEVE MINIMAL SIGNAL LOSS AND OPTIMAL SIGNAL STRENGTH. THE DAS MUST BE CAPABLE OF REMOTE MONITORING SOLUTION THAT PROVIDES ALARMING, DIAGNOSTICS AND PROVISIONING AND CONTROL OF THE SYSTEM. THE DAS MUST MEET ALL TECHNICAL REQUIREMENTS REFERENCED UL2524, IFC - SECTION 510 AND ALL SUBSECTIONS, NFPA 72, AND NFPA 1221.
4. THE VENDOR IS TO PROVIDE ALL EQUIPMENT, WIRING AND INSTALLATION TO RESULT IN A TURN-KEY SOLUTION INCLUDING, BUT NO LIMITED TO, THE FOLLOWING:
   a. PRE- AND POST-INSTALLATION TESTING - BY AN FCC-GROL LICENSED VENDOR
   b. SYSTEM DESIGN
   c. ENGINEERING SERVICES
   d. BI-DIRECTIONAL AMPLIFIER INFRASTRUCTURE
   e. INDOOR AND OUTDOOR ANTENNA SYSTEMS (INCLUDING DIRECTIONAL COUPLERS AND/OR SPLITTERS)
   f. COAXIAL CABLING AND CONNECTORS (MEETING OR EXCEEDING MANUFACTURER'S LINE LOSS SPECIFICATIONS)
   g. LIGHTNING/SURGE SUPPRESSION
   h. ANCILLARY EQUIPMENT
   i. MISCELLANEOUS HARDWARE AND INSTALLATION SUPPLIES (AS REQUIRED FOR JOB COMPLETION)
5. THE DAS, AS DESIGNED AND INSTALLED, SHALL COVER ALL FREQUENCIES UTILIZED BY LOCAL EMERGENCY SERVICES, AND SHALL NOT CAUSE INTERFERENCE TO OTHER LICENSED FCC USERS IN THE 700 TO 900 MHZ BAND. THE DAS SHALL BE CAPABLE OF MODIFICATION OR EXPANSION IN THE EVENT FREQUENCY CHANGES ARE REQUIRED BY THE FCC OR ADDITIONAL FREQUENCIES ARE MADE AVAILABLE BY THE FCC TO LOCAL EMERGENCY SERVICES.
6. APPROVAL PRIOR TO INSTALLATION: AMPLIFICATION SYSTEMS CAPABLE OF OPERATING ON FREQUENCIES LICENSED TO ANY PUBLIC SAFETY AGENCY BY THE FCC SHALL NOT BE INSTALLED WITHOUT THE PRIOR REVIEW AND APPROVAL OF THE FIRE DEPARTMENT OFFICIAL.
7. FCC COMPLIANCE: THE EMERGENCY RESPONDER RADIO COVERAGE SYSTEM INSTALLATION AND COMPONENTS SHALL ALSO COMPLY WITH ALL APPLICABLE FEDERAL REGULATIONS INCLUDING, BUT NOT LIMITED TO, FCC47 CFR, PART 90.219.
8. VENDOR SHALL PROVIDE A COMPLETE WRITTEN DESIGN OF THE PROPOSED DAS NETWORK AND SHALL INCLUDE SPECIFICATIONS SHEETS FOR ALL PROPOSED EQUIPMENT. INCLUDED WITHIN THE DESIGN WILL BE A WRITTEN DESCRIPTION OF LIGHTNING AND SURGE SUPPRESSION MEASURES.
9. INSTALLATION SERVICES SHALL INCLUDE TUNING, ALIGNMENT, AND OPTIMIZATION, AS NEEDED. INSTALLATION SERVICES SHALL ALSO INCLUDE SYSTEM TESTING & APPROVAL IN ASSOCIATION WITH THE LOCAL FIRE MARSHAL'S OFFICE.
10. THE DAS SHALL MEET THE FOLLOWING TECHNICAL REQUIREMENTS:
    a. 95% COVERAGE IN ALL AREAS OF THE BUILDING. THE BUILDING SHALL BE CONSIDERED TO HAVE ACCEPTABLE EMERGENCY RESPONDER RADIO COVERAGE WHEN SIGNAL STRENGTH MEASUREMENTS IN 95% OF ALL AREAS ON EACH FLOOR OF THE BUILDING.
    b. MINIMUM SIGNAL STRENGTH: THE DAS SHALL BE CAPABLE OF ENHANCING THE RADIO SIGNAL OF THE 911 SYSTEM SO THAT A SIGNAL STRENGTH OF -95 DBM OR GREATER IS RECEIVED WITHIN AND TRANSMITTED FROM 95% OF ALL AREAS ON EACH FLOOR OF THE BUILDING.
    c. SECONDARY POWER: EMERGENCY RESPONDER RADIO COVERAGE SYSTEMS SHALL BE PROVIDED WITH AN APPROVED SECONDARY POWER SOURCE. THE SECONDARY POWER SOURCE SHALL BE CAPABLE OF OPERATING THE EMERGENCY RESPONDER RADIO COVERAGE SYSTEM FOR A MINIMUM OF 24 HOURS. WHEN PRIMARY POWER IS LOST, THE SYSTEM SHALL AUTOMATICALLY TRANSFER TO THE SECONDARY POWER SOURCE.
    d. SIGNAL BOOSTER REQUIREMENTS:
    i. ALL SIGNAL BOOSTER COMPONENTS SHALL BE CONTAINED IN A NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION NEMA-4 TYPE WATERPROOF CABINET.
    ii. BATTERY SYSTEMS (IF APPLICABLE) USED FOR EMERGENCY POWER SOURCE SHALL BE CONTAINED IN A NEMA-4 TYPE WATERPROOF CABINET.
    iii. THE SIGNAL BOOSTER AND SECONDARY POWER SOURCE SHALL BE ELECTRICALLY SUPERVISED AND MONITORED WITH A LOCAL ANNUNCIATOR AND MONITORED FOR SUPERVISORY SIGNAL BY THE BUILDING FIRE ALARM CONTROL PANEL (FACP).
    iv. EQUIPMENT SHALL HAVE FCC CERTIFICATION PRIOR TO INSTALLATION.
    e. CELLULAR CARRIER AND EXPANSION REQUIREMENTS: ALL PATHWAYS AND PROVISIONS SHALL BE CAPABLE OF EXPANSION TO INCLUDE A FUTURE INDEPENDENT DAS FOR ALL COMMERCIAL CELLULAR BANDS. THE BACKBONE OF THE SYSTEM SHALL ALSO BE CAPABLE OF EXPANSION TO OTHER BUILDINGS AND/OR LOCATIONS ON CAMPUS.
11. THE DESIGN AND IMPLEMENTATION OF AN OPTIMALLY FUNCTIONAL DAS REQUIRES SUBSTANTIAL RADIO FREQUENCY (RF) TECHNICAL KNOWLEDGE. VENDORS MUST HAVE THE REQUIRED QUALIFICATIONS WITHIN THEIR ORGANIZATIONS IN ORDER TO BE CONSIDERED. AS FOLLOWS ARE THE MINIMUM REQUIREMENTS FOR CONSIDERATION:
    a. A VALID FCC-ISSUED GENERAL RADIO OPERATOR'S LICENSE
    b. CERTIFICATION OF IN-BUILDING SYSTEM TRAINING, ISSUED BY A NATIONALLY RECOGNIZED ORGANIZATION, SCHOOL OR A CERTIFICATE ISSUED BY THE MANUFACTURER OF THE EQUIPMENT BEING PROPOSED
    c. ELECTRONICS TECHNICIANS ASSOCIATION SENIOR LEVEL CERTIFICATION
    d. REFERENCES FROM AT LEAST TWO PREVIOUS AND STILL OPERATING DAS INSTALLATIONS OF SIMILAR SIZE AND SCOPE, WITHIN A 100-MILE RADIUS OF JOBSITE.
    e. AT LEAST 25 YEARS COMBINED EXPERIENCE IN THE RF INSTALLATION AND SERVICES INDUSTRY, CLEARLY DEMONSTRATING ADEQUATE SKILLS AND EXPERIENCE TO PERFORM REQUIRED INSTALLATION, PERFORMANCE TESTING AND SUBSEQUENT SERVICES.
    f. 24/7 ON-CALL SERVICE WITH AN ON-SITE RESPONSE TIME OF NO MORE THAN TWO HOURS FROM RECEIPT OF INITIAL SERVICE CALL.
12. UPON INSTALLATION COMPLETION, SYSTEM ACCEPTANCE TESTING SHALL BE PERFORMED IN ACCORDANCE WITH SECTION 510.5.3 (IFC). TESTING SHALL BE CONDUCTED USING A CALIBRATED PORTABLE RADIO OF THE LATEST BRAND AND MODEL USED BY THE AGENCY TALKING THROUGH THE AGENCY'S RADIO COMMUNICATION SYSTEM. A SPECTRUM ANALYZER OR OTHER SUITABLE TESTING EQUIPMENT SHALL BE UTILIZED TO ENSURE NO SPURIOUS OSCILLATIONS ARE BEING GENERATED BY THE SYSTEM'S SIGNAL BOOSTER. WRITTEN TESTING REPORTS SHALL BE SUBMITTED TO THE OWNER AND ENGINEER.

FACILITY SYSTEMS CONSULTANTS, LLC
713 South Central Street, Suite 101
Knoxville, Tennessee 37902
Phone: (865) 246-0164
Fax: (865) 246-1084

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

Drawing Title:
ELECTRICAL NOTES

Date: 11.26.2025

Project No. 25026

Sheet No. E0.2

PANEL: A1 | **VOLTAGE: 120/208** | **PHASE: 3** | **WIRE: 4** | **BUS: CU**
BUS RATING: 600 AMP | **MAIN: NO - MLO** | **ENCLOSURE: NEMA 1**
LUGS/PHASE: 2#350MCM | **ENTRY: BOTTOM** | **AIC: 22,000** | **MOUNTING: SURFACE**

#	AMP	POLE	DESCRIPTION	LOAD	A	B	C	LOAD	DESCRIPTION	POLE	AMP	#
1	30	3	AHU-2	2,784.0	7,008.0	-	-	4,224.0	AHU-5	3	45	2
3	|	|		2,784.0	-	7,008.0	-	4,224.0		|	|	4
5	|	|		2,784.0	-	-	7,008.0	4,224.0		|	|	6
7	20	2	HP-2	1,082.0	2,821.0	-	-	1,739.0	HP-5	3	30	8
9	|	|		1,082.0	-	2,821.0	-	1,739.0		|	|	10
11	30	3	AHU-3	2,784.0	-	-	4,523.0	1,739.0		|	|	12
13	|	|		2,784.0	6,944.0	-	-	4,160.0	R - RANGE	2	50GF	14
15	|	|		2,784.0	-	6,944.0	-	4,160.0		|	|	16
17	20	2	HP-3	999.0	-	-	2,331.0	1,332.0	HP-4	2	25	18
19	|	|		999.0	2,331.0	-	-	1,332.0		|	|	20
21	30	3	AHU-4	2,786.0	-	4,436.0	-	1,650.0	EH-1	2	20	22
23	|	|		2,786.0	-	-	4,436.0	1,650.0		|	|	24
25	|	|		2,786.0	2,786.0	-	-	0.0	PROVISIONAL SPACE	1		26
27			PROVISIONAL SPACE	0.0	-	0.0	-	0.0	PROVISIONAL SPACE	1		28
29			PROVISIONAL SPACE	0.0	-	-	0.0	0.0	PROVISIONAL SPACE	1		30
31			PROVISIONAL SPACE	0.0	0.0	-	-	0.0	PROVISIONAL SPACE	1		32
33			PROVISIONAL SPACE	0.0	-	0.0	-	0.0	PROVISIONAL SPACE	1		34
35	20	1	SPRINKLER MONITORING PANEL	500.0	-	-	500.0	0.0	PROVISIONAL SPACE	1		36
37	50	3	TVSS	0.0	0.0	-	-	0.0	SPARE	3	100	38
39	|	|		0.0	-	0.0	-	0.0		|	|	40
41	|	|		0.0	-	-	0.0	0.0		|	|	42

		A	B	C	
FURNISH AND INSTALL A 50kA PER MODE, 100kA PER PHASE TVSS, UL 1449, 4TH EDITION. PROVIDE 150A/3P SUB FEED BREAKER FOR PANEL "A2". PROVIDE 200A/3P SUB FEED BREAKER FOR PHP-1.	SUB TOT	21,890.0	21,209.0	18,798.0	VA
	SUB FED	26,092.0	26,205.0	25,232.0	VA
	TOTAL	47,982.0	47,414.0	44,030.0	VA
	TOTAL CONNECTED			139,426.0	VA

PROVIDE RED "LOCK-ON" DEVICE ON CKT. #35.
*GF INDICATES GROUND-FAULT TYPE BREAKER.

PANEL: A2 | **VOLTAGE: 120/208** | **PHASE: 3** | **WIRE: 4** | **BUS: CU**
BUS RATING: 225 AMP | **MAIN: NO - MLO** | **ENCLOSURE: NEMA 1**
LUGS/PHASE: #1/0 | **ENTRY: TOP** | **AIC: 22,000** | **MOUNTING: SURFACE**

#	AMP	POLE	DESCRIPTION	LOAD	A	B	C	LOAD	DESCRIPTION	POLE	AMP	#
1	20	1	R - MECH. 114 & STORAGE 115	360.0	900.0	-	-	540.0	R - STOR. 107 & ENTRY 100	1	20	2
3	20	1	R - WATER COOLER	540.0	-	1,440.0	-	900.0	R - OFFICE 105	1	20	4
5	20	1	R - HALL 106 & HIST. CENTER 110	540.0	-	-	740.0	200.0	L - HALL 106	1	20	6
7	20	1	R - RESTROOMS	540.0	780.0	-	-	240.0	L - KITCHEN 116	1	20	8
9	20	1	R - LODGE 113	540.0	-	1,065.0	-	525.0	L - RMS 115,114,112,111,118	1	20	10
11	20	1	R - LODGE 113	720.0	-	-	990.0	270.0	L - COMP. LAB 117	1	20	12
13	20	1	R - LODGE 113	720.0	1,250.0	-	-	530.0	L - RMS 101,102,103,104	1	20	14
15	20	1	R - LODGE 113	540.0	-	1,100.0	-	560.0	L - LODGE 113	1	20	16
17	20	1	R - COMPUTER LAB 117 QUADS	720.0	-	-	1,220.0	500.0	L - DECORATIVE WOOD WALL	1	20	18
19	20	1	R - COMPUTER LAB 117 QUADS	720.0	1,320.0	-	-	600.0	UNDERCOUNTER DISHWASHER	1	20GF	20
21	20	1	R - COMPUTER LAB 117	720.0	-	1,248.0	-	528.0	RECIRC PUMP	1	20	22
23	20	1	R - RECEPTION 102	900.0	-	-	3,150.0	2,250.0	WH-1	2	30	24
25	20	1	R - CLINIC LOBBY 101	360.0	2,610.0	-	-	2,250.0		|	|	26
27	20	1	R - EXAM ROOM 104	720.0	-	1,520.0	-	800.0	R - KITCHEN REFRIG.	1	20GF	28
29	20	1	R - EXAM ROOM 103	720.0	-	-	1,520.0	800.0	R - KITCHEN REFRIG.	1	20GF	30
31	20GF	1	R - MICROWAVE	1,200.0	2,000.0	-	-	800.0	R - KITCHEN UPRIGHT FREEZER	1	20GF	32
33	20GF	1	R - MICROWAVE	1,200.0	-	2,200.0	-	1,000.0	R - KITCHEN COUNTER	1	20	34
35	20	1	SPARE	0.0	-	-	500.0	500.0	R - KITCHEN COUNTER	1	20	36
37	20	1	SPARE	0.0	720.0	-	-	720.0	R - KITCHEN	1	20	38
39	20	1	L - EXTERIOR OF BLDG	400.0	-	1,120.0	-	720.0	TTB-A	1	20	40
41	20	1	PARKING POLE LIGHTS	600.0	-	-	600.0	0.0	SPARE	1	20	42

		A	B	C	
FED-THRU FROM PANELBOARD A1.	TOTAL	9,580.0	9,693.0	8,720.0	VA
	TOTAL CONNECTED			27,993.0	VA

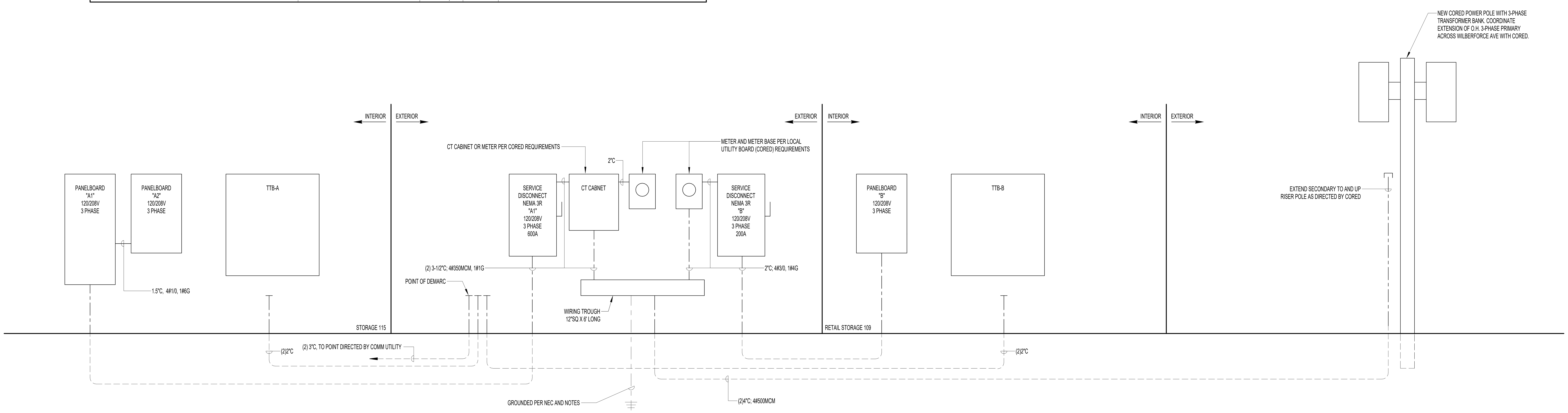
FURNISH AND INSTALL AN (6) RELAY LIGHTING CONTROL PANEL WITH INTEGRAL ASTRONOMIC TIMECLOCK. WIRE CKTS 39 & 41 THROUGH LCP.

PANEL: B | **VOLTAGE: 120/208** | **PHASE: 3** | **WIRE: 4** | **BUS: CU**
BUS RATING: 200 AMP | **MAIN: NO - MLO** | **ENCLOSURE: NEMA 1**
LUGS/PHASE: #3/0 | **ENTRY: BOTTOM** | **AIC: 22,000** | **MOUNTING: SURFACE**

#	AMP	POLE	DESCRIPTION	LOAD	A	B	C	LOAD	DESCRIPTION	POLE	AMP	#
1	45	3	AHU-1	4,227.0	4,947.0	-	-	720.0	R - RETAIL 108	1	20	2
3	|	|		4,227.0	-	4,947.0	-	720.0	R - RETAIL 108	1	20	4
5	|	|		4,227.0	-	-	4,947.0	720.0	R - RETAIL 108	1	20	6
7	30	3	HP-1	1,739.0	2,459.0	-	-	720.0	TTB	1	20	8
9	|	|		1,739.0	-	1,996.0	-	257.0	LIGHTING	1	20	10
11	|	|		1,739.0	-	-	1,739.0	0.0	SPARE	1	20	12
13	20	1	L - EXTERIOR CANOPY	400.0	400.0	-	-	0.0	SPARE	1	20	14
15	20	1	FUTURE SIGNAGE	1,000.0	-	1,000.0	-	0.0	SPARE	1	20	16
17	20	1	SPARE	0.0	-	-	0.0	0.0	SPARE	1	20	18
19	20	1	SPARE	0.0	0.0	-	-	0.0	SPARE	1	20	20
21	20	1	SPARE	0.0	-	0.0	-	0.0	SPARE	1	20	22
23	20	1	SPARE	0.0	-	-	0.0	0.0	SPARE	1	20	24
25	20	1	SPARE	0.0	0.0	-	-	0.0	SPARE	1	20	26
27	20	1	SPARE	0.0	-	0.0	-	0.0	SPARE	1	20	28
29	20	1	SPARE	0.0	-	-	0.0	0.0	SPARE	1	20	30
31	20	1	SPARE	0.0	0.0	-	-	0.0	SPARE	1	20	32
33	20	1	SPARE	0.0	-	0.0	-	0.0	SPARE	1	20	34
35	20	1	SPARE	0.0	-	-	0.0	0.0	SPARE	1	20	36
37	50	3	TVSS	0.0	0.0	-	-	0.0	SPARE	1	20	38
39	|	|		0.0	-	0.0	-	0.0	SPARE	1	20	40
41	|	|		0.0	-	-	0.0	0.0	SPARE	1	20	42

		A	B	C	
FURNISH AND INSTALL A 50kA PER MODE, 100kA PER PHASE TVSS, UL 1449, 4TH EDITION.	TOTAL	7,806.0	7,943.0	6,686.0	VA
	TOTAL CONNECTED			22,435.0	VA

FURNISH AND INSTALL A 2-POLE ASTRONOMIC TIMECLOCK TO CONTROL EXTERIOR CKTS 13 & 15.



POWER RISER DIAGRAM
NOT TO SCALE

ELECTRICAL RISER DIAGRAM NOTES

1. THIS SERVICE HAS BEEN DESIGNED FOR 120/208V, THREE PHASE, FOUR WIRE, SOLIDLY GROUNDED WYE.
2. CONTRACTOR SHALL PROVIDE GROUNDING ELECTRODE SYSTEMS AND CONDUCTORS PER NEC 250.50 AT A MINIMUM AND ADDITIONAL REQUIREMENTS AS SHOWN ON THE RISER DIAGRAM.
3. CONTRACTOR SHALL PROVIDE GROUNDING ELECTRODE CONDUCTOR(S) TO ALL METALLIC PIPING SYSTEM(S) AND BOND AS PER NEC 250.52 (A)(1).
4. CONTRACTOR SHALL PROVIDE GROUNDING ELECTRODE CONDUCTOR TO METAL BUILDING FRAME AND BOND PER NEC 250.52 (A)(2).
5. CONTRACTOR SHALL PROVIDE CONCRETE ENCASED ELECTRODE IN BUILDING FOUNDATION AND BOND PER NEC 250.52 (A)(3).
6. CONTRACTOR SHALL PROVIDE GROUNDING ELECTRODE RODS (3/4" BY 10' COPPER) AND BOND PER NEC 250.52 (A)(5)(a).
7. GROUND ALL DRY TYPE TRANSFORMERS IN ACCORDANCE WITH NEC 250.30.
8. ALL UNDERGROUND AND ENCASED CONNECTIONS SHALL BE EXOTHERMIC. EXPOSED CONNECTIONS MAY BE COMPRESSION TYPE.
9. ALL COMPONENTS SHALL BE COPPER UNLESS NOTED OTHERWISE.

FACILITY SYSTEMS CONSULTANTS, LLC

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

STATE OF TENNESSEE
No. 104449
AGRICULTURE COMMERCE
11/26/2025

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

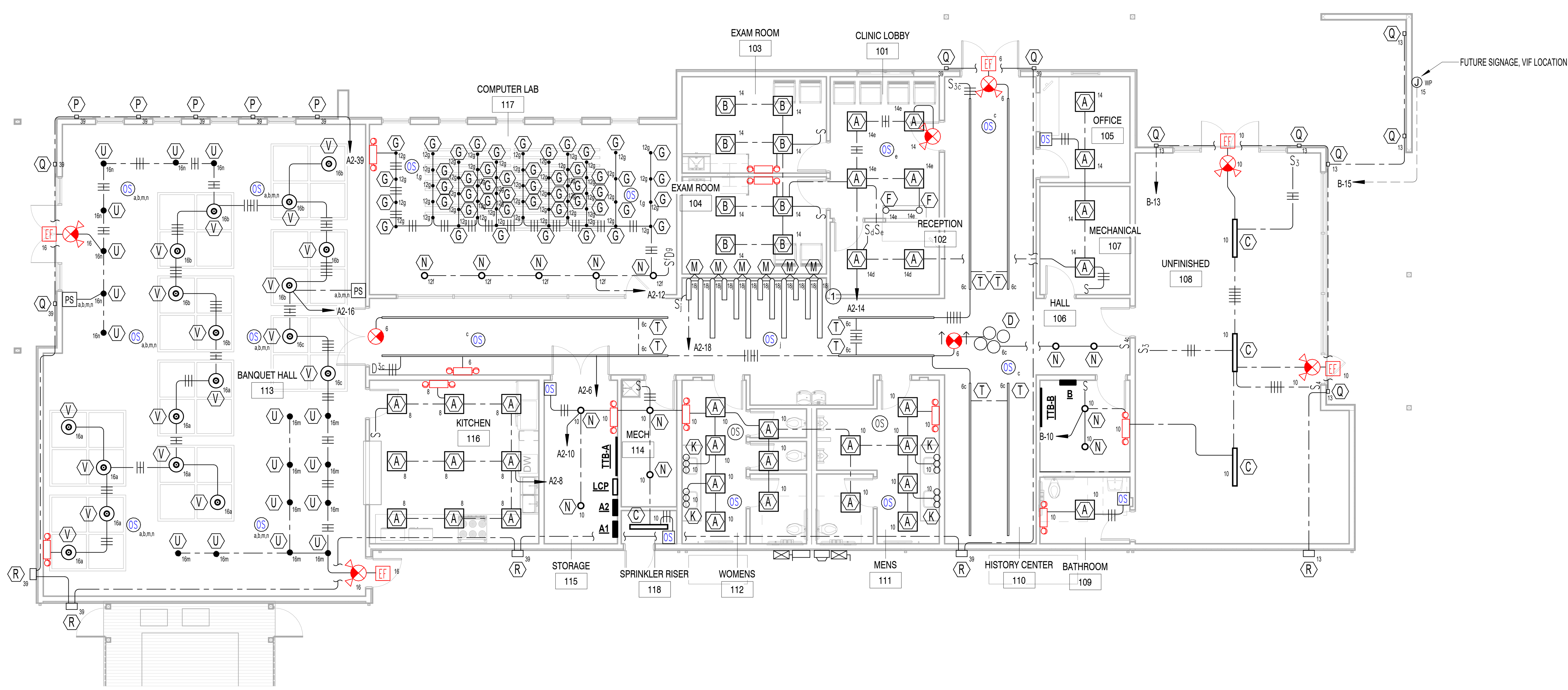
No.	Date

Drawing Title:
ELECTRICAL SCHEDULES

Date: 11.26.2025

Project No. 25026

Sheet No. E0.3



2 LIGHTING PLAN
1/8" = 1'-0"

REFERENCE NOTES:

1. PROVIDE 1 DIMMER BOX AND 1 POWER SUPPLY PER EVERY (4) "M" LIGHTS. INSTALL RECEPTACLES IN ACCESSIBLE LOCATION FOR POWER SUPPLIES. FURNISH AND INSTALL SLAT LIGHTS. SEE ARCHITECT'S RCP & ELEVATION NO. 5 ON F2.1. PROVIDE ONE REMOTE DRIVER PER (4) RUNS OF LIGHT, EACH DRIVER ABOVE CEILING WITH IN-LINE DIMMER, PRESET LEVEL PER OWNER'S DIRECTIVE IN FIELD; CONTROL ALL SEGMENTS TOGETHER. UNITS MOUNT IN BOTH CEILING AND WALL SLATS.

FACILITY SYSTEMS CONSULTANTS, LLC

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:
ELKS COMMUNITY ENRICHMENT CENTER
262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

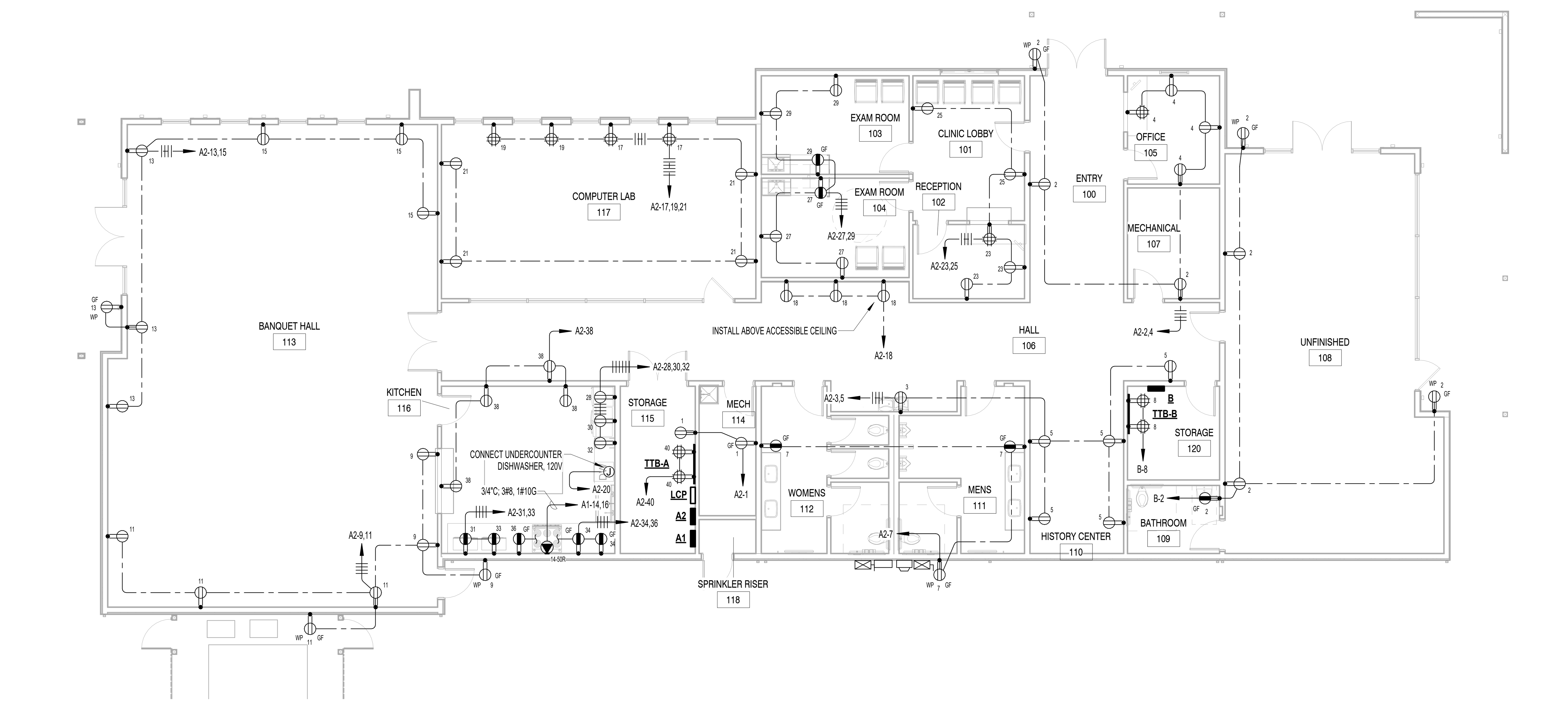
No.	Date

Drawing Title:
LIGHTING PLAN

Date: 11.26.2025

Project No. 25026

Sheet No. E1.1



1 POWER PLAN
1/8" = 1'-0"

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:

ELKS COMMUNITY ENRICHMENT CENTER

262 WILBERFORCE AVE
OAK RIDGE, TN 37830

STATE OF TENNESSEE
AGRICULTURE
COMMERCE
No. 104449
11/26/2025

THIS DRAWING IS ISSUED FOR:

CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

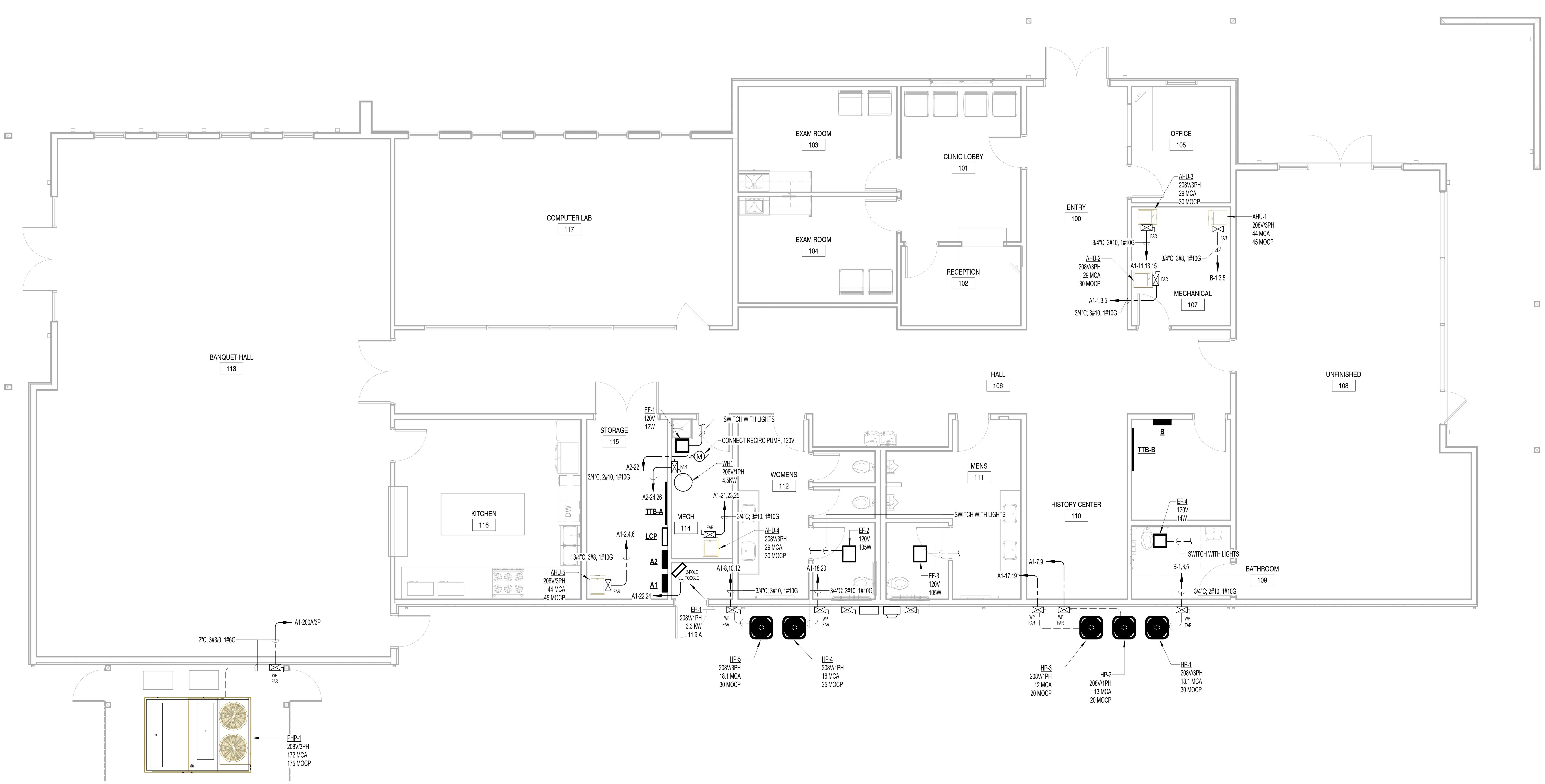
FACILITY SYSTEMS CONSULTANTS, LLC

Drawing Title:
POWER PLAN

Date: 11.26.2025

Project No.
25026

Sheet No.
E2.1



1 HVAC POWER PLAN
3/16" = 1'-0"

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:

ELKS COMMUNITY ENRICHMENT CENTER

262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

Drawing Title:
HVAC POWER PLAN

Date: 11.26.2025

Project No. 25026

Sheet No. E3.1

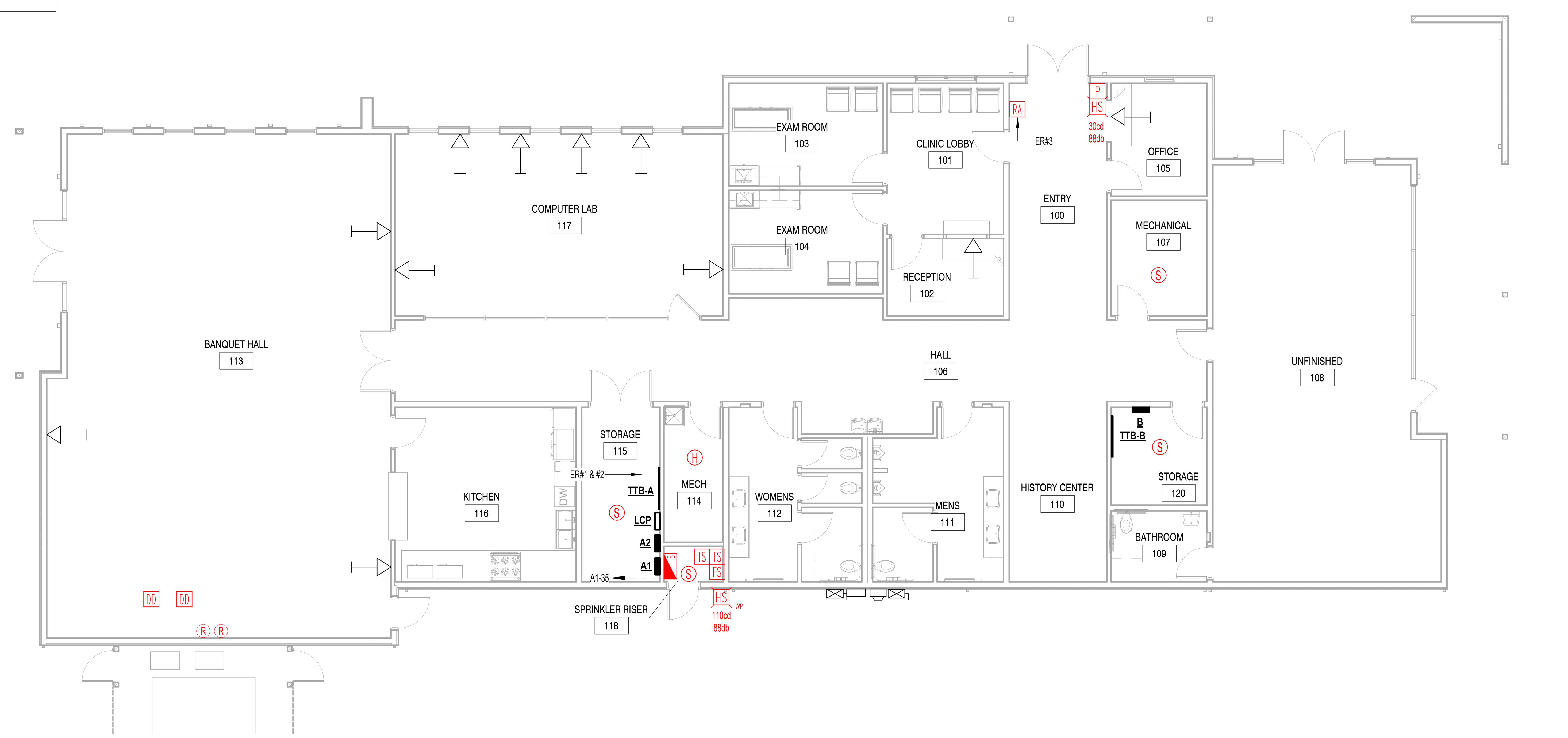
FACILITY SYSTEMS CONSULTANTS, LLC

ERRC SYSTEM GENERAL NOTES:

1. REFER TO ERRC SYSTEM REQUIREMENTS FOR PERFORMANCE CRITERIA OF SYSTEM.
2. SYSTEM SHALL CONSIST OF BI-DIRECTIONAL AMPLIFIERS (BDA) AND DISTRIBUTED ANTENNA SYSTEM (DAS) COMPONENTS AS NECESSARY TO PROVIDE PUBLIC SAFETY SIGNALS THROUGHOUT THE FACILITY BETWEEN 700 AND 900 MHZ.
3. CONFIRM ALL REQUIREMENTS WITH STATE AND LOCAL AHJ'S.
4. ALL ERRC SYSTEM BDA/DAS HEADEND AND DOWNSTREAM AMPLICATION EQUIPMENT SHALL BE INSTALLED IN COMMUNICATION/ELECTRICAL ROOMS WITHIN THE BUILDING. CABLES CAN UTILIZE EXISTING COMMUNICATIONS ROOMS AND CONDUITS FOR BACK-BONE CABLES (COAX/FIBER) BETWEEN ROOMS (SEE MAXCELL REQUIREMENT IN REFERENCE NOTES). ERRC SYSTEM EQUIPMENT SHALL NOT BE INSTALLED IN WORKING SPACE OF OTHER SYSTEMS.
5. ALL ERRC SYSTEM CABLING SHALL BE INSTALLED USING ITS OWN SUPPORTS, WITH J-HOOKS BEING THE PRIMARY SUPPORT MECHANISM. ERRC CABLING SHALL NOT BE INSTALLED IN BUILDING DATA OR AV CABLE TRAYS. HOWEVER, J-HOOKS MAY BE ATTACHED TO CABLE TRAY ALL-THREAD HANGERS AND WALLS ALONG SIDE THE CABLE TRAY.
6. PROVIDE 12O VOLT POWER TO ALL BDA/DAS EQUIPMENT AS REQUIRED, UTILIZE SPARE 20 AMPERE CIRCUITS WITHIN EXISTING PANELBOARDS.
7. ERRC SYSTEM INSTALLERS SHALL VISIT THE SITE AND OBSERVE ALL EXISTING CONDITIONS PRIOR TO SUBMITTING A PRICE TO FURNISH AND INSTALL THE SYSTEM.
8. CONTRACTOR SHALL PROVIDE (1) 2" CONDUIT TO ROOF FOR DIRECTIONAL OUTDOOR DONOR ANTENNA, CONFIRM PATH AND LOCATION WITH ERRC SYSTEM INSTALLER. VERIFY AS PART OF FIELD SURVEY PRIOR TO SUBMITTING PRICING. ROOF PENETRATION SHALL BE COORDINATED WITH ROOFING SUB TO MAINTAIN ROOF WARRANTY.
9. ERRC SYSTEM SHALL UTILIZE 2 HOUR RATED CIRCUIT INTEGRITY (CI) FOR BACKBONE COAX AND FIBER, PLENUM RATED CABLES/COAX FOR INTERIOR ANTENNA CABLES AND WET-LOCATION/SUNLIGHT RESISTANT CABLES ON EXTERIOR DONOR ANTENNA. PROVIDE NFPA 780 COMPLIANT LIGHTNING ARRESTORS/SPD DEVICES ON ALL CABLES EXITING THE BUILDING. ALL CABLES SHALL BE INSTALLED CONCEALED AT OR NEAR SAME HEIGHT AS EXISTING CABLE TRAYS.
10. DIRECTIONAL OUTDOOR DONOR ANTENNA WITH HIGH-GAIN SIGNAL ISOLATION SHALL HAVE LIGHTNING ARRESTOR BONDED TO BOTH BUILDING GROUNDING SYSTEM MAIN BUS (TMGB) AND BUILDING STEEL BY WAY OF A #4/0 COPPER CONDUCTOR INSTALLED IN CONDUIT.
11. VERIFY ALL CEILING TYPES AT DAS ANTENNA LOCATIONS IN THE FIELD PRIOR TO SUBMITTING PRICE; PLAN ROUTING AND COVERAGE ACCORDINGLY.
12. SYSTEM SHALL MEET ALL REQUIREMENTS OF CSFM, FCC, IBC/IFC, NFPA 70, 72, 101, 1221, AND UL 2524. SYSTEM SHALL HAVE 24 HOUR FULL BATTERY BACK-UP AND SHALL ANNUNCIATE SYSTEM STATUS AS WELL AS PROVIDE CONTACT CLOSURE SIGNALS FOR ALARM AND TROUBLE FUNCTIONS.
13. FURNISH AND INSTALL ALL NECESSARY ANCILLARY COMPONENTS FOR A COMPLETE SYSTEM INCLUDING INTERNAL AND EXTERNAL ANTENNAS, CONNECTORS, POWER, COAX DIVIDERS/TAPS/COUPLERS, AND LIGHTING ARRESTORS/SPD.

ERRC SYSTEM REFERENCE NOTES - ER#:

1. EMERGENCY RESPONDER RADIO COVERAGE (ERRC) SYSTEM AMPLIFIER AND/OR DISTRIBUTION EQUIPMENT MAY BE LOCATED IN THIS ROOM.
2. PROVIDE ALL SLEEVES AS MAY BE REQUIRED FOR ERRC SYSTEM CABLING; FIRE STOP AFTER COMPLETION AND TESTING OF ERRC SYSTEM.
3. INSTALL ERRC SYSTEM REMOTE ANNUNCIATOR NEXT TO FACP REMOTE ANNUNCIATOR. CONNECT FACP TO MONITOR THE SYSTEM FOR TROUBLE, POWER LOSS, ANTENNA FAILURE, CHARGER TROUBLE, AND LOW BATTERY. PROVIDE FA MONITOR MODULES AND PROGRAMMING, AS NECESSARY.



1 COMM & FIRE ALARM PLAN
1/8" = 1'-0"

TAC
The Architecture Collaborative

6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:

ELKS COMMUNITY ENRICHMENT CENTER

262 WILBERFORCE AVE
OAK RIDGE, TN 37830

THIS DRAWING IS ISSUED FOR:
CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

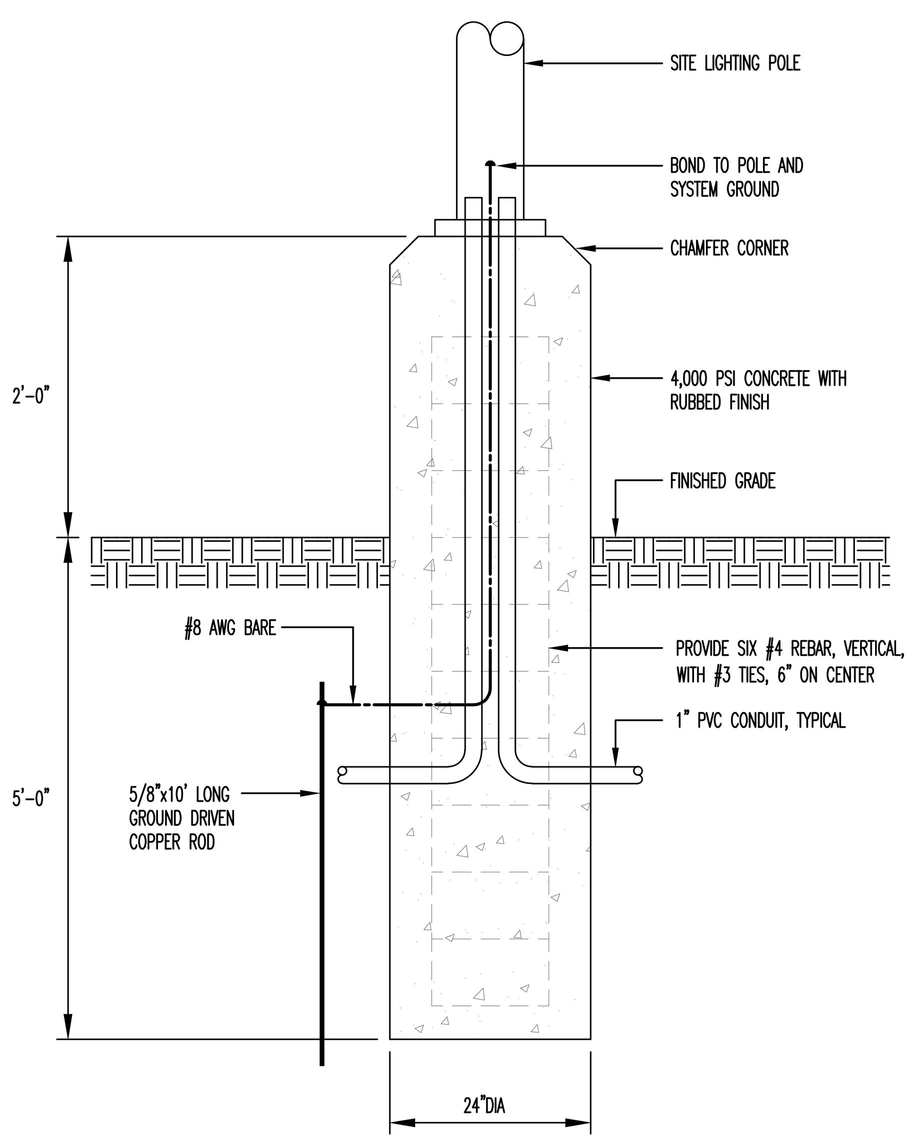
Drawing Title:
COMM & FIRE ALARM PLAN

Date: 11.26.2025

Project No. 25026

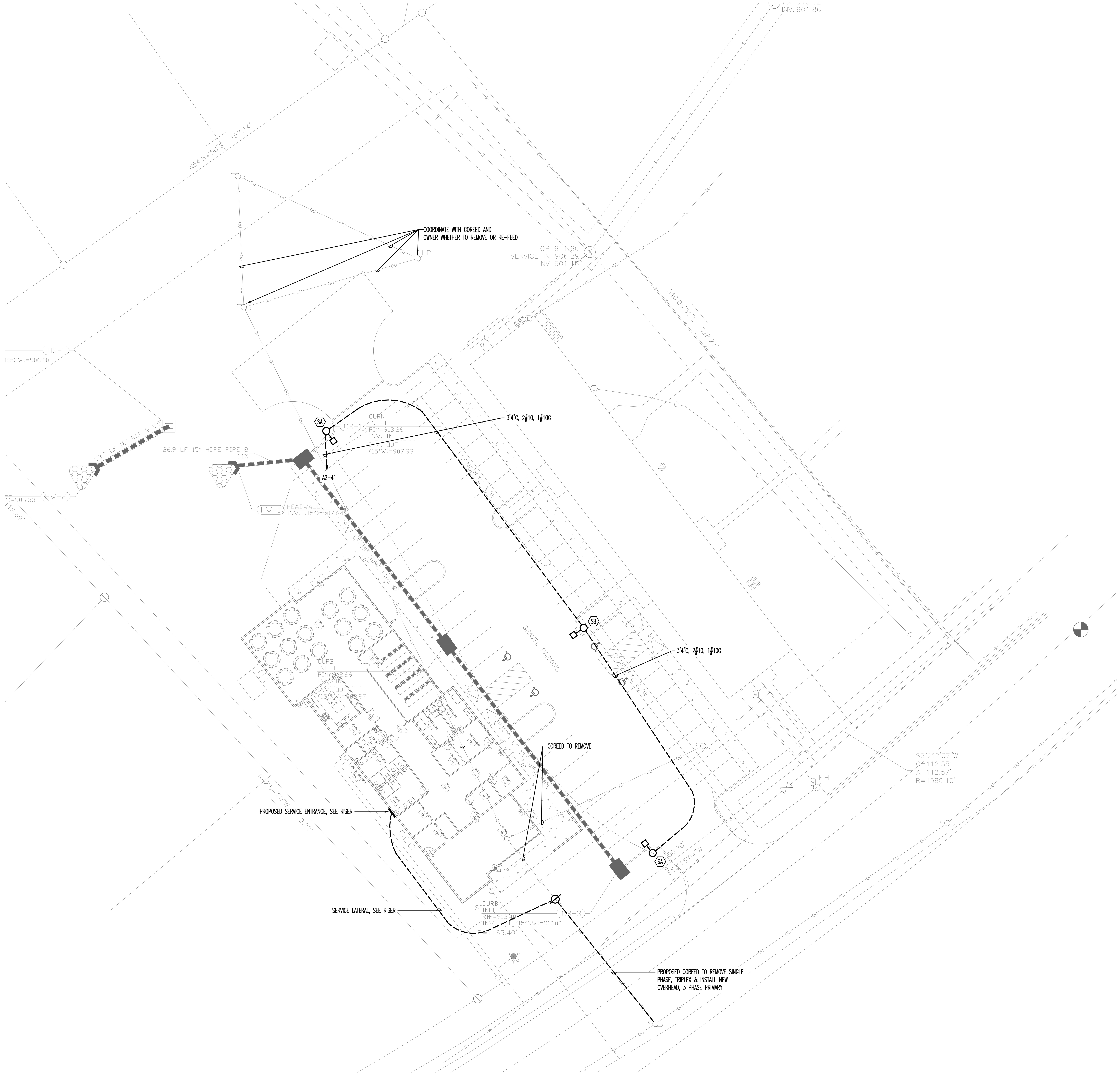
Sheet No. E4.1

FACILITY SYSTEMS CONSULTANTS, LLC



POLE BASE DETAIL

NO SCALE USE FOR SA & SB



ELECTRICAL SITE PLAN

SCALE: 1"=20'-0"

GENERAL SHEET NOTES
1. ALL SITE LIGHTING SHALL BE CERTIFIED DARK SKY COMPLIANT, FULL CUT-OFF IN ACCORDANCE WITH THE CITY OF OAK RIDGE ZONING REQUIREMENTS.

FACILITY SYSTEMS CONSULTANTS, LLC
713 South Central Street, Suite 101
Knoxville, Tennessee 37902
Phone: (865) 246-0164
Fax: (865) 246-1084

TAC
The Architecture Collaborative
6215 Kingston Pike
Knoxville, TN 37919
www.tac-45.com
865.342.7505

A NEW DEVELOPMENT FOR:

ELKS COMMUNITY ENRICHMENT CENTER

262 WILBERFORCE AVE
OAK RIDGE, TN 37830

STATE OF TENNESSEE
AGRICULTURE
COMMERCE
No. 104449

11/26/2025

THIS DRAWING IS ISSUED FOR:

CONSTRUCTION DOCUMENTS

Revisions:

No.	Date

Drawing Title:
ELECTRICAL SITE PLAN

Date: 11.26.2025

Project No. 25026

Sheet No. ES0.1